

# **INTRODUCTION TO WELL LOGGING AND INTERPRETATION**

**(PGG434)**

**Full Course Slides - Draft**

**O. Olamigoke, PhD;**

Dept. of Chemical & Petroleum Engineering

([oolamigoke@unilag.edu.ng](mailto:oolamigoke@unilag.edu.ng))

Thursdays: 8 – 10 am

# Course Description (Student Handbook)

- Driller's log
- Sample logs
- Mud logging
- Electric logging
- Radioactivity logging
- Miscellaneous logging devices
- Wire line logs
- Well logging interpretation

# Course Contents

| Content                                                                                                                                                   | Hrs | Instructor |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------|
| <b>1. Introduction</b> – History, Present and future of a science and technology of Well Logging. Fundamental problems and way forward                    | 0.5 | Olamigoke  |
| <b>2. Reservoir Rocks and Properties</b> – Reservoir rock types, Porosity, Saturation, Permeability, Core Analysis, Wettability and Capillarity           | 1.5 | Olamigoke  |
| <b>3. Overview of Well Logging</b> – Principle of log measurement, Classification of Logging methods, The tool (probe) and the environment of measurement | 2   | Olamigoke  |
| <b>4. Well Logging Methods</b> – Electrical and electromagnetic logs, Nuclear logs, Acousticlog, Sonic log, NMR measurements Imaging Techniques           | 2   | Olamigoke  |
| <b>5. Introduction to Well Log Interpretation</b> – Overview, Quick Look Methods, Quantitative Interpretation: Shale Content                              | 6   | Olamigoke  |
| <b>6. Some Applications in Water Exploration</b> – Water well in an unconsolidated formation, Mineral water well in a fractured carbonate formation       | 1   | Olamigoke  |
| <b>7. Examples and Exercises</b> – Oil-bearing Sandstone, Shaly sand profile, Mixed lithology Carbonate rock                                              | 2   | Olamigoke  |
| <b>8. Field Management &amp; Data Acquisition Planning and Execution</b> – Logging Programs, Operational Challenges, Value of Information, Miscellaneous  | 4   | Olamigoke  |
| <b>9. Revision and Contingency</b>                                                                                                                        | 4   | Olamigoke  |

# Learning Outcomes

- By the end of this course students should be able to:
  - describe relevant rock properties to this course
  - explain in basic terms how petrophysical data is obtained from well logs
  - interpret basic wireline logs
  - quantitatively evaluate basic petrophysical properties from well logs.



# Required Background

- **PGG 322 – Formation Evaluation and Applied Geophysical Methods**
- PGG 313 – Basic Petroleum Reservoir Engineering
- PGG 308 – Drilling Methods
- PGG 310 – Petroleum Engineering Lab. II
- Knowledge from other courses might come in handy

# References

- Prof. Dr. Jurgen Schon, *Basic Well Logging and Formation Evaluation*, 2015  
bookboon.com
- Baker Atlas. *Introduction to Wireline Log Analysis*, 2002
- Toby Darling, *Well Logging and Formation Evaluation*, 2005 Elsevier Inc
- Krygowski DA, Texas A, *Guide to Petrophysical Interpretation*, 2003

# Grading System

| <u>Components</u>               | <u>Marks</u>       |
|---------------------------------|--------------------|
| Mid Semester Test(s)            | 20                 |
| Group Presentation / Term Paper | 20                 |
| Final Examination               | <u>60</u>          |
|                                 | Total = <u>100</u> |

# Ground Rules

- Keep your phones on vibration or silent mode
- You are free to note typographical errors on the slides or and inconsistencies. **Please take note of the slide number and notify me after the class.**
- Some slides appear to be repetition of earlier slides or material from PGG322. **IT IS FOR EMPHASIS.**
- If you have a question, raise your hand. Note that all question can't be entertained in class.
- The number of attendees at each class would be taken. This would be used to confirm the day's attendance list.

# **Introduction – Lecture 1**

# History, Present and Future of the Science & Technology of Well Logging

- The genesis of electric well logging resides with Conrad Schlumberger, who while a physics professor at the Ecole de Mines de Paris, France, conceived the idea of prospecting for metal ore deposits by using their electrical conductivity to distinguish them from their less conductive surroundings.
- One of the first tests, according to Schlumberger historians, was performed in his bathtub, which was filled with various rocks for the experiment.
- Working with his brother Marcel Schlumberger, Conrad began a series of test surface surveys in Europe, Africa and North America over a 3-year period. Their discoveries included an oil-productive salt dome in Romania, a precursor of things to come.

# History, Present and Future of the Science & Technology of Well Logging

- In 1926, the brothers formed Societe de Prospection Electrique and began to develop the theory that adding resistivity information from deeper formations would increase the effectiveness of their surface prospecting.
- By lowering an electric sonde down a 1,600-ft (488-m) well in France's **Pechelbronn** field Sept. 5, 1927, the brothers created the first well log. This log was painstakingly recorded point by point, meter by meter, using makeshift equipment and then plotted by stitching together the successive readings.
- The technology worked simply. Three electrodes – A, M and N – are lowered to the bottom of the wellbore on three insulated wires. Current from electrode A passes through the drilling mud and spreads out into the formation.

# History, Present and Future of the Science & Technology of Well Logging

- The potentials measured at M and N are transmitted to the surface where they are measured. By measuring the potential difference between M and N, and the strength of current from A, the apparent formation resistivity is calculated.
- The fundamental design of this new technology called “electrical coring” or “electrolog” is still valid today.
  - Create a continuous plot of a measured property (e.g. resistivity) as a function of the (measured) depth and call it “the log”
  - With the advantage of representing continuous information this log must be transformed into information for reservoir characterization (porosity, saturation) by a process called interpretation
-



# History, Present and Future of the Science & Technology of Well Logging

- For quantitative measurement petrophysical knowledge is necessary. The first step was done Archie's famous equation in 1942 describing the correlation between specific electrical resistivities (measured properties), porosity, water saturation (derived properties) and empirical parameters:

$$S_w = \left( \frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left( \frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

- Where  $S_w$  is water saturation,  $R_t$  is the specific resistivity of the formation,  $R_o$  is the specific resistivity of the formation at 100% water saturation,  $n$  is Archie's empirical saturation coefficient and  $m$  is the cementation exponent.

# History, Present and Future of the Science & Technology of Well Logging

- A formation or reservoir characterization cannot be derived from one type of logs alone – it needs a combination of various physical parameters in order to derive a consistent model of the formation.
- The historical development of borehole geophysics is therefore characterized by the development of various systems with defined sensitivity. Cornerstones of the first period are:

**Table 1: Early Logs Developed**

| Log Type                  | Parameter of Interest                                |
|---------------------------|------------------------------------------------------|
| Resistivity Log           | Water Saturation determination from clean rocks      |
| Spontaneous Potential Log | Separation of sand (clean rock, reservoir) and shale |
| Acoustic Log              | The “Porosity log” based on Wyllie’s equation        |

# Evolution of Borehole Geophysics Tools

**Table 2: Logs after the advent of Neutron Measurements**

| Log Type                             | Parameter of Interest                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural GammaLog                     | Estimation of Shale Content                                                                                                                          |
| Gamma Gamma Density<br>(Density Log) | More accurate determination of porosity<br>and mineral composition calculation<br>using sophisticated combined<br>techniques (with the Acoustic Log) |
| Neutron Log                          |                                                                                                                                                      |

- All tools were developed for the elimination of caliper effects, better resolution and increased depth of investigation

# Evolution of Borehole Geophysics Tools

- Focusing tools (laterologs) and Dual-System tools (Density, Neutron) are important levels of development and have been refined – based on the possibilities of digital processing – to different array systems.
- At the same time new techniques (sensor, data transmission) allow spectral measurements mainly for the nuclear techniques and for full-waveform-registration of acoustic systems with digital techniques.
- With the Nuclear Magnetic Resonance (NMR) technique a major step forward was done in order to derive pore space properties with respect to a permeability estimate and to realize fluid characterization. Permeability is derived using the Coates Equation

# Evolution of Borehole Geophysics Tools

- Permeability is derived using the Coates Equation (Coates and Dumanoir 1974; Coates et. al. 1999):

$$k_{Coates} = \left( \frac{\phi}{C} \right)^m \cdot \left( \frac{BVM}{BVI} \right)^n$$

where

$\phi$  is porosity in percent,

$BVM$  is bulk volume fluid movable,

$BVI$  is bulk volume fluid non-movable (irreducible),

$m, n$  are empirical parameters and approximately equal to 4 and 2,

$C$  is also an empirical parameter generally between 6 and 15.

# Early Quantitative Models

- Coates equation reflects the philosophy of Archie's equation with a simple, physically understandable structure (based on a capillary model) and empirical parameters covering the complicated geometry and structure of the pore space in a fascinating way.
- It implies that laboratory core data is needed to derive empirical parameters for a specific formation i.e. tools need to be calibrated with laboratory core data.
- In the 1980s to 1990s, imaging tools were developed: Resistivity Imaging Tools, Induction Imaging tools,

# Early Quantitative Models

- The specific environment related to the development of well logging and interpretation tools is: The tool is surrounded by mud in an (open) well typically referred to as the “wireline technique”. Logs are also obtained in “Cased-hole” environments – for control of completion and quality (e.g. Cement Bond Logs) and monitoring fluid saturation (Cased hole saturation Logging)
- The modern techniques of Measurement While Drilling (MWD) and Logging While Drilling (LWD) reflect the dream to “see” in real-time what is penetrated with the bit; such a measurement is important for an optimized and safe drilling operation also in deviated and horizontal wells.

# History of Logging Tools & Interpretation

- **1920s:** First Electrical log recorded in 1927 by Schlumberger brothers – Marcel and Conrad in France using stationed resistivity method. The first commercial electrical resistivity tool in 1929 was used in Venezuela, USA and Indonesia
- **1930s:** SP, Short Normal, Long Normal and Long Lateral combined, Core Sample Taker. Schlumberger developed the first continuous recording in 1931.
- **1940s:** Gamma ray and Neutron, 3-arm Dipmeter using SP, then electrical measurements, Induction tool.
- **1950s to 1960s:** Microlog, Laterolog, Sonic log, Formation Tester, Formation Density tool
- **1970s:** Dual Spacing Neutrons, Advanced Dipmeter, Computerized Computer Systems, Repeat Tester Tools, Electromagnetic Propagation tool
- **1980s:** Resistivity Imaging Tools, Advanced Sonic Tools
- **1990s:** Advanced Formation Testing Tools, Induction Imaging tools, Azimuthal Laterolog Tools, Ultrasonic Imaging tools, Epithermal Porosity tools, Magnetic Resonance Tools.



# Fundamental problems & the way we go

- Borehole geophysical measurement are an important groups of methods for the solution of fundamental problems in
  - a) Hydrocarbon exploration and production
  - b) Exploration of water and geothermal resources
  - c) Mineral exploration
  - d) Geotechnical investigations
  - e) And a variety of general problems in earth science

# Evolution of Borehole Geophysics Tools

- All tools were developed for the elimination of caliper effects, better resolution and increased depth of investigation

**Table 2: Logs after the advent of Neutron Measurements**

| Log Type                             | Parameter of Interest                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Natural GammaLog                     | Estimation of Shale Content                                                                                                                 |
| Gamma Gamma Density<br>(Density Log) | More accurate determination of porosity and mineral composition calculation using sophisticated combined techniques (with the Acoustic Log) |
| Neutron Log                          |                                                                                                                                             |
| Nuclear Magnetic Resonance (NMR)     | Derive pore space properties with respect to a permeability estimate; realize fluid characterization                                        |

# Well Log: Definition

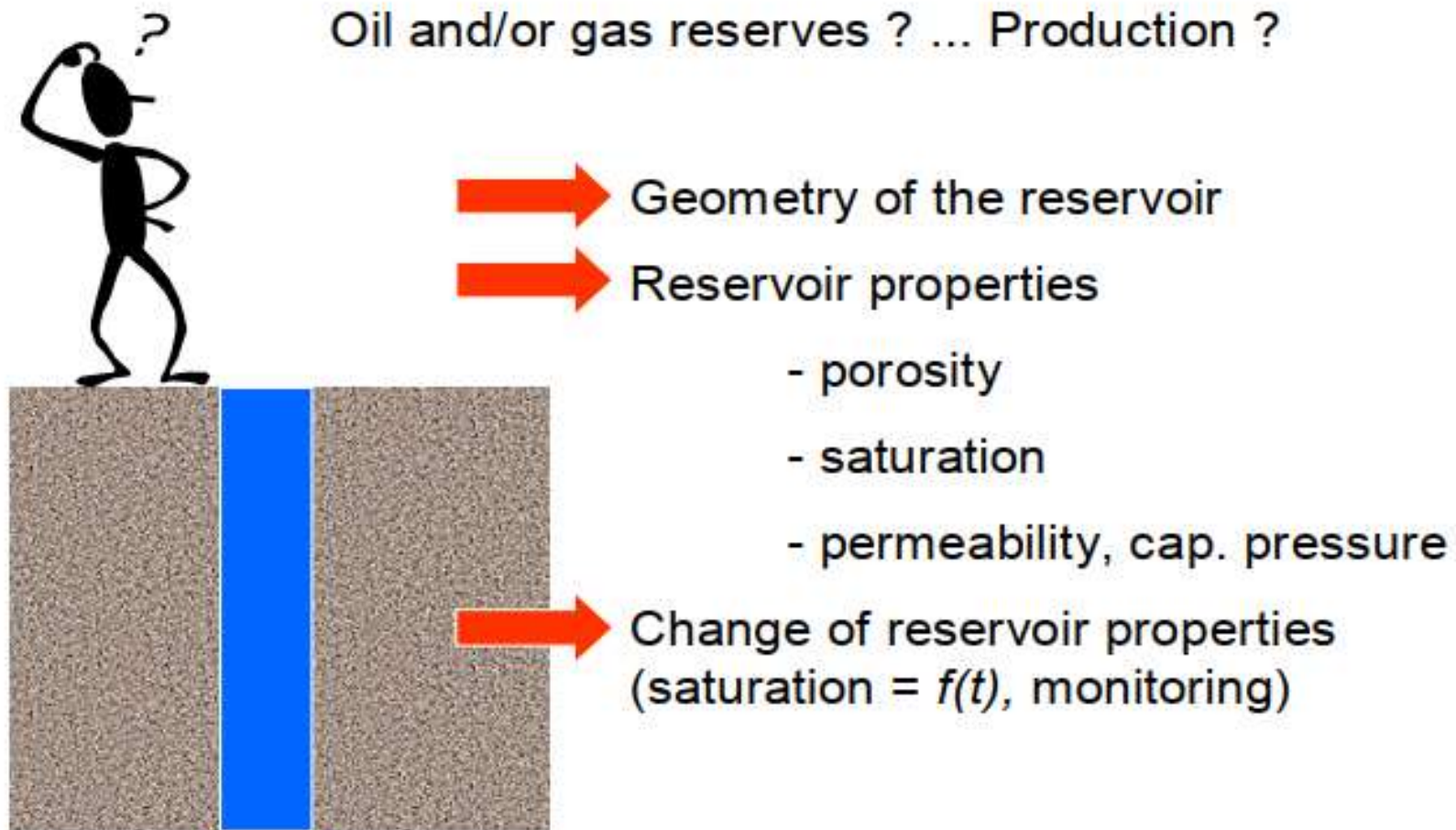
- A well log is a continuous record of downhole data organized and interpreted by depth and represented on a graph plotted to scale.
- A continuous record as a function of depth, usually graphic and plotted to scale on a narrow paper strip, of observations made on the rocks and fluids of the geologic section exposed in the well-bore (*4th Edition of J. A. Jackson's Glossary of Geology*).

# How are Well Logs measurements taken?

- Logs are taken in the Open Hole (while drilling or after the hole section has been drilled) or in the Cased Hole.
- Logs can be obtained while drilling by running tools downhole (MWD) with the drilling assembly (complimented by Surface Logging Systems).
- Well Logging tools run downhole after the hole section has been drilled are either (i) drillpipe-conveyed; (ii) wireline-conveyed (iii) Coiled tubing – conveyed (iv) via well tractors; depending on the well trajectory,

# Why are Well Logs Important?

- The drilled well helps to confirm the following:



# General Purpose of Well Logs

## Measurements

- Lithology with exact depth of formations / rock boundaries
- Rock properties and rock composition (mineralogy)
- Fractures and tectonic elements
- Mechanical rock properties for stability, sanding problems, frac operations
- Indications or content and properties of other substances than hydrocarbons (water, thermal water, ores, coal, salt, etc)
- Seismic-relevant parameters for integrated interpretation of seismic and log measurements
- Changes of properties; this is important particularly with respect to change in fluid content / saturation during production (monitoring, time lapse measurements)

# Some Questions Addressed By Log Interpretation

## • Geophysicist / Geologist

- Are the tops as predicted?
- Are potential zones porous?
- Formation intervals?
- Lithology? Hydrocarbons?
- What type of hydrocarbons?
- **Commercial quantities?**

## • Drilling Engineer

- Hole volume for cementing?
- Are there any Key-Seats or severe Dog-legs in the well?
- Where can you get a good packer seat for testing?
- Where is the best place to set a Whipstock?

## • Reservoir Engineer

- How thick is the pay zone?
- How homogeneous is the zone?
- Will the well pay-out?
- **How long will it take?**

## • Production Engineer

- Which zone(s) to complete?
- What production rates?
- Any water production?
- Is zone hydraulically isolated?
- Will well need stimulation?
- **What stimulation would be best?**

# Property Measurement in the Well: Implications

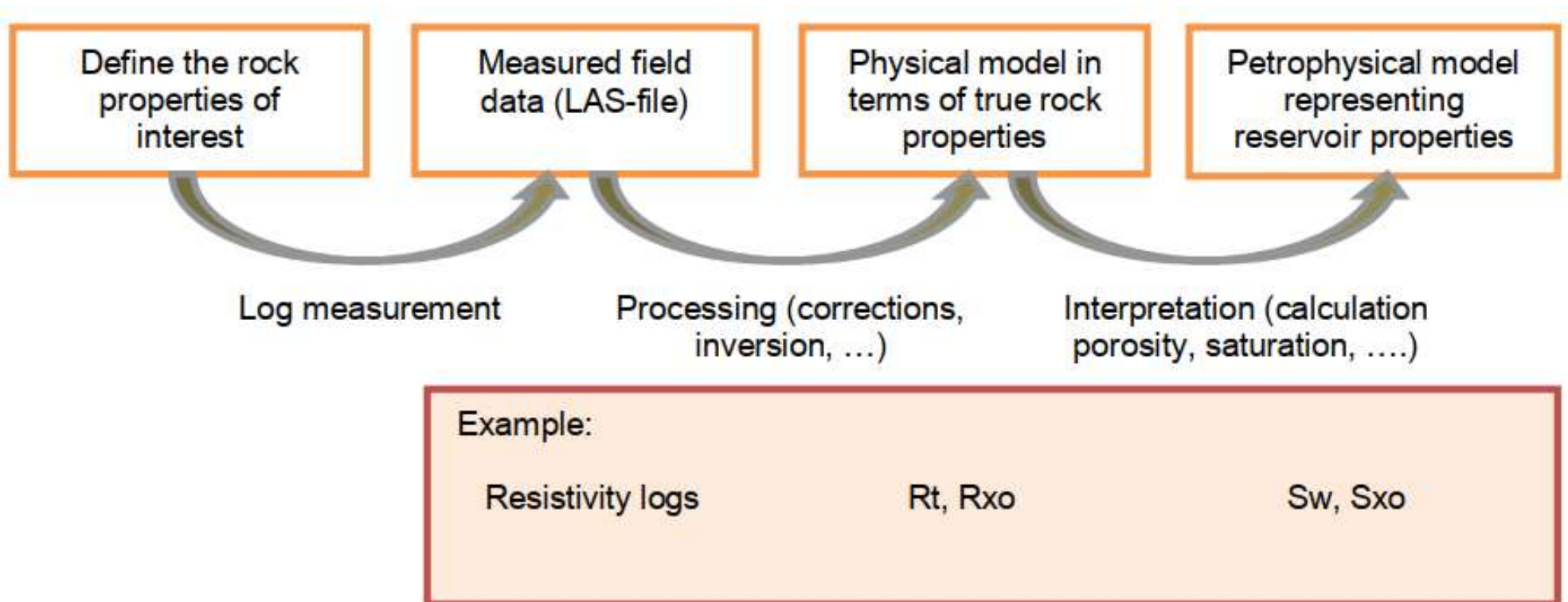
- Well logging involves taking measurements along the borehole in terms of physical properties which is presented as a continuous documentation of the whole well profile and a physical characterization of the individual layers and sections in terms of properties like resistivity, nuclear radiation etc
- Sophisticated interpretation methods transform the measured data into reservoir properties and other properties of interest. There are two types of properties:
- Reservoir properties – properties of primary interest (porosity, saturation, permeability)
- Log-measured properties – properties directly measured by well logging tools (resistivity, acoustic travel times, nuclear cross section, natural gamma radiation)



# Transforming Property Measurement in the Well into Reservoir Properties

How can we derive equations & algorithms to transform measured properties to reservoir properties?

- Empirically using experiments.
- Theoretically using models.
- Combination of Theoretical and Empirical results



# The Process of Borehole Measurements

- 1) Define the parameters of interest for the lithological type of the section. Design your logging program based on sensitivity of the individual tools.
- 2) Take the corresponding log measurements. The result is a dataset for each method  $i$ . With regard to the resistivity measurement, this could be Microlog, Laterolog-shallow, and Laterolog-deep
- 3) Derive a physical model in terms of the measured physical property  $p_i$  using processing algorithms. Regarding resistivity measurement, this results in  $R_{xo}$  and  $R_t$ . This step called “processing” has strong reference to the tool characteristics
- 4) Transform measured data into reservoir properties using interpretation methods.

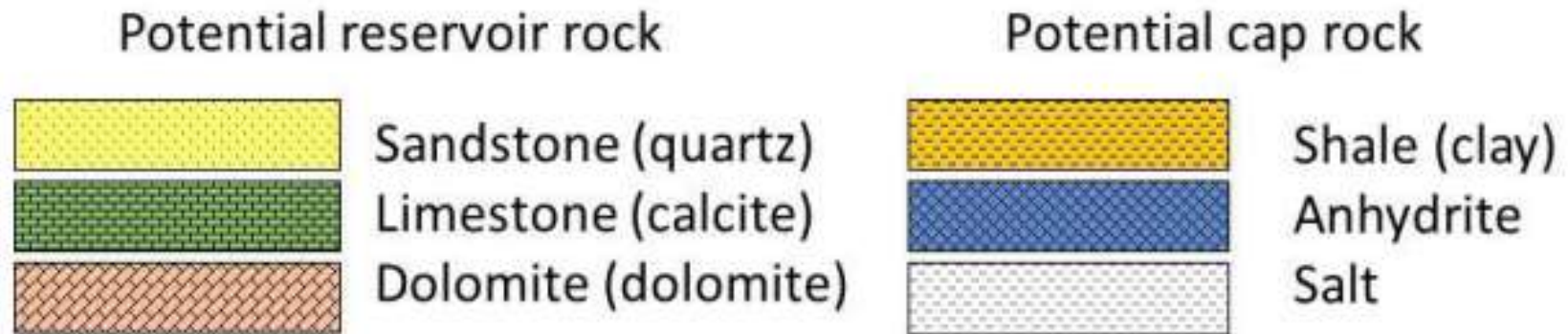
# **Properties for Reservoir Characterization – Lecture 2**

# Basic Reservoir Rock-Fluid Properties

- Fluids (oil, gas but also water) are accumulated in the pore space of the reservoir rock. Therefore the fundamental reservoir rock properties are:
  - **Porosity:** How much space is available in the rock?
  - **Fluid Type:** Is it oil, gas or water?
  - **Saturations:** Which volume fraction of the pore space is occupied by oil, gas and water?
  - **Permeability:** At what rate can I produce or inject a fluid?
- Reservoir Rocks can be classified into two major types:
  - **Clastic Rocks:** (Sandstone, siltstone, claystone, and shale)
  - **Carbonate Rocks:** (Limestone, Dolomite)
- These two types have different pore properties, abundances, & importance for the world's hydrocarbon production

# Basic Lithologies considered in Interpretation

- For a reservoir, the “protecting or sealing formation” (caprock) is also of critical importance. Thus the main subject of interpretation are the lithologies present below :



# The Influence of Clay

- Clay has strong influence on all rock properties; it decreases the effective pore spaces and reduces permeability. Let's avoid confusion between “clay” and “shale”. Please note:
  - Clay is defined as a particle size ( $< 0.002$  mm),
  - Clay minerals are a group of phyllosilicates with specific properties (Cation Exchange Capacity CEC),
  - Shale is a rock type (high amount of clay minerals, but also fine grained Felspars and Quartz). The clay fraction in shale makes up about 40 – 90 %. In many shale physical properties the properties of clay minerals are reflected.

# The Influence of Clay

- For the effect of clay in a reservoir rock, not only the volume fraction (clay content) and the clay mineralogy, but also the clay distribution is important.
  - Laminar clay: thin clay layers alternating with sand,
  - Dispersed clay: clay in the pores (also authigenic clay),
  - Structural clay: clay forms grains and is a rock-building element.



**dispersed**

Formed within the sandstone framework when clay crystals precipitate from pore fluids as result of changing chemistry, pressure, and temperature



**laminated**

Detrital origin, i.e. are formed outside the sandstone framework



**structural**

Diagenetic origin, i.e. are formed within the sandstone framework; deposit of clay clasts

# Porosity - Definitions

- **Porosity** is defined as the ratio of the pore volume to the bulk volume of a substance. In oil and gas reservoirs, the pore volume is the space available for the storage of the hydrocarbons and water. Porosity is normally expressed as a percentage of bulk volume and is symbolised by  $\phi$ .

$$\text{Porosity, } \phi = \frac{\text{Pore Volume}}{\text{Bulk Volume}} \times 100$$

$$\text{Porosity, } \phi = \frac{\text{Bulk Volume} - \text{Grain Volume}}{\text{Bulk Volume}} \times 100$$

$$\text{Porosity, } \phi = \frac{\text{Pore Volume}}{\text{Pore Volume} + \text{Grain Volume}} \times 100$$

- **Total Porosity:** is defined as the ratio of the volume of all the pores to the bulk volume of a material, regardless of whether or not all of the pores are interconnected.
- **Effective Porosity:** is defined as the ratio of the interconnected pore volume to the bulk volume of a material, i.e. it does not include dead-end pore-space.
- **Water of Hydration:** The water of hydration of crystallisation of the constituent minerals of a reservoir rock is defined as a portion of the grain volume. It is not a portion of the pore volume. This can pose problems when comparing certain log porosities with core-derived data and in the derivation of fluid saturations on these rocks.



# Porosity - Texture

- Porosity in sands and sandstones varies primarily with grain size distribution, grain shape, packing arrangement, cementation, and clay content. These parameters control the overall pore geometry as well as the porosity value. The porosity of typical hydrocarbon productive sandstones ranges between 3 and 38 percent in gas reservoirs and 10 to 38 percent in oil reservoirs.
- Porosity in carbonate rocks can be much more variable in magnitude than it is in sandstones. In some carbonates, such as reef build-ups and chalks, it is very high, in a few cases exceeding 50 percent. However, the fractures commonly encountered in carbonate rocks contribute little to the porosity.
- The development of vugs and fractures as found in carbonate reservoir rocks is termed **Secondary Porosity** and is a function of the depositional history and diagenesis of the rocks. Often, carbonate reservoir rock's porosity can be correlated with the degree of dolomitisation, as the dolomitisation of limestone can generate up to 12% additional porosity due to shrinkage of the crystalline lattice. While vugular porosity can be large, caverns of some tens of metres size having been encountered in some rare cases, fractures, which make up the other major component of secondary porosity tend to be of a smaller aperture.

# Fluid Saturations

- Where porosity is the capacity to hold fluids, saturation is the percentage or fraction of this total capacity that actually holds any particular fluid. Porosity, hydrocarbon saturation, the thickness of the reservoir rock and the areal extent of the reservoir rock all contribute to the total hydrocarbons in place. These establish the economic potential of any given reservoir.
- Saturation of any given fluid in a pore space is the ratio of the volume of the fluid to the total pore space volume. For example, a water saturation of 10% means that 1/10 of the pore space is filled with water.
- The saturation  $S_i$  describes the volume fraction of a fluid  $i$  in a porous rock as:  $S_i = (\text{volume of fluid } i) / (\text{volume of pores})$
- The saturation of the fluids in a porous rock must sum up to 1:
$$S_{\text{gas}} + S_{\text{water}} + S_{\text{oil}} = 1$$
- In addition the parameter, “bulk volume of the fluid” is used which is the product of the porosity and the fluid saturation.

$$BVW = \frac{\text{volume of water}}{\text{rock volume}} = \phi \cdot S_w$$

# Water Saturation

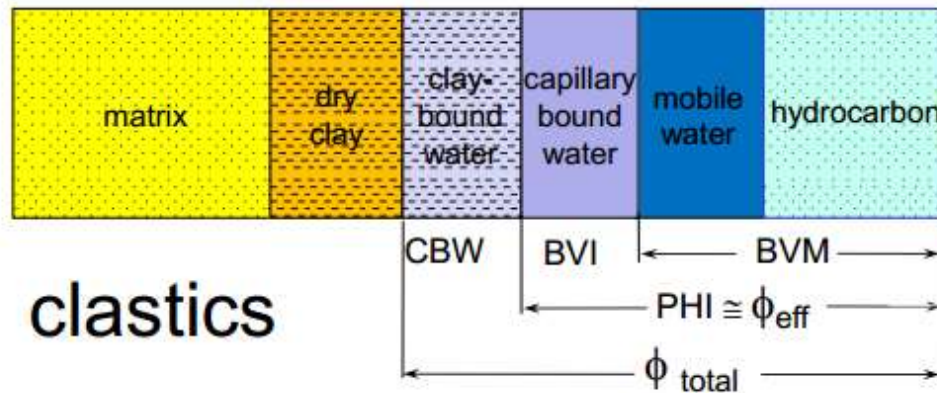
- In formation evaluation we are most often concerned with the water saturation ( $S_w$ ) as this is the phase that carries electrical current from the logging tools. By necessity,  $1-S_w$  is the hydrocarbon saturation of the pore-space. The  $S_w$  commonly derived is the portion of the porosity that is water and the Bulk Water Content is the product of porosity,  $\phi$  and  $S_w$ .
- The  $S_w$  of the rock is dependent upon the balance between the gravity forces and the adhesive forces of the water–rock system. The water saturation of a reservoir rock is therefore a function of Capillary Pressure ( $P_c$ ), which in turn is controlled by pore-geometry, wettability and the height of the hydrocarbon column.
- A major goal of Formation Evaluation is to define and use reservoir saturation - height relationships. These are determined through developing relationships between porosity, permeability, lithology and saturation, as a function of height above Free Water Level. Most water saturations are determined from the formation's electrical properties.

**Note:** When averaging saturation data, the bulk water should be summed and divided by the sum of the porosity thickness products.

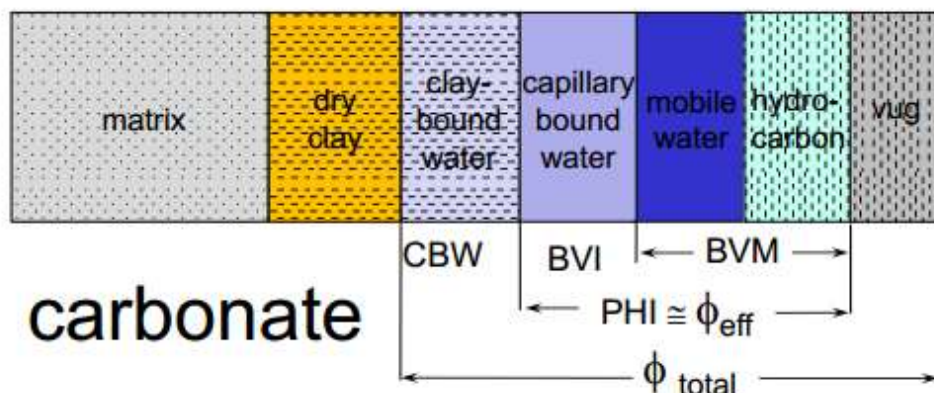
$$\overline{S_w} = \frac{\sum S_w \cdot \phi \cdot h}{\sum \phi \cdot h}$$

# Water Saturation

- In a water-wet porous rock, the water, depending on its interaction with minerals and bonding type is presented as:
  - Free movable water in the pore space (Bulk Volume Movable BVM)
  - Capillary bound water, connected with the grain surface [Bulk Volume Immovable (BVI)]
  - Clay bound water (CBW), with its strong clay-water effects
- These water types have different physical properties and effects



These water types have different physical properties & effects (e.g. with respect to permeability or electrical resistivity)



Volumetric description of a hydrocarbon-bearing clastic and carbonate reservoir

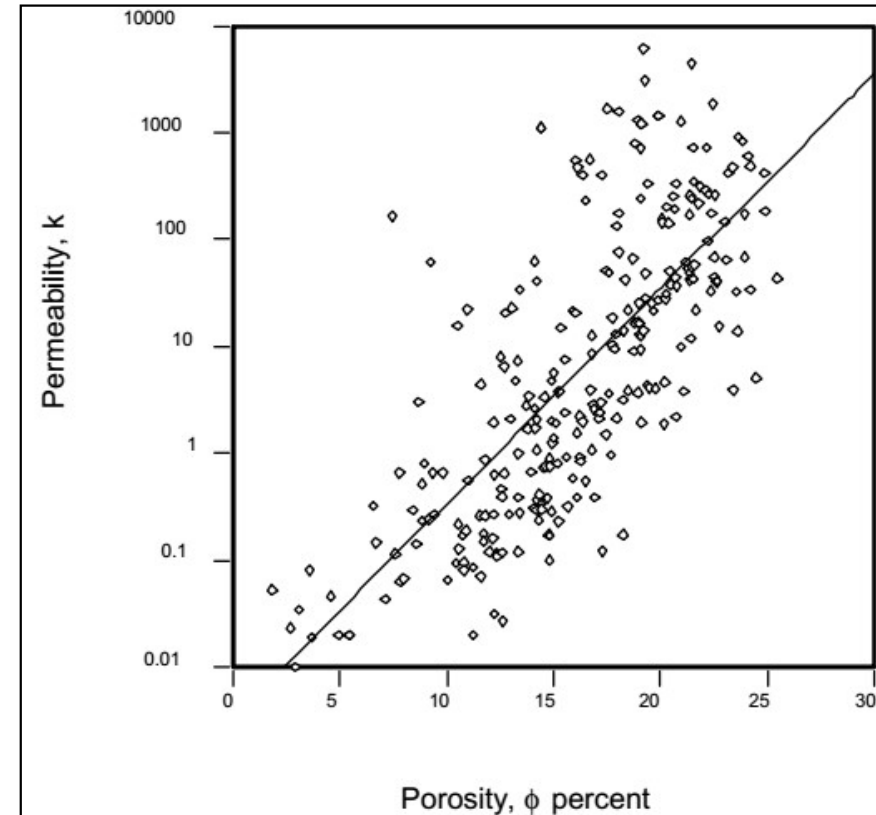
# Permeability - Definition

- **Permeability**

- Characterises the ability of the rock to transmit fluid; it connects the fluid flow rate with the applied gradient and the fluid viscosity.
  - Is controlled by the connected passages of the pore space (pore throats),
  - Is a tensorial property and exhibits, in many cases, an anisotropy (mostly expressed by horizontal ( $k_h$ ) and vertical permeability ( $k_v$ ))
- Permeability is a pore space property that is mainly controlled by: which is a constant when the following boundary conditions are met:
    - 1) Interconnected Porosity (clay decreases permeability by several orders of magnitude)
    - 2) Pore size and fracture width

# Permeability and Porosity Relationships

- The Geological environment and depositional factors influencing porosity also influence permeability, and often there can be an obvious relationship between the two.
- The relationship varies with formation and rock type, and reflects the variety of pore geometry present.
- Typically, in sandstone reservoirs, increased permeability is accompanied by increased porosity.
- Constant permeability accompanied by increased porosity indicates the presence of more numerous but smaller pores.

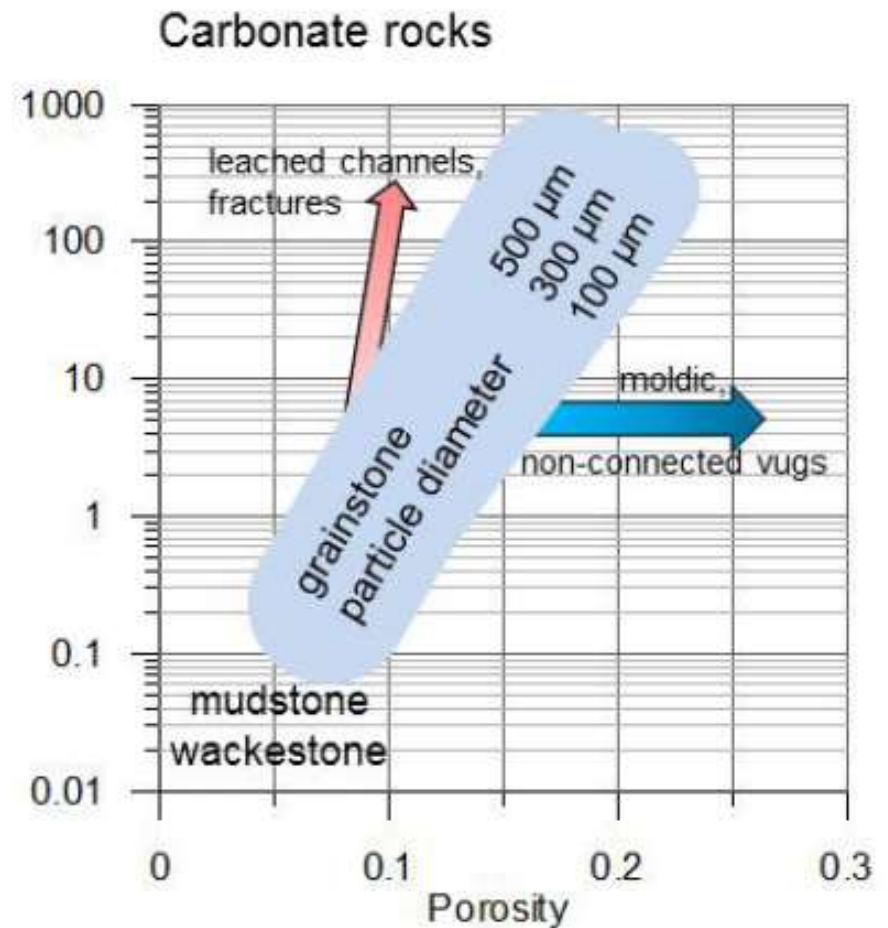
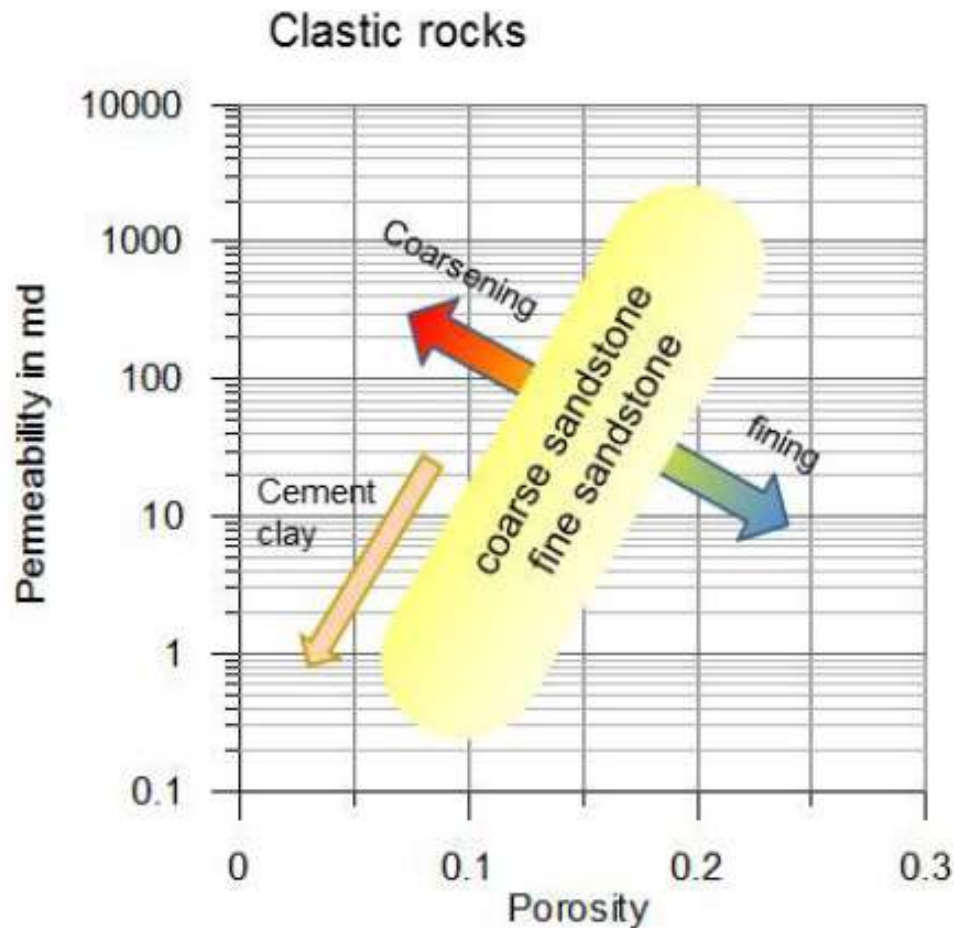


A typical  $k - \phi$  plot



# Permeability and Porosity Relationships

- The porosity-permeability relationship with tendencies for clastic and carbonate rocks is shown in the figure below.



# Directional Permeability

- Permeability is a directional quantity and should be truly considered a tensor property.
- Water-borne deposition of sand occurs with alignment of the long axis of the grains parallel to the current. In Aeolian sands the long axis may be at right angles to the paleo-wind direction. The greatest cross-sectional area of the grains lies in a horizontal plane. Permeability is highest parallel to the long axis of the grains.
- Vertical permeability (perpendicular to bedding planes) is usually less than horizontal permeability, due to platy minerals lying flat along bedding planes. This vertical permeability may be further reduced by shale laminations in sands, or stylolites in carbonates. In reservoir description the ratio between  $k_v$  and  $k_h$  is an important factor in understanding sweep efficiency and recovery.
- Fractures or joint trends existing in carbonates and hard, low porosity sands, result in widely varying directional permeability. This anisotropy is important in understanding reservoir behaviour.



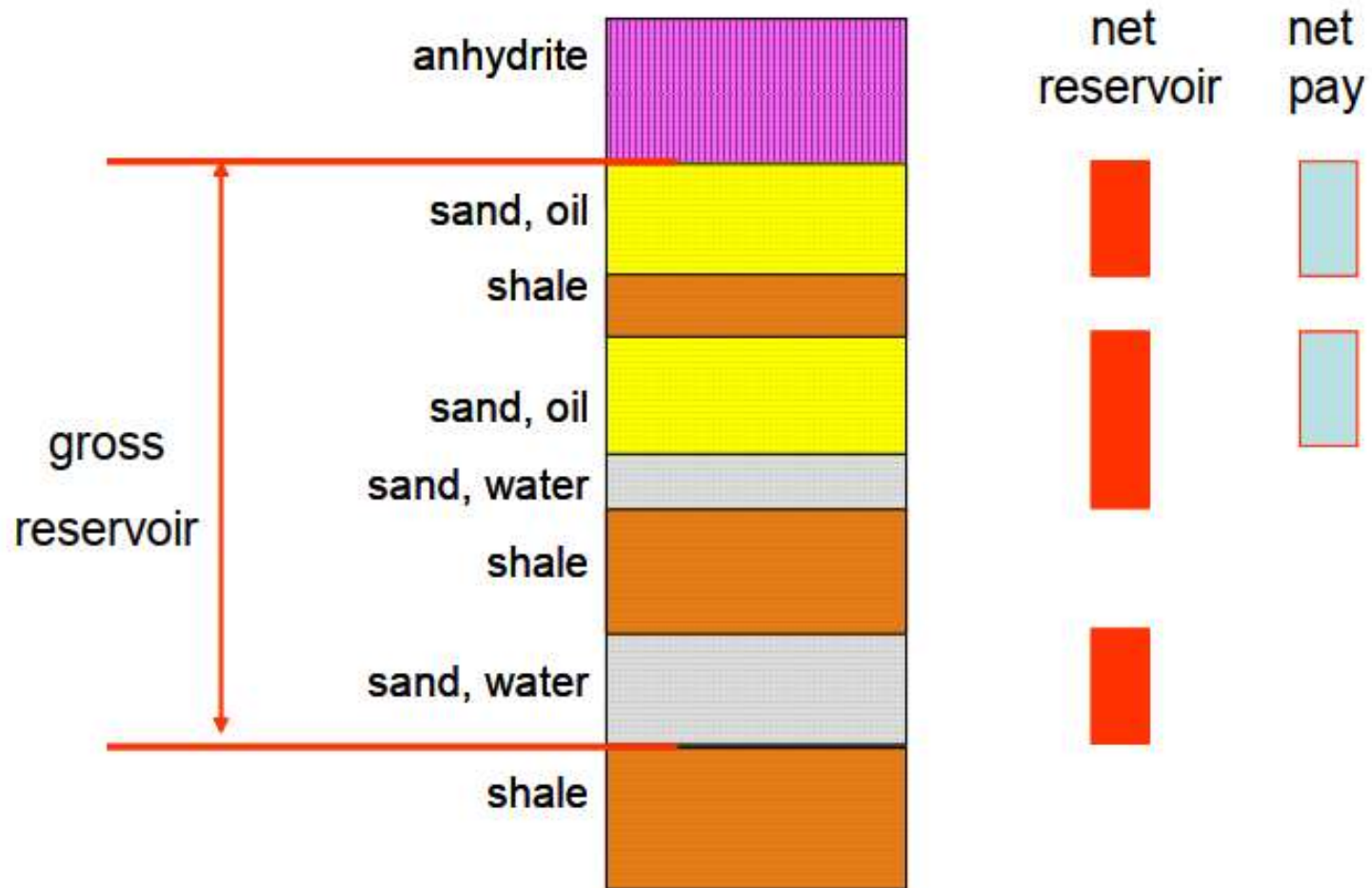
# Hydrocarbon In Place Volumes

- Static models are concerned with characterizing and quantifying the structure prior to any production from the field. Hence, key parameters that the models aim to determine are:
  - STOIP = stock tank oil initially in place; usually measured in stock tank barrels (stb)
  - GIIP = gas initially in place; usually measured in billion standard cubic feet (Bcf)
  - GBV = gross bulk volume; the total rock volume of the reservoir containing hydrocarbon
  - NPV = net pore volume; the pore space of the reservoir
  - HCPV = hydrocarbon pore volume; the pore space actually containing hydrocarbon
  - $\phi$  = porosity; the proportion of the formation that contains fluids
  - $k$  = permeability; usually expressed in millidarcies (md)
  - $S_w$  = water saturation; the proportion of the porosity that contains water
  - $S_h$  = hydrocarbon saturation; the proportion of the porosity that contains hydrocarbon
  - FWL = free water level; the depth at which the capillary pressure in the reservoir is zero; effectively the depth below which no producible hydrocarbons will be found

# Hydrocarbon In Place Volumes

- Static models are concerned with characterizing and quantifying the structure prior to any production from the field. Hence, key parameters that the models aim to determine are:
  - HWC = hydrocarbon/water contact; the depth below which the formation is water bearing as encountered in a particular well. Likewise, OWC for oil and GWC for gas
  - GOC = gas oil contact; the depth below which any gas in the reservoir will be dissolved in the oil
  - Gross thickness = the total thickness of the formation as encountered in a particular well; it is not related to the fluids in the formation
  - Net thickness = the part of the gross thickness that contains porous rock subject to given cutoff criteria
  - Pay thickness (Net Pay) = the part of the net thickness that is considered to be capable of producing hydrocarbons in a particular well
- Because of inherent uncertainties in all the parameters used to determine STOIIP or GIIP, geologists will usually develop probabilistic models, in which all the parameters are allowed to vary according to distribution functions between low, expected, and high values.

# Net Pay Definition

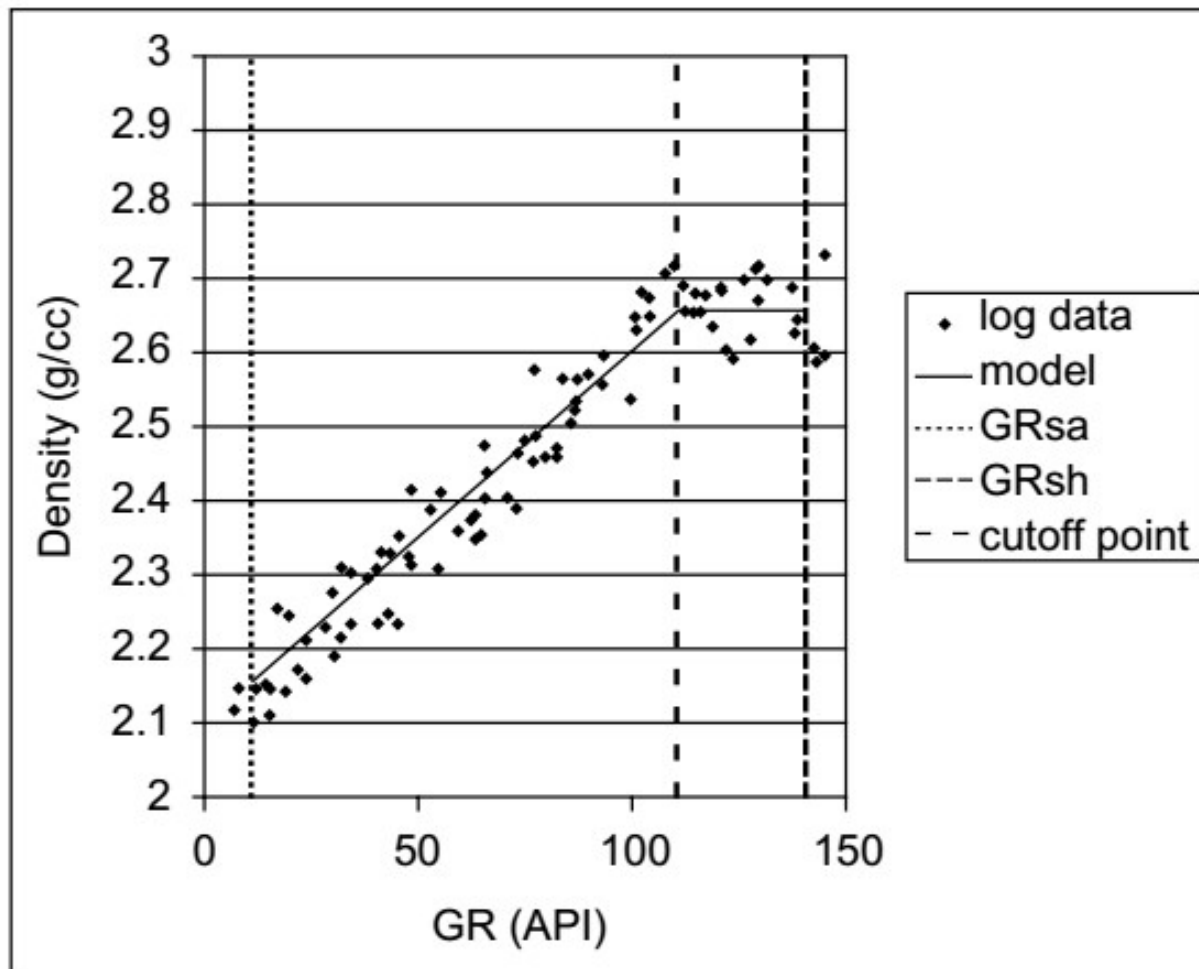


# Reservoir Cutoffs

Cutoffs are defined in literature using different criteria. Examples are:

- Bigelow (2002) defined it thus:  $\phi < 15\%$ ,  $V_{sh} > 30\%$ ,  $S_w > 50\%$ ,  $k < 50 \text{ md}$
- Darling (2005) formulated thus, “Generally, the cutoff point should be set such that net sands don’t contain porosities much below a level corresponding to a permeability of 1 md for oil zones and 0.1 md for gas zones.”
- Where core data are not available, it is sometimes helpful to plot the gamma ray (GR) vs. the density log to help to establish the best point to discriminate net from non-net from the GR log. This method would not be appropriate if radioactive minerals are present in the sand. In such formations it is recommended to use purely a porosity cutoff.
- In the case of thinly laminated sands, it is possible that the entire formation interval will be designated as non-reservoir using a  $V_{sh}$  or porosity cutoff. In this situation it is recommended to not apply any cutoffs whatsoever and the Archie approach will no longer be appropriate.

# Reservoir Cutoffs Using a GR-Density Crossplot



The correct cutoff point is therefore the point at which the gradient changes, corresponding to zero effective porosity.

- As shale becomes dispersed in the pore space (increasing GR), the density will rise until the point at which the pore space available for free fluids becomes zero. Beyond this point, the amount of shale may still increase until the formation becomes 100% shale, but the density will change only slightly (depending on variation in density between quartz and shale).

# Production, Recovery & Reserves

- In order to predict the hydrocarbons that may be actually produced from a field (the reserves), it is necessary to construct a dynamic model of the field. This will generate production profiles for individual wells, subject to various production scenarios. Additional terminology that comes into play includes:
  - Reserves = the part of the STOIIP or GIIP that may be actually produced for a given development scenario. Oil companies have their own rules for how reserves are categorized depending on the extent to which they are regarded as proven and accessible through wells. Terms frequently used are proven reserves, developed reserves, scope for recovery reserves, probable reserves, and possible reserves.
  - Remaining reserves = that part of the reserves that has not yet been produced
  - Cumulative production = that part of the reserves that has already been produced
  - UR = ultimate recovery; the total volume of reserves that will be produced prior to abandonment of the field
  - NPV = net present value; the future economic value of the field, taking into account all future present value costs and revenues
  - RF = recovery factor; the reserves as a proportion of the STOIIP (or GIIP)
  - B<sub>o</sub> = oil formation volume factor; the factor used to convert reservoir volumes of oil to surface (stock tank) conditions. Likewise B<sub>g</sub> for gas.

# HCIIP & Recovery Factor

- In order to predict the hydrocarbons that may be actually produced from a field (the reserves), it is necessary to construct a dynamic model of the field. This will generate production profiles for individual wells, subject to various production scenarios. Additional terminology that comes into play includes:

The equation for calculating gas-in-place is:

$$G = \frac{43,560 Ah \phi (1 - S_{wi})}{B_{gi}}$$

with:

$$B_{gi} = 0.02827 \frac{Z_i T}{p_i} \text{ ft}^3/\text{scf}$$

where:

$G$  = gas-in-place, scf  
 $A$  = area of reservoir, acres  
 $h$  = average reservoir thickness, ft  
 $\phi$  = porosity  
 $S_{wi}$  = water saturation  
 $B_{gi}$  = gas formation volume factor at initial pressure  $p_i$ , ft<sup>3</sup>/scf.

The equation for Oil Initially in Place,  $N$  in STB:

$$N = \frac{7,758 V_b \phi (S_{oi})}{B_{oi}} = \frac{7,758 Ah \phi (1 - S_{wi})}{B_{oi}}$$

The constant 7,758 is the number of barrels in each acre-ft,  $V_b$  is bulk volume in acre-ft,  $\phi$  is the porosity, ( $\phi V_b$  is pore volume),  $S_{oi}$  is the initial oil saturation,  $B_{oi}$  is the initial oil formation volume factor in reservoir barrels per stock tank barrel,  $A$  is area in ft<sup>2</sup>.

- Ultimate Recovery = (Hydrocarbons Initially in Place) X (Recovery Factor)

# **An Overview of Well Logging Practice – Lecture 3**



**SURFACE LOGS – AS  
COMPLEMENTARY TO WELL LOGS**

# Surface Logging

- Surface Logging Services encompass measurement technologies, data acquisition platforms, analysis and reporting systems required from accurate and timely monitoring of lithology and pore fluid composition of the formations being drilled to support critical decisions that help drill wells more efficiently into potential pay zones.
- Specialized surface-located sensors measure mechanical, hydraulic, and engineering data while drilling providing the basis for calculating parameters such as mud pit volumes, mud flow rate, mud weight in the shale shaker area, mud-gas level, and return flow rate.
- This information helps us detect underbalanced formation fluid pressures, optimize the drilling process, and maintain a safer operating environment. (**Source:** *Schlumberger, Geoservices*)

# Recall - Surface Logging

- The term “**Surface Logging Services**” is often used to encompass a range of enhanced mud logging services that incorporate advanced sensor and computing technology to provide monitoring for well stability and early kick detection. This was made possible as additional sensors brought more data into the mud logging unit.

Geoservices Surface Logging Workflow



# Surface Logging Services

- **Weatherford**

- Mud Logging Services
- Advanced Cutting Analysis
- Advanced Gas Analysis
- Advanced Wellbore Cleaning Analysis
- Pore Pressure Prediction

- **Baker Hughes**

- Geologic Services
- Advanced Cutting Evaluation Service
- Gas Measurement and Monitoring
- Advanced Gas Analysis
- Fluids Monitoring
- Operations Monitoring

What are the Surface Logging Services that Schlumberger offers?

# Drill Site Information Systems

- Graphics and gauge displays with real-time access to Rig equipment data, MWD, LWD, Directional Drilling, Geosteering and Mud Logging data through a secure web-based portal
- All modern rigs and Coiled Tubing Units have digital integrated drilling information systems and they are easy to fit to older ones.
- Electronic drilling recorders (EDR) have replaced the earlier mechanical strip-chart recorders “Geolograph” due to higher resolution and the usefulness of the digital format.

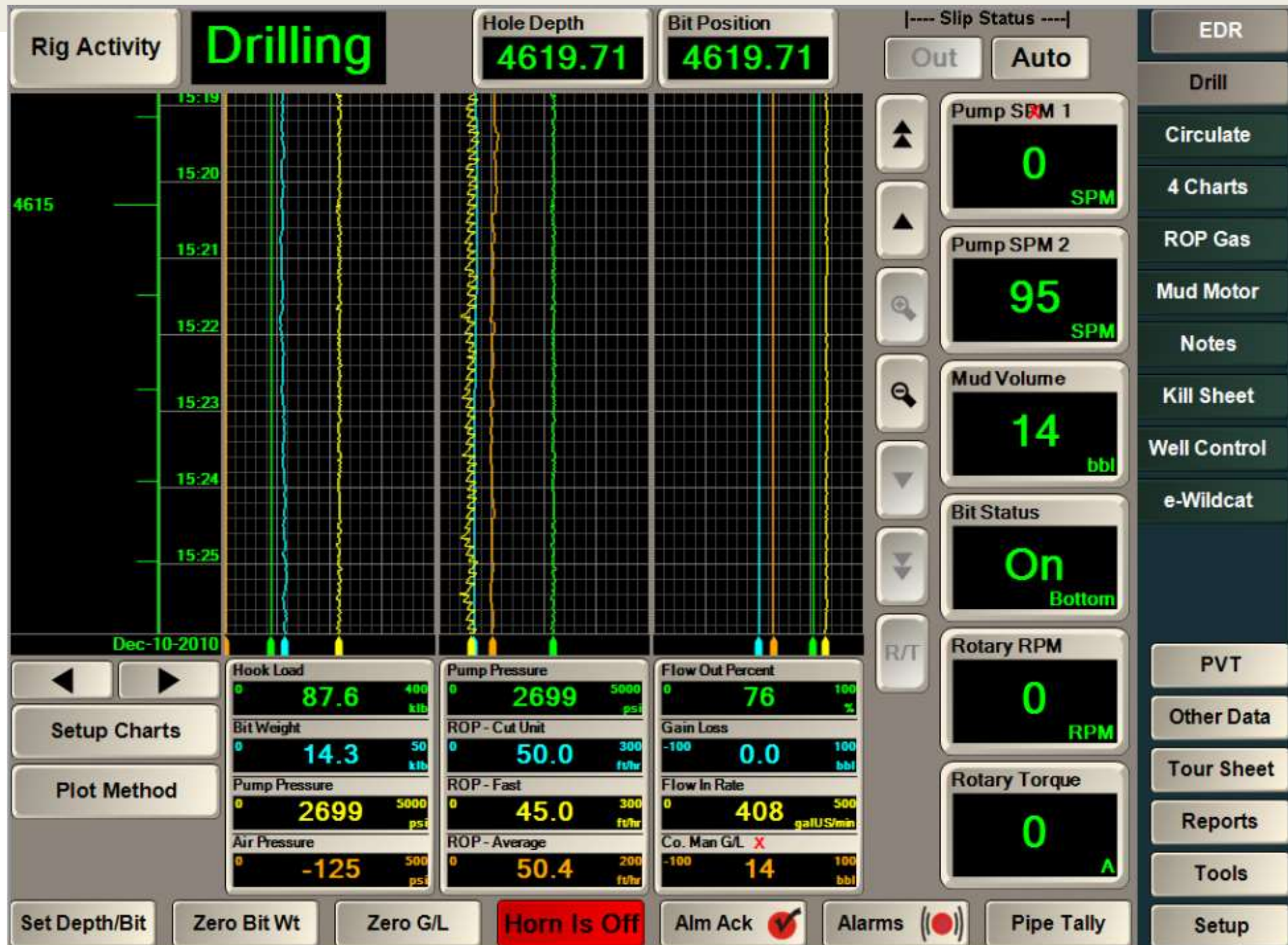
# Geolograph (For Reference Purposes)

- A geolograph is a machine that records many functions of a rig unto paper e.g.
  - Rate of Penetration
  - Connection speed; Tripping speed
  - Flow rate,
  - Torque, Weight of string
  - Stand pipe pressure; Pump pressure etc
- A weight transducer measures the load of the drill string on the rig.
- A movement sensor provides information on movement of the hook.
- An internal clock provides time signals.

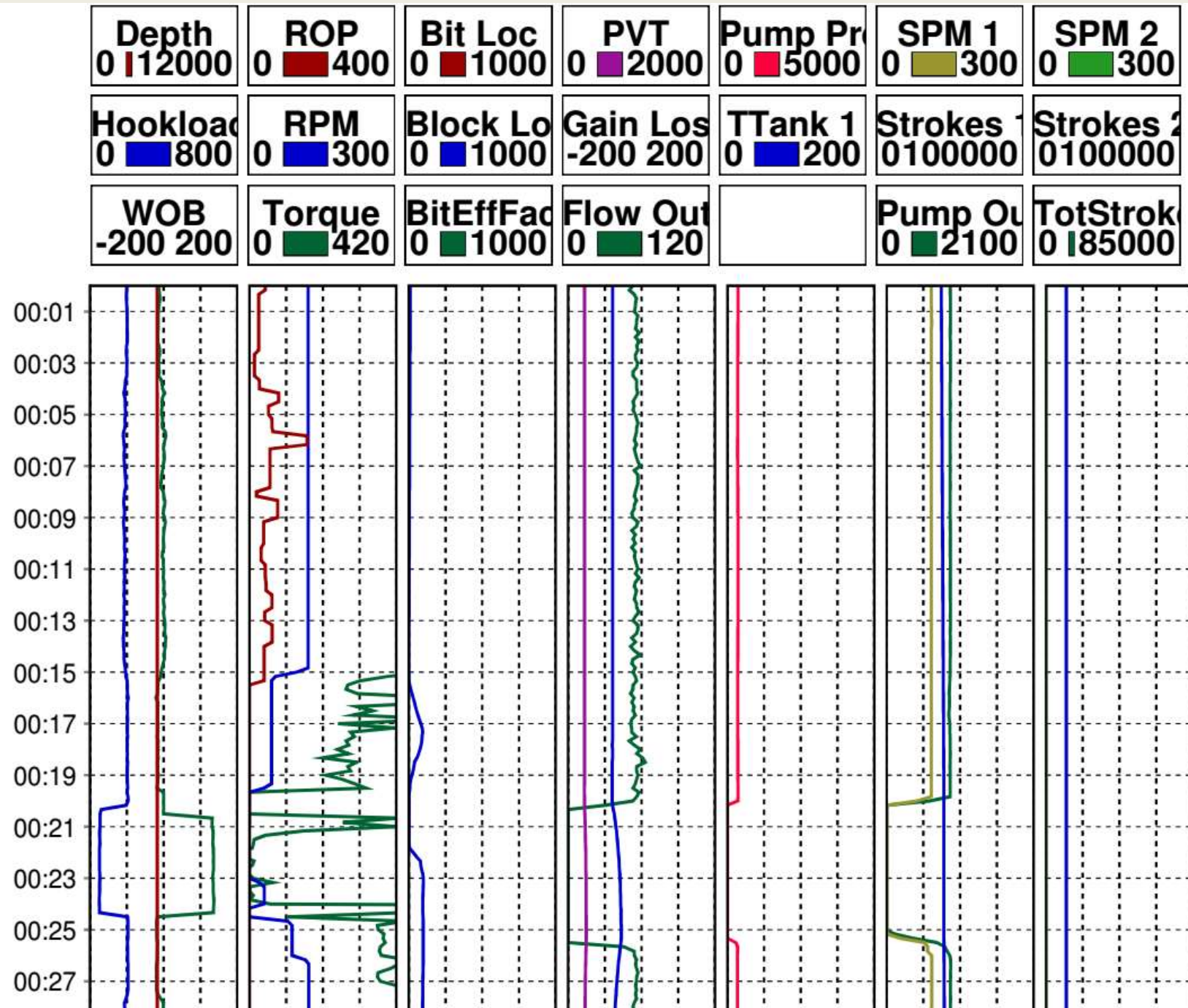




# Electronic Data Recorder



# Drilling Parameters Log





# What is the Drilling Parameters logs for?

- The primary use is for the driller to optimize drilling and to make it easier for him to spot early signs of problems like hole cleaning and fluid losses to minimize wellsite risk.
- Location of loss zones is important as they may have to be cemented off in the cased portion of the well and in the open-hole interval the losses indicate the future production zones.
- The log is very valuable in determining the causes of drilling problems.
- Changes from established “trends” usually indicate downhole problems. Commonly monitored parameters for kick detection are:

# Application of Drilling Parameter Logs - Warning Signs of Kicks from Drilling Parameters

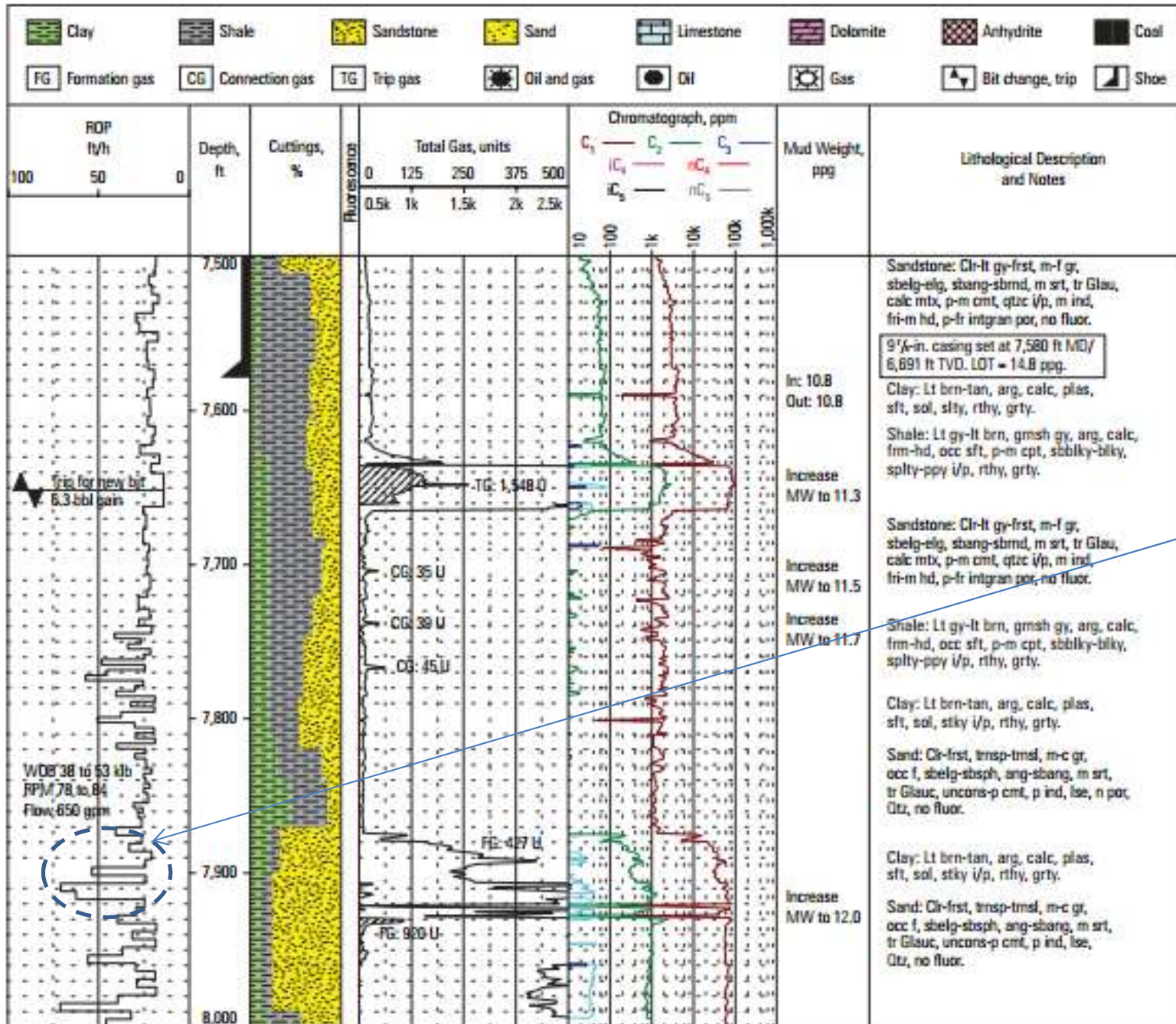
- Flow Indicator:
  - Flow rate at given pump speed
  - Flow with pump shut down
- Pumps:
  - Pump pressure changes
  - Pump speed
- Rotary:
  - Rate of Penetration whether a formation or milling
  - Torque/drag
  - String weight
  - Fill
- Pits:
  - Changes in Pit Level during any activity
  - Hole fill during trip in
  - Displacement during trip out

# **THE MUD LOG – AS COMPLEMENTARY TO WELL LOGS**

# Correlation of ROP With Other Logs

- ROP plot if plotted with values increasing from right should match up with an SP or GR log
- The mud log's ROP and total gas curves exhibit a remarkable correspondence to gamma ray and resistivity curves, respectively
- Typically a shale baseline is established through a thick interval of slow, consistent drilling – the inference can be verified through analysis of formation cuttings.
- Deviations from this ROP baseline may indicate changes in lithology or other downhole variables.

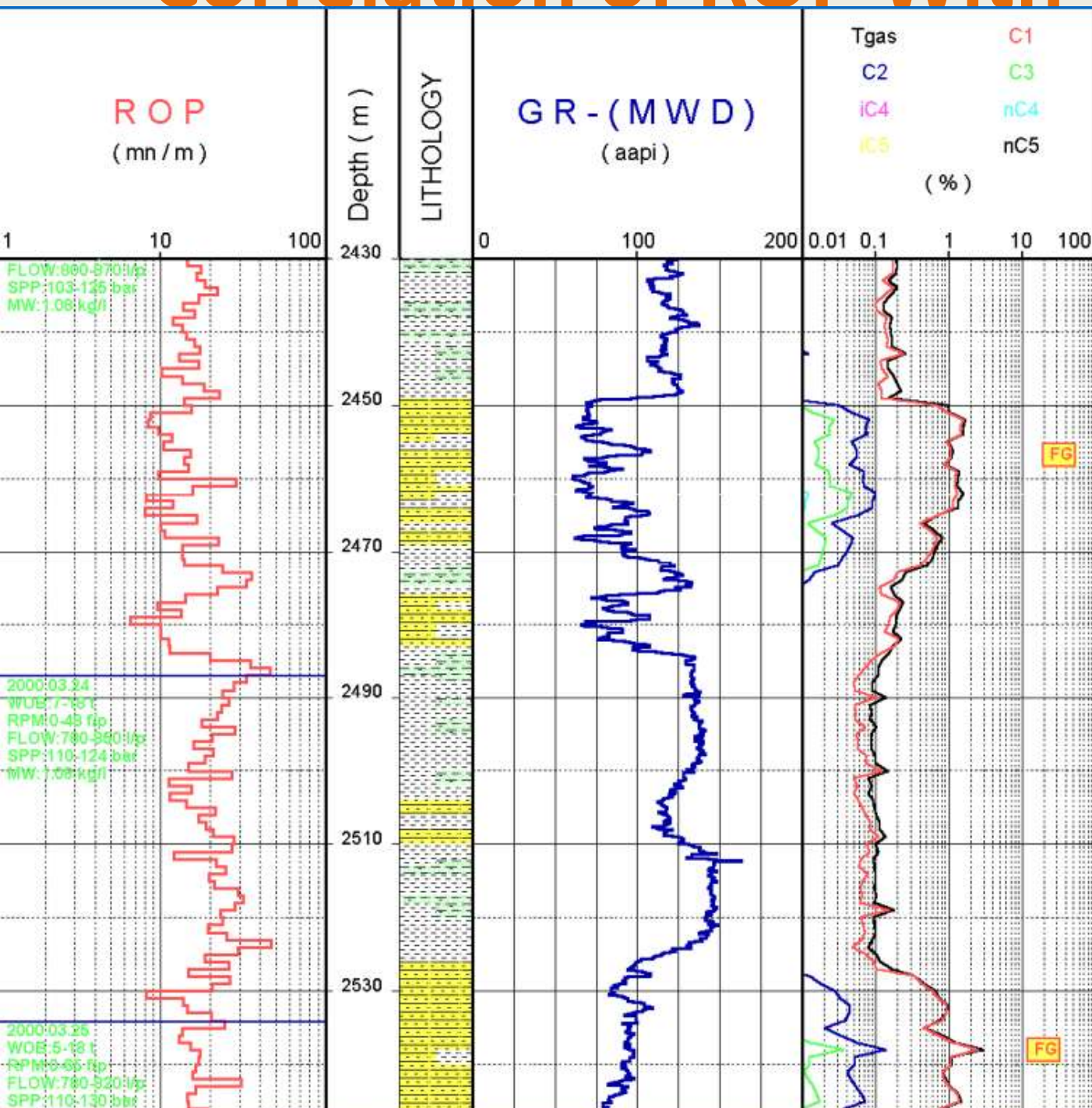
# Correlation of ROP With Other Logs



Increase in ROP at 7,890ft signified a drilling break which was accompanied by increasing sand content and a gas show, which reached a peak of 920 units of formation gas.



# Correlation of ROP With Other Logs



# Correlation of ROP With Other Logs

γ ray

Res.

N&D

THC

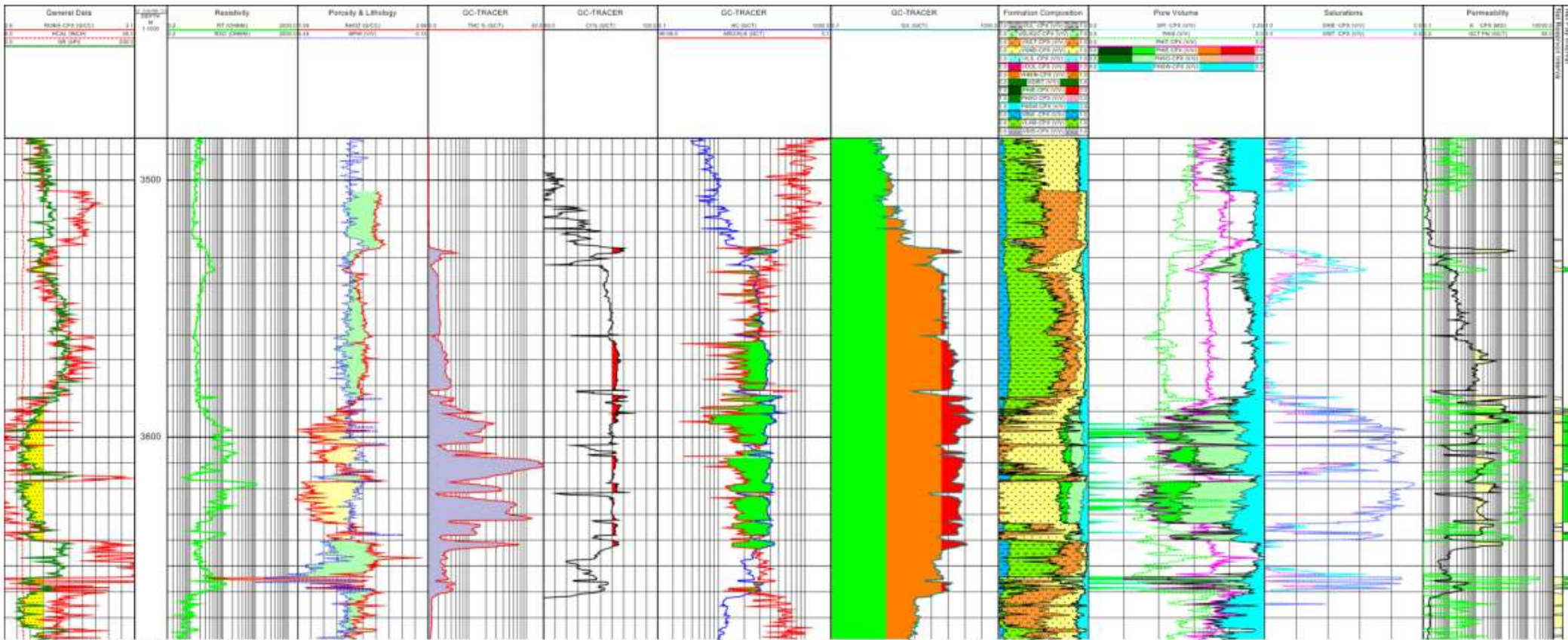
C1%

HC

ARO/ALK

G/L

LHR



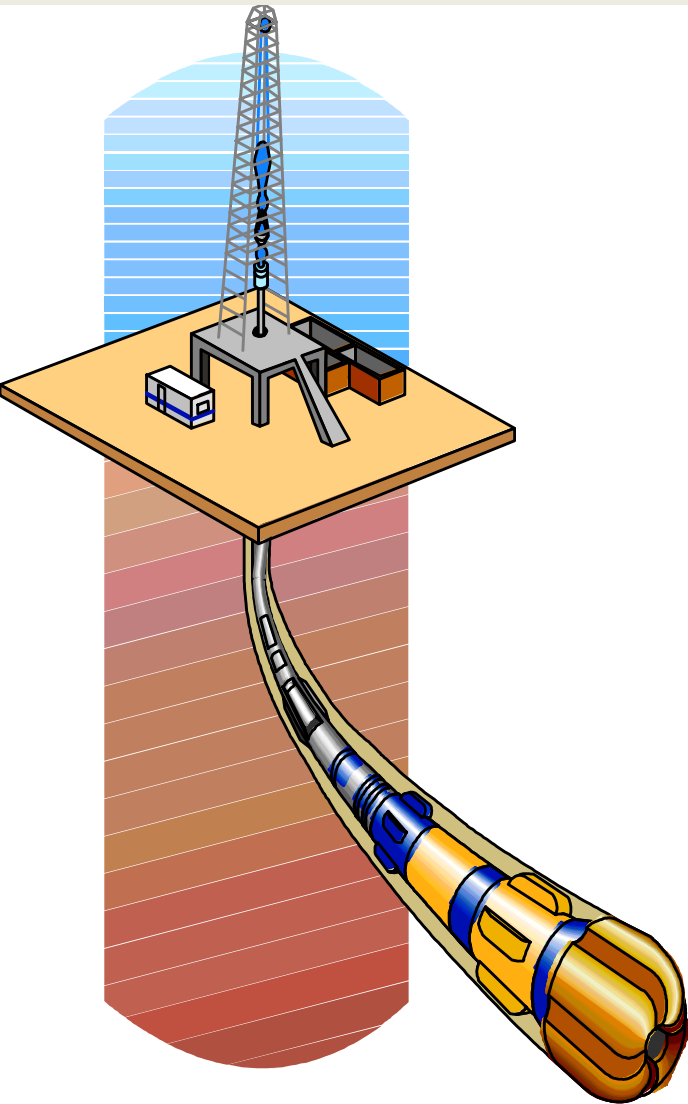
# **LOGGING WHILE DRILLING (LWD)**



# Measurement While Drilling (MWD)

- To assist drillers in the complex task of a rotary drilling operation, a number of types of information like the downhole weight on bit and the downhole torque at bit are desirable in real time. To respond to this need, a type of service known as measurement while drilling (MWD) began to develop in the late 1970s.
- A typical MWD system consisted of a downhole sensor unit close to the drill bit, a power source, a telemetry system, and equipment on the surface to receive and display data. The telemetry system was often a mud pulse system that used coded mud pressure pulses to transmit at a few bits per second the measurements from the downhole subassembly. The power source was a combination of a generating turbine, deriving its power from the mud flow, and batteries.

# Measurement While Drilling (MWD)



- MWD tools
  - Measure the direction & inclination of the wellbore
  - Allow drilling tools to be oriented
  - Provide mechanism for transmitting downhole data to the surface
- MWD comprises
  - Downhole telemetry and sensor system
  - Mud pulse transmission
  - Surface receiver decoding/display/ archive
- Depth
  - Drill line encoder
  - Measurements in time
  - Regular TIME sampling

To “convey” wireline logging tools through the complex geometry of the non-vertical well, another technology , referred to as LWD and provides, in addition to drilling-specific MWD measurements, a family of measurements entirely analogous to the measurements of traditional wireline logging.

# Logging While Drilling (LWD)

- One of the major drawbacks of wireline information is that it is received several hours to several weeks after the borehole is drilled.
- During this time period, the formation can undergo significant alteration, especially in its fluid saturation, effective porosity, and relative permeability.
- LWD allows wireline-type information to be available as near as real-time as possible.
- Logging While Drilling (LWD) is a technique of conveying well logging tools into the well borehole downhole as part of the bottom hole assembly (BHA).
- The LWD tools are all built into heavy thick-walled drill collars – the special portion of the drill string used to counter buoyancy and provide stiffness to the lower segments of the drill string.

# Logging While Drilling (LWD)

- These tools may transmit data as pulses in the mud column in real time while drilling to the surface or store the data in a downhole memory from which it may be downloaded when the assembly is brought back to the surface.
- In a typical operation, both modes will be used, with the memory data superseding the pulsed data once the tool is retrieved. The use of both modes is limited by:
  - Drilling mode: Data may be pulsed only if the drillstring is having mud pumped through it.
  - Battery life: Depending on the tools in the string, tools may work in memory mode only between 40 and 90 hours.
  - Memory size: Most LWD tools have a memory size limited to a few megabytes. Once the memory is full, the data will start to be overwritten. Depending on how many parameters are being recorded, the memory may become full within 20–120 hours.
  - Tool failure: It is not uncommon for a fault to develop in the tool such that the pulse data and/or memory data are not transmissible/recordable.

# Logging While Drilling (LWD)

- Another important consideration with LWD tools is how close to the bit they may be placed in the drilling string. While the petrophysicist will obviously want the tools as close to the bit as possible, there may be limitations placed by drilling, whose ability to steer the well and achieve a high ROP is influenced by the placement of the LWD toolstring.
- As the drill collar is rotated, data can be acquired from multiple azimuths around the borehole, something not often achievable with a wireline.
- Unlike wireline tools that are generally of a standard diameter, many of the LWD tools come in families of sizes (e.g., 4, 6, and 8 in.). This is to accommodate popular drilling bit sizes and collar sizes since the LWD device must conform to the drilling string.

# Logging While Drilling (LWD)

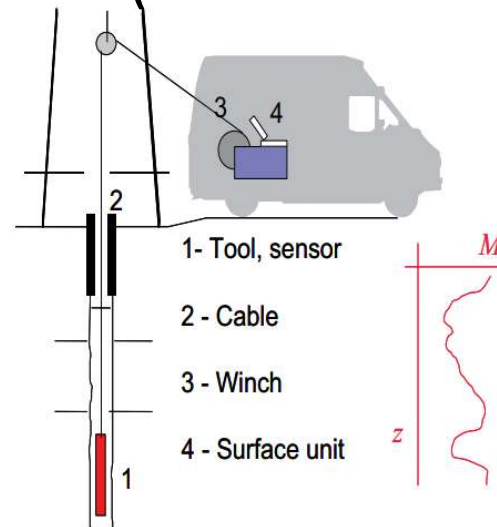
- However, LWD tools present a complication for drilling, as well as additional expense. However, their use may be justified when:
  - Real-time information is required for operational reasons, such as steering a well (e.g., a horizontal trajectory) in a particular formation or picking of formation tops, coring points, and/or casing setting depths
  - Acquiring data prior to the hole washing out or invasion occurring
  - Safeguarding information if there is a risk of losing the hole
  - The trajectory is such as to make wireline acquisition difficult (e.g., in horizontal wells)

# WIRELINING LOGGING

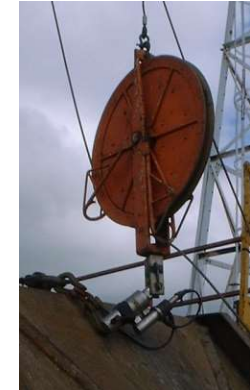
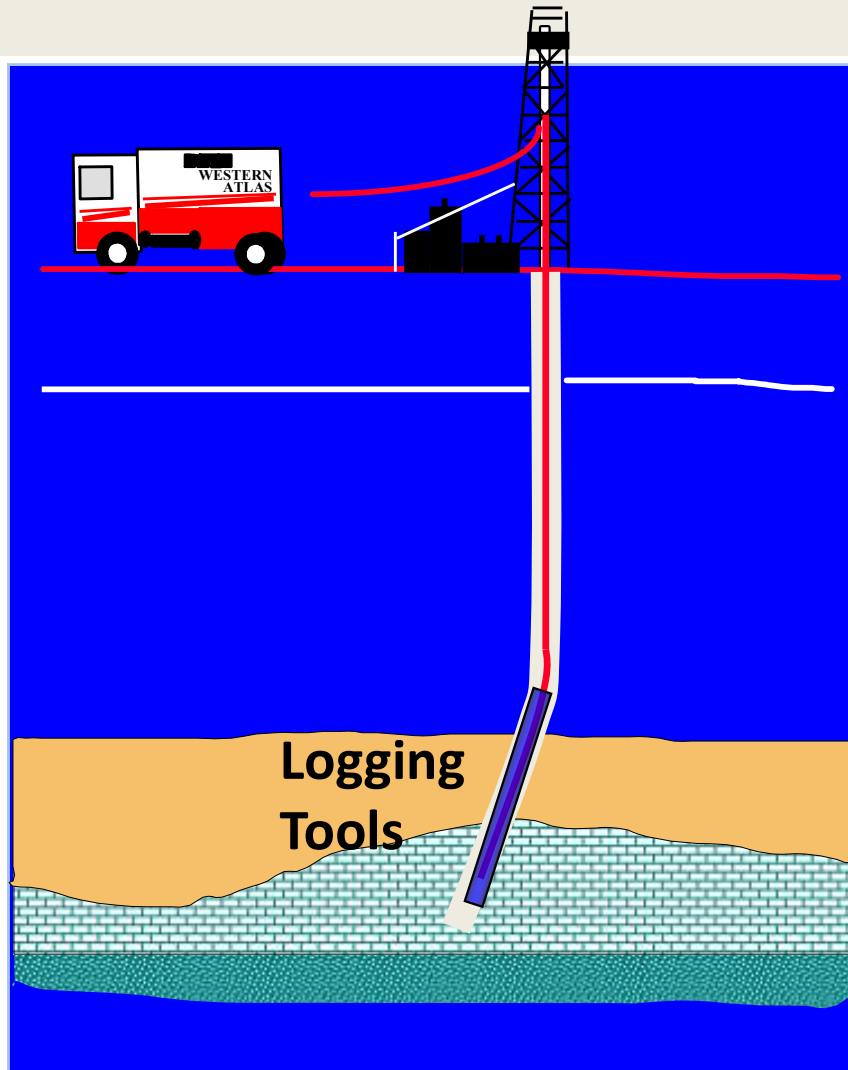


# Wireline Logging

- A Wireline Logging Setup has the following main components
  - Tool or probe with sensors, transmitters, sources
  - Cable connected with the probe by a cable-connector. The cable gives depth information about tool position, transmits the energy downward and the measured data upward
  - Winch with depth counter
  - Surface unit for controlling the measured process, visualizing and storing measured data (Mobile Laboratory)



# Wireline Logging



The mobile laboratory provides the downhole power to the instrument package. It provides the cable and winch for the lowering and raising of the sonde, and is equipped with computers for data processing, interpretation of measurements, and permanent storage of the data.

# Wireline Logging – Sensor System

- Of primary interest is the measurement device, or sonde. There are over fifty different types of these logging tools developed in order to meet various information needs and functions.
- Superficially, sondes all resemble one another. They are generally cylindrical devices with an outside diameter on the order of 4 in. or less; this is to accommodate operation in boreholes as small as 6 in. in diameter. Their length varies depending on the sensor array used and the complexity of associated electronics required. It is possible to connect a number of devices concurrently, forming tool strings as long as 100 ft.

# Wireline Logging – Sensor System

- Some of them are passive measurement devices; others exert some influence on the formation being traversed (active measurement devices). Their measurements are transmitted to the surface by means of the wireline.
- This tool is lowered into mud-filled hole on logging cable to measure the electrical, acoustic, and radioactive properties of the formation
- A depth calibration wheel records the length of cable in the hole.
- Survey is normally done from the bottom up. As the sonde is pulled up the hole, a continuous measurement signal is sent to the surface where the data is processed and recorded as a curve.

# Wireline Logging – Cable & Depth Control

- Wireline refers to the armoured cable by which the measuring devices are lowered and retrieved from the well and, by a number of shielded insulated wires in the interior of the cable, provide for the electrical power of the device and a means for the transmission of data to the surface.
- The mobile laboratory provides the downhole power to the instrument package. It provides the cable and winch for the lowering and raising of the sonde, and is equipped with computers for data processing, interpretation of measurements, and permanent storage of the data.
- A depth calibration wheel records the length of cable in the hole.

# Basic Data Differences between LWD & Wireline

## LWD

- Almost always Recorded Down
- Time sampled
  - Resolution depends on ROP
  - Sensor spacing
- Real time data density depends on ROP and acquisition rate
- Memory – full dataset
- Downhole – storage/data limited
- Sensors fixed in drill collar
- Typically recorded with minimal invasion effects;
- Small borehole enlargement
- Affected by stick/slip
- Logging rate of slowest tool (e.g. SoundTrak™ – 15 sec acq.)
- Can log in any direction
- More capable in tough environment

## Wireline

- Almost always Recorded Up
- Depth sampled
  - Surface controlled
  - High resolution possible
- All data telemetried while logging, but after drilling; Tools lowered in borehole on cable
- Future Memory options – slickline
- Storage limit is surface disk size
- Sensors positioned as required (cent/decent/ pads)
- Recorded after invasion (susceptible to hole condition)
- Significant to moderate borehole effects
- Affected by stick/pull
- Large combinations may affect logging speed



# LWD versus Wireline





# Measurements Made By MWD / LWD Tools

| Tool   | Measurement                     | Approximate vertical resolution (cm)      |
|--------|---------------------------------|-------------------------------------------|
| ADN    | Bulk density<br>Porosity        | 15<br>30                                  |
| CDN    | Bulk density<br>Porosity        | 61<br>34                                  |
| CDR    | Resistivity<br>Gamma ray        | 66 - 330 (deep), 30 - 254 (shallow)<br>46 |
| ISONIC | Velocity                        | 61                                        |
| NMR    | Nuclear magnetic resonance      | 15                                        |
| RAB    | Resistivity-at-bit<br>Gamma ray | 5 - 8<br>46                               |
| MWD    | Drilling parameters-at-bit      | NA                                        |

The takeaway is that for virtually every measurement and wireline tool, there is an equivalent LWD tool.

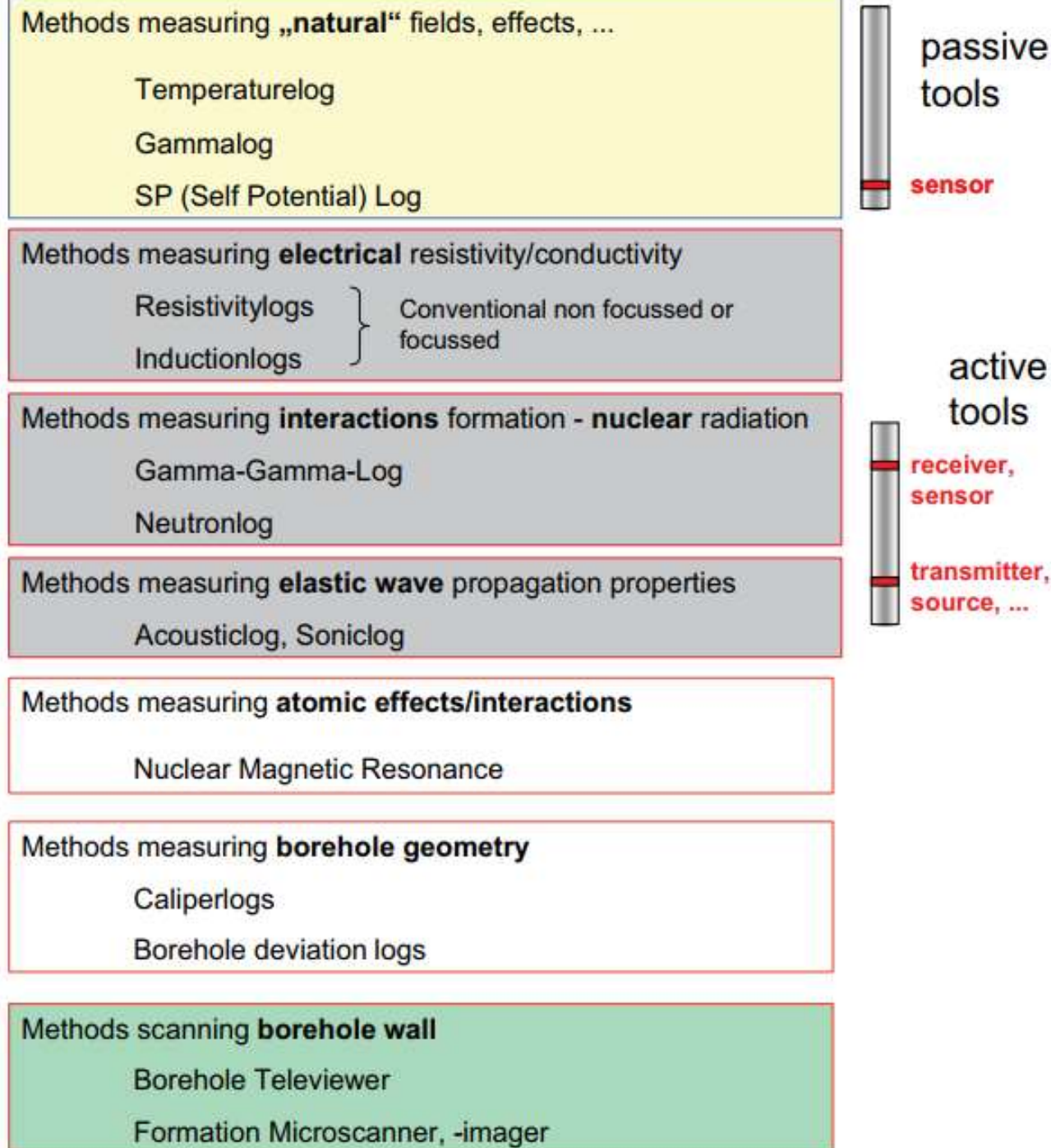
Notable exceptions were Fluid Sampling and Sidewall Coring\*\*\*\*

# Classification by Principle of Measurement

- Passive Tools: measure properties or parameters delivered by the formation or by interaction with the formation and the borehole fluid without any source.
  - Caliper, Natural Gamma Measurements, Spontaneous Potential (SP)
- Active Tools: measure the response to a signal, pulse, radiation, current i.e. the result of an interaction with the formation in the vicinity of the tool (typically they have one source and one or more detectors)
  - Acoustic measurements  
 $\Delta t_c, \Delta t_s, A_c, A_s$
  - Nuclear measurements  
 $\rho_b, \phi_N, Pe, \tau_1, \tau_2$
  - Resistivity / Electromagnetic  
 $R, t_{pL}, EATT$



# Classification by Principle of Measurement



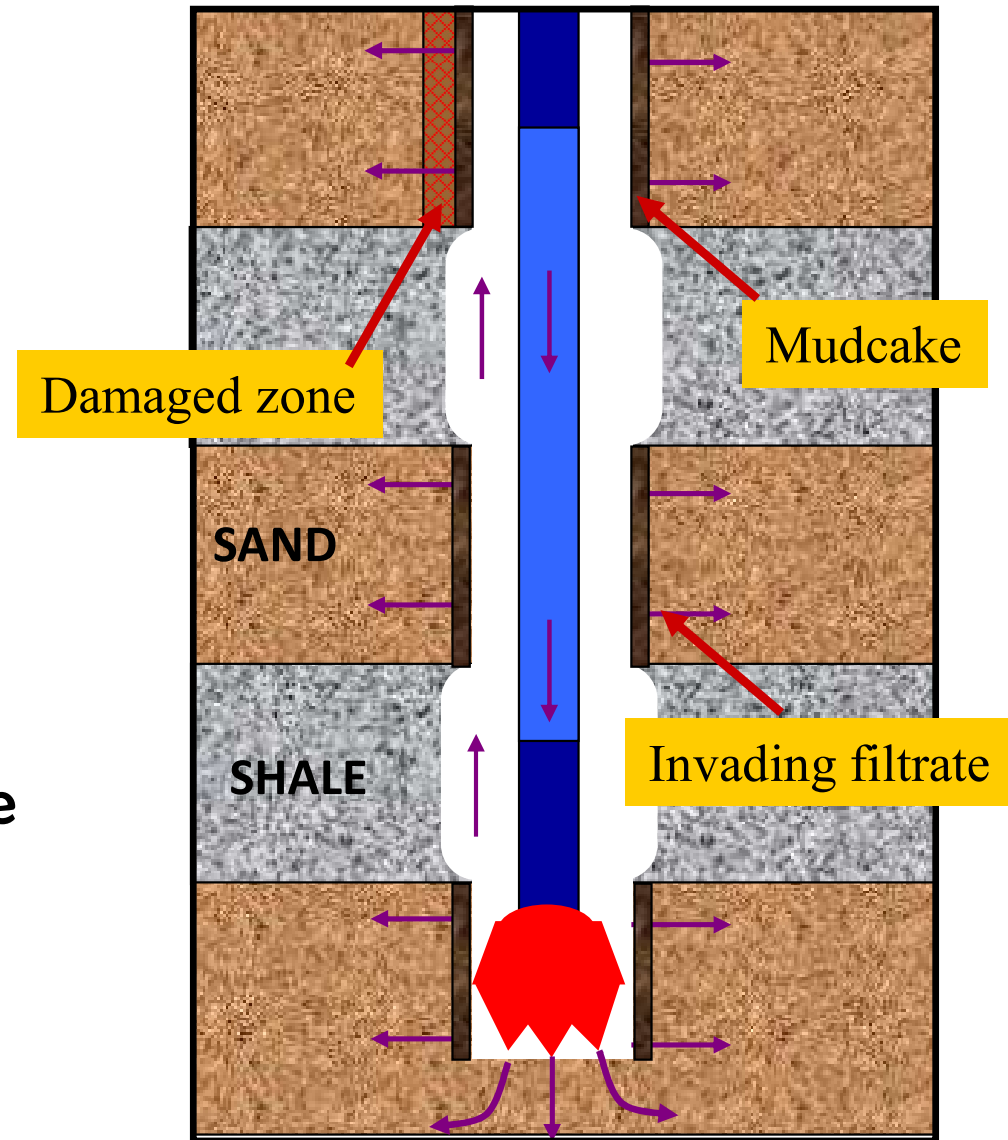
# THE WELLBORE ENVIRONMENT

# The Borehole Environment

- Some characterization of the borehole environment can be made using the following set of generalizations.
  - The temperature, at full depth, ranges between 100°F and 300°F.
  - Well depths are ordinarily between 1,000 and 20,000 ft, with diameters ranging from 5 to 15 in. Of course, larger ones can exist.
  - A truly vertical hole is rarely encountered, and generally the deviation of the borehole is between 0° and 5°. More deviated wells, between 20° and 60° are often encountered offshore.
  - Since the early 1990s an increasing number of horizontal wells have been drilled. These are drilled at a suitable deviation down to near the top of the reservoir, at which point the deviation is increased until they penetrate the reservoir within a few degrees of horizontal. They are then maintained within 5° of horizontal between 1,000 and 5,000 ft.
  - The drilling fluid, or mud, ranges in density between 9 and 16 lb/gal; weighting additives such as barite ( $\text{BaSO}_4$ ) or hematite are added to ensure that the hydrostatic pressure in the wellbore exceeds the fluid pressure in the formation pore space to prevent disasters such as blowouts.
  - The salinity of the drilling mud ranges between 1,000 and 200,000 ppm of NaCl.

# Drilling Disturbs Formation

- Drilling and rock crushing
  - ✓ Damaged zone
- Mud systems and invasion
  - ✓ Oil based mud
    - Small conductivity mud
    - Shallow invasion
    - Thin cake
  - ✓ Water based mud
    - Moderate to very conductive mud
    - Shallow to deep invasion
    - Thin to thick cake



Degradation of the formation during and after drilling. Overpressured mud is indicated to be invading porous and permeable sand formations with the formation of a mudcake. The mud circulation also causes borehole washout in the shale zone

# The Invasion Process

- The generally overpressured wellbore causes invasion of a porous and permeable formation by the drilling fluid.
- In the permeable zones, due to the imbalance in hydrostatic pressure, the mud begins to enter the formation but is normally rapidly stopped by the buildup of a mudcake of the clay particles in the drilling fluid. This initial invasion is known as the spurt loss.
- As the well is drilled deeper, further invasion occurs slowly through the mudcake, either dynamically, while mud is being circulated, or statically when the mud is stationary. In addition, the movement of the drill string can remove some mudcake, causing the process to be restarted. Thus, while a typical depth of invasion at the time of wireline logging is 20 in., the depth can reach 10 ft or more in certain conditions.



# Invasion Process

- To account for the distortion which is frequently present with electrical measurements, a simplified model of the borehole/formation in vertical wells with horizontal beds has evolved. It considers the invaded formation of interest, of resistivity  $R_t$ , to be surrounded by “shoulder” beds of resistivity  $R_s$ .
- The invasion is represented by the profile shown schematically in Fig. 2.1, along with the regions and parameters of interest, starting with the mudcake of thickness  $h_{mc}$  and resistivity  $R_{mc}$ .
- The next annular region of diameter  $d_i$  is the flushed zone whose resistivity is denoted by  $R_{xo}$ , determined principally by the resistivity of the mud filtrate. Beyond the invaded zone lies the uninvaded or virgin zone with resistivity  $R_t$ .
- A transition zone separates the flushed zone from the virgin zone. The transition may be smooth, but when hydrocarbons are present its resistivity can be significantly lower than either  $R_{xo}$  or  $R_t$ . This condition is known as an annulus and occurs mainly when the oil or gas is more mobile than the formation water, so that the formation water displaced from the flushed zone accumulates in the transition zone while the oil or gas is displaced beyond it. The annulus disappears with time, but can still exist at the time of logging.

# Non Anc

- Resistivity of zone
- Resistivity of the water in the zone
- △ Water saturation in the zone

## Borehole

$R_m$  : Borehole mud resistivity;

$R_{mc}$  : Mudcake resistivity

## Invaded zone

$R_{mf}$  : Mud filtrate resistivity

$R_{xo}$  : Invaded zone resistivity;

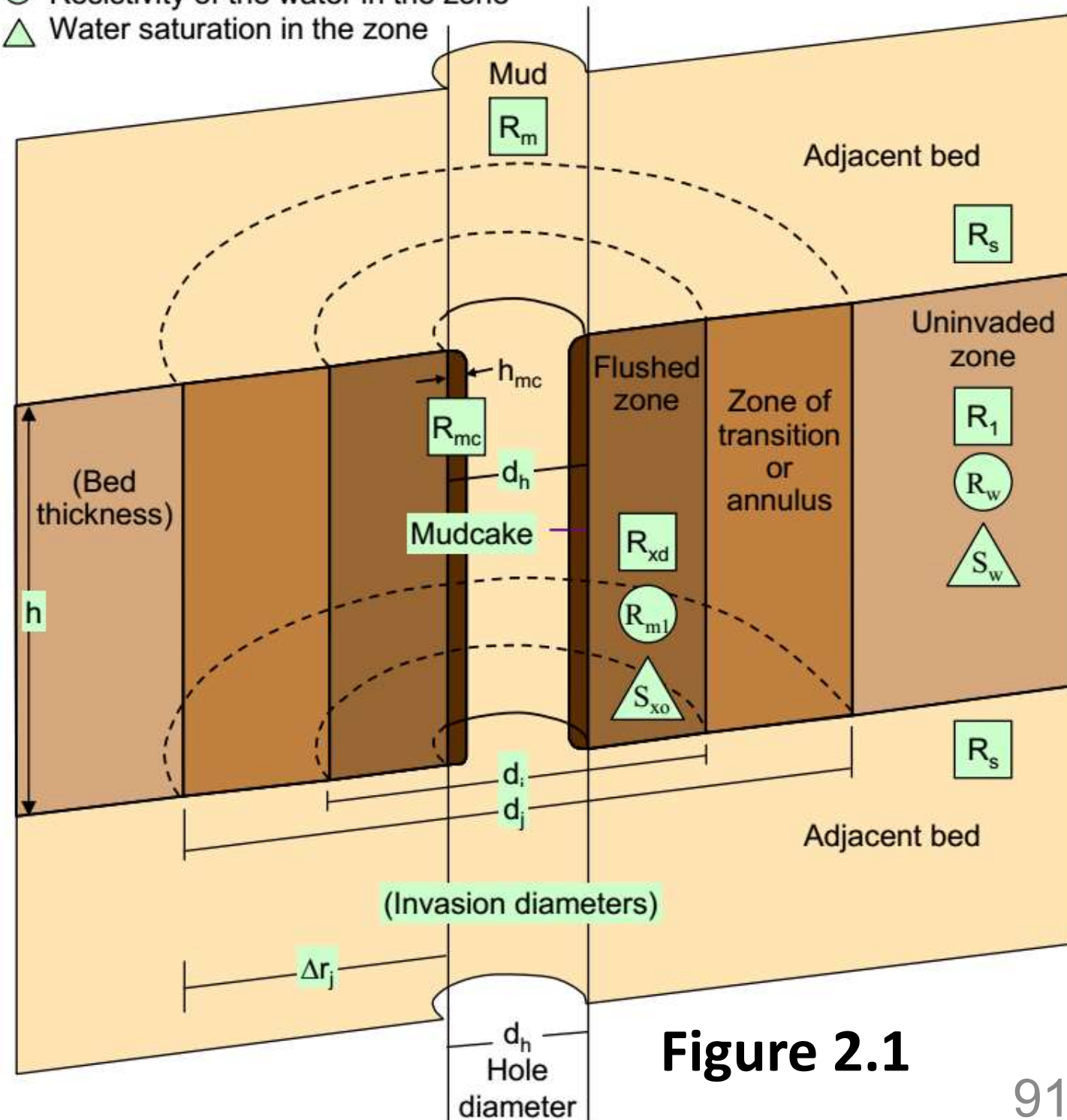
$S_{xo}$  : Invaded zone water saturation

## Un-invaded zone

$R_w$  : Interstitial water resistivity;

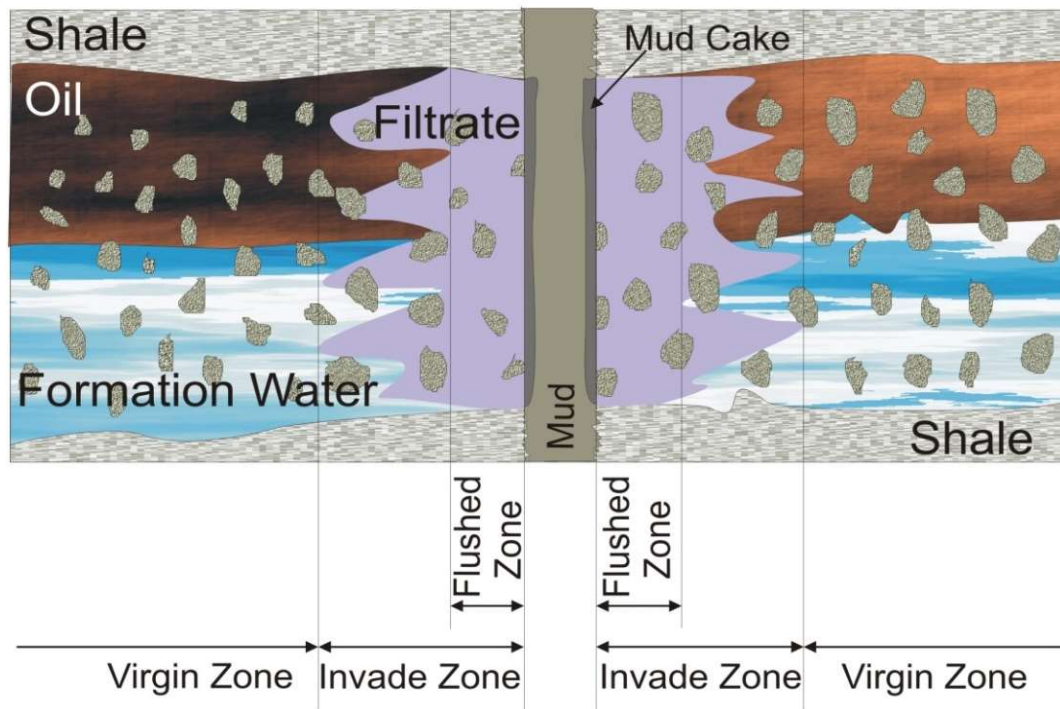
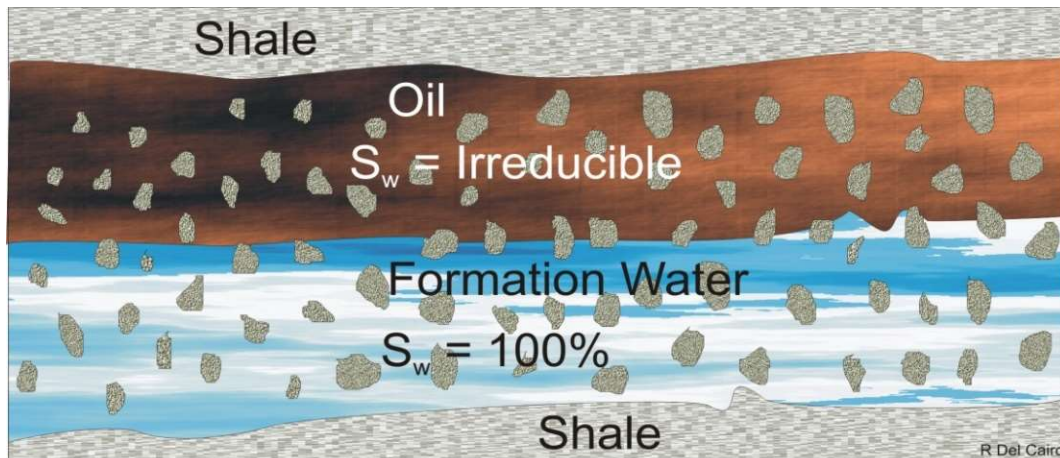
$R_t$  : Un-invaded zone resistivity;

$S_w$  : Un-invaded zone water saturation



**Figure 2.1**

# Invasion Profile



- **Mudcake:** Mud filtrate infiltrates the formation due to pressure differential between the mud column in the borehole and the formation. This causes mud to be deposited on the borehole wall where it forms *mudcake*. Mudcake usually has well low permeability and once developed considerably reduces the rate of further mud filtrate invasion.
- **Flushed zone:** the zone just beyond the mudcake. Original formation water and hydrocarbons have been flushed by Mud Filtrate to residual saturations.
- **Transition zone:** Further out into the wellbore where there is less displacement of mud filtrate. The flushed and transition zones make up the **Invaded zone**.
- **True zone:** the undisturbed zone beyond the transition zone where the fluid type we are interested in measuring is.

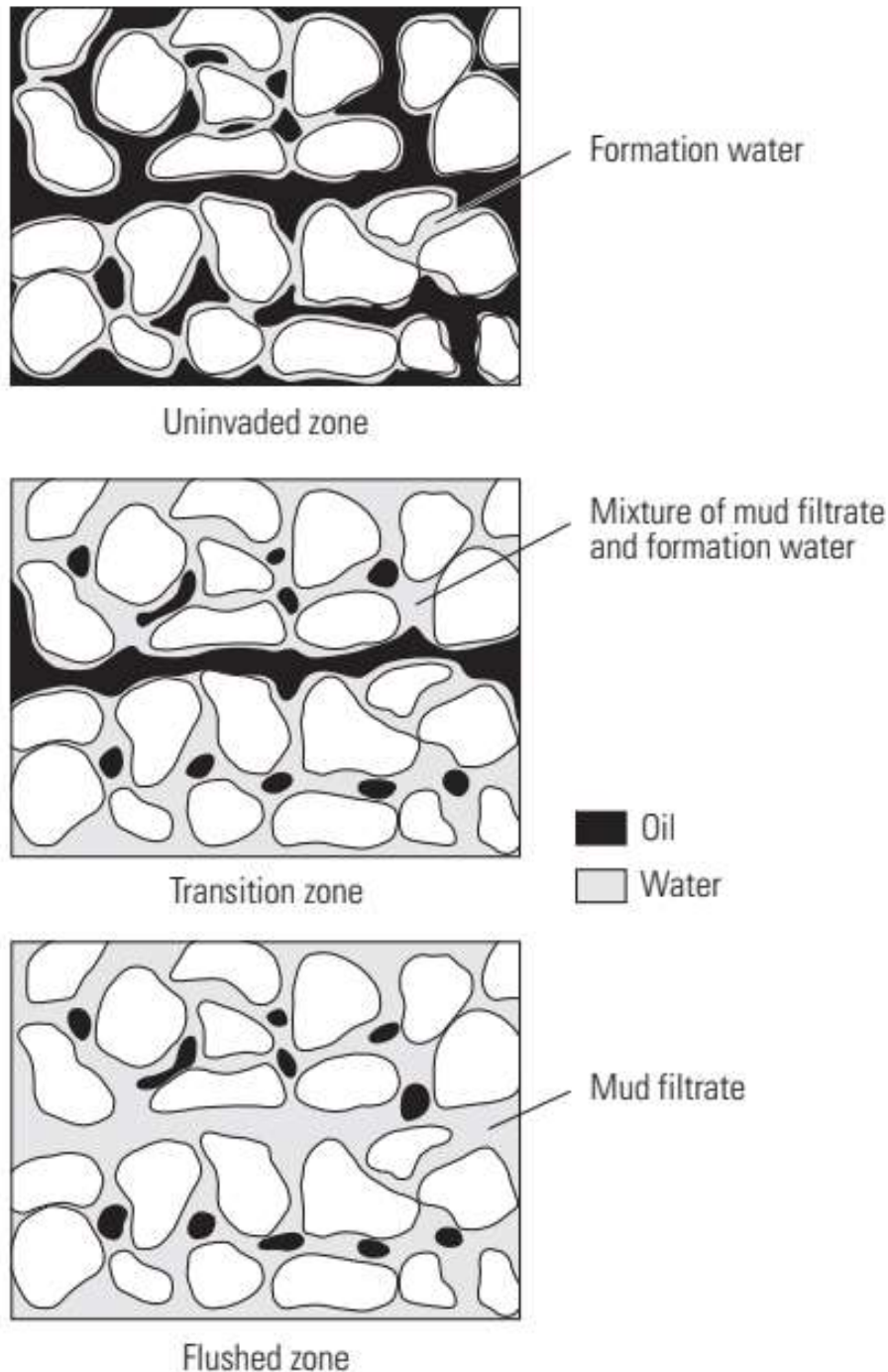
# The Invasion Process

- The simplest model, known as the step-profile model, ignores the transition zone and describes the invaded zone in terms of just two parameters, the resistivity  $R_{xo}$  and the diameter  $d_i$ .
- Figure 2.2 indicates schematically the distribution of pore fluids in the uninvaded, transition, and flushed zones. This model also assumes azimuthal symmetry around the borehole. In a horizontal well gravity can cause the heavier mud filtrate to sink below the well, leaving more of the lighter oil or gas above it. Gravity effects can also affect the fluid distribution around deviated wells or in highly dipping beds.
- Figure 2.2 is valid for both wireline and LWD logs. LWD logs are normally recorded a few hours after a formation is drilled, and therefore encounter less invasion than that seen by the wireline logs, which may be recorded several days after drilling. However this is not always the case: some LWD logs are recorded later while the drill string is being run out of the hole from a deeper total depth.

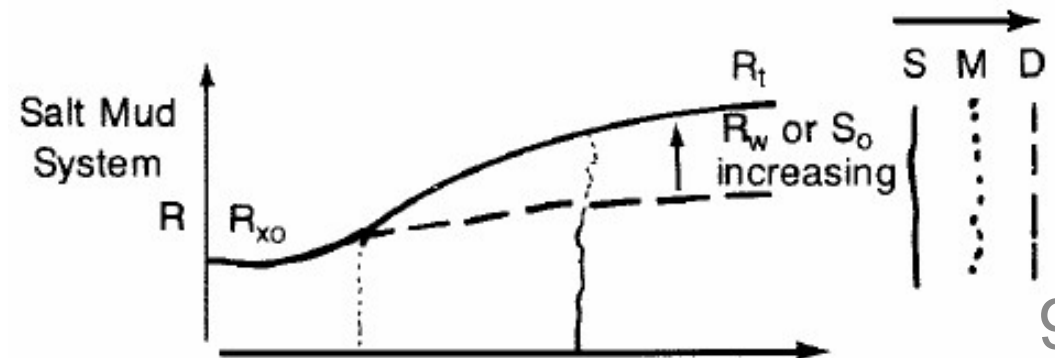
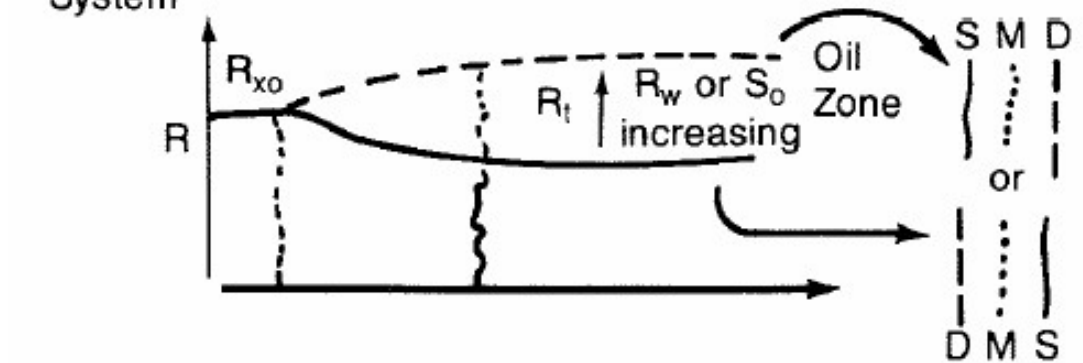
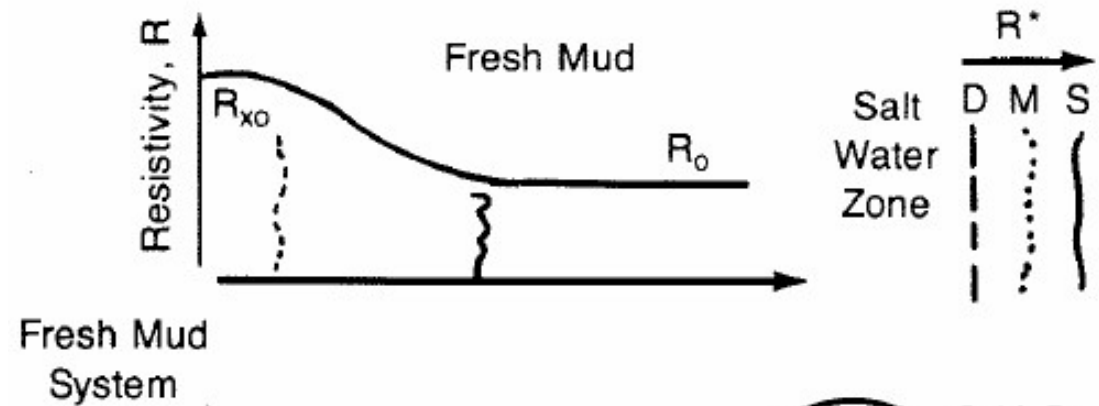
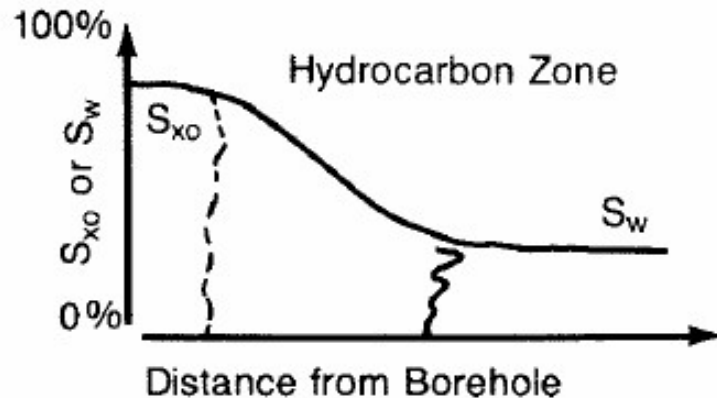
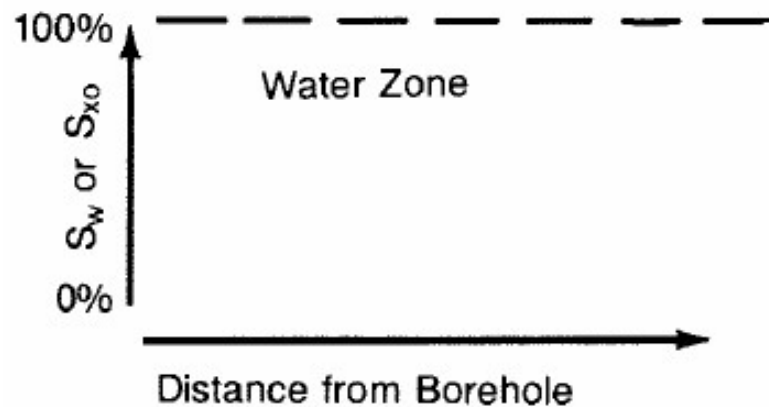
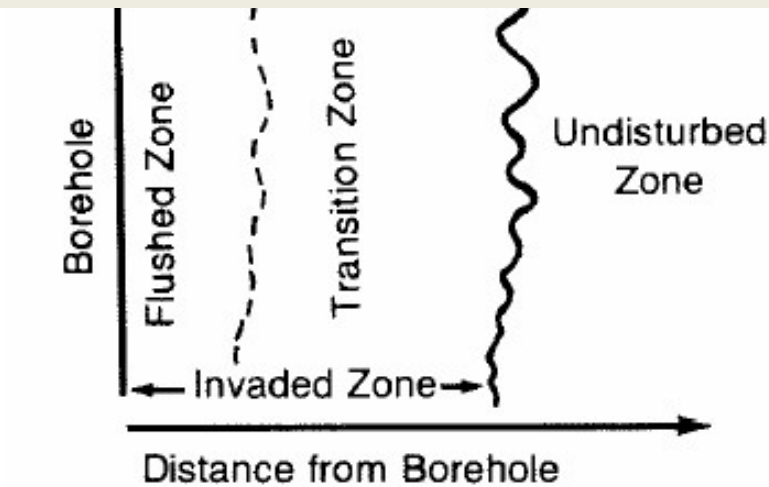


# The Invasion Process

- **Figure 2.2:** Distribution of pore fluids in zones around a well which initially contained hydrocarbons. From Dewan [2].



# Invasion Profile for different Mud Systems



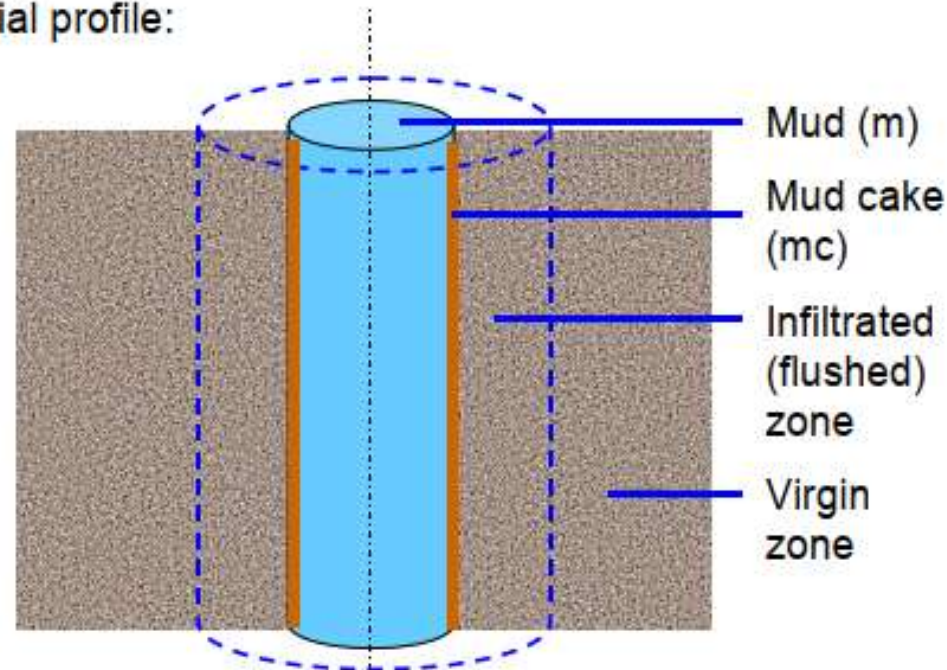
# Relationship between Mud Filtrate Invasion and Formation Porosity

- The amount of invasion which takes place is primarily dependent on the permeability of the zone and early formation of mud cake. However, before the formation of impervious mud cake, diameter of invasion will be greatest in low porosity rocks.
- Mud cake is the sealing agent which slows down invasion. As a result, high permeability zones which allow quick build-up of mud cake, invade the least and low permeability zones invade the most or deepest.
- Non-permeable zones are not invaded. This sounds counter-intuitive, but that is what happens with a good quality drilling mud. Poor quality or native mud may not form a mud cake, and invasion will proceed in proportion to the permeability.
- The diameter of invasion is often expressed as a ratio:  $d_i/d_h$  where  $d_h$  is borehole diameter.



# Relationship between Mud Filtrate Invasion and Formation Porosity

The radial profile:



**Figure 3-6:** Invaded and virgin zone in a permeable section of a borehole.

For the ratio  $d_i/d_h$ , i.e. the ratio of the diameter of the invasion  $d_i$  to the diameter of borehole  $d_h$  (see also Figure 4-6), Asquith and Krygowsky (2004) give the following values as orientation:

|                |                               |
|----------------|-------------------------------|
| $d_i/d_h = 2$  | for high porous rocks         |
| $d_i/d_h = 5$  | for intermediate porous rocks |
| $d_i/d_h = 10$ | for low porous rocks          |

# **THE TOOL CHARACTERISTICS**

# Radius (Depth) of Investigation & Vertical Resolution

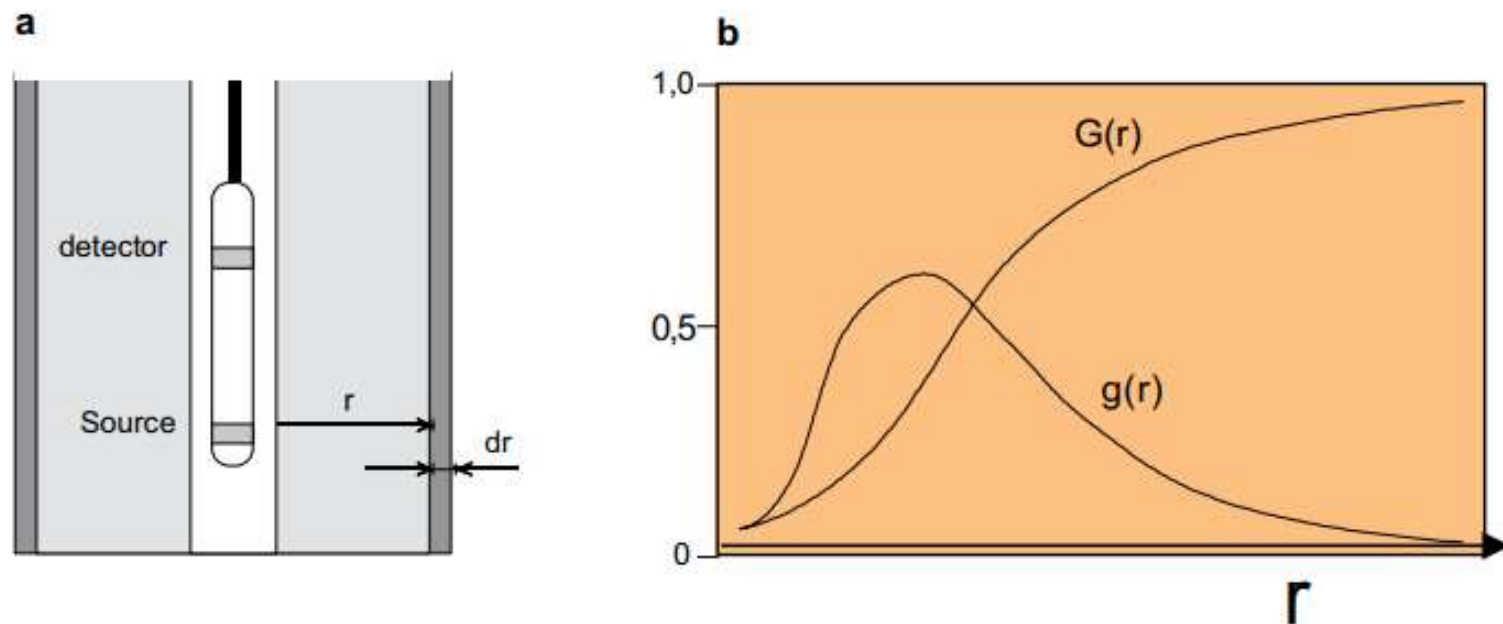
- The tool characteristic is the response of a tool controlled by the design of its elements. The tool characteristics are described by its response function and/or its vertical resolution, and the radius (depth) of investigation.
- Illustrating this for a simple transmitter-receiver tool, the differential response  $g(r)$  is the normalized contribution of a cylinder with radius  $r$  to the measured signal. Thus, the maximum can be considered as the radial distance with the strongest influence on the measured magnitude.
- As cumulative presentation the integral response  $G(r)$  describes the radial buildup of the total measured signal. For tool characterization the radius of  $G(r) = 0.5$  (50%) is frequently used as “radius of investigation 50%”. Radius of investigation 50% ( $r_{50}$ ) implies that 50% of the total signal response is originated in a radial distance below  $r_{50}$ , and 50% originates from the space outside this cylinder.

# Radius (Depth) of Investigation & Vertical Resolution

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- The radius of investigation 50% implies that 50% of the total signal response is originated in a radial distance below  $r_{50}$ , and 50% originates from the space outside this cylinder.

# Relationship between Mud Filtrate Invasion and Formation Porosity

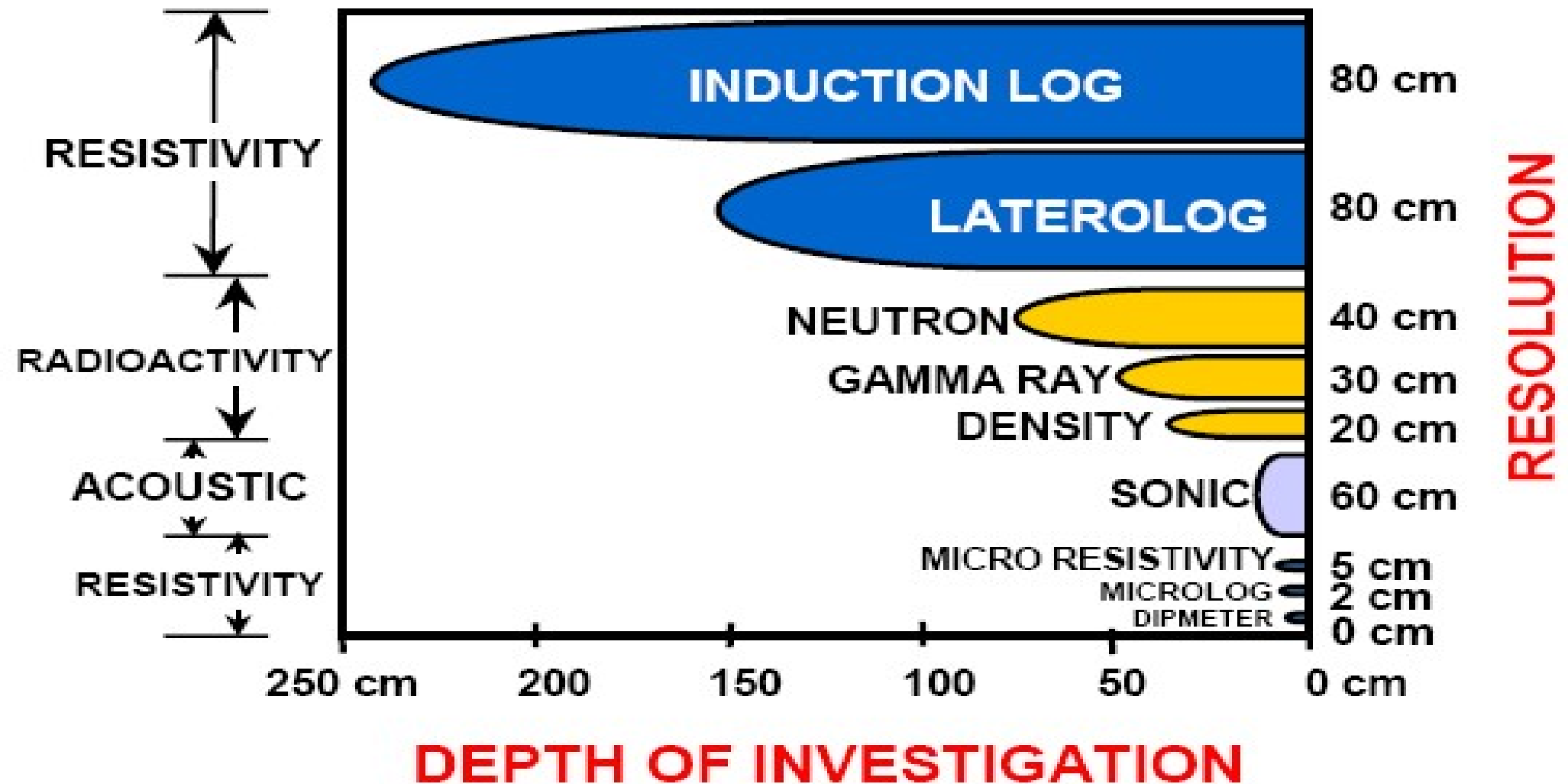
- The specific resolution of the tool, describes the ability of the tool to detect and separate thin layers individually
- In permeable zones – depending on the radial characteristics of the tool and the depth of invasion – the device measures contributions from the invaded and non-invaded zone.



**Figure 3-7:** Radial characteristic and response, after Tittman (1986) ( $g(r)$  Differential Response;  $G(r)$  Integral Response).

# Depth of Investigation vs. Resolution of Logging Tools

## Logging Tools



# **OPERATIONAL CONSIDERATIONS FOR WIRELINE LOGGING**



# Choosing A Logging Tool

It is necessary to choose the right tool to get the desired measurement.

## Considerations:

- Type of well (wildcat or development)
- Hole conditions (depth, deviation, hole size, mud type)  
Examples:
  - Oil based mud : Induction tool
  - Water based salty mud : Laterolog Tool
- Formation fluid content (fresh/salt connate water)
- Economics (cost of the job, rig time involved)

# Types of Logs To Be Run

- Logging suites generally include one resistivity and one porosity device
- The logging string will also have other tools like the gamma ray, SP and caliper tools
- However, logging suites usually have two porosity devices to give more information about rock type, hydrocarbon type and porosity
- Other considerations – to estimate permeability or to take fluid samples – require other special tools like the formation testers

# Tool Calibration & Log Quality Control

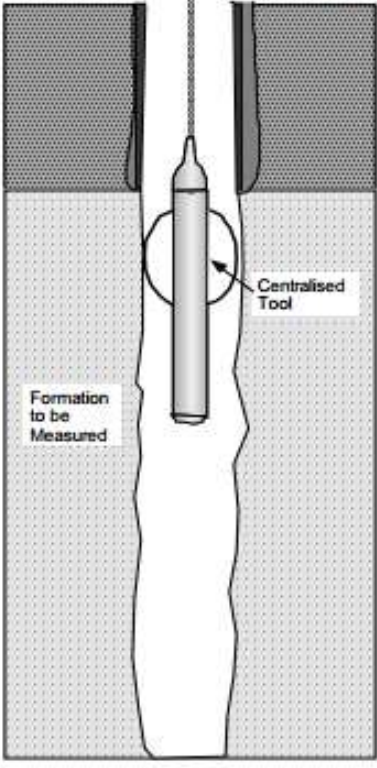
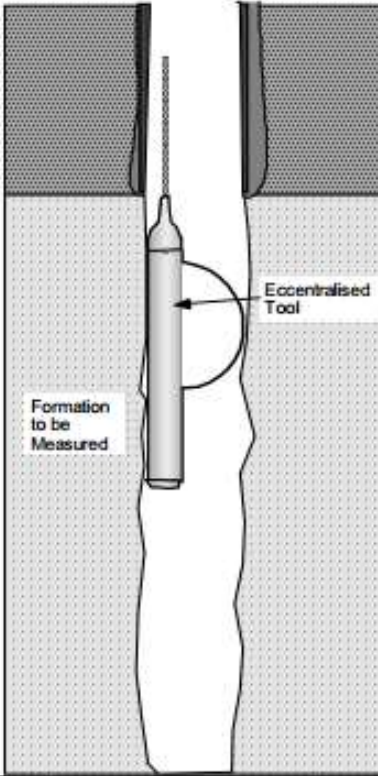
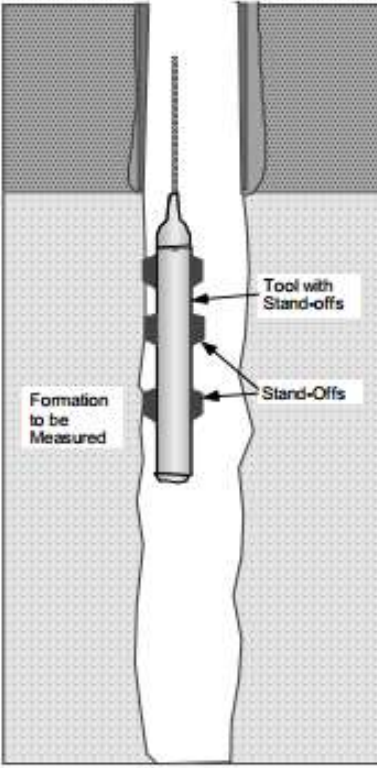
- A logging tool collects data that are converted to porosity, resistivity, and other values
- Each tool is calibrated to an industry standard. This ensures that each tool, irrespective of the type of tool or tool history or service company, reads the same value when logging the same formation (normalization may still be required between logs)
- Check tool calibrations before & after a logging job to ensure good quality log data. Check all calibrations before and after job
- Record a repeat section of about 200 ft to ensure validity of data and to explain abnormal curve response
- Compare log response with offset well logs
- Keep hole conditions (hole size, mud type, tool centralization) in mind when interpreting log data
- Ensure that logging speeds are as recommended by the service company.

# Effect of Borehole Fluids

- Additives such as barite affect density, gamma ray and photoelectric effect measurements.
- Foam and air muds will not transmit sonic signals. Neutron tools are also affected.
- Oil based mud will not allow current to pass so electrical logs will not work
- Mud salinity affects electrical and induction tools in different manners.
- The mud type, salinity and additives must be known so that the appropriate corrections can be made.

# Tool Positioning

- The position of the tool in the hole depends on the type of measurement. Tools are positioned as shown below:

| Centralized Tool Positioning                                                                                                                                             | Eccentralized Tool                                                                                                                                                    | Standoff Tool                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        |                                                                                    |                                                                  |
| <p>Some tools are run centralized in the borehole in order to measure properly. These include laterolog and sonic devices. Special centralizers are put on the tool.</p> | <p>Some tools are run eccentred, pushed, against the borehole wall. In some cases this is done with an eccentraliser. In other cases a caliper arm does this job.</p> | <p>Some tools are run with “stand-offs” to position them at a fixed distance from the wall. The induction family are usually run in this manner.</p> |

- Knuckle-joints act as crossovers in long combination tools with some tools eccentered and some with stand-offs.

# Tool Failures

- The standard procedure if a tool fails is to replace it with a backup and carry on as before. In general, if the program has specified that a tool is to be run over a specified interval (particularly a reservoir interval), then it is best to ensure that good-quality data are acquired, even if it means making additional runs. However, the following situations may arise:
- When logging with LWD and in the hole drilling near the end of a section, it may be far more cost-effective, if the data are not critical, to simply carry on drilling and re-acquire the data during a check trip. In some cases, the memory data may still be usable.
- If a wireline tool starts to act erratically, it may not justify re-running the tool, since the data may not be critical over the particular interval and can sometimes be corrected by post-processing the raw tool data at the surface.

# Tool Failures

- If an advanced tool fails, it sometimes happens that no backup tool is available on the rig. The choices then are to either try and repair the tool, arrange a backup tool to be sent from another location, or replace the tool with an earlier version with less capability. Most commonly, the latter option is chosen.
- In the event of failures occurring with potential safety implications (e.g., explosive charges going off accidentally), it is normal for operations to be suspended until a full investigation has been carried out to establish the cause.

# Stuck Tools

- Fairly regularly during logging operations, tools get stuck in the hole, either temporarily or permanently. There are often indications of bad hole conditions during drilling, and the logging program may be adjusted to take this into consideration. In general it is found that the longer a hole is left open, the greater the likelihood of hole problems occurring. Three sorts of sticking are found to occur:
  - differential sticking,
  - key seating, and
  - holding up.
- Differential sticking occurs when either the cable or the toolstring becomes embedded in the borehole wall and gets held in place by the differential pressure between the mud and the formation. In such a situation, it is impossible to move the toolstring either up or down. The usual procedure is to alternately pull and slack off on the tool, pulling up to 90% of the weak point of the cable (that point at which the cablehead will shear off the top of the toolstring). This procedure is often successful, and a tool may become free after 30 minutes or so of cycling.



# Stuck Tools

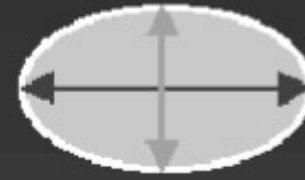
- Key seating occurs when a groove is cut into one side of the borehole, which allows the cable, but not the toolstring, to pass upward. The toolstring is effectively locked in place at a certain depth. Unfortunately this often means that when the weak point is broken, the toolstring will drop to the bottom of the hole and may be hard to recover or may be damaged.
- Holding up occurs when a constriction, blockage, dogleg, or shelf occurs in the borehole such that the toolstring may not pass a certain depth, although it can be retrieved. The usual practice in such a situation is to pull out of the hole and reconfigure the tool in some way, making the toolstring either shorter or, in some cases, longer in an attempt to work past the holdup depth.
- Once it is no longer possible to recover a tool on wireline, there are two options: cutting-and-threading or breaking the weak point. In the cut-and-thread technique, the cable is cut at the surface. The drillpipe (with a special fishing head called an overshot) is run into the hole with the cable being laboriously threaded through each stand of pipe.

# Stuck Tools

- At a certain depth it may be possible to install an Side-Entry Sub (SES), which means that when the toolstring is being recovered, log data may be acquired in a similar fashion to pipe-conveyed logging. If the weakpoint has been broken (either accidentally or on purpose), cutting-and-threading is not possible. Then the pipe is run into the hole until the toolstring is tagged, although often the tool will drop to the bottom of the hole before it can be engaged by the overshot.
- Most oil companies will specify that they do not wish to break the weak point on purpose, even if the cut-and-thread technique is much slower. This is particularly true where nuclear sources have been lost in the hole, and every effort should be made to ensure that they are recovered undamaged at the surface. In the event of nuclear sources being lost irretrievably in the hole, which occasionally happens (particularly with LWD tools), there are usually special procedures to be followed involving notification of government bodies and steps taken to minimize the risk of and monitor any potential nuclear contamination that may occur.

# **Well Logging Methods – Lecture 4**

Dual Axis (x,y)



**CALIPER LOG**

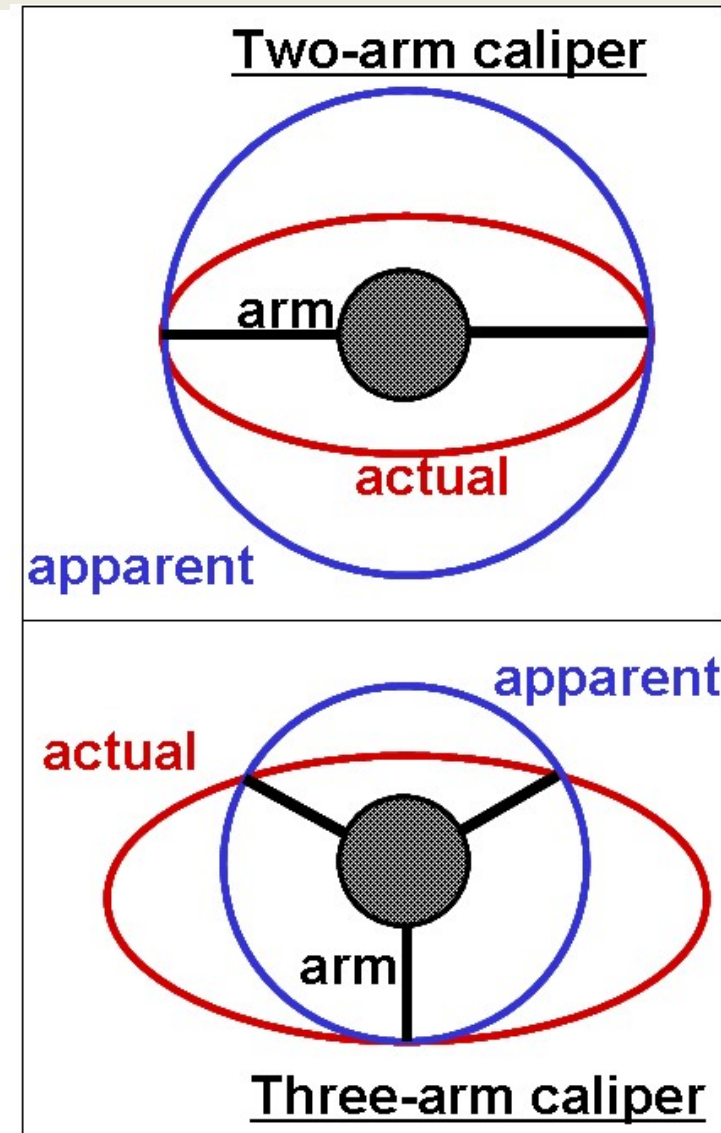
# Calipers

- **The Caliper Log**

- is a continuous profile of the borehole wall showing variations in borehole diameter.

- **Uses**

- Indication of Hole volume and diameter: required for mud and cement volume in hole
- Environmental corrections: hole diameter from the caliper is used in charts for other logs and to indicate thickness of the mud cake for  $R_{xo}$  tool corrections.
- Qualitative indication of permeability: due to development of mudcake in association with gamma ray log.
- Correlation for lithological assessment
- Selection of consolidated formations for wireline pressure tests, fluid sampling, packer seating and for determining casing setting depth
- Log quality control: Indication that the hole is rough is a warning that the measurements are from tools pushed against the borehole wall such as Density, Neutron, Micro-resistivity curves, may not be reliable. Thus it is an indicator of Hole Quality.



# Calipers

- **Physics of Measurement**

- For wireline tools, the physical movement of arms on the tool (*Mechanical Systems*) is converted to a diameter measurement through electrical circuitry. The angle between the pad mechanism and the tool axis is a measure of pad distance from the tool.
- In some MWD tools the caliper curve is generated based on the response of the detectors as the tool rotates (gyro system). Others use ultrasonic sensors which measure the two-way travel time taken for an acoustic pulse from the sensor to the formation wall and back. Conversion into distance via mud wave velocity gives a detailed azimuthal caliper. Some tools come with built in calipers by design.

- **Properties**

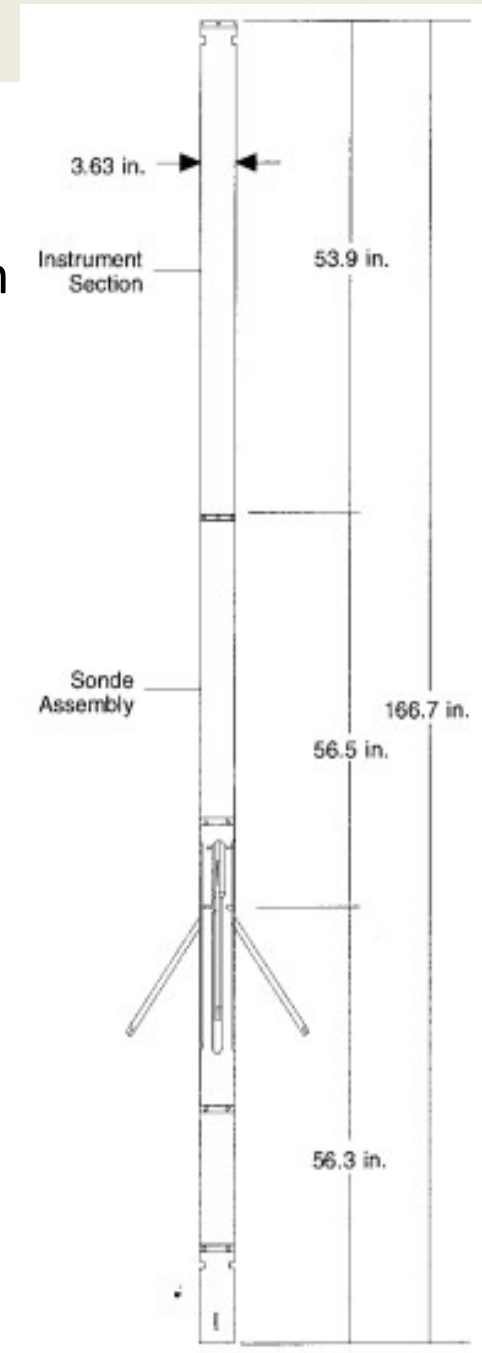
- two, three, four arms or more; linked or independent

- **Operational Constraints**

- In Open hole, the tool is run centered
- In Cased hole, the tool is run eccentric
- Can be run in air, water or oil-based mud

- **Secondary Effects**

- In highly deviated holes, the caliper mechanism may not be strong enough to support the weight of the logging tool thus may not indicate actual hole diameter
- Mud cake indications can be masked by a wash out



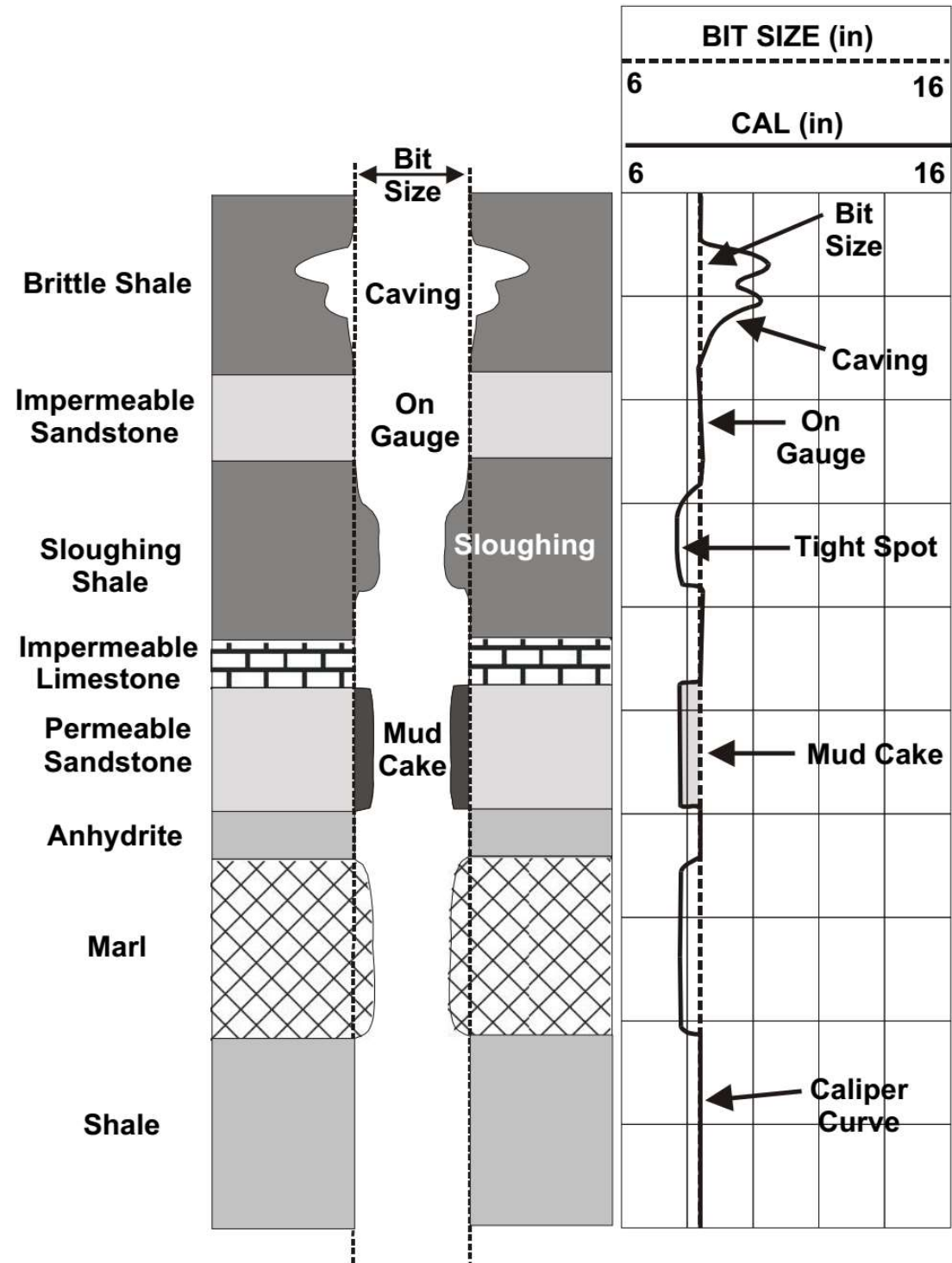
4-arm Caliper Tool

# Caliper log

- **Log Presentation**

- The caliper logs are plotted in track 1 with the drilling bit size for comparison, or as a differential caliper reading,
- The scale is generally given in inches, which is standard for measuring bit sizes..

***N.B.:** Check that the caliper is reading correctly in casing and in non-permeable zones that are not washed out (when a hole is the same diameter as the bit-size it is called “on gauge”).*



Typical caliper responses to various lithologies



# Factors affecting Caliper tool responses

| Hole Diameter         | Cause                                                                                                         | Possible Lithologies                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| On Gauge              | Well consolidated formations<br>Non-permeable formations.                                                     | Massive sandstones<br>Calcareous shales<br>Igneous rocks<br>Metamorphic rocks                     |
| Larger than Bit Size  | 1. Formation soluble in drilling mud.<br>2. Formations weak and cave in.                                      | 1. Salt formations drilled with fresh water.<br>2. Unconsolidated sands, gravels, brittle shales. |
| Smaller than Bit Size | 1. Formations swell and flow into borehole.<br>2. Development of mudcake for porous and permeable formations. | 1. Swelling shales.<br>2. Porous, permeable sandstones.                                           |

# **THE SPONTANEOUS POTENTIAL (SP) LOG**

# Spontaneous Potential (SP)

- is a record of small direct current (DC) voltage differences between the naturally occurring potential of a moveable electrode in the water-based mud filled well bore, and the potential of a fixed electrode located at the surface measured in **millivolts**. The SP is mostly measured simultaneously with the open hole resistivity log.
- The origin of the SP is the charge exchange process driven by the difference of concentration (salinity) between formation water  $C_w$  and the mud filtrate  $C_{mf}$ . The concentration difference (in most cases  $C_w > C_{mf}$ ) results in an ion motion controlled primarily by **electrochemical properties** of the:
  - Non-permeable shales creating a positive **membrane potential**  $E_m$  as a result of anion-blocking (ion-selectivity) of clay minerals i.e. selective to sodium ions (Na+) but impervious to chloride ions (Cl-).
  - **Liquid-junction potential**  $E_j$  : which arises due to net transfer of negative electric charges from the more concentrated sodium chloride solution to the less concentrated at the boundary between the invaded and un-invaded zones in the formation

# Spontaneous Potential - Uses

- Correlation of formations from well to well: Curves are scanned for similarities in shape and magnitude.
- Gross Lithology: Reservoir are shown as deflections (either positive or negative) from a relatively stable (and arbitrary) shale baseline. The direction of the deflection is determined by the relative salinities (resistivities) of the formation water ( $R_w$ ) and the mud filtrate ( $R_{mf}$ ), and is not directly related to formation porosity or permeability.
- As a rule of thumb the following relationships are true:
  - If  $R_{mf} > R_w$ , then SP deflection is negative
  - If  $R_{mf} = R_w$ , then SP deflection is zero
  - If  $R_{mf} < R_w$ , then SP deflection is positive
- Qualitative indication of permeability: The presence of an SP (positive or negative) opposite a bed indicates permeability. Only a minimal amount of permeability is required to develop an SP and therefore there is no technique to determine the magnitude of permeability from SP as permeability may be ionic not hydraulic.

# Spontaneous Potential - Uses

- Identification of depositional environments: Depo. environment can be inferred from the shape of the SP. It however can be ambiguous, thus it should be used in support of other data in an area of interest.
- Estimate of Clay Content: This method assumes a constant  $R_w$  for all zones. It also assumes that the response of SP to shaliness is linear.

Shale/clay volume equation:

$$V_{clay} = V_{shale} = \left( \frac{SP_{clean} - SP_{log}}{SP_{clean} - SP_{shale}} \right)$$

$V_{clay} = V_{shale}$  = Shale or clay volume.

$SP_{log}$  = SP in the zone of interest (read from the log).

$SP_{clean}$  = maximum SP deflection from a nearby clean wet zone in the same well.

$SP_{shale}$  = SP value at the shale baseline (often considered to be zero).

# Spontaneous Potential - Uses

- Estimate of formation water resistivity: Since the electrolyte's specific electrical resistivity and concentration are inverse with a good approximation, the ratio  $C_w/C_{mf}$  can be replaced with the ratio  $R_{mf}/R_w$ .
- The resistivity of the formation water  $R_w$  is a fundamental input for the calculation of water saturation. The equation offers the possibility of a determination from the SP-log at the formation temperature with a water sample.
- For  $R_w$  determination,  $R_{mf}$  – the mud resistivity (at formation temperature), SSP – the difference between the SP level in a clean zone and the shale baseline, and T – the formation temperature are used as input.
- A monogram is frequently used for  $R_w$  determination.
- A good estimate of  $R_w$  at formation temperature can be obtained from the following equation: ( $R_{mf}$  @ formation temp.)

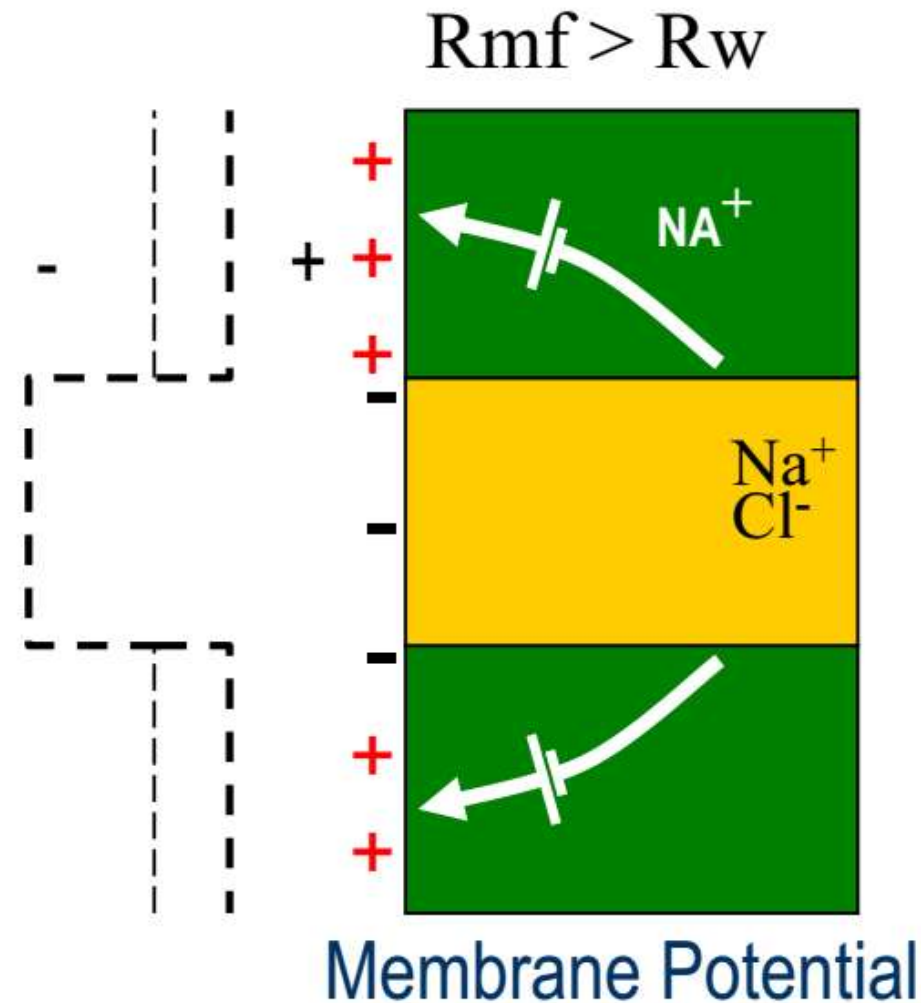
$$SSP = k_{SP} \cdot \log\left(\frac{C_w}{C_{mf}}\right) = k_{SP} \cdot \log\left(\frac{R_{mf}}{R_w}\right)$$

$$R_w = 10^{(K \cdot \log(R_{mf}) + SP) / K}$$

# Spontaneous Potential - Physics of Measurement

- **Shale membrane potential  $E_m$ :**

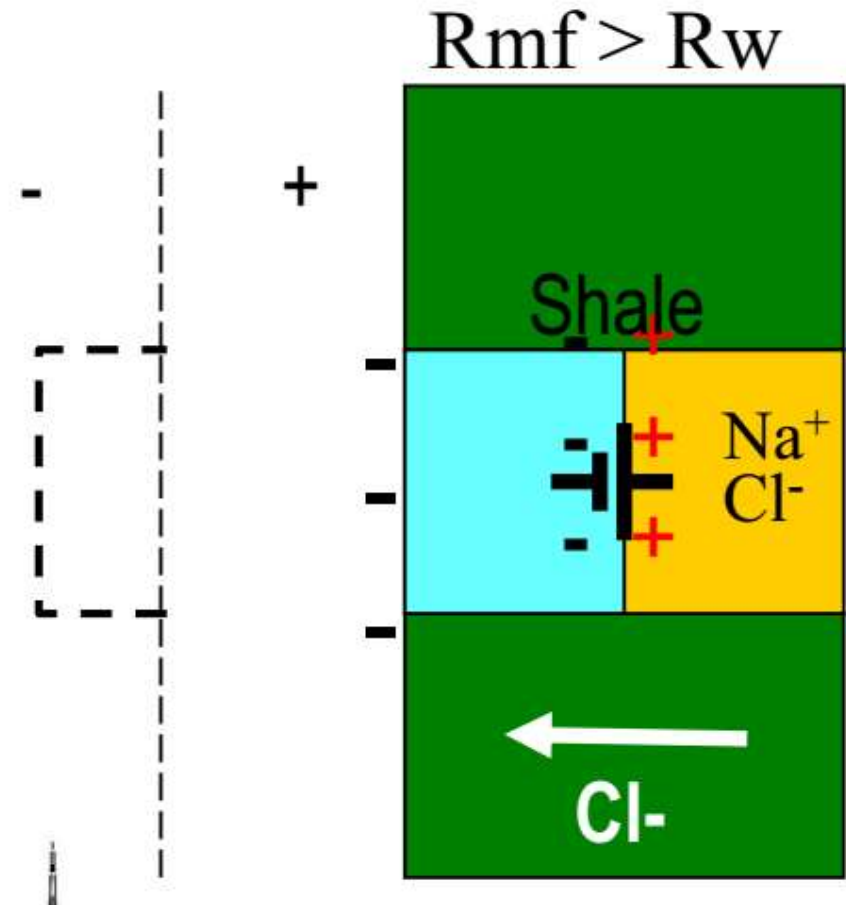
- Ion selective membranes allow ions with certain electrical charges (positive or negative) to pass through them. Shales act like ion selective membranes as they allow only cations (positively charged ions) to pass through. This flow of charged particles is an electric current.
- The Membrane Potential graphic illustrates the current flow for the case where  $R_{mf} > R_w$  (fresh mud). The flow would be reversed if  $R_{mf} < R_w$  (salt mud).
- Membrane potential is normally 5 times larger than the liquid junction potential





# Spontaneous Potential - Physics of Measurement

- **Liquid-junction potential  $E_j$ :**
  - Ions will move between liquids of differing salinities. The ions will normally move from higher salinity to lower. The most abundant ions in a wellbore are sodium ( $\text{Na}^+$ ) and chlorine ( $\text{Cl}^-$ ).
  - Because chlorine is a smaller atom it moves easier and faster through the formation (permeable sandstone).
  - Current flow will be from the lower salinity to the higher salinity. In this case from the invaded zone to the true zone.



# Spontaneous Potential

- **Operational Constraints**

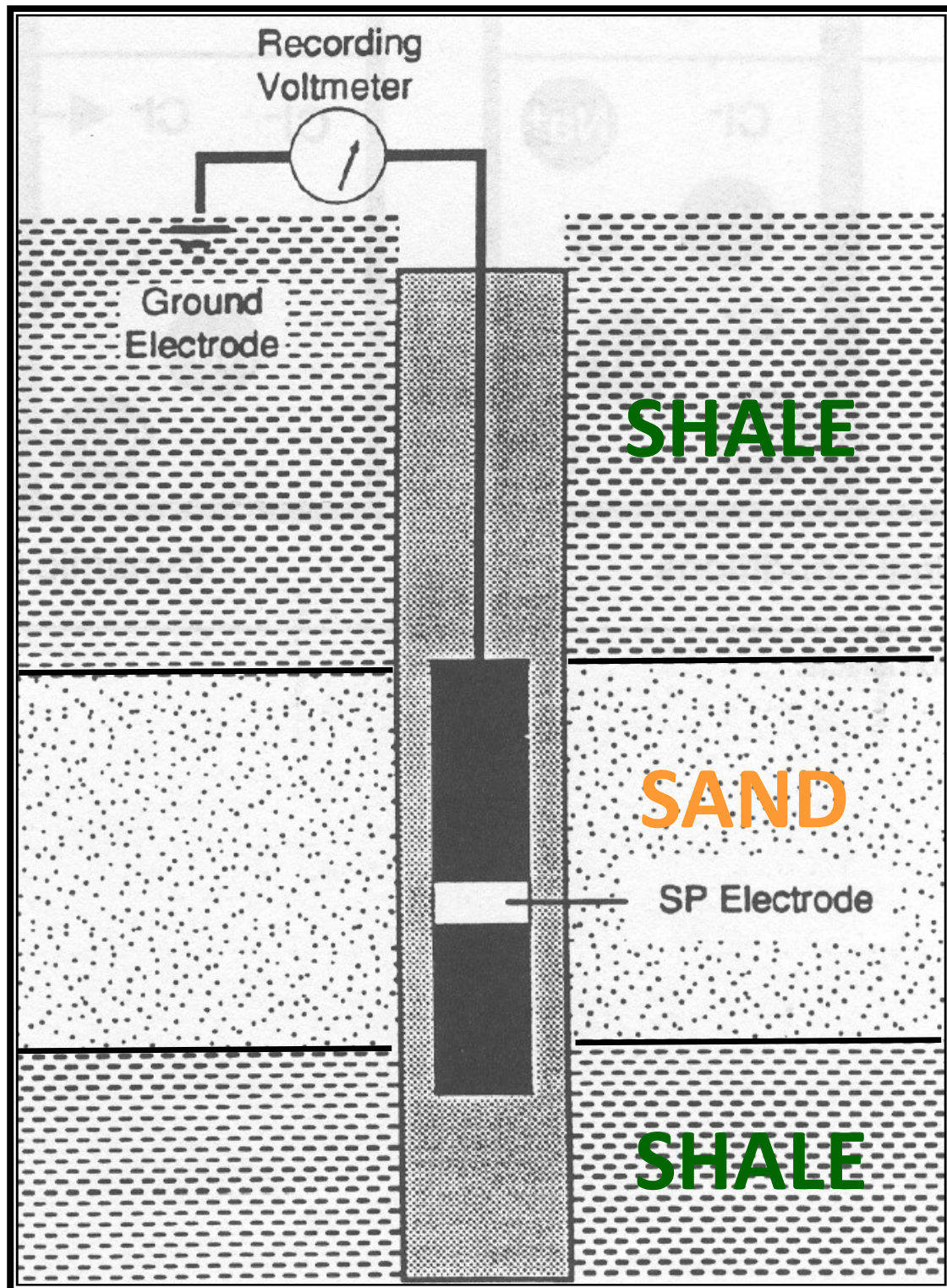
- Only run in Open hole with the tool centered
- Only run in water-based mud. Because a conductive fluid is needed in the borehole for the SP log to operate, it **cannot be used** in non-conductive (i.e. oil-based) drilling muds.
- There is no separate SP tool; the SP band is usually run with induction logs and other electric logs

- **Environmental Effects**

- In highly deviated holes, the caliper mechanism may not be strong enough to support the weight of the logging tool thus may not indicate actual hole diameter
- Mud cake indications can be masked by a wash out

- **Caution as regards interpretation**

- Unlike other logging tools which are displayed on a specific scale with a specified reference value, the SP has no specified origin and values used for computation are referenced to deflection from nearby shale baseline established by the interpreter.



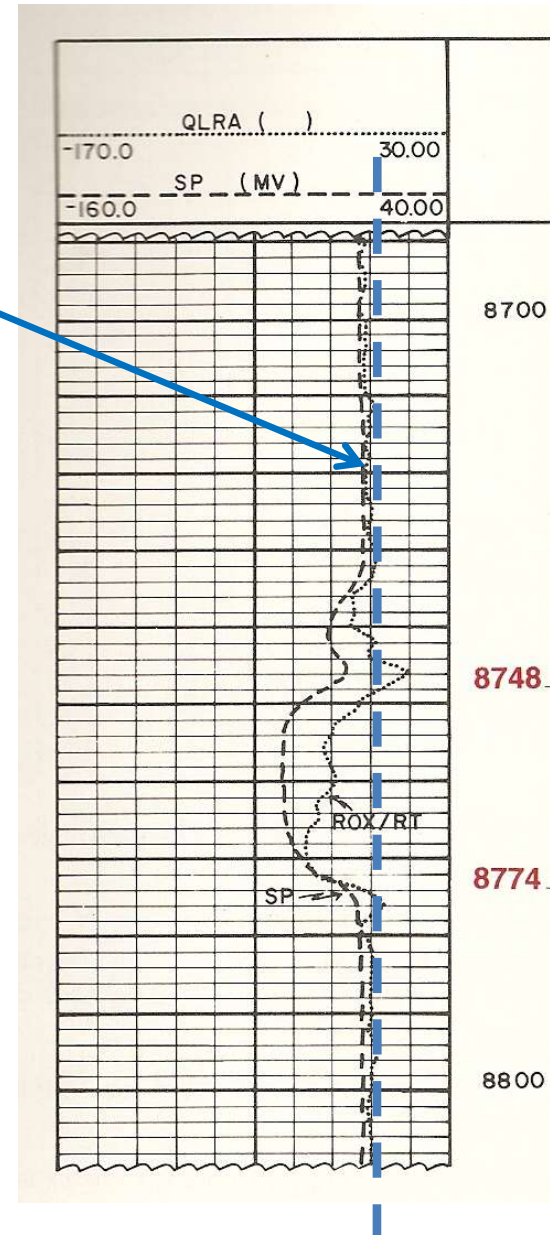
## THE SP TOOL

---

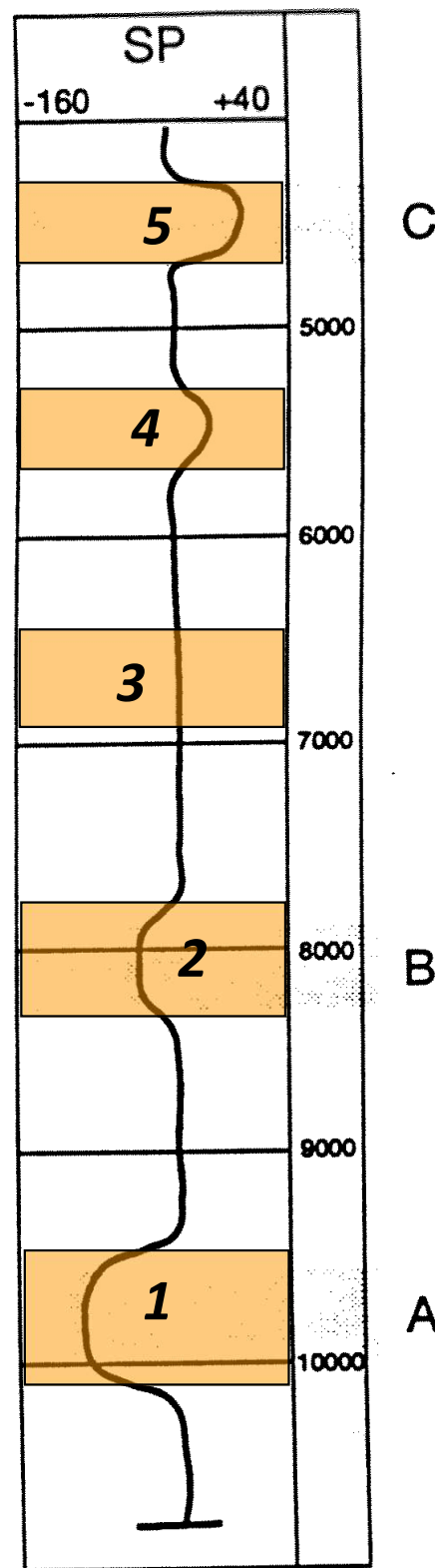
- One movable electrode
- Insulators on either side
- Surface ground electrode – at a stable potential

# Shale Base Line

- The **SP response of shales** is relatively **constant** and follows a straight line called **a shale baseline**.
- SP curve **deflections** are measured from this shale baseline.
- *Permeable zones are indicated where there is **SP deflection** from **the shale baseline**.*





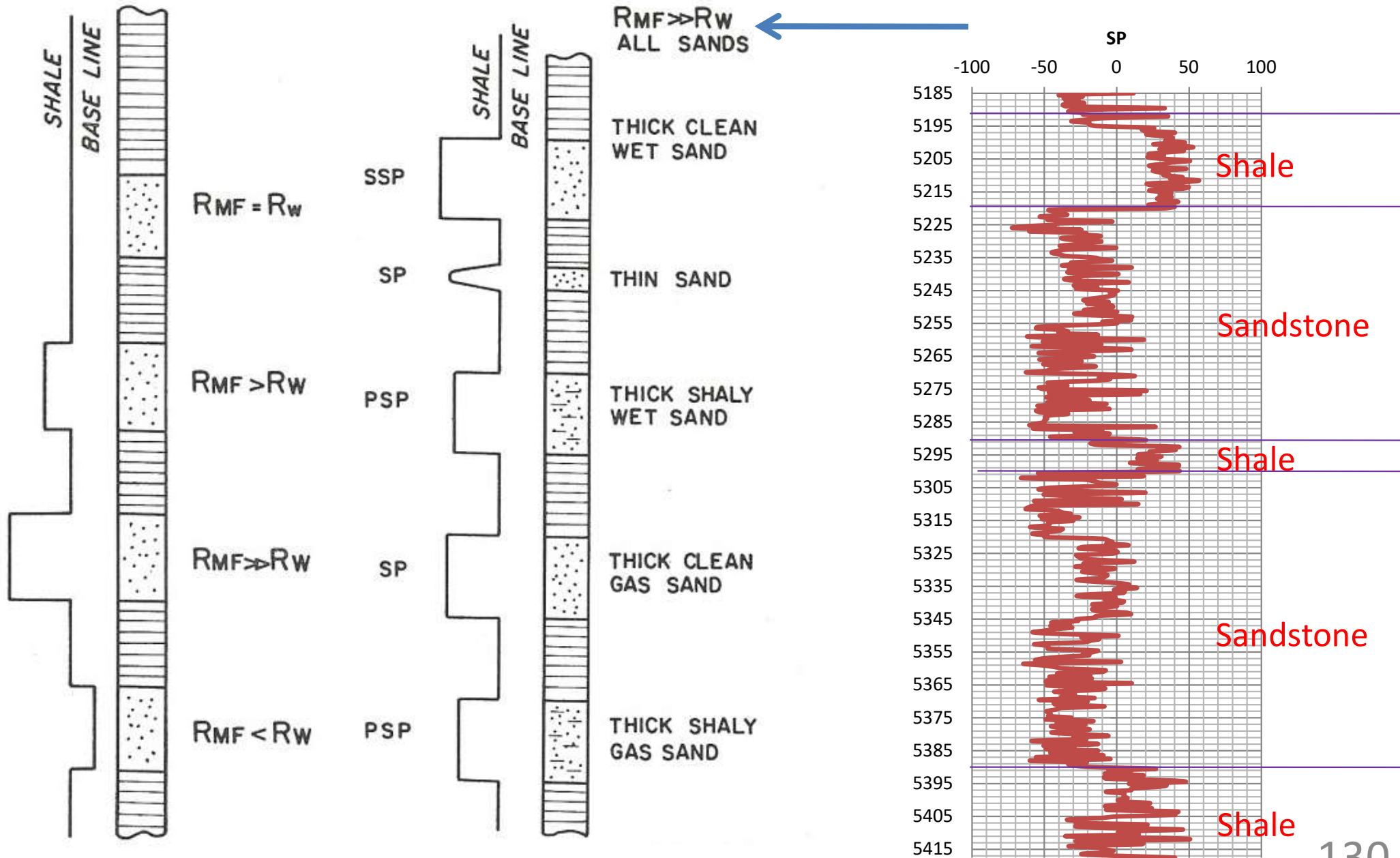


# TYPICAL SP RESPONSES

– BASED ON THE DIFFERENCE BETWEEN  $R_w$  and  $R_{mf}$ .

1.  $R_{mf} \gg R_w$  - Amplitude large and negative
2.  $R_{mf} > R_w$  - Amplitude negative but not large
3.  $R_{mf} = R_w$  - No SP deflection
4.  $R_{mf} < R_w$  - Amplitude positive but not large
5.  $R_{mf} \ll R_w$  - Amplitude large and positive

# SP deflection



# Static spontaneous potential (SSP)

- The concept of **static spontaneous potential (SSP)** is important because SSP represents **the maximum SP** that a thick, shale-free, porous and permeable formation can have for a given ratio between  $R_{mf} / R_w$ .
- It is the maximum difference between the most negative and most positive level defined as:

$$SSP = |E_D| + |E_M| = k_{SP} \cdot \log\left(\frac{C_w}{C_{mf}}\right)$$

- SSP is determined by formula or chart and is a necessary element for determining accurate values of  $R_w$  and **volume of shale**.



# Static spontaneous potential (SSP)

- The parameter  $k_{sp}$  is controlled by temperature, type of dissolved salts (electrolyte), and membrane activity of shale; it is about 71 mV at 25°C (or °F) for NaCl. For practical application, the following empirical equation can be used:
  - $k_{SSP}(\text{in mV}) = 65 + 0.24 * T (\text{in } ^\circ\text{C})$
  - $k_{SSP}(\text{in mV}) = 61 + 0.133 * T (\text{in } ^\circ\text{F})$
- The SP curve is influenced by several effects (correction charts eliminate most effects):
  - Bed thickness and borehole size
  - Depth of invasion
  - Hydrocarbons:  $S_w < 1$  causes a reduced magnitude of the SP in the reservoir
- Frequently a “shale baseline shift” can be observed as a result of change of clay mineralogy, water mineralization and/or temperature with depth.

**Resistivity of Formation Water  
( $R_w$ ) Calculates from the SP Curve**

# Resistivity of Formation Water ( $R_w$ )

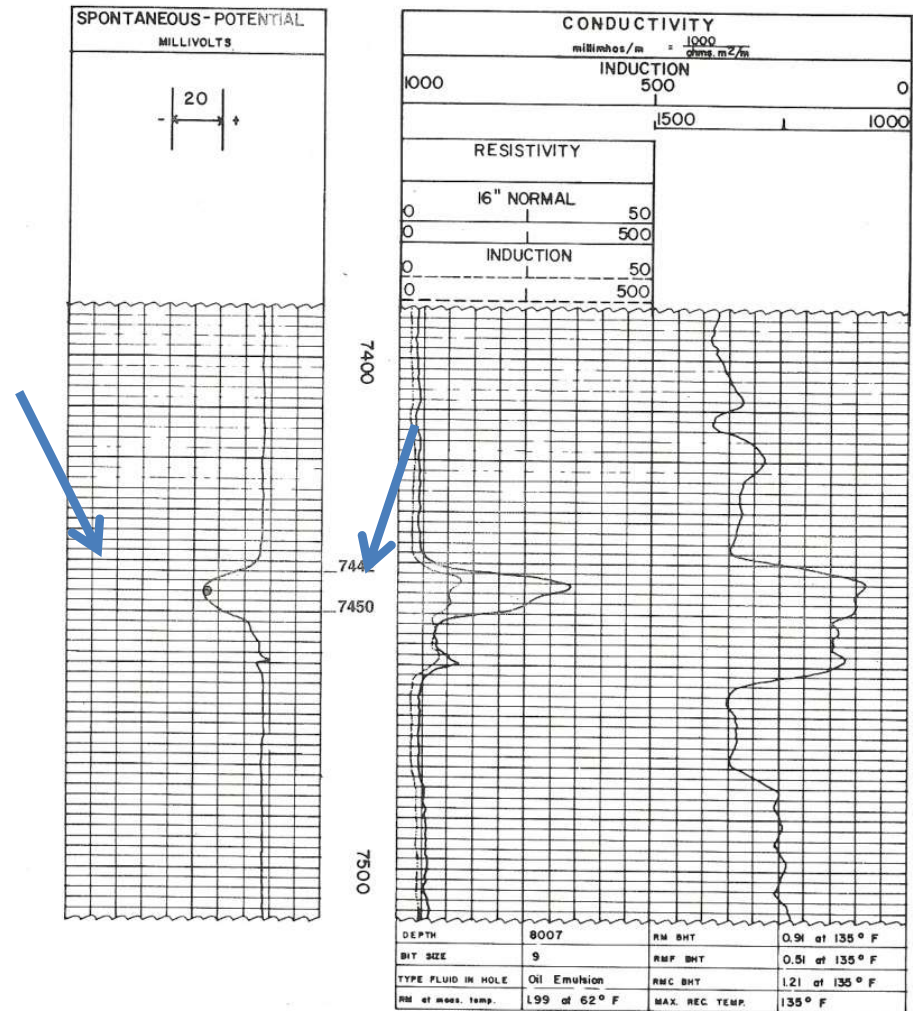
## Calculated from the SP Curve

### Step 0:

- Setup Shale-base-line
- read **SP log recording** and **16" Normal Resistivity log reading** every 2 ft from 7430 to 7460

### Step 1:

- determine the formation temperature ( $T_f$ )



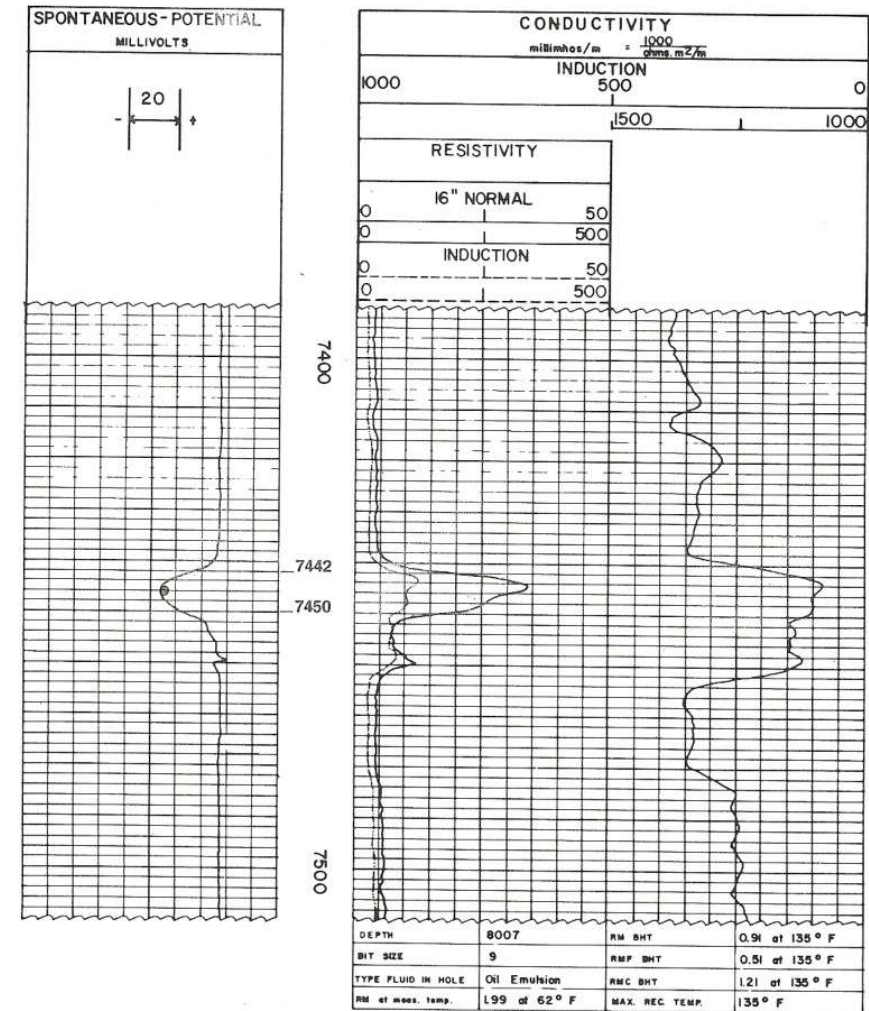
# Resistivity of Formation Water ( $R_w$ ) Calculated from the SP Curve (Cont.)

## Step 2:

- correct the resistivities of the mud filtrate ( $R_{mf}$ ) and drilling mud ( $R_m$ ) to formation temperature ( $T_f$ )

## Step 3:

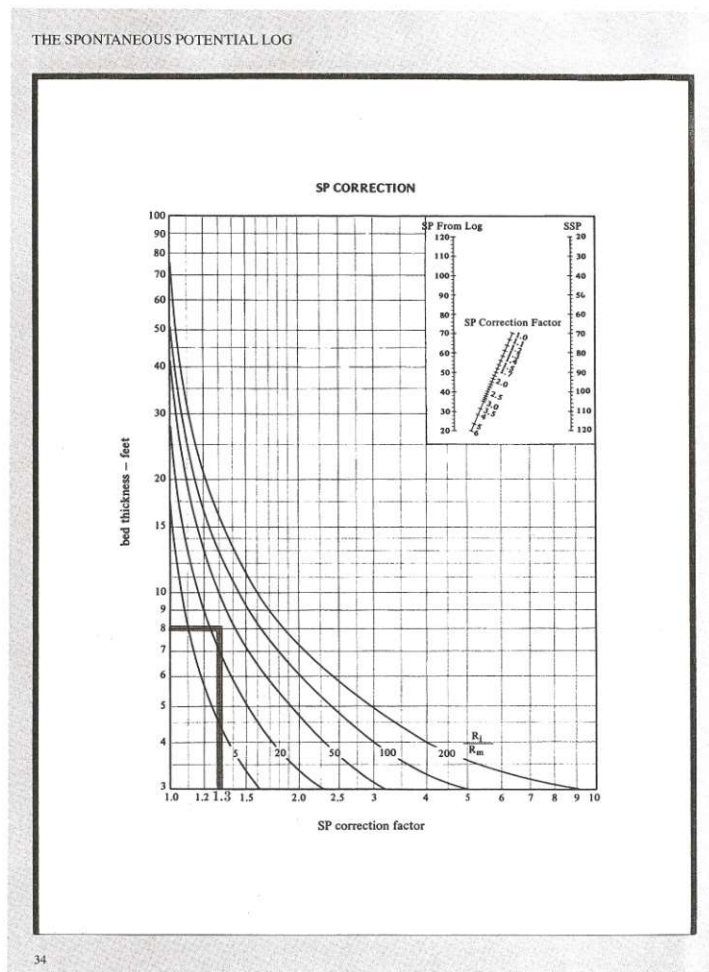
- Determine the SP reading (in water-bearing zone or wet-formation)



# Resistivity of Formation Water ( $R_w$ ) Calculated from the SP Curve (Cont.)

## Step 4:

- Correct SP to SSP

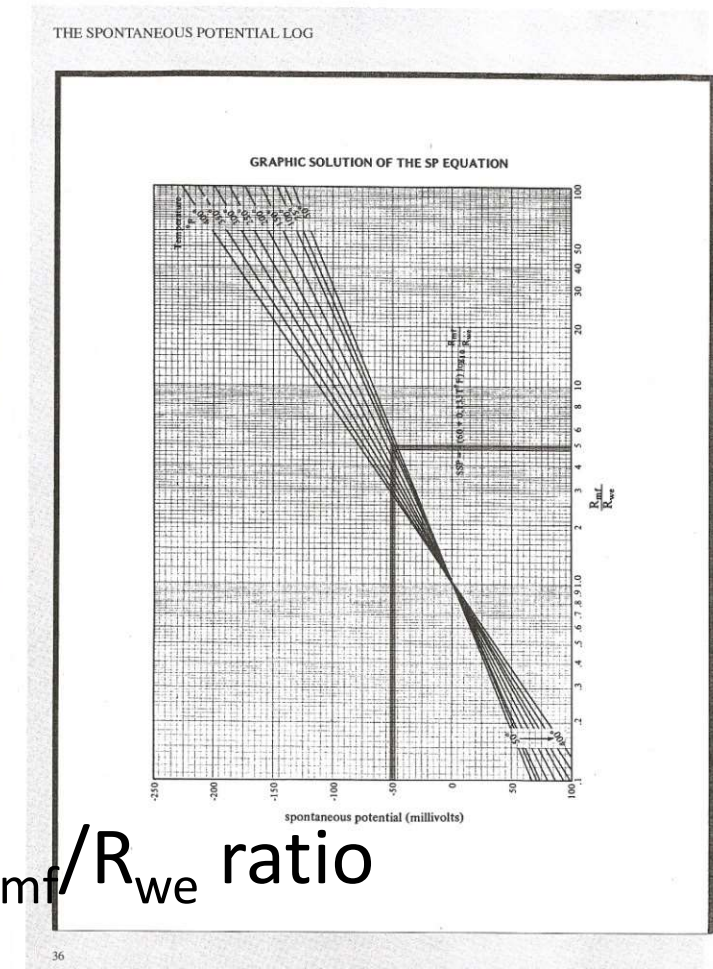


## Step 5:

- Determine  $R_{mf}/R_{we}$  ratio

## Step 6:

- Determine  $R_{we}$



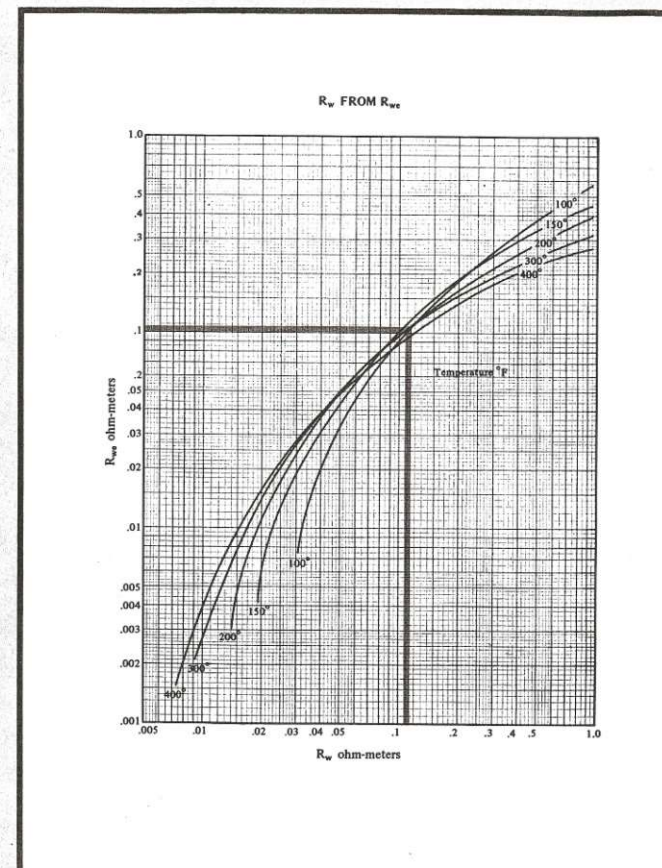


# Resistivity of Formation Water ( $R_w$ ) Calculated from the SP Curve (Cont.)

## Step 7:

- Correct  $R_{we}$  to  $R_w$

THE SPONTANEOUS POTENTIAL LOG



# Volume of Shale Calculation

- The SP log can be used to calculate the volume of shale in a permeable zone by the following formula:

$$V_{sh} = 1.0 - \frac{PSP}{SSP}$$

- The volume of shale in a sand can be used in the evaluation of shaly sand reservoirs.

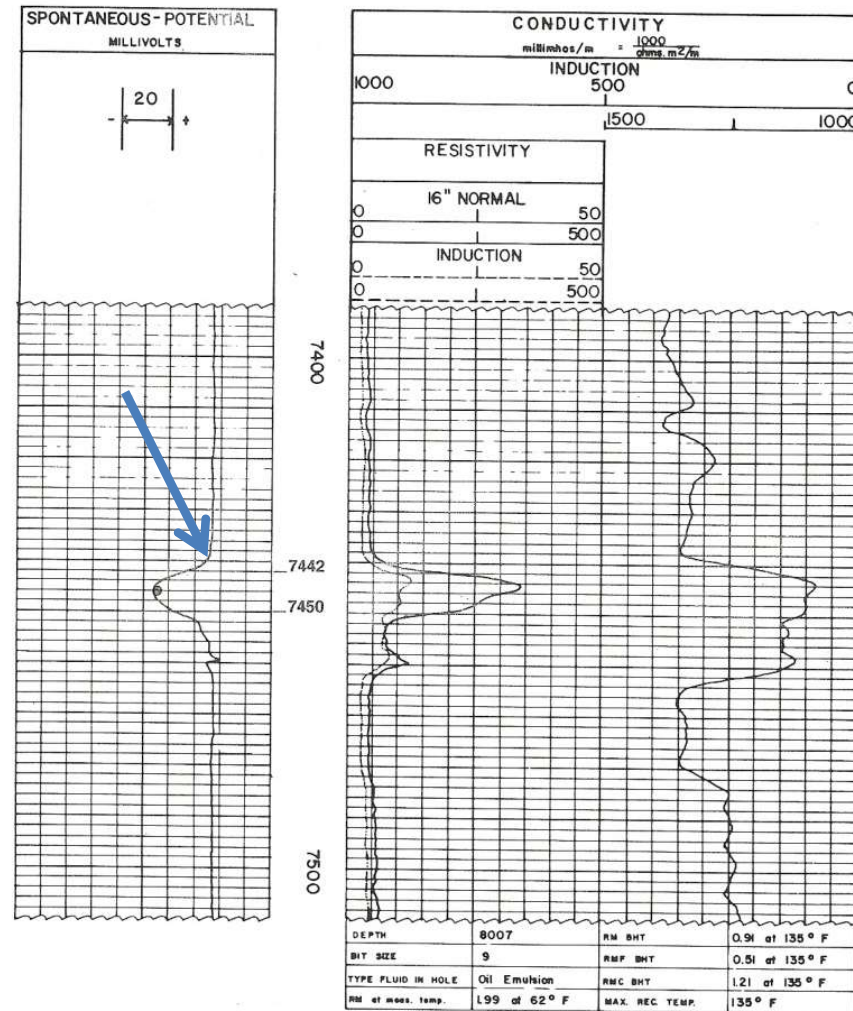


# Calculate Vsh from the SP Curve

- Use **SP log recording (PSP)** every 2 ft from 7430 to 7460

- SSP** is from step 4

$$V_{sh} = 1.0 - \frac{PSP}{SSP}$$



# **THE GAMMA RAY (GR) LOG**

# Natural Radioactivity

- Natural radioactivity is a spontaneous decay of a certain isotope into another isotope, characterized by emission of radiation. The three types of radiation ( $\alpha$ ,  $\beta$ ,  $\gamma$ ) have different ability to penetrate a material:
  - ✓  $\alpha$ ,  $\beta$  : are particle radiation with a very shallow penetration or high absorption.
  - ✓  $\gamma$  : is an electromagnetic wave with high penetration. Therefore,  $\gamma$ -radiation is preferred for practical application.
- Natural gamma activity of rocks and minerals is originated by:
  - ✓ Uranium-radium series (half-life time of  $4.4 \cdot 10^9$  years)
  - ✓ Thorium series (half-life time of  $1.4 \cdot 10^9$  years)
  - ✓ Potassium  $K^{40}$  (half-life time of  $1.3 \cdot 10^9$  years) – mono-energetic radiation
- Occurrence and behaviour of these components in rocks are different. The abundance of these elements or isotopes in a formation controls the intensity of detectable natural radioactivity.
- In most cases the “API-unit” (American Petroleum Institute) is applied as measurement unit.

# Gamma Ray Measurements

- The spectral character of decay results in two techniques of measurement:
  - ✓ **integral measurement**: measures all incoming counts above an energy threshold.
  - ✓ **spectral measurement**: measures in energy-windows and delivers count rates (and concentration) for the three components K (in %), U and Th (in ppm).
- With an integral gamma measurement the summarized contributions of all three origins are measured:
  - ✓  $I_{GR} = k * (a * K + U + b * Th)$
- The counts originated by each source-element are converted into equivalent U concentration by factors  $a$  and  $b$  is a factor characterizing the sensitivity of the sonde (determined by calibration).
- If a spectral gamma measurement is available, it is recommended to use a Gamma Ray Index calculated only from K and Th. The two components are dominated by the mineral composition, whereas U is controlled more by the environment of sedimentation and organic components. The spectral measurement helps us to detect k-feldspar, mica or glauconite as origin of radiation.
- N.B.: The spectral measurements are necessary for carbonate series as the integral gamma intensity is very often a poor clay indicator.

# Gamma Ray Log

- **Types**

- Natural (Integral); -- Spectral (Thorium, Uranium, Potassium)

- **Physics of Measurement**

- The number of naturally occurring gamma rays from potassium, uranium, thorium and associated daughter products, is counted by the detector in both natural and spectral gamma ray tools.
- Spectral tools also measure the energy of each detected gamma ray. The range, or spectrum, of energy detected is divided into windows, or limited energy ranges, which indicate the elemental gamma ray source (i.e. specific isotopes of potassium, uranium or thorium)

- **Uses** (both Natural & Spectral Gamma Ray logs)

- Correlation: Curves are scanned for similarities in shape and magnitude. Lithological correlation from well to well if some wells are logged in a field
- Gross Lithology indicator; exploration for radioactive materials: In general, reservoir are less radioactive than shales. However some sandstones and dolomites can be radioactive.
- Evaluation of shale content: The magnitude of the gamma ray in the formation of interest (relative to that of nearby clean and shale zones) is related to the shale content of the formation
- Depth Adjustment: Discrepancies are caused for example by cable stretch. One GR log is chosen as a reference log for data quality. The other GR logs are matched to the reference using an automatic routine. Resulting depth shifts are applied to other logs on the tool string.

# Shale Content from GR Log

- **Step 1:** Calculate Gamma Ray Shale Index

$$I_{GR} = \frac{GR - GR_{cn}}{GR_{sh} - GR_{cn}}$$

where

- GR = log reading at depth of interest
- $GR_{cn}$  = Gamma Ray value in nearby clean zone
- $GR_{sh}$  = Gamma Ray value in nearby shale

- **Step 2: Apply an empirical relationship between  $I_{GR}$  & Shaliness,  $V_{shale}$**

- Linear relationship;  $V_{sh} = I_{GR}$
- Non-Linear relationships

Steiber:

$$V_{shale} = \frac{I_{GR}}{3.0 - 2.0 \cdot I_{GR}}$$

Larionov (Tertiary rocks):

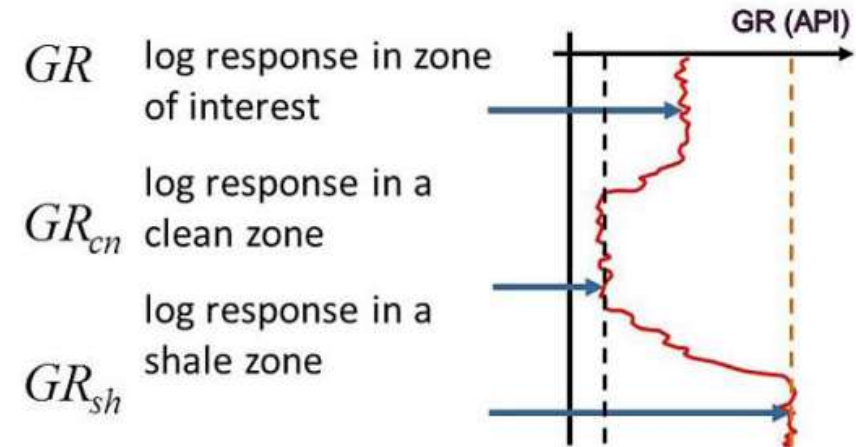
$$V_{shale} = 0.083 \cdot (2^{3.7 \cdot I_{GR}} - 1)$$

Clavier:

$$V_{shale} = 1.7 \cdot [3.38 \cdot (I_{GR} + 0.7)^2]$$

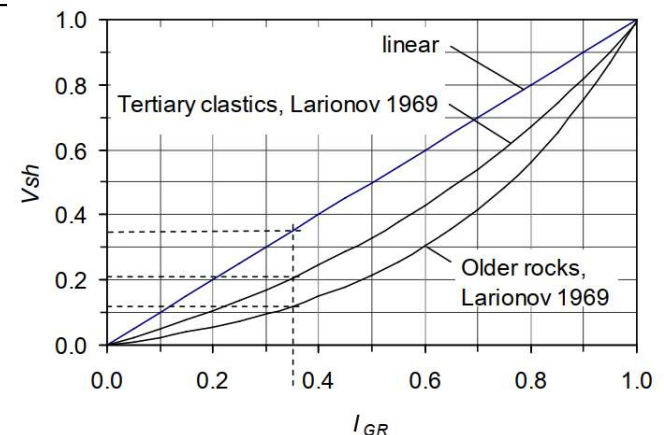
Larionov (older rocks):

$$V_{shale} = 0.33 \cdot [(2^{2 \cdot I_{GR}}) - 1.0]$$



This normalization of the logs results in  $I_{GR} = 0$  for clean rocks (0% shale) &  $I_{GR} = 1$  for pure shale (100% shale)

An empirical equation is recommended in most cases because the Gamma Ray Index is not identical with the volumetric determined shale content from core analysis



# Indications for Carbonates from Spectral Gamma Ray Measurements

- Pure Carbonate (Chemical origin) which has a Th and K level near zero. If U is zero too, this carbonate was precipitated in an oxidizing environment.
- If there is variable U content, the carbonate could either have been deposited in a reducing environment, or it corresponds to a carbonate with stylolithes or to phosphate bearing layers .
- If Th and K are present along with U, this indicates a certain clay content of the carbonate
- If K is present with or without U it can correspond to a carbonate of algal origin or carbonate with glauconite.
- A kind of clay typing is possible because clay minerals show different Th/K ratio i.e. clay minerals give different slopes on Th(ppm) vs K(%) plot
- The high U content of organic components is an indicator for source rocks and gives link to Total Organic Content (TOC). As a rule of thumb, the Th/U ratio is recommended as an indicator for environment.
  - $\text{Th/U} > 7$  continental, oxidizing
  - $\text{Th/U} < 7$  marine, grey...green shales
  - $\text{Th/U} < 2$  marine, black shales, phosphates

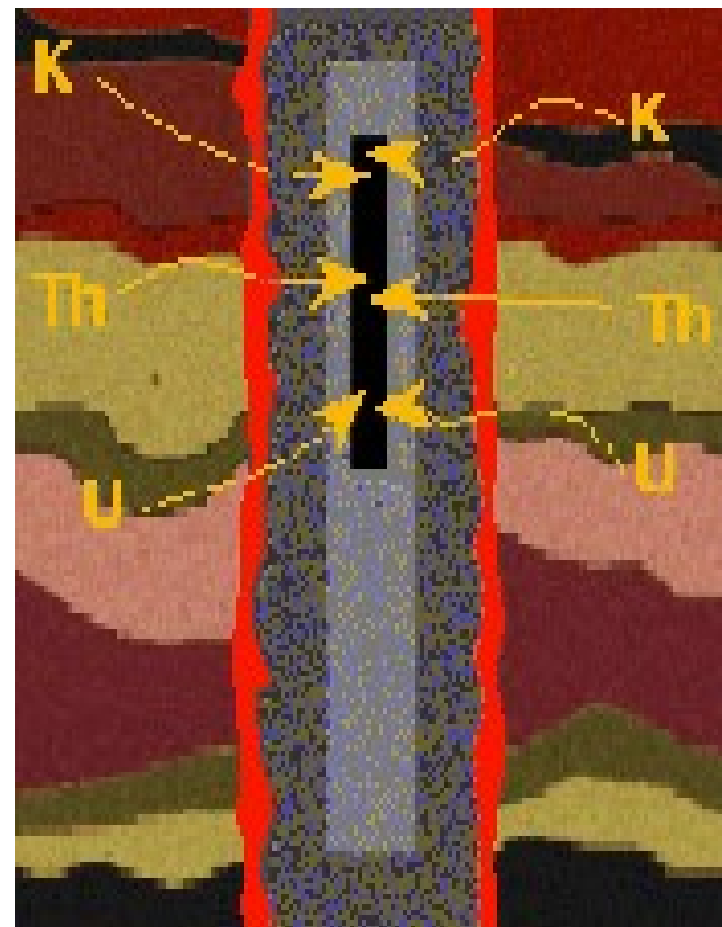


# Spectral Gamma Ray Log

- It is displayed as three curves from all sources of thorium, uranium and potassium.

## Applications:

- Source Rock identification: Consistently high uranium readings in shales may indicate high source rock potential due to the uranium compounds associated with organic material. As a rule of thumb, the Th/U ratio is recommended as an indicator for environment.
- Clay typing: The uranium free curve is often a good indicator of shaliness
- Fracture detection: Spikes to higher values of uranium may indicate fractures due to deposition of soluble uranium compounds in fractures during reservoir fluid movement.



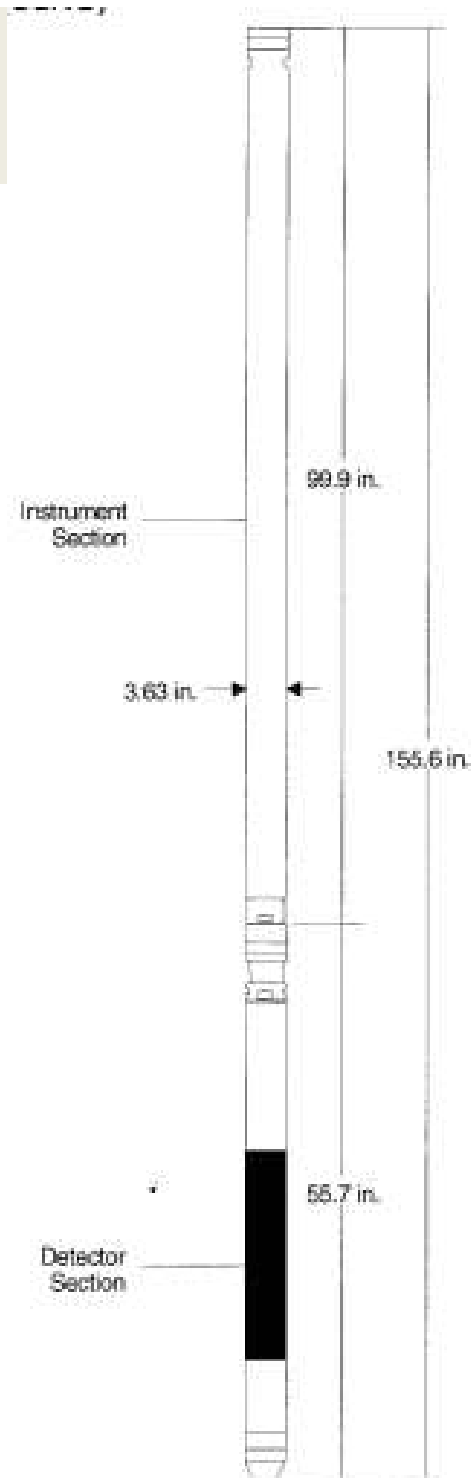
# Gamma Ray Log

- **Operational Constraints**

- In Open hole, the tool is run centered
- In Cased hole, the tool is run eccentric
- Can be run in air, water or oil-based mud
- Logging speed: For standard gamma ray, it is constrained by other tools while for spectral gamma ray  $\approx 10$  feet/minute

- **Secondary Effects**

- Hole Size: increasing hole size decreases count rates
- Mud weight: increasing mud weight decreases count rates
- Mud type: KCl mud increases potassium count rates in spectral tools; barite tools affect all count rates
- Centering: centering the tool decreases count rates



# Gamma Ray Log Interpretation Summary

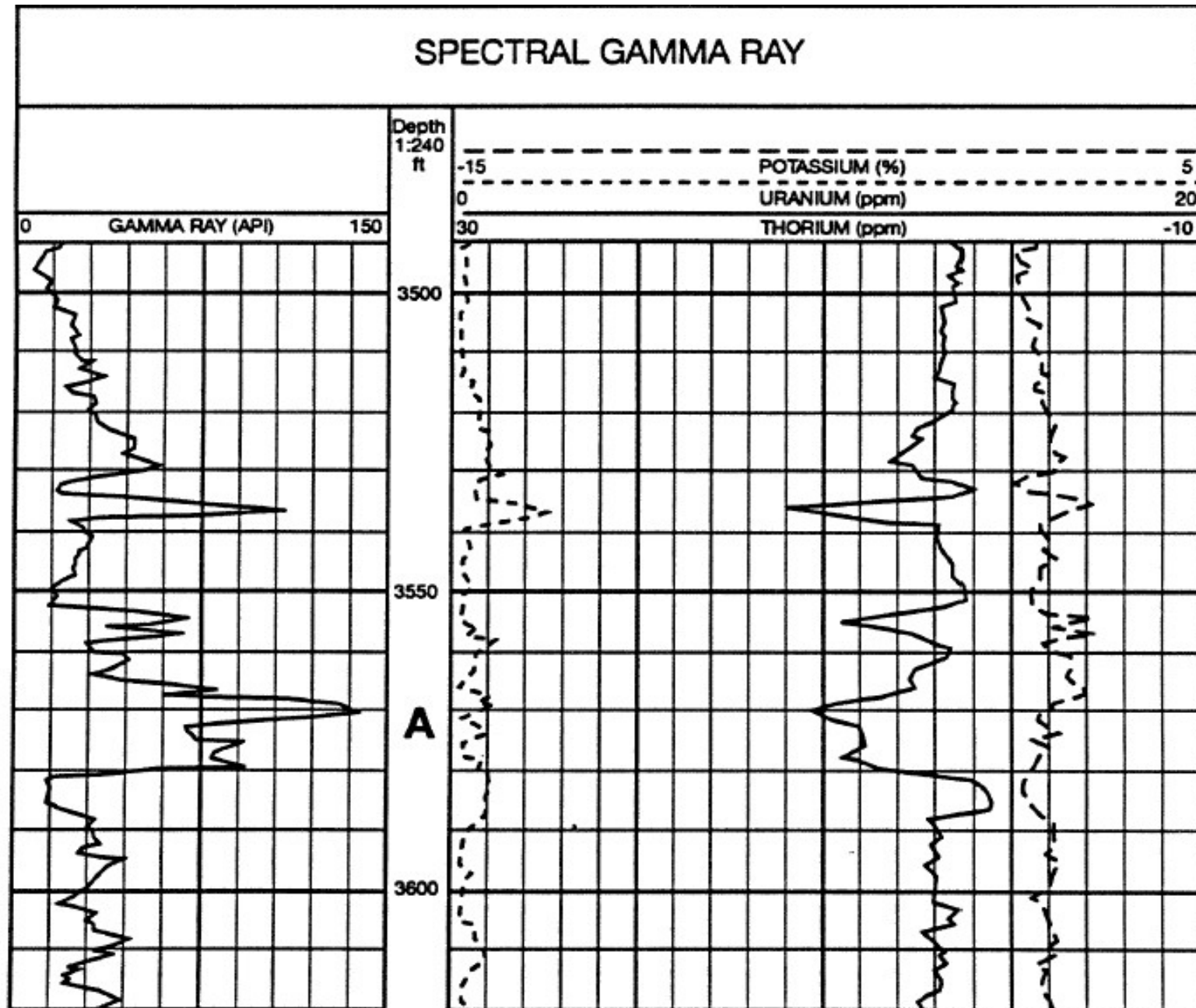
- Clean sands, sandstones, limestones, and dolomites have **low radioactivity**. Anhydrite, salt, lignite and coal have also a low radioactivity.
- Ordinary shales have a much **higher radioactivity** than the rocks listed above. Shales are sufficiently high in radioactivity and can generally be easily distinguished from the other rocks on a gamma ray log.
- Sandstones and dolomites may be occasionally radioactive and respond as shales. A Density-Sonic crossplot may help distinguish between radioactive (“hot”) reservoirs from shales.

## Curves Displayed

*(Curves are listed by generic name, common mnemonics (if any) and measurement units.)*

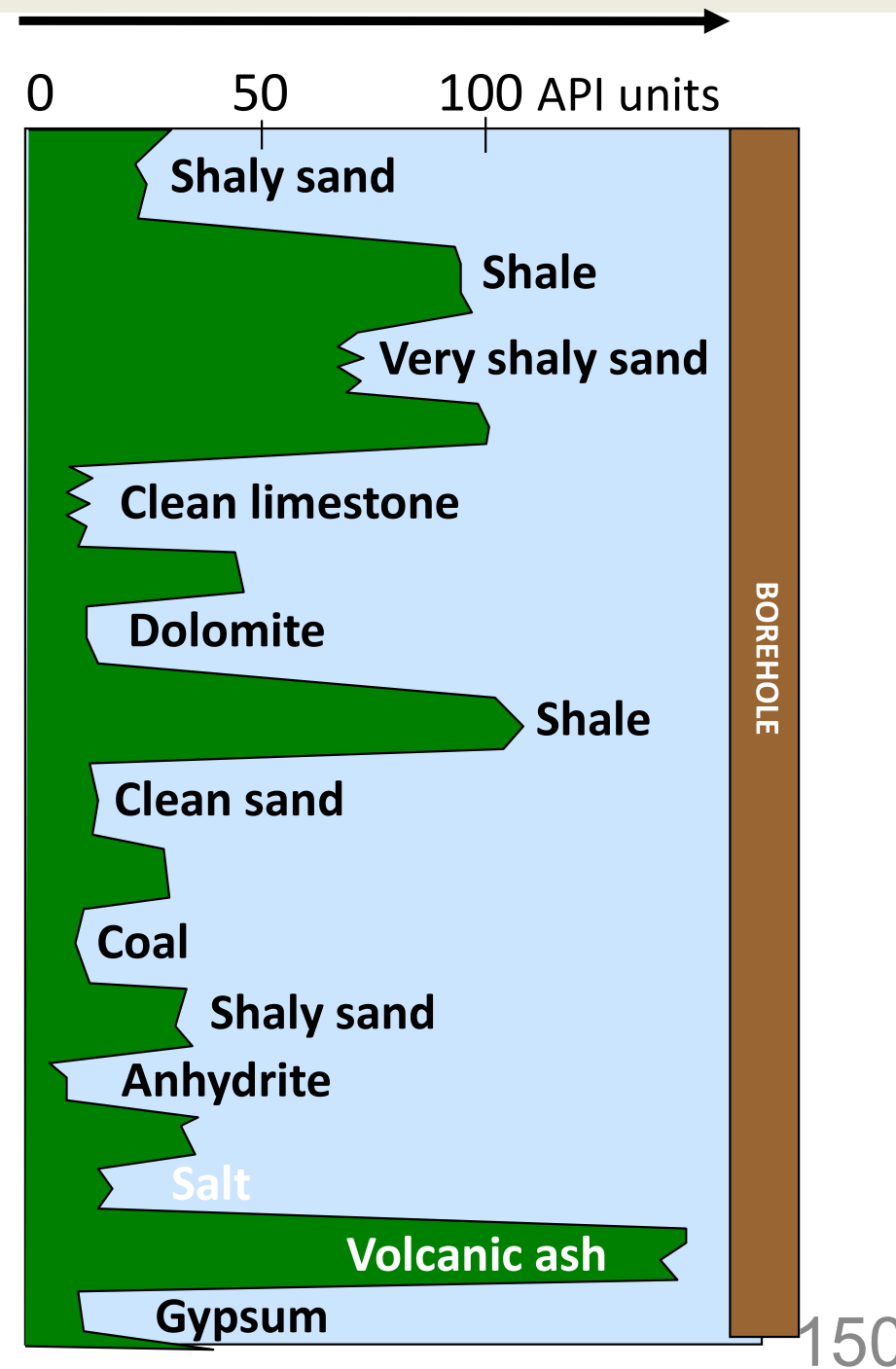
| <b>Curve Name</b>          | <b>Mnemonics</b> | <b>Units of Measurement</b> |
|----------------------------|------------------|-----------------------------|
| Gamma Ray, Total Gamma Ray | GR               | API Units                   |
| Uranium-Free Gamma Ray     | GRS, SGR, KTH    | API Units                   |
| Potassium                  | POTA, K          | Percent                     |
| Uranium                    | URAN, U          | ppm                         |
| Thorium                    | THOR, TH         | ppm                         |

# Spectral Gamma Ray



# GR Response In Common Formations

- Shales often radioactive
  - Clays
  - Trace and heavy minerals
- Sandstones may be radioactive
  - Non-clay minerals, e.g., mica, feldspar
  - Clays
- Units
  - GR calibrated to standard
  - Response in “mid-continent shale” equals 200 API units
  - Calibration pits



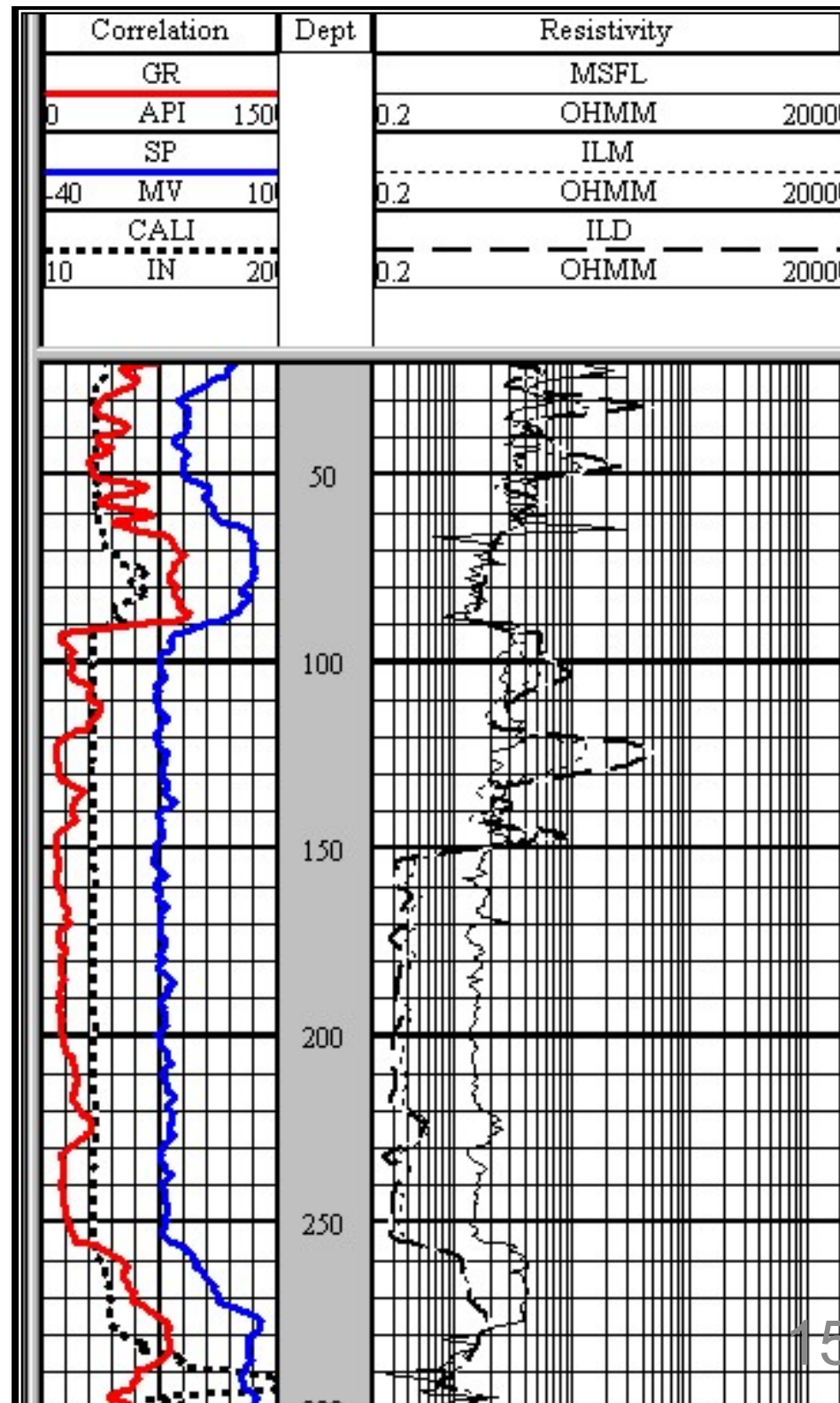
# Gamma Ray Log – Quality Control

- The gamma ray should agree with other shale indicators except in radioactive beds.
- The uranium-free curve should always be less than or equal to the total gamma ray curve.
- The uranium curve should never be negative.
- Shale values should be similar to those of nearby wells.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Note that corrections are usually made for the borehole, mud weight and borehole salinity.



# PASSIVE LOG CORRELATION

- GR, SP, and CAL
  - often correlate
  - different measurements
  - different reasons
- Correlation helps
  - GR instead of SP in OBM
  - Easier detection of shales
  - Facilitates “zonation”





# **Porosity Measurements**

## **– Lecture 7**

# Sources of Porosity Measurement

- Density Logs (ZDL). – Nuclear Logs
- Compensated Neutron Logs (CN). – Nuclear Logs
- Acoustic (Sonic) Logs (AC, XMAX).
- Nuclear Magnetic Resonance (NMR) MRIL.

These logs do not measure porosity directly. To accurately calculate porosity, the analyst must know:

- Formation lithology
- Fluid in pores of sampled reservoir volume

# Nuclear Logs

- Are important group of log measurements using nuclear radiation. Measurements are possible in both open & cased hole.
- Basic types of nuclear methods are:
  - Measurement of natural gamma radiation of the formation (Gammalog)
  - Measurement of the radiation as a result of an interaction of the source-emitted radiation with the formation
    - Gamma-Gamma-Logs (uses a gamma source)
    - Neutronlogs (uses a chemical neutron source or neutron generator)
- Interactions of radiation with the target material (formation) are processes connected with the atomic structure. Radiation energy is given in Electronvolt (eV): 1 eV is the energy needed to move one electron with the charge of  $1.6 \cdot 10^{-19}$  C (Coulomb) through a potential difference of 1 Volt
- Nuclear processes are statistically determined. The statistical variation obeys a Poisson distribution with the standard deviation  $\sigma$  equal to the square root of the total number of counts  $N$ .
$$\sigma = \sqrt{N}$$
- It follows that the precision of a nuclear measurement depends on the number of counts recorded – therefore statistical precision can be improved by logging slower and/or using sensors with high sensitivity.

# Fundamentals of Gamma Ray Interactions

- Gamma-Gamma methods use the interaction of gamma radiation produced by a source (as part of the tool) with the formation. The main component of a Gamma-Gamma probe are:
  - ✓ Gamma source (Cs or Co),
  - ✓ One or more gamma detectors detecting the back scattered radiation,
  - ✓ A shield between the source and detectors to suppress direct radiation.
- There are three types of gamma ray interactions in earth formations that are of interest:
  - ✓ the Photoelectric Effect,
  - ✓ Compton scattering, and
  - ✓ Pair Production.
- The probability of a specific gamma ray interaction occurring will depend on the atomic number of the material and the energy of the gamma ray.
- The ordering of these three interactions in the following discussion reflects the change of the dominant process as the gamma ray energy increases.

# Fundamentals of Gamma Ray Interactions

- The **photoelectric effect** results from interaction of a gamma ray (GR) with an atom in the material. In this process the incident GR disappears and transfers its energy to a bound electron. If the incident GR energy is large enough, the electron is ejected from the atom and begins interacting with the adjacent material. Normally the ejected electron is replaced by another less tightly bound electron with the accompanying emission of a characteristic fluorescence x-ray with an energy which is dependent on the atomic number of the material. For most earth formations, the PE effect becomes the dominant process for GR energies below about 100 keV.
- Moving up the gamma ray energy scale, the next dominant process is **Compton scattering (CS)** which involves interactions of gamma rays and individual electrons. It is a process in which only part of the gamma ray energy is imparted to the electron; the remaining gamma ray is of reduced energy. Unlike the *PE* effect, the cross section for CS changes relatively slowly with energy. The attenuation of gamma rays due to Compton scattering will be a function of the bulk density  $\rho_b$  and the ratio  $Z/A$ . The fact that  $Z/A$  is constant ( $\approx 1/2$ ) for most elements of interest is the basis for the determination of bulk density from gamma ray scattering devices.
- The third and final gamma ray interaction is **pair production**. Like the *PE* process, pair production is one of absorption rather than scattering. In this case the gamma ray interacts with the electric field of a nucleus, and if the gamma ray energy is above the threshold value of 1.022 MeV, it disappears and an electron-positron pair is formed. The onset of this interaction corresponds to the rest mass energy of the electron and positron. The subsequent annihilation of the positron (positively charged electron) results in the emission of two gamma rays of 511 keV each. The nuclear cross section of this process is zero below the threshold energy of 1.022 MeV and rises quite rapidly with increasing energy. It is also dependent on the charge of the nucleus, varying approximately as  $Z^2$ .

# Fundamentals of Gamma Ray Interactions – Photoelectric Effect

- The photoelectric effect is the mechanism by which one type of well logging tool is made sensitive to the lithology of the formation. This tool measures the so-called photoelectric absorption factor (or Index) ,  $Pe$ , which is proportional to the photoelectric cross section per electron.
- Since  $PE$  is very sensitive to the average atomic number of the medium ( $Z$ ), it can be used to obtain a direct measurement of lithology or rock type. This is due to the fact that the principal rock matrices (sandstone, limestone, and dolomite) have different average atomic numbers and thus, considerably different photoelectric absorption characteristics, and that the pore fluids play only a minor role because of their low average atomic numbers.
- In the case of several minerals in a formation, the  $Pe$  index is not a linear function of the  $Pe$ -values of the mineral components, a second parameter  $U$  (volumetric photoelectric adsorption index) was introduced:

$$U = PE * \rho$$

- In a multimineral lithology this index is the volume weighted index of the individual contributions.

# Fundamentals of Gamma Ray Interactions – Compton Scattering

- The Compton effect is a scatter reaction with the orbiting electrons of the target material. Therefore the probability of an interaction is controlled by the number of electrons, which is equal to Z. For the most common earth materials, the ratio  $Z / A$  is constant.
- Since rock bulk density  $\rho_b$  is dominated by Atomic Mass A, the following relationship connects the “electron density”  $\rho_e$  as seen by the Compton effect via this ratio:

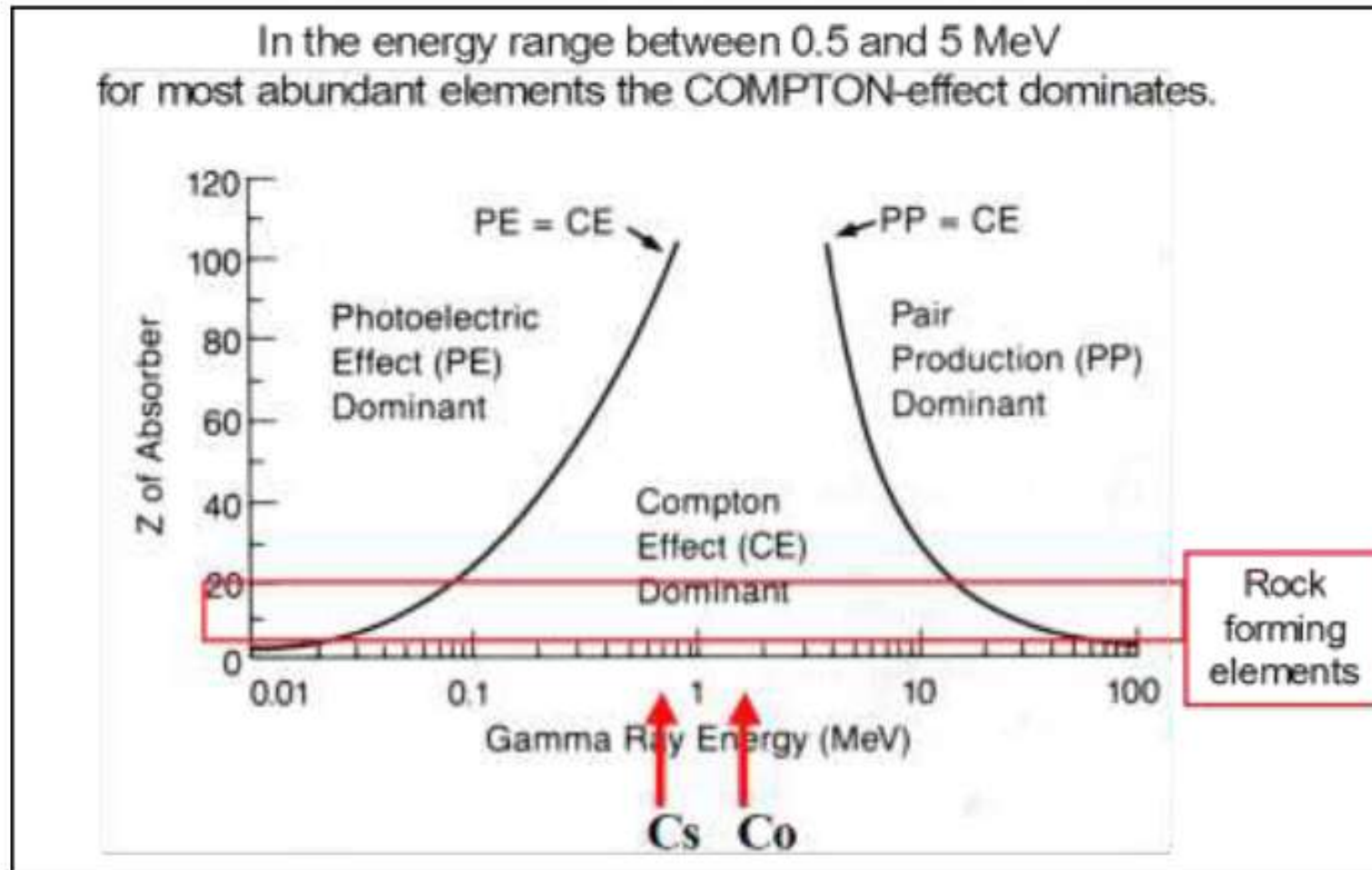
$$\rho_b = 2 * (Z / A) * \rho_b \approx \rho_e$$

| Mineral, fluid     | $\rho$ | $(Z/A)_{\text{eff}}$ | $\rho_{\text{GG}}$ | Pe   | U     |
|--------------------|--------|----------------------|--------------------|------|-------|
| Quartz             | 2,654  | 0,499                | 2,650              | 1,81 | 4,79  |
| Calcite            | 2,710  | 0,500                | 2,708              | 5,08 | 13,77 |
| Dolomite           | 2,850  | 0,499                | 2,864              | 3,14 | 9,00  |
| Anhydrite          | 2,960  | 0,499                | 2,957              | 5,05 | 14,95 |
| Gypsum             | 2,320  | 0,511                | 2,372              | 3,42 | 8,11  |
| Halite             | 2,165  | 0,479                | 2,074              | 4,65 | 9,65  |
| Montmorillonite    | 2,12   | 0,500                | 2,12               | 2,04 | 7,28  |
| Illite             | 2,65   | 0,496                | 2,63               | 3,55 | 10,97 |
| Kaolinite          | 2,44   | 0,500                | 2,44               | 1,84 | 6,14  |
| Barite             | 4,50   | 0,446                | 4,01               | 267  | 1070  |
| Water (fresh)      | 1,000  | 0,555                | 1,110              | 0,36 | 0,40  |
| Water (120000 ppm) | 1,086  | 0,545                | 1,185              | 0,81 | 0,96  |
| Oil (mean)         | 0,850  | 0,558                | 0,948              | 0,12 | 0,11  |

Mean Values of density in  $10^3 \text{ kg/m}^3$ , electron density, or gamma-gamma density, effective ratio  $Z / A$ , PE index in barn/electron & volumetric PE absorption index U in barn/cm<sup>3</sup>



# Fundamentals of Gamma Ray Interactions



- Regions of Dominance of the three principal gamma ray scattering / absorption mechanisms as a function of energy and atomic number of the scattering material.

# DENSITY LOG

# DENSITY LOGS – Introduction

- Another important property of an earth formation is bulk density. Its use in seismic interpretation is well known, but more important bulk density is linearly dependent on formation porosity, a key ingredient for the interpretation of electrical measurements.
- Gamma rays are typically used to measure bulk density since their scattering and transmission are strongly affected by this material property.
- At very low energies the transmission of gamma rays is influenced additionally by the formation chemical composition. This additional absorption is related to the atomic number,  $Z$ , of the absorber and thus provides a third application of gamma rays.
- The average electron density in the volume of formation probed by the tool controls the scattered gamma ray count rates at the detectors. The average electron density, in turn, correlates strongly (but not perfectly) with bulk density. Formation Bulk density,  $\rho_b$ , is dependent upon the (i) Lithology (ii) Porosity (iii) Density and saturation of fluids in pores.

# Density Log

- **Uses**

- Porosity determination: from bulk density
- Lithology Identification: Lithology is determined by comparison of bulk density with Sonic and Neutron data in cross plots, in Matrix identification (MID) plots and in M-N (A-K) plots. The charts may vary by Neutron tool type, Sonic equation response type, and by service company. The Photoelectric effect (PEF) can be used alone to determine single lithology, or in combination with bulk density, or bulk density and Neutron curves to determine mixed lithologies.
- Gas Indication: with the Neutron.
- Synthetic Seismograms: Sonic compressional and Density data are used to determine acoustic impedance of the formations along the borehole, and reflection coefficients at the bed boundaries.
- Formations mechanical properties: with the Sonic
- Clay content (shaliness): with the Neutron .
- Borehole size: from an attached caliper.

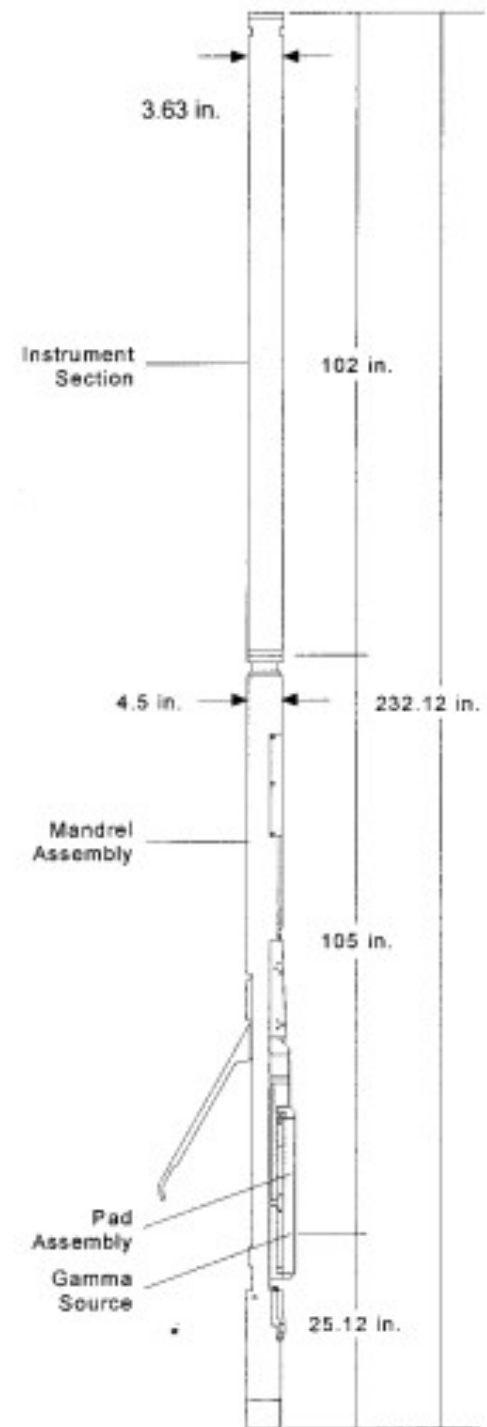
# Density Logging Tool

- **Physics of Measurement**

- High energy gamma rays are emitted from a chemical source (usually Cesium-137) of 0.66MeV and interact with the electrons of the elements in the formation. Two detectors in the tool count the number of returning (back-scattered) gamma rays which are related to the formation bulk density. For most earth materials of interest, the electron density is related to the formation bulk density through a constant.
- The number of returning gamma rays are measured at two energy ranges. The higher energy gamma rays (from Compton Scattering) determine bulk density, and therefore porosity, while the lower energy gamma rays (due to photoelectric effect) are used to determine formation lithology. The lower energy gamma rays are related to the lithology of the formation and show little dependence on porosity or fluid type. The average atomic number (which correlates with rock type) sets the amount of PE absorption that a formation exhibits.

- **Operational Constraints**

- In Open hole or Cased hole (in special conditions), the tool is run eccentric
- Can be run in air, water or oil-based mud



Spectral Density Tool

# DENSITY LOGS – Tool Physics

- The source and both detectors are heavily shielded (collimated) to ensure that the radiation only goes into the mudcake and formation, and that detected gamma rays only come from the mudcake or formation. The leading edge of the shield is fashioned into a plough which removes part of the mudcake as the tool is pulled up the well.
- The tool is pressed against one side of the borehole using a servo-operated arm with a force of 800 pounds force. Under this pressure and the pulling power of the wireline winch the plough can make a deep impression in the mudcake. The large eccentricing force also means that there is much wear of the surface of the tool that is pressed against the borehole wall. The heavy shielding also doubles as a skid and a wear plate that protects the source and detectors and can be replaced easily and cheaply when worn down (see Figure DL1).
- The density logging tool detects gamma rays that have been supplied to the formation by the source on the tool and have undergone scattering. The natural gamma rays from the formation however confuse the measurement. A background gamma ray count is therefore carried out so that the gamma rays coming from the formation can be removed from the measurement.

# DENSITY LOG – Tool Physics

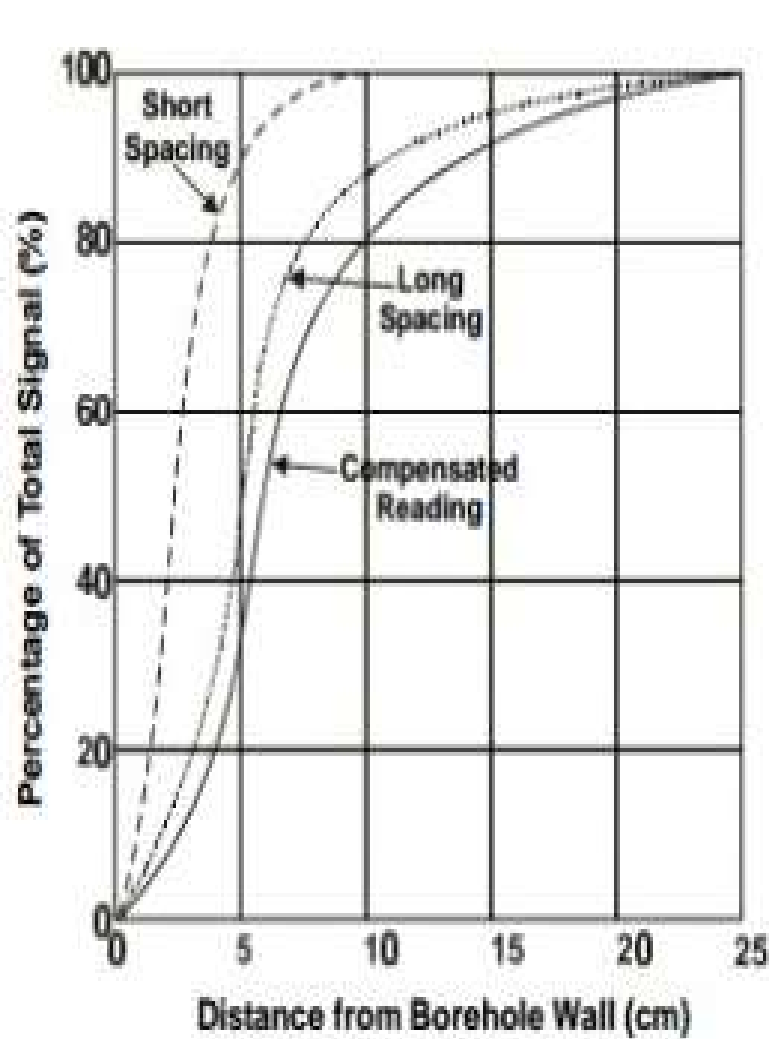


Figure DL2

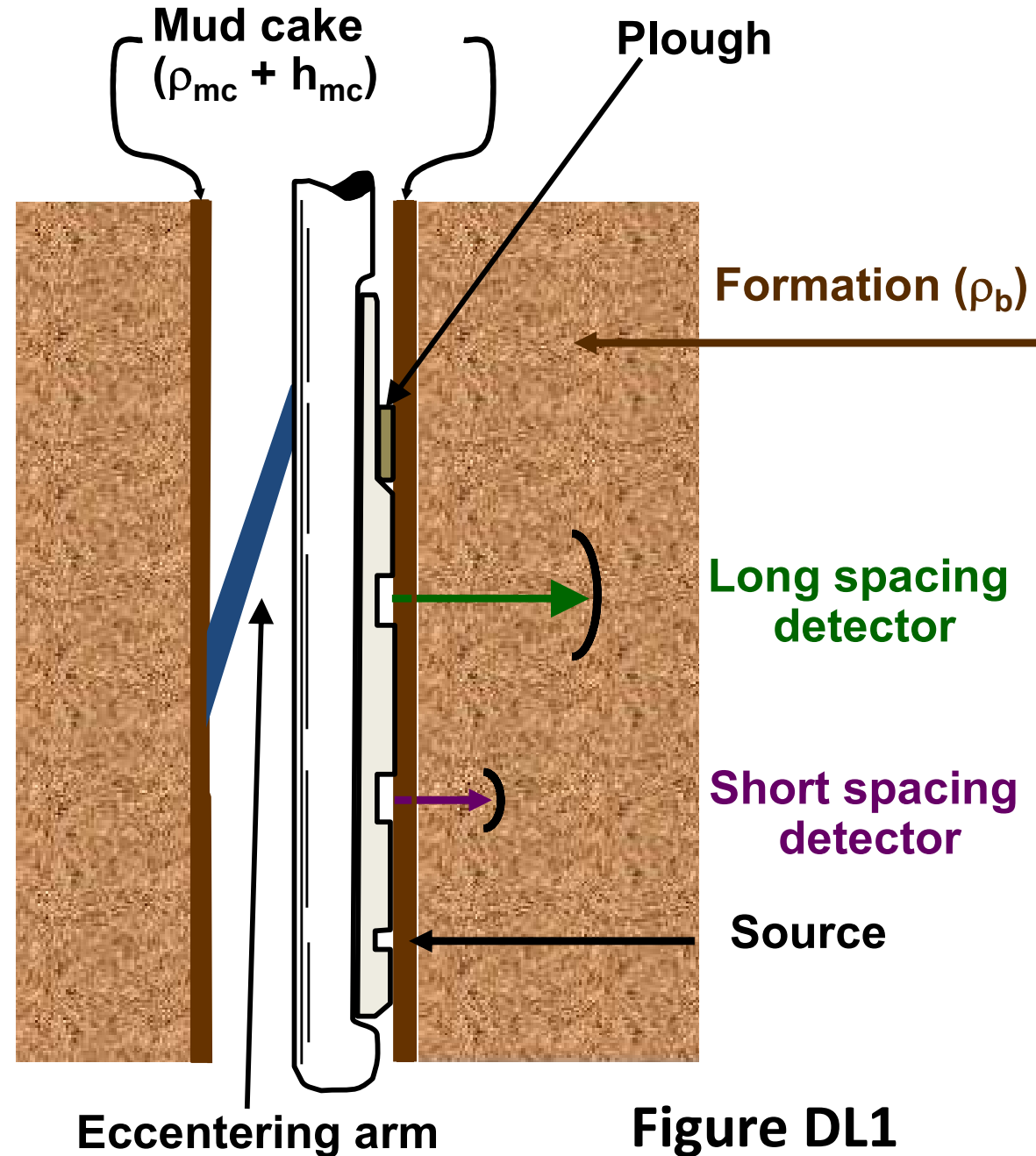


Figure DL1



# DENSITY LOGS – Tool Physics

- Note in Figure DL2 that for the short spacing detector over 80% of its signal comes from within 5 cm of the borehole wall, which is commonly mainly mudcake. About 80% of the long spacing signal comes from within 10 cm of the borehole wall. Therefore the tool has a shallow depth of investigation. When the mudcake compensation has been carried out, one can see from Figure DL2 that the depth of investigation has been improved, and less signal comes from the first 5 cm region. The depth of investigation of a density tool decreases with increasing density.

## The tool consists of:

- **A radioactive source:** This is usually caesium-137 or cobalt-60, and emits gamma rays of medium energy (in the range 0.2 – 2 MeV). For example, caesium-137 emits gamma rays with a energy of 0.662 MeV.
  - **A short range detector:** This detector is very similar to the detectors used in the natural gamma ray tools, and is placed 7 inches from the source.
  - **A long range detector.** This detector is identical to the short range detector, and is placed 16 inches from the source.
- Mudcake and tool standoff have particularly strong effects on this measurement. Again, all the bulk average effects accrue, with special problems posed by barium-weighted mud.

# DENSITY LOGS – Tool Physics

- The distance between the near and far detectors sets the vertical resolution, approximately 10 in. typically. Correction is by the spine-and-ribs technique. The spine is the normal calibrated relationship between the density measured by the near-spaced and far-spaced detectors in the absence of any gap between the tool and the borehole wall. Gaps cause departures from this spine and lead to density corrections that are applied to the density from the long-space detector.
- This correction is presented as a curve with the density log. Above an inch or so of standoff, the compensation fails. There is no universal correction threshold, but corrections greater than 0.05 to 0.1 g/cm<sup>3</sup> are suspicious. Unfortunately, a low correction does not guarantee a good measurement. Connecting a particular rib back to the correct location on the spine (and, hence, to the correct bulk density) requires that the density pad be parallel to the borehole wall. Likewise, very small water-filled gaps can give rise to large corrections that are perfectly correct.

# DENSITY LOGS – Interpretation Effects

## The Effect of Fluid Density:

- The porosity may also be in error if the fluid density is misjudged. The fluid existing in the zone of the rock measured by the formation density tool is usually mud filtrate. The density of these fresh and salt waters is approximately 1.0 & 1.1 g/cm<sup>3</sup>, respectively.
- However, these vary with temperature and composition, so accurate values for the actual reservoir formation water at the relevant reservoir temperatures should be used wherever possible. Such data can be obtained from samples of reservoir fluid from RFT analysis, or from the analysis of mud filtrate.
- Bearing in mind that the tool measures the invaded zone, the relevant fluid density to use in the porosity calculations is most often the mud filtrate density. Mud filtrate densities are now corrected automatically for temperature and pressure in most petrophysical software. However, the correction used to be carried out with the aid of a nomogram.

# DENSITY LOGS – Interpretation Effects

## The Effect of Gas:

- If gas is present in the formation, porosities can be overestimated. Gas density is very low ( $\approx 0.0001 \text{ g/cm}^3$ ) compared to aqueous fluids. If the formation is gas bearing a significant amount of gas is always left in the invaded zone which will reduce the mean fluid density of the invaded zone, and cause porosity overestimation if a fluid density of  $1.0 - 1.1 \text{ g/cm}^3$  is used.

## The Effect of Oil:

- The density of oil ( $\approx 0.7 \text{ g/cm}^3$ ) is less than that of aqueous fluids. However the presence of oil-bearing formations rarely causes problems in porosity calculation because both the oil and water in the invaded zone, which the tool measures, is replaced by mud filtrate.

## The Effect of Shale:

- The density of shales varies greatly, and if present as a proportion of a lithology (such as a shaly sandstone) can make the derivation of a reliable porosity inaccurate. If we have zones of clean lithology and zones of shale with a shaly sandstone in between (or a full fining-up or coarsening-up sequence), we can use the bulk densities in the shale and clean sandstone together with  $V_{sh}$  to obtain a corrected density for the shaly sandstone at any given depth.

# DENSITY LOGS – Environmental Effects

## Enlarged Borehole (> 9 inches):

- $RHOB < \text{formation bulk density}$  ( $DPHI > PHI_{\text{actual}}$ ).

## Rough Hole:

- $RHOB < \text{formation bulk density}$  ( $DPHI > PHI_{\text{actual}}$ ). This is due to the sensor pad losing contact with the borehole wall. Other indications of a rough hole will be a variable Caliper curve, and a high-valued density correction (DRHO) curve. There are no environmental corrections than can be applied to correct for loss of pad contact.
- DRHO is a significant measure of log quality (QC):
  - Correction curve DRHO should be near zero in smooth holes,
  - $DRHO > 0.05$ ; Data quality may be questionable due to loss of pad contact,
  - $DRHO > 0.10$ ; indicates that the density value is not reliable

## Barite Muds:

- $RHOB > \text{formation bulk density}$  ( $DPHI < PHI_{\text{actual}}$ ), and  $PEF > PEF_{\text{actual}}$

# BULK DENSITY

$$\rho_b = \rho_{ma} (1 - \phi) + \rho_f \phi$$

Matrix

Fluids in  
flushed zone

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

Porosity equation

We usually assume the fluid density ( $\rho_f$ ) is between 1.0 and 1.1. If gas is present, the actual  $\rho_f$  will be  $< 1.0$  and the calculated porosity will be too high.

$\rho_{mf}$  is the mud filtrate density, g/cc

$\rho_h$  is the hydrocarbon density, g/cc

$S_{xo}$  is the saturation of the flush/zone, decimal

- Measures electron density of a formation
- Strong function of formation bulk density
- Matrix bulk density varies with lithology
  - Sandstone 2.65 g/cc
  - Limestone 2.71 g/cc
  - Dolomite 2.87 g/cc

# POROSITY FROM DENSITY LOG

## Fluid density equation

$$\rho_f = \rho_{mf} S_{xo} + \rho_h (1 - S_{xo})$$

## Working equation (hydrocarbon zone)

$$\begin{aligned} \rho_b = & \phi S_{xo} \rho_{mf} + \phi (1 - S_{xo}) \rho_{hc} \\ & + V_{sh} \rho_{sh} + (1 - \phi - V_{sh}) \rho_{ma} \end{aligned}$$

|                               |   |                                  |
|-------------------------------|---|----------------------------------|
| $\rho_b$                      | = | Recorded parameter (bulk volume) |
| $\phi S_{xo} \rho_{mf}$       | = | Mud filtrate component           |
| $\phi (1 - S_{xo}) \rho_{hc}$ | = | Hydrocarbon component            |
| $V_{sh} \rho_{sh}$            | = | Shale component                  |
| $1 - \phi - V_{sh}$           | = | Matrix component                 |



# DENSITY LOGS

- If minimal shale,  $V_{sh} \approx 0$
- If  $\rho_{hc} \approx \rho_{mf} \approx \rho_f$ , then
- $\rho_b = \phi \rho_f - (1 - \phi) \rho_{ma}$

$$\phi = \phi_d = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

$\phi_d$  = Porosity from density log, fraction

$\rho_{ma}$  = Density of formation matrix, g/cm<sup>3</sup>

$\rho_b$  = Bulk density from log measurement, g/cm<sup>3</sup>

$\rho_f$  = Density of fluid in rock pores, g/cm<sup>3</sup>

$\rho_{hc}$  = Density of hydrocarbons in rock pores, g/cm<sup>3</sup>

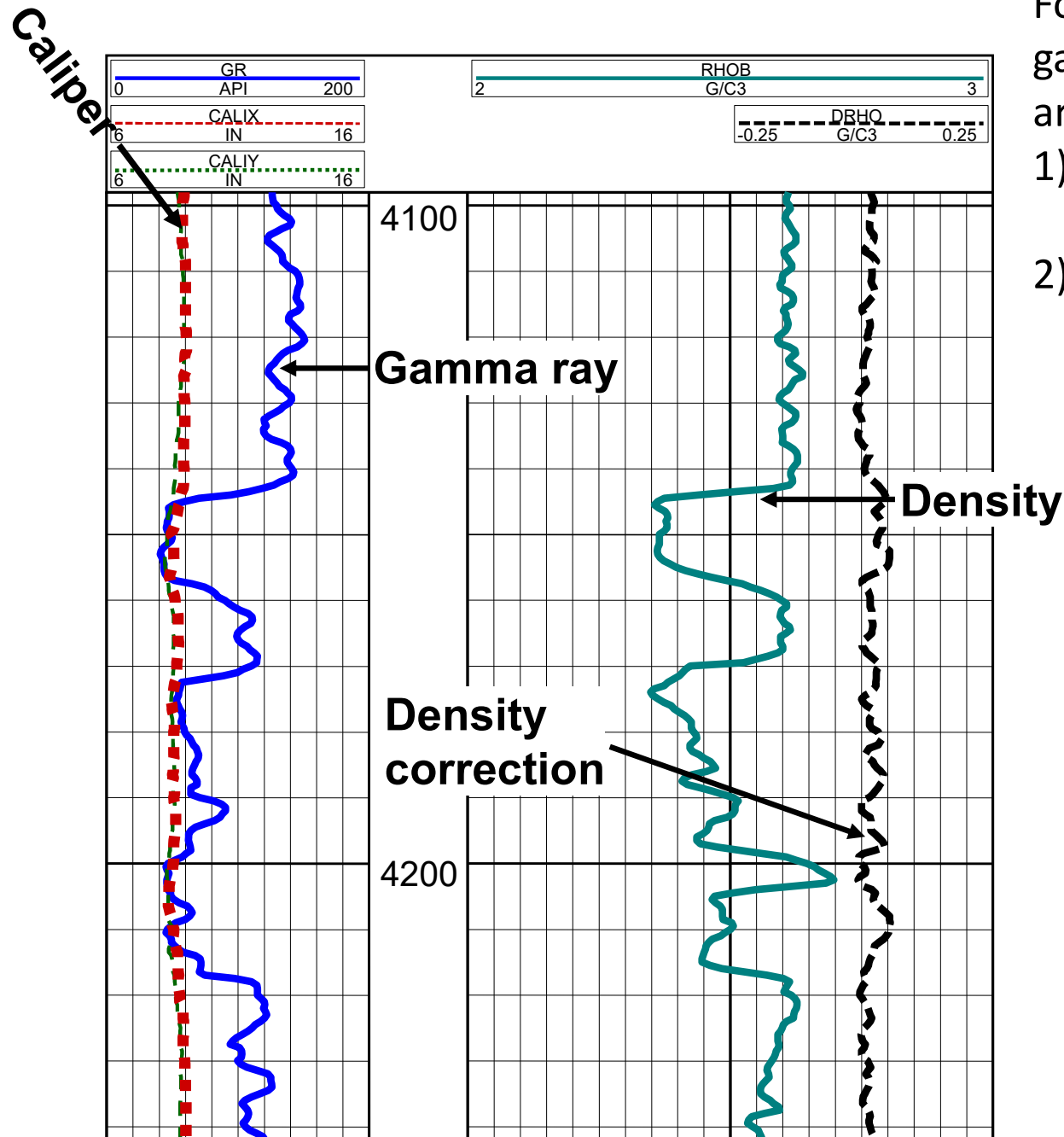
$\rho_{mf}$  = Density of mud filtrate, g/cm<sup>3</sup>

$\rho_{sh}$  = Density of shale, g/cm<sup>3</sup>

$V_{sh}$  = Volume of shale, fraction

$S_{xo}$  = Mud filtrate saturation in zone invaded by mud filtrate, fraction

# DENSITY LOG

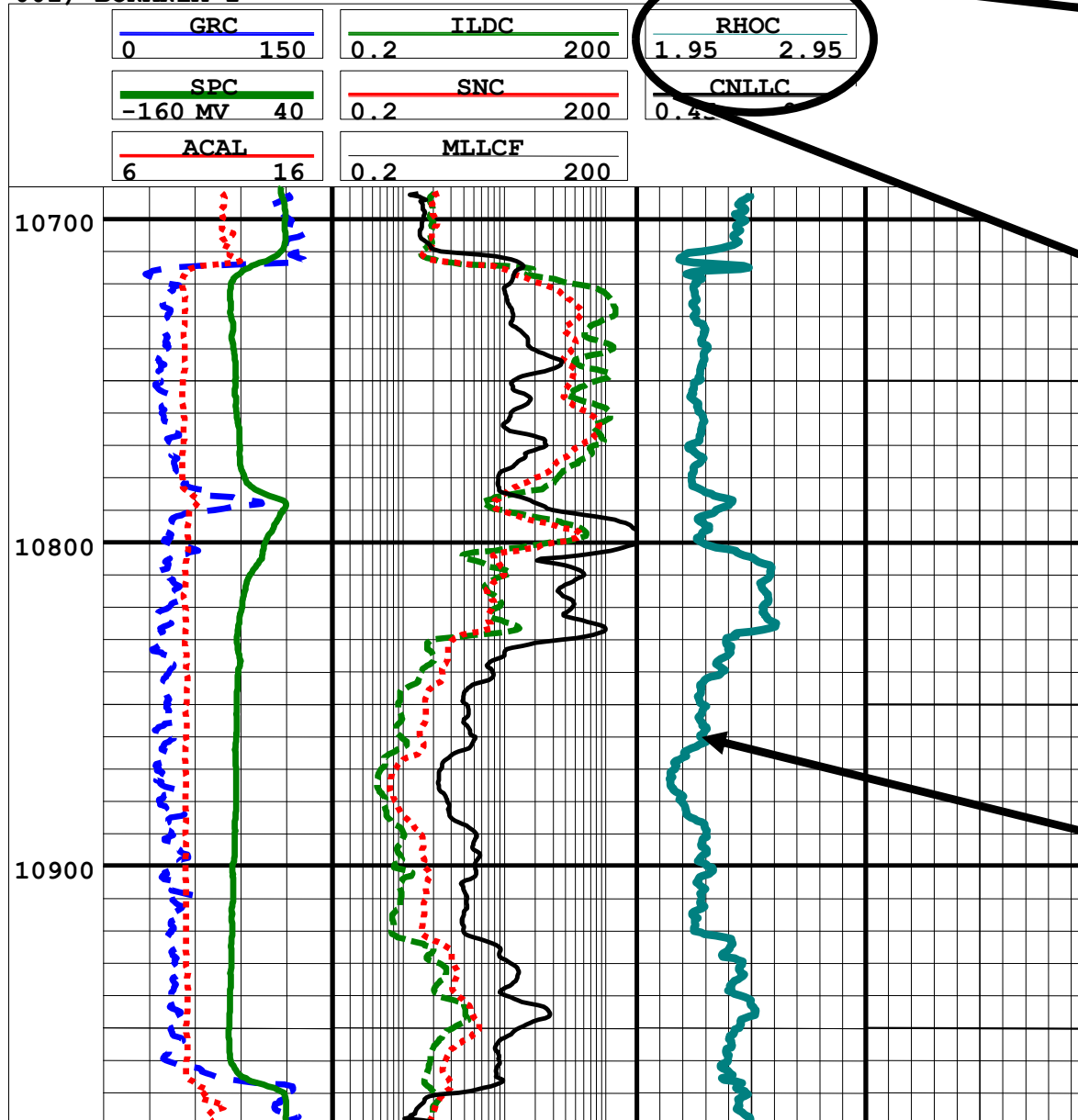


For porosity calculation from gamma-gamma-density two-input parameters are necessary:

- 1) Matrix density (or information about mineralogy).
- 2) Fluid density: The depth of investigation for gamma-gamma measurements is very shallow; thus, the measurement is controlled by the flushed zone in most cases and the density of mud filtrate can be used. Only in the case of shallow gas invasion density is lowered, for calculation an average fluid density of  $0.7 \text{ g/cm}^3$  is recommended.

# BULK DENSITY LOG

001) BONANZA 1



**RHOC**  
1.95 2.95

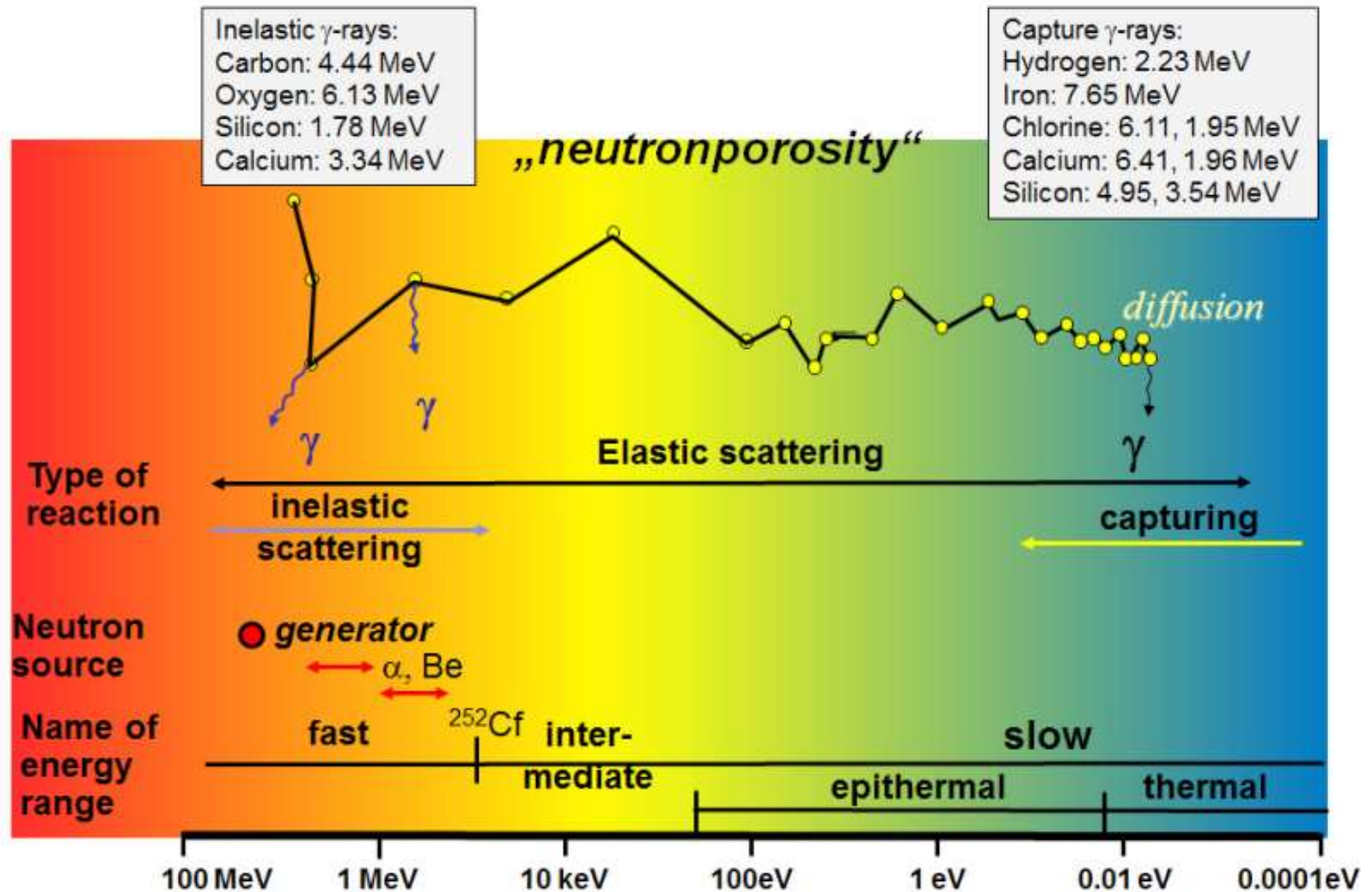
Bulk Density  
Log

# NEUTRON LOG

# Fundamentals of Neutron methods

- Neutron methods use the interaction of neutron radiation from a source (as part of the tool) with the formation. The main component of the neutron probe are:
  - ✓ Neutron source (Chemical source or Neutron Generator),
  - ✓ One or more detectors (neutron and gamma) detecting neutron radiation after slowing down processes and induced gamma radiation,
  - ✓ A shield between the source and detectors to suppress direct radiation.
- Neutrons are classified by their energy:
  - ✓ Fast neutrons have energies greater than 500 KeV;
  - ✓ Intermediate neutrons: 1 to 500 KeV;
  - ✓ Slow neutrons  $< 1$  KeV (epithermal neutrons have energies above 0.1 to 1 KeV; thermal neutrons  $< 0.1$  eV (at 25 °C).
- Neutron as a particle radiation (a neutron has approximately the mass as a proton) react with the nuclei of the target material.
- There are different types of interactions:
  - ✓ Moderating interactions result in energy decrease (slowing down effect);
    - (a) Inelastic scattering                      (b) Elastic scattering
  - ✓ Capturing interactions;

# Fundamentals of Neutron methods



- Neutron phenomena as a function of energy. The neutron sources and the energy regions are indicated at the bottom. At the top are reactions and processes plotted.

# Neutron Scattering

- **Inelastic Scattering:** Neutrons collide with the nucleus and transfer a fraction of their kinetic energy to the nucleus which results in excitation of the nucleus followed by the emission of a characteristic gamma photon (de-excitation). The emitted gamma radiation are characteristic of the target material. An application is the Carbon/Oxygen log.
- **Elastic Scattering:** Neutrons collide with the nucleus and lose kinetic energy. The maximum energy loss results for a target material with the same atomic mass as the neutron; this is the case for hydrogen. The other rock forming elements have a lower slowing down effect and are different. This leads to the conclusion:
  - ✓ Electron scattering is dominated by hydrogen in the formation; there a strong combination is expected from the fluids with a high H-concentration (water & oil) and somewhat a lower contribution from gas with lower H-concentration. In general elastic scattering is dominated by pore fluids
  - ✓ Other elements have a distinctly smaller effect; but this effect is different for each element. Therefore a different neutron response for different materials (quartz, calcite and dolomite) is expected



# Neutron Capture

- Neutrons can be captured when they are thermalized. Thermal neutrons have small energies associated with the random kinetic motion at room temperatures. With this process a compound nucleus in an excited state is formed. When it returns to the ground state, it emits gamma radiation with an exactly defined energy. This capturing effect is used with the Thermal neutron diffusion / Decay measurements (TDT) log.

| Element | Gamma energy in MeV<br>- inelastic scattering - | Gamma energy in MeV<br>- capturing - |
|---------|-------------------------------------------------|--------------------------------------|
| H       |                                                 | 2.2                                  |
| O       | 6.1, 7.0                                        |                                      |
| C       | 4.43                                            |                                      |
| Si      | 1.78                                            | 3.5, 4.9                             |
| S       | 2.2                                             | 3.2, 4.9, 5.4                        |
| Ca      | 3.7                                             | 2.0, 4.4, 6.4                        |
| Al      | 2.2                                             | 7.7                                  |
| Mg      |                                                 | 3.9                                  |
| Fe      |                                                 | 6.0, 7.3, 7.6                        |

Inelastic scattering and capturing – Some characteristic energies of emitted gamma radiation

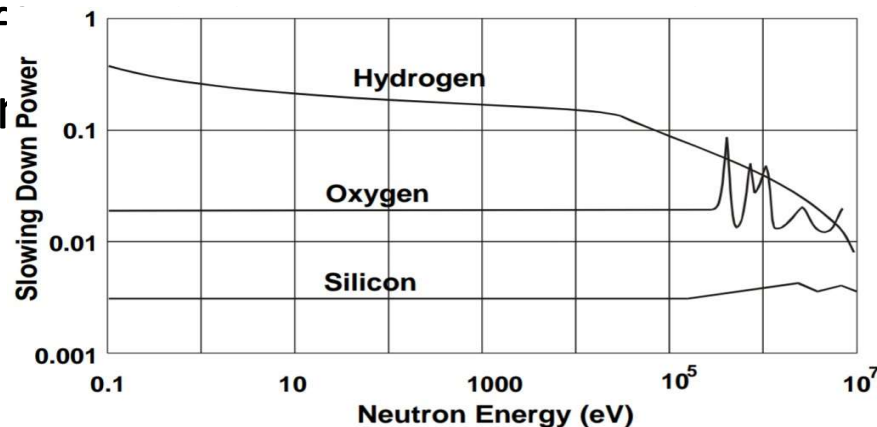
# Neutron Logging Processes

- In neutron logging there are three processes of interest:
  - ✓ (i) Neutron Emission
  - ✓ (ii) Neutron Scattering
  - ✓ (iii) Neutron Absorption
- Neutron Emission:
  - The neutron tool emits high energy (4.5 MeV) neutrons from a radioactive source. They move very fast, and their energy is related to their speed. They are called fast neutrons.
  - The neutron sources used in logging are a mixture of two elements;
    - (i) a source of alpha radiation such as radium, plutonium or americium, and
    - (ii) beryllium-9.
  - The alpha particles from the radium, plutonium or americium interact with the beryllium-9 in an atomic reaction that produces carbon-12, a fast neutron and gamma rays.



# Neutron Logging Processes

- Neutron Scattering: The fast neutrons interact with the nuclei of atoms within the formation. The interaction is a form of elastic scattering involving the neutrally charged neutron and a stationary positively charged nucleus. At each interaction the neutron loses some energy and slows down, and the nucleus of the atom in the formation material gains energy. Such collisions occur with nuclei of all elements.
- However, the process of energy transfer is most efficient when the masses of the neutron and the nucleus are the same, and becomes much less efficient when the nuclei of the formation material are more massive than the neutron. The neutron has approximately the same mass as hydrogen nuclei (the lightest element). Hence the neutrons lose energy by elastic scattering most efficiently by interaction with hydrogen nuclei, and lose energy much less efficiently by interaction with massive nuclei such as silicon or oxygen.



This figure shows the efficiency of several elements in slowing down fast neutrons for a clean sandstone

# Neutron Logging Processes

- The initially fast neutrons ( $>0.5$  MeV) quickly lose their energy and become slower, passing through stages called intermediate neutrons ( $10^2$  to  $10^5$  eV), epithermal neutrons ( $0.1$  to  $100$  eV), and finally thermal neutrons ( $<0.1$  eV). In solid materials containing reasonable amounts of low atomic mass elements, this process can happen very quickly for a given neutron (of the order of microseconds).
- However, the time taken to slow down to a given energy will vary from neutron to neutron, depending on the chance collisions with nuclei. Thermal neutrons are so called because they are energies possessed by a particle as a result of its existence at room temperatures. In other words, they only have the small energies associated with the random kinetic motion associated with room temperatures.

# Neutron Logging Processes

- When the neutrons attain epithermal or thermal energies, collisions occur much less frequently because the neutrons are moving from nucleus to nucleus much more slowly. Within a few microseconds of being exposed to the fast neutron source, the formation has slowed the incoming neutrons down to epithermal and thermal levels, and a cloud of these thermal neutrons exists in the formation surrounding the source. Collisions continue resulting in little further loss of energy and the slow diffusion of the thermal neutrons from the zone around the detector. During this process the neutrons are absorbed by the formation nuclei.

# Neutron Logging Processes

- Neutron Absorption: Thermal (and to some extent epithermal) neutrons can be absorbed by the nuclei of the formation atoms. The efficiency of neutron absorption varies from element to element. The only element which exhibit significant neutron absorbing behaviour and exist in reasonable quantities in rocks are hydrogen and chlorine. The process of neutron absorption by hydrogen and chlorine are:



- In neutron logging, some tools measure the epithermal neutrons, some the thermal neutrons and some the gamma rays emitted when a neutron is absorbed.

# Neutron Log

- **Uses**

- Porosity determination: display directly on the log
- Lithology Identification: with the Sonic and/or Density.
- Gas Indication: with the Density.
- Clay content (shaliness): with the Density .
- Correlation: especially in cased holes.
- Characterization of mineralogical composition of the formation via Elemental Analysis based on Spectral Nuclear Measurements

|                         | thermal                                                               | epithermal                   | neutron-gamma       |
|-------------------------|-----------------------------------------------------------------------|------------------------------|---------------------|
| Radius of investigation | 6 in = 15 cm (50 %)                                                   |                              | 8 in = 20 cm (50 %) |
| Vertical resolution     | 36 in = 84 cm;<br><br>20 in = 51 cm enhanced<br>resolution processing | 30 ... 44 in = 76 ... 112 cm | 20 in = 51 cm       |
| Precision               | 0.4 p.u                                                               | 1 p.u.                       |                     |
| Logging speed           | < 60 feet/min = 18 m/min<br>Lower for enhanced resolution processing  |                              |                     |



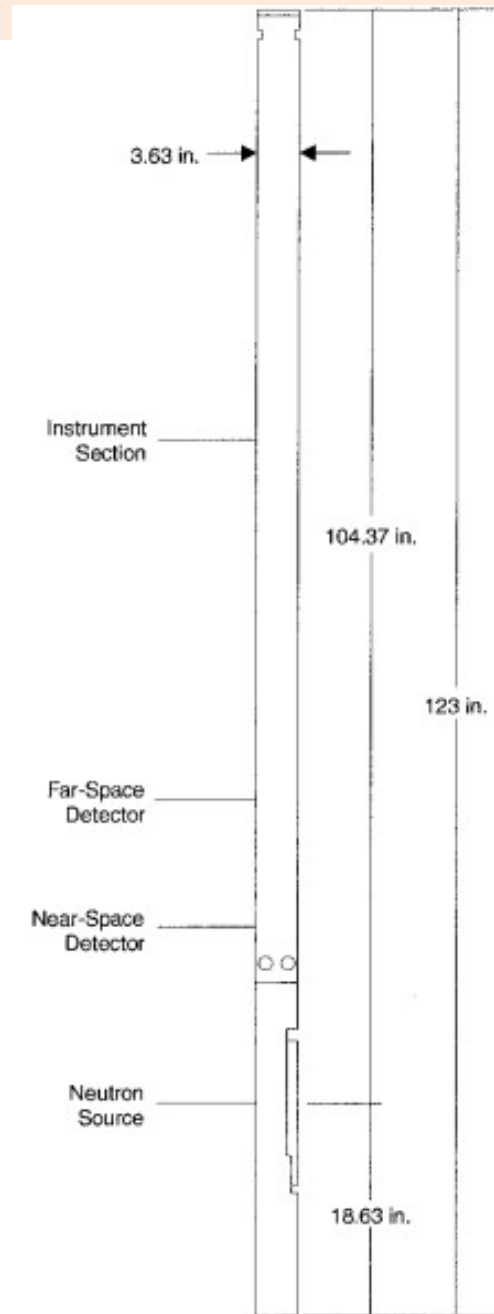
# Neutron Logging Tool

- **Physics of Measurement**

- A chemical source (alpha particles from the radium, plutonium or americium interacting with beryllium-9) or a particle accelerator (which forces deuterium and tritium collisions at high energy levels) emit fast, high energy which are slowed down by the formation nuclei. Two detectors in the tool count the number of returning captured gamma rays or neutrons (depending on the tool type). The detector count rates are inversely proportional to the amount hydrogen in the formation (“Hydrogen index”). By assuming that .
- “Gamma-ray-neutron” tools detect gamma rays and thermal neutrons; “sidewall” tools detect epithermal neutrons, “compensated” tools detect thermal neutrons.

- **Operational Constraints**

- In Open hole or Cased hole (in special conditions), the tool is run eccentric
- Can be run in air, water or oil-based mud



Neutron Tool

# Neutron Tool – Hydrogen Index

- We can define a partial concentration of hydrogen per unit mass  $(C_H)_{mass}$  of a material as the mass of hydrogen atoms in the material divided by the mass of all the atoms of all elements in the material.

$$(C_H)_{mass} = \frac{n_H A_H}{\sum_i n_i A_i + n_H A_H}$$

- $A_H$  = the atomic mass of hydrogen atoms in the material
  - $A_i$  = the atomic mass of non-hydrogen element “i”
  - $n_H$  = the number of hydrogen atoms in a molecule of the material
  - $n_i$  = the number of non-hydrogen atoms of element “i” in a molecule of the material
- Thus, for pure water ( $H_2O$ ), where the atomic mass of hydrogen is 1.0 and the atomic mass of oxygen is 16.0, the partial concentration of hydrogen is

$$(C_H)_{mass} = \frac{2 \times 1.0}{(1 \times 16.0) + (2 \times 1.0)} = \frac{1}{9}$$

# Neutron Tool – Hydrogen Index

- Hence, the partial concentration of hydrogen per unit volume is  $1/9$ , because the density of pure water is  $1.000 \text{ g/cm}^3$ . The Hydrogen Index of a material is defined as the partial concentration of hydrogen per unit volume relative to water. So, if the hydrogen index of water is constrained by the definition to be unity, and water has a partial concentration of hydrogen per unit volume of, the hydrogen index of a material is

$$HI = \frac{9n_H A_H}{\sum_i n_i A_i + n_H A_H} \rho_b$$

- Note the following:
  - If the tool is in 100% water, the  $HI = 1$ . This is equivalent to a rock of 100% porosity saturated with water. So we have a fixed point that  $HI = 1$  represents porosity,  $\phi = 1$ .
  - If the tool is in a pure limestone rock with zero porosity, the  $HI = 0$ , because there are no hydrogen atoms in the solid matrix of calcite. So we have a second fixed point where  $HI = 0$  represents  $\phi = 0$ .
  - If the tool is in a pure limestone with a given porosity  $f$ , the hydrogen index will be directly proportional to the amount of water in the formation. Hence  $HI = \phi$ , for completely water saturated limestone.

# Neutron Tool – Hydrogen Index

Hydrogen Index Calculations for some reservoir minerals & fluids

| Compound   | Formula                              | $A_i$      | $n_i$   | $n_H$ | $b$   | $HI$            |
|------------|--------------------------------------|------------|---------|-------|-------|-----------------|
| Pure water | H <sub>2</sub> O                     | 16         | 1       | 2     | 1.000 | 1.000           |
| Oil        | (CH <sub>2</sub> ) <sub>x</sub>      | 12         | 1       | 2     | 0.780 | 1.003           |
| Methane    | CH <sub>4</sub>                      | 12         | 1       | 4     | $m$   | $2.25 \times m$ |
| Gas        | C <sub>1.1</sub> H <sub>4.2</sub>    | 12         | 1.1     | 4.2   | $g$   | $2.17 \times g$ |
| Quartz     | SiO <sub>2</sub>                     | 28, 16     | 1, 2    | 0     | 2.654 | 0.000           |
| Calcite    | CaCO <sub>3</sub>                    | 40, 12, 16 | 1, 1, 3 | 0     | 2.710 | 0.000           |
| Gypsum     | CaSO <sub>4</sub> .2H <sub>2</sub> O | 40, 32, 16 | 1, 1, 6 | 4     | 2.320 | 0.4855          |

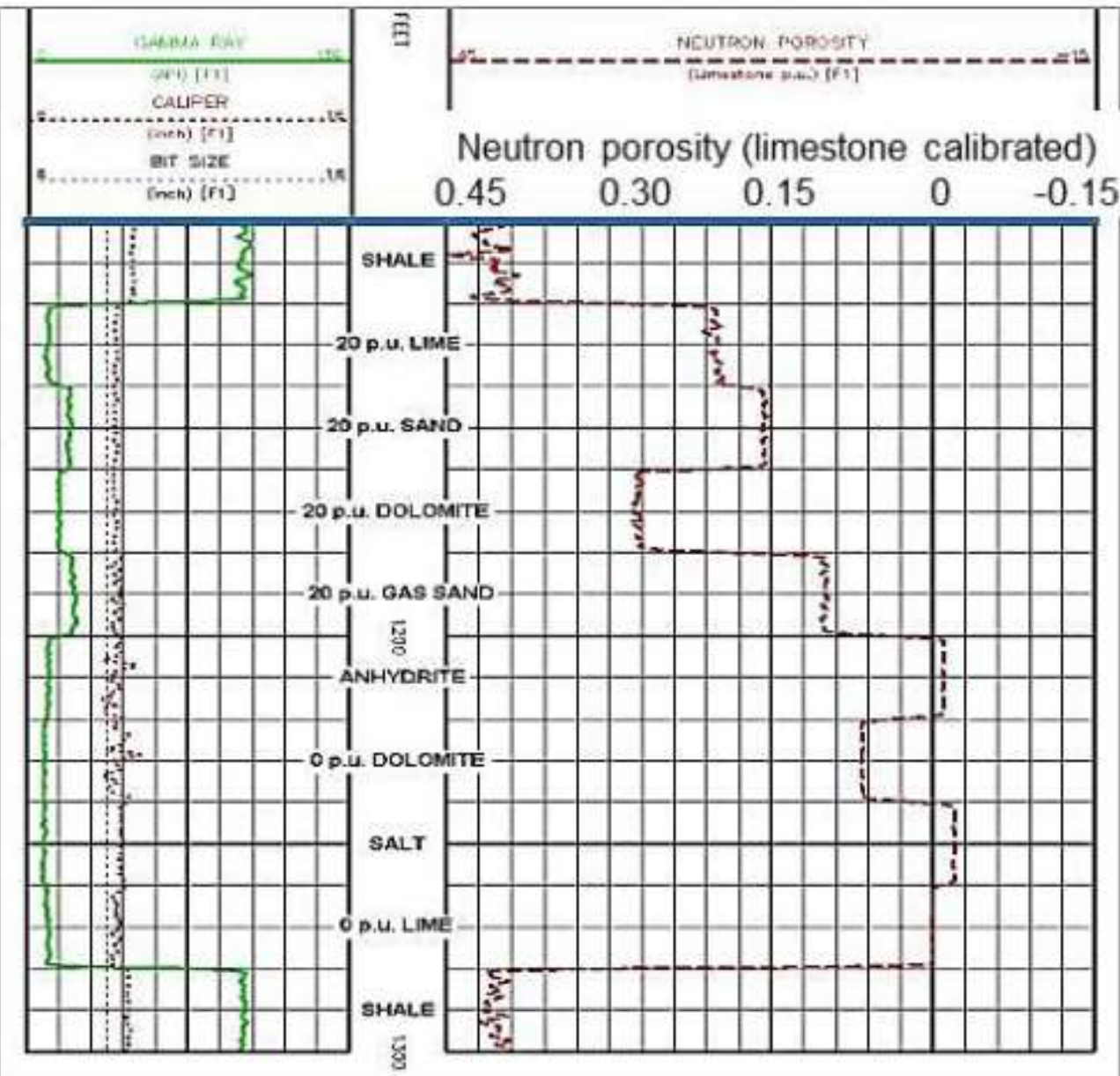
# Neutron Response



| Component                    | $\phi_N$      | Component      | $\phi_N$      |
|------------------------------|---------------|----------------|---------------|
| Water, fresh                 | 1             | Calcite        | 0             |
| Water, 200 000 ppm NaCl      | 0.9           | Dolomite       | 0.01 ... 0.02 |
| Oil, average                 | 0.96 ... 1.02 | Quartz         | - 0.02        |
| Gas, average, 15 °C, 0.1 MPa | 0.0017        | Gypsum         | 0.49          |
| Gas, average, 93 °C, 48 MPa  | 0.54          | Shale, average | 0.2 ... 0.4   |



# Neutron Response – Influence of Mineralogy and Pore Fluid



## Comments

Shale: indicated by high Neutron-porosity and gamma

Limestone,  $S_w = 1$ ; shows exact porosity

Sandstone,  $S_w = 1$ ; shows lower Neutronporosity

Dolomite,  $S_w = 1$ ; shows higher Neutronporosity

Sandstone shows lower porosity (gas effect)

Anhydrite; small negative deviation from porosity = 0; matrix effect

Dolomite; positive deviation from porosity = 0; matrix effect

Anhydrite; negative deviation from porosity = 0; matrix effect

Limestone porosity = 0; exact reading

Shale: indicated by high neutronporosity and gamma

# Interpretation Effects

- **The Hydrocarbon Effect**

- As with the formation density tool, the zone of investigation of the neutron tool is constrained to the flushed zone. Hence, the tool will measure the hydrogen index associated mainly with the mud filtrate, together with whatever remains of the hydrocarbons and formation water in the formation. Imagine, for example that we are in an oil zone in a limestone, the mud filtrate almost completely replaces the formation fluids such that there is a saturation of mud filtrate  $S_{XO}$ , and a residual saturation of hydrocarbons  $(1 - S_{XO})$ . The porosity read by the neutron tool is related to the actual porosity in the formation by

$$\phi_N = \phi [HI_{mf} S_{XO} + HI_{hc} (1 - S_{XO})]$$

- If the hydrocarbon is oil, this equation reduces to  $\phi_N \approx \phi$ , because it is observed from Table 1 that the hydrogen index of water and oil is similar;  $HI_{mf} \approx HI_{hc}$  whether the mud is oil based or water based.



# Interpretation Effects

- **Chlorine Effect**

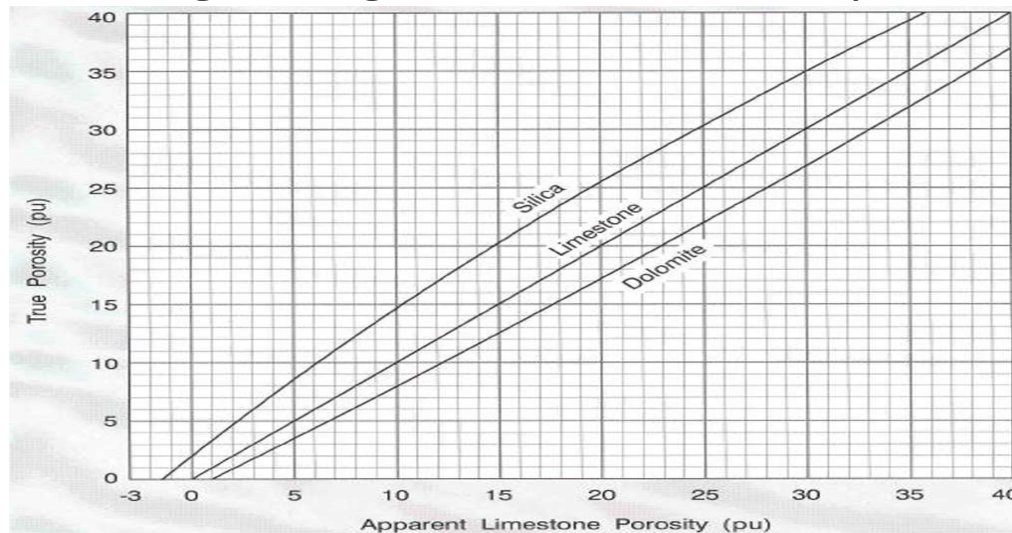
- Some types of neutron tool measure the thermal neutrons and gamma rays produced during the capture of neutrons. There are only two elements that are found in reservoirs that contribute significantly to neutron absorption; hydrogen and chlorine. The presence of hydrogen in the fluids is what we want to measure, so this is not a problem.
- However, if the drilling mud, mud filtrate or formation fluids contain a significant amount of dissolved chloride ions, as is often the case, the tool will measure a lower flux of neutrons and hence overestimate the porosity. This is called the chlorine effect, and is present in wells drilled or logged in the presence of drilling mud containing dissolved chlorine or in formations where the formation waters are particularly salty.

- **Shale Effect**

- Shale contains clays that have a significant amount of surface absorbed (bound) water. Hence shale can contain a significant proportion of hydrogen despite being low porosity. The apparent porosity read from the neutron tool in shale formations is therefore always significantly higher than it really is. This is called the shale effect or the bound-water effect.

# Calibration and Correction

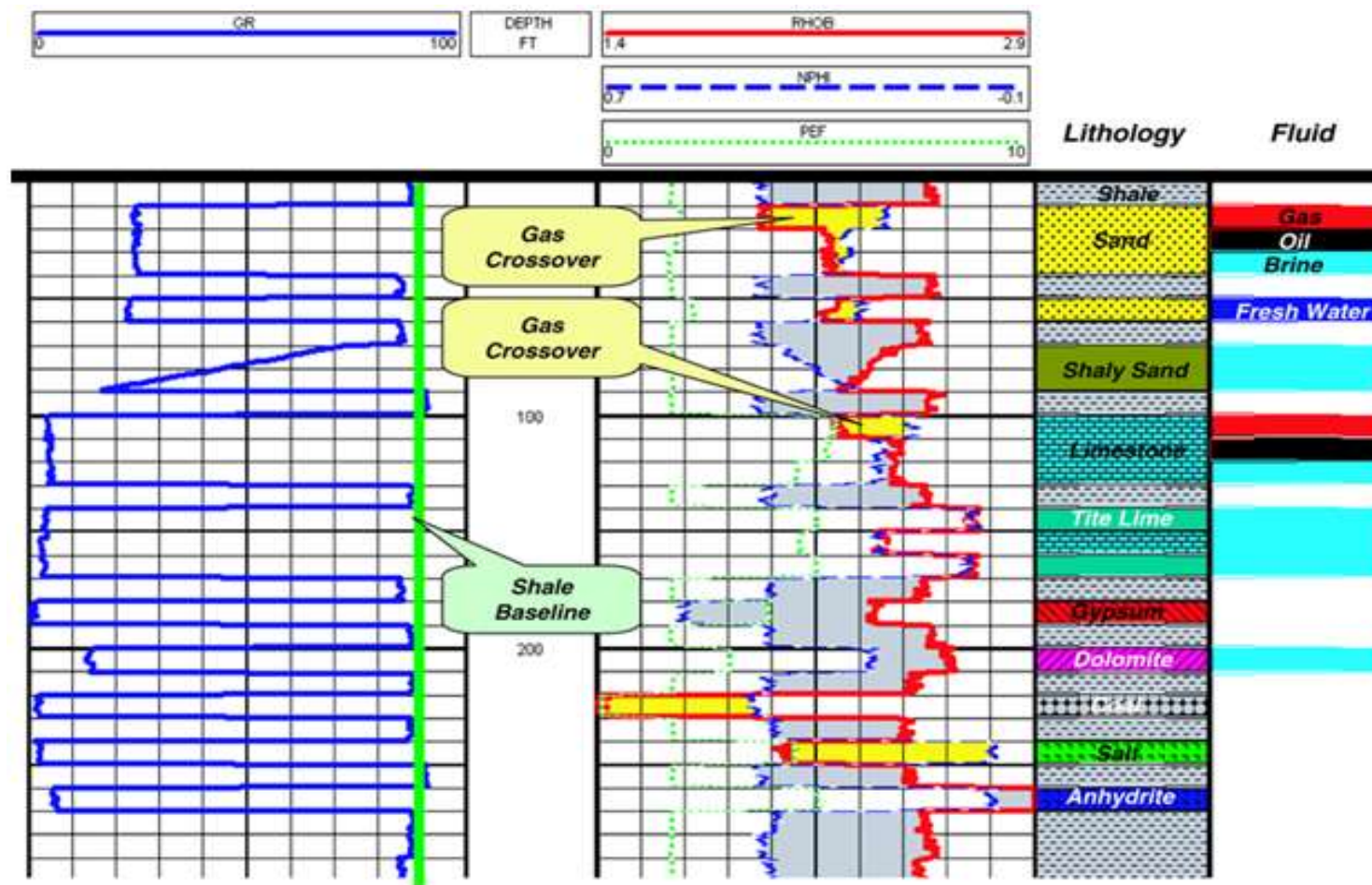
- This tool is calibrated in blocks of limestone, sandstone and dolomite of high purity and accurately known porosity in test pits. The tools are calibrated, not to give readings in API neutron porosity units, but to give the porosity directly in percent. Limestone is chosen as the calibrating lithology, so the readings are given in limestone porosity units.



- The common use of limestone porosity units as the calibrating unit for these logs means that the reading of porosity in a limestone formation will be exact, but the reading of porosity in these units in any other formation lithology will need correction.
- The calibration of the CNL tool is checked at the well site before and after each logging run by the use of a neutron source of accurately known activity placed a standard distance from each detector.

# Log Interpretation

- Used in triple- and quad-combo toolstrings, the combination of the CNL and other porosity logs results in more accurate porosity values and lithology identification, including the evaluation of shale content.



CNL Response for Common Formations

# Porosity from CNL Log

- At fundamental level, the neutron log records the amount of hydrogen in the formation regardless of if it occurs in a gas, solid, or liquid. However, the commercial application of this measurement is to assess the porosity in potential reservoir zones.
- The neutron log is anchored to two calibration points: the matrix and the fluid. The matrix choices are calcite, dolomite, or quartz. These have slightly different neutron moderation characteristics, so that the zero porosity point differs slightly, depending on whether the scale selected is limestone-equivalent, sandstone equivalent, or dolomite equivalent. The units can either be in percent (often written as "porosity units" or "p.u.") or in decimal units (zero to one).
- Because the neutron log is actually a response to the hydrogen in the formation, regardless of its presence in a gas, liquid or solid, hydrogen-rich lithologies will show high neutron porosities, even though there is no effective pore space. Examples include;
  - Shales

# Porosity from CNL Log

- Water of Crystallization: some evaporitic minerals, such as gypsum have water of crystallization. Gypsum is a hydrated form of calcium sulfate with a formula of  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ . The gypsum response exceeds 60% equivalent limestone porosity on a neutron log.
- Hydroxyls: minerals that have a hydroxyl component (OH). Goethite is an example, with its formula of  $\text{FeO}(\text{OH})$ , results in a neutron porosity reading in excess of 60%.
- Coal: the high neutron porosities of coals are caused by their high moisture content. Their neutron porosity varies by rank so that anthracite has an expected value of about 38%, while bituminous coals have much higher values.
- In contrast with these high hydrogen sources, the occurrence of gas is marked by anomalously lower neutron porosities. The neutron porosity scale is calibrated to a pore content of water. Both water and oil have a high concentration of hydrogen per unit volume of pore space. However, hydrogen is relatively diffuse in a gas such as methane, so that the recorded neutron porosity is less than the actual pore volume. Together with the density porosity, which becomes anomalously higher with gas saturation, the "gas effect" has commercial implications in the recognition of gas zones.

# NEUTRON LOG

- For any porous rock composed of different minerals components, shale, fluids, the porosity derived from a limestone-freshwater calibrated device results in:

$$\phi_N = \phi \cdot \phi_{N, fluid} + (1 - \phi) \cdot \left[ (1 - V_{shale}) \cdot \phi_{N, matrix} + V_{shale} \cdot \phi_{N, shale} \right]$$

|                    |                                  |
|--------------------|----------------------------------|
| $\phi$             | = Rock porosity                  |
| $\phi_N$           | = Measured neutron porosity      |
| $\phi_{N, fluid}$  | = Neutron response of the fluid  |
| $\phi_{N, matrix}$ | = Neutron response of the matrix |
| $\phi_{N, shale}$  | = Neutron response of the shale  |
| $V_{shale}$        | = Shale Content                  |



# NEUTRON LOG

## Theoretical equation

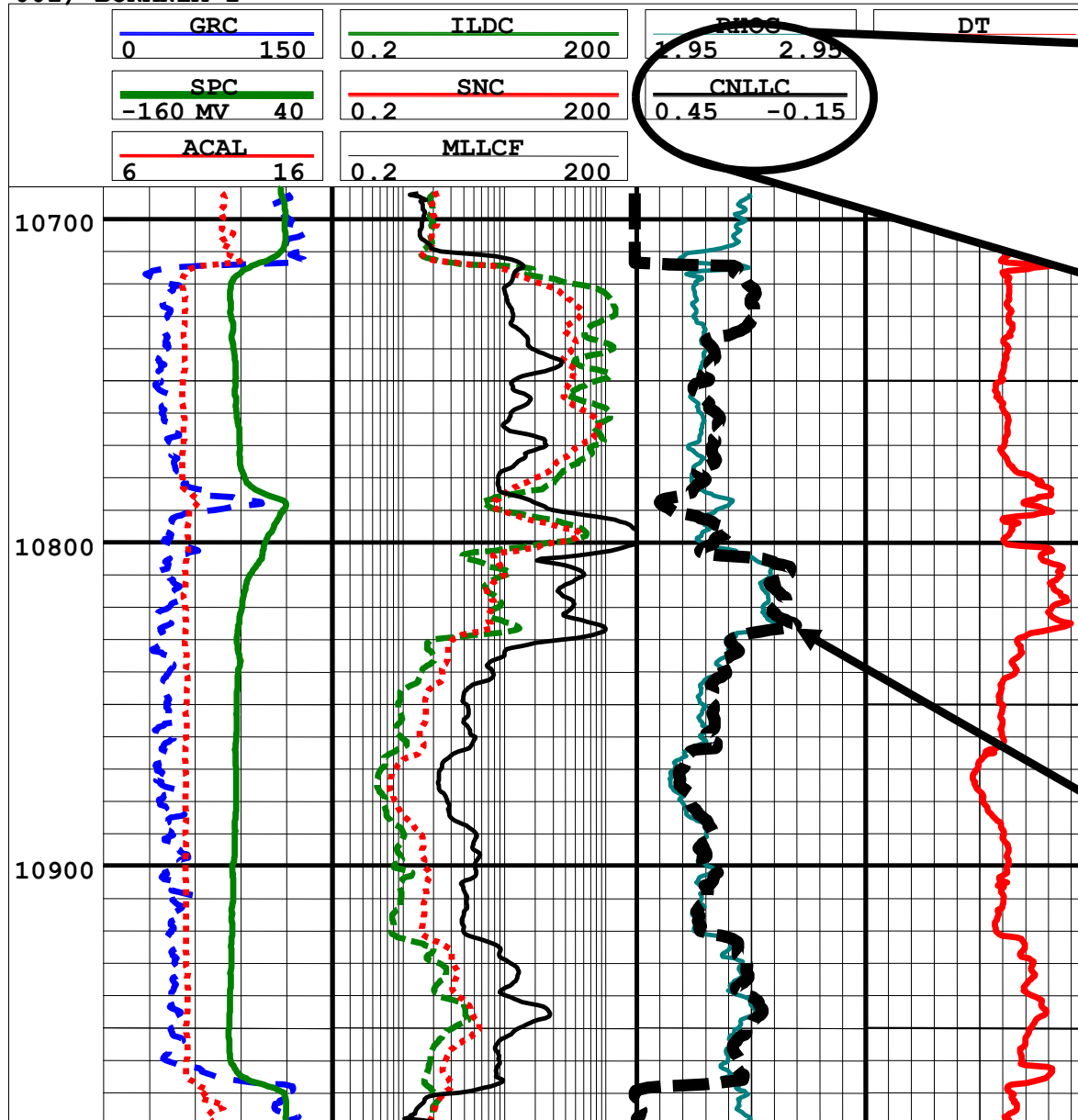
$$\phi_N = \phi S_{xo} \phi_{Nmf} + \phi (1 - S_{xo}) \phi_{Nhc} + V_{sh} \phi_{sh} + (1 - \phi - V_{sh}) \phi_{Nma}$$

|                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\phi_N$ = Recorded parameter<br>$\phi S_{xo} \phi_{Nmf}$ = Mud filtrate portion<br>$\phi (1 - S_{xo}) \phi_{Nhc}$ = Hydrocarbon portion<br>$V_{sh} \phi_{sh}$ = Shale portion<br>$(1 - \phi - V_{sh}) \phi_{Nhc}$ = Matrix portion where $\phi$ = True porosity of rock<br>$\phi_N$ = Porosity from neutron log measurement, fraction | $\phi_{Nma}$ = Porosity of matrix fraction<br>$\phi_{Nhc}$ = Porosity of formation saturated with hydrocarbon fluid, fraction<br>$\phi_{Nmf}$ = Porosity saturated with mud filtrate, fraction<br>$V_{sh}$ = Volume of shale, fraction<br>$S_{xo}$ = Mud filtrate saturation in zone invaded by mud filtrate, fraction |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



# POROSITY FROM NEUTRON LOG

001) BONANZA 1



# **SONIC (ACOUSTIC) LOG**

# Sonic Logging

- **Elastic Properties**

- Compressional wave with velocity  $v_p$ . (velocity in  $\text{m s}^{-1}$  or  $\text{ft s}^{-1}$ )
- Shear wave with velocity  $v_s$ .
- Additionally travelling along the borehole wall are boundary waves: Stoneley and Rayleigh waves

- Wave properties can be expressed in terms of elastic moduli and density of the rock.

Compressional wave: 
$$v_p = \sqrt{\frac{M}{\rho}} = \sqrt{\frac{E}{\rho} \cdot \frac{1-\nu}{(1+\nu) \cdot (1-2\nu)}} = \sqrt{\frac{k + \frac{4}{3} \cdot \mu}{\rho}}$$

Shear wave: 
$$v_s = \sqrt{\frac{\mu}{\rho}} = \sqrt{\frac{E}{\rho} \cdot \frac{1}{2 \cdot (1+\nu)}}$$

- Where

- $E$  Young's modulus defined as the ratio of stress to strain in a uni-axial stress state,
- $M$  Compressional wave modulus defined as ratio of stress to strain in a uni-axial strain state,
- $k$  Bulk Compressional modulus defined as the ratio of hydrostatic stress to volumetric strain,
- $\mu$  Shear modulus, defined as the ratio of shear stress to shear strain, (N.B.: Moduli in Pascal (Pa))
- $\nu$  Poisson's ratio, defined as ratio of lateral strain to axial strain (uni-axial stress state), (Poisson's ratio is dimensionless)
- $\rho$  Bulk density.

# Sonic Logging

- Instead of the velocity the slowness  $\Delta t$  is frequently used; it is the reciprocal of velocity.
  - Compressional slowness  $\Delta t_p = v_p^{-1}$  ( $\mu\text{s m}^{-1}$ )
  - Shear slowness  $\Delta t_s = v_s^{-1}$
- The velocity of elastic waves in rocks – particularly in reservoir rocks - shows complex dependence on various influences. The following influences are of interest for practical application:
  - Velocities  $v_p$  and  $v_s$  decrease with increasing porosity (porosity effect)
  - Different lithology (sandstone, limestone, dolomite) show different magnitudes of velocity for the same porosity value (matrix effect)
  - A change of pore fluid from gas to water results in strong increase of  $v_p$  but decrease of  $v_s$  (pore fluid effect)
  - Velocities increase nonlinearly with increasing effective pressure (pressure effect)

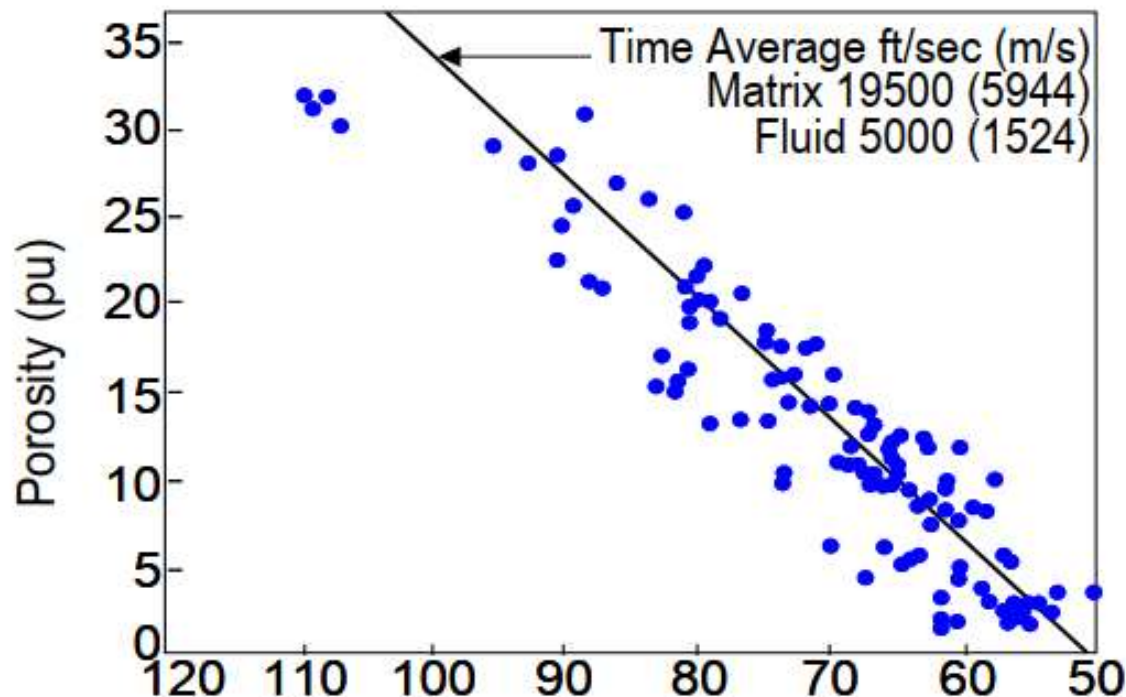
# Time Average Relationship

From the strong correlation between compressional wave slowness and porosity, Wyllie et. al (1956) derived their famous time-average equation. Written in terms of slowness:

$$\Delta t_p = \phi \Delta t_{fl} + (1 - \phi) \Delta t_{p,ma}$$

$$\phi = \frac{\Delta t_p - \Delta t_{p,ma}}{\Delta t_{fl} - \Delta t_{p,ma}}$$

This equation works best for water-saturated and well-compacted porous rock, particularly sandstones. The presence of gas can give erroneous results



- $\Delta t_p$  compressional wave slowness of the porous rock
- $\Delta t_{p,ma}$  compressional slowness of the matrix material
- $\Delta t_{p,ma}$  compressional slowness of the pore fluid (mostly assumed water)
- $\phi$  is the porosity

# Mean Input Parameters for Wyllie's Equation & Corrections

| Rock type              | $\Delta t_{p,ma}$ $\mu\text{s}/\text{ft}$ | $\Delta t_{p,ma}$ $\mu\text{s}/\text{m}$ | Fluid                    | $\Delta t_f$ $\mu\text{s}/\text{ft}$ | $\Delta t_f$ $\mu\text{s}/\text{m}$ |
|------------------------|-------------------------------------------|------------------------------------------|--------------------------|--------------------------------------|-------------------------------------|
| Sandstone $\phi > 0.1$ | 55.5                                      | 182                                      | fresh water mud filtrate | 189                                  | 620                                 |
| Sandstone $\phi < 0.1$ | 51.2                                      | 168                                      |                          |                                      |                                     |
| Limestone              | 47.6                                      | 156                                      | Saltwater mud filtrate   | 185                                  | 607                                 |
| Dolomite               | 43.5                                      | 143                                      |                          |                                      |                                     |

Anhydrite:  $50.0 \mu\text{s m}^{-1}$ , Halite:  $66.7 \mu\text{s m}^{-1}$ .

## Compaction Correction

- Poor consolidation or poor effective stress results in high slowness values and therefore an overestimate of porosity. In this case a “compaction correction” is recommended. The slowness in an adjacent shale bed is used as “compaction reference”. If measured slowness is  $\Delta t_{shale} > 100 \mu\text{s m}^{-1}$ , the following equation for a compaction-corrected Wyllie porosity is:

$$\phi_{corrected} = \phi_{Wyllie} \cdot \frac{1}{C_p} = \frac{\Delta t - \Delta t_{ma}}{\Delta t_{fl} - \Delta t_{ma}} \cdot \frac{100}{\Delta t_{shale}}$$

# Correction in Carbonate Rocks

## Compaction Correction

- For Carbonates with intergranular porosity, Wyllie's equation can be applied. For Carbonates with secondary porosity, vugs, fractures with much larger dimensions than primary pores the wave propagation is still controlled mainly by primary porosity. Therefore the equation results in porosities that are too low by the amount of secondary porosity. Thus an estimate of secondary porosity  $\varphi_{\text{sec}}$  can be given by the difference between total porosity (density or neutron)  $\varphi_t$ , and the sonic derived porosity  $\varphi_{sv}$  based on Wyllie's equation:  $\varphi_{\text{sec}} = \varphi_t - \varphi_{sv}$

## Compaction Correction

- In deep and invaded reservoir zones (and in water zones), compressional wave propagation is controlled by mud filtrate as pore fluid and the magnitude of  $\Delta t_{\text{shale}}$  is approximated by water slowness. For hydrocarbon bearing zones at shallow invasion an empirical correction is required.

$$\phi_{\text{corr}} = \phi_{\text{wyllie}} \times 0.7 \text{ for gas}$$

$$\phi_{\text{corr}} = \phi_{\text{wyllie}} \times 0.9 \text{ for oil.}$$



# Sonic Log

- Acoustic logs or Sonic logs measure the velocity of elastic waves (or slowness, the inverse property) of the formation. Therefore the general elastic properties of the formation are investigated with the measurements
- **Uses**
  - Porosity determination: from interval transit time, DT
  - Lithology Identification: with the Density and/or Neutron.
  - Synthetic Seismograms: with the Density.
  - Formation mechanical properties: with the Density.
  - Detection of abnormal formation pressures
  - Permeability identification: (with waveforms)
  - Cement bond quality
  - Borehole size: (from an attached caliper)

# Sonic Logging Tool

- **Principle of Measurement**

- A high frequency acoustic (ultrasonic) pulse from a transmitter is detected at two or more receivers. From the borehole wall, incident wave propagates as reflected and refracted waves. The Acoustic log uses the refracted wave.
- If the wave velocity of the formation is greater than that of the mud, the refraction wave moves in the direction of the boundary. At the critical angle, a refracted wave propagates with the velocity of the formation along the borehole. The wave generates secondary waves passing the mud and arriving at the two receivers. Difference of the arrival time divided by receiver distance gives the slowness of the formation (if the tool is centered).
- The first detection of the transmitted pulse at each receiver is processed to produce an interval transit time (DT or  $\Delta t$ ). The delta t is the transit time of the wave front over one foot of formation. If the entire acoustic waveform is captured, arrival times and attenuations (energy decrease) of several portions of the waveform can be measured including: compressional, shear and Stoneley.
- “Compensated” tools use multiple transmitter-receiver pairs to minimize the effects of borehole size changes. “Array” tools usually have 4 or more receivers, and the data from all receivers is processed to determine the arrival times.

- **Operational Constraints**

- In Open hole, to minimize signal attenuation, the tool should be run centered in boreholes < 16in with a standoff to reduce road noise.
- Can be run in water or oil-based mud

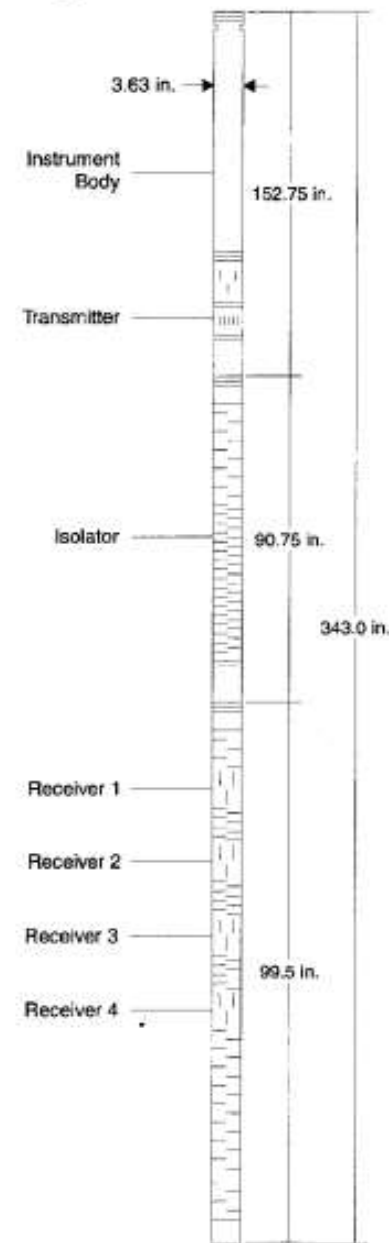
# Sonic Logging Tool

- **Physics of Measurement**

- A high frequency acoustic (ultrasonic) pulse from a transmitter is detected at two or more receivers. From the borehole wall, incident wave propagates as reflected and refracted waves. The time of the first detection of the transmitted pulse at each receiver is processed to produce an interval transit time (DT or  $\Delta t$ ). The delta t is the transit time of the wave front over one foot of formation. If the entire acoustic waveform is captured, arrival times and attenuations (energy decrease) of several portions of the waveform can be measured including: compressional, shear and Stoneley.
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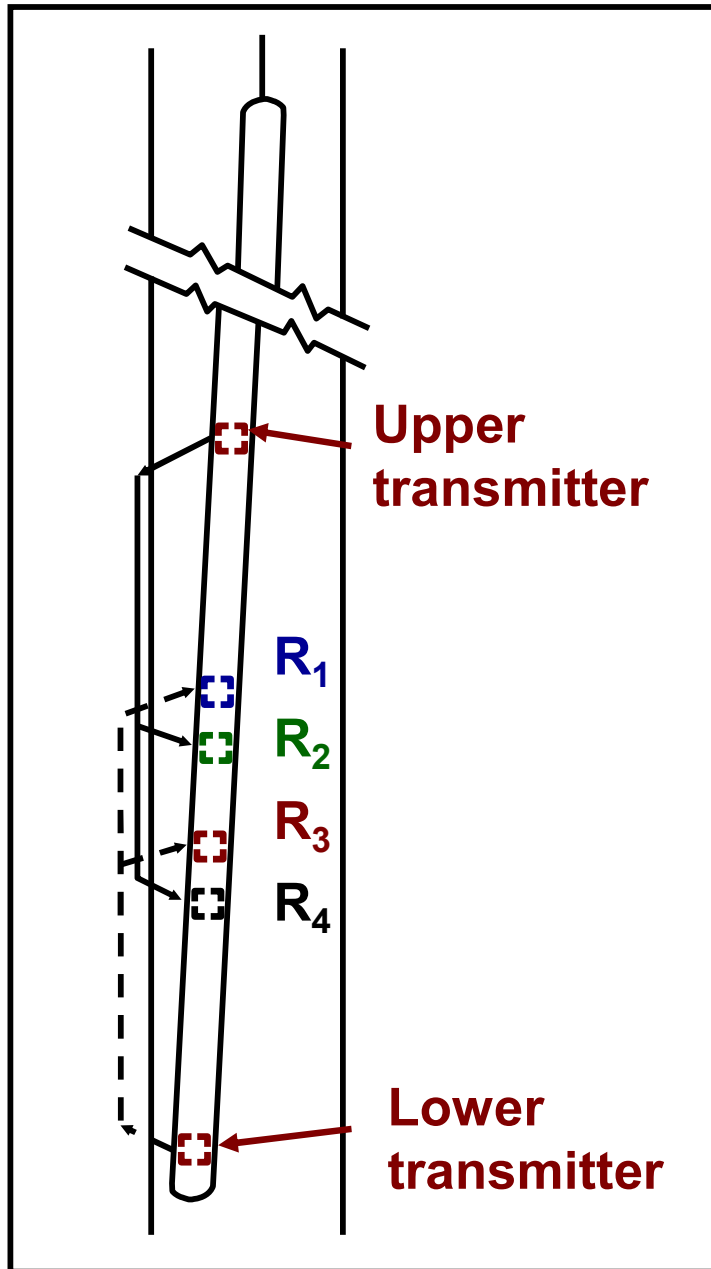
- **Operational Constraints**

- In Open hole, to minimize signal attenuation, the tool should be run centered in boreholes < 16in with a standoff to reduce road noise.
- Can be run in water or oil-based mud



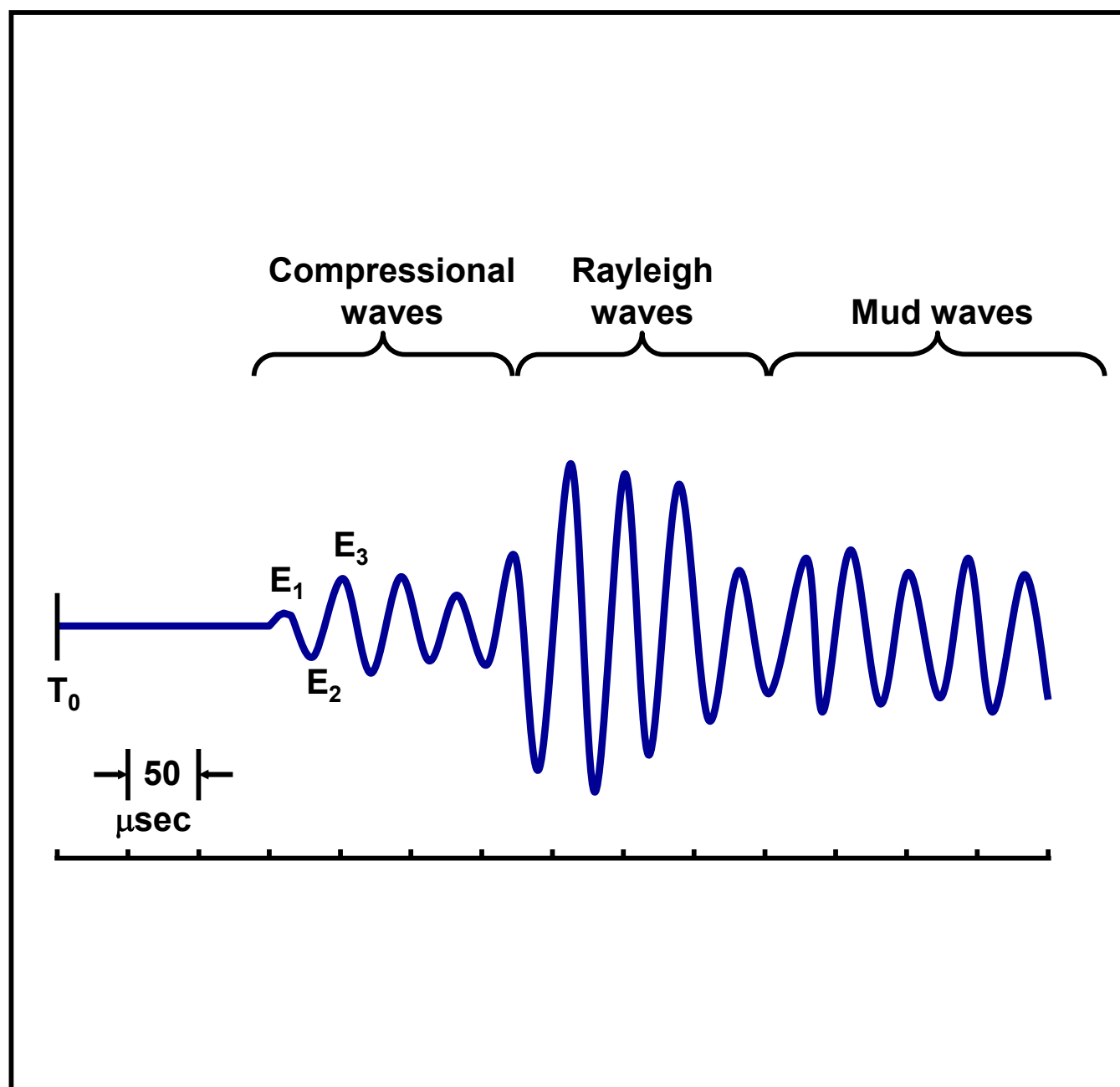
Full Wave Sonic Tool

# ACOUSTIC (SONIC) LOG



- Sonic tools are usually borehole compensated (BHC), which substantially reduces spurious effects at hole size changes as well as errors due to sonde tilt. The BHC transmitters are pulsed alternately, and the  $\Delta t$  readings are averaged. In this way, the tool is compensated for tilt.
- If the travel time for the matrix is known, then porosity can be calculated.
- The speed of sound in the sonic sonde and mud is less than that in the formations. Accordingly, the first arrivals of sound energy the receivers corresponds to the sound-travel paths in the formation near the borehole wall.
- Tool usually consists of one sound transmitter (above) and two receivers (below)
- Sound is generated, travels through formation
- Elapsed time between sound wave at receiver 1 vs receiver 2 is dependent upon density of medium through which the sound traveled

This figure illustrates sonic log response to an acoustic wave transmitted through a compacted formation. The time,  $T_0$ , at which the acoustic wave was initiated at the transmitter, is shown for reference. The first arrival at the receiver is the compressional wave. The Rayleigh wave, traveling at a slower rate, arrives later and is superimposed on the compressional wave. Following the Rayleigh waves are the slower mud waves, transmitted through the mud column and the tool.



# ACOUSTIC (SONIC) LOG

## Working equation

$$\Delta t_L = \phi S_{xo} \Delta t_{mf} + \phi (1 - S_{xo}) \Delta t_{hc} \\ + V_{sh} \Delta t_{sh} + (1 - \phi - V_{sh}) \Delta t_{ma}$$

$\Delta t_L$  = Recorded parameter, travel time read from log

$\phi S_{xo} \Delta t_{mf}$  = Mud filtrate portion

$\phi (1 - S_{xo}) \Delta t_{hc}$  = Hydrocarbon portion

$V_{sh} \Delta t_{sh}$  = Shale portion

$(1 - \phi - V_{sh}) \Delta t_{ma}$  = Matrix portion

# ACOUSTIC (SONIC) LOG

- If  $V_{sh} = 0$  and if hydrocarbon is liquid (i.e.  $\Delta t_{mf} \approx \Delta t_f$ ), then
- $\Delta t_L = \phi \Delta t_f + (1 - \phi) \Delta t_{ma}$

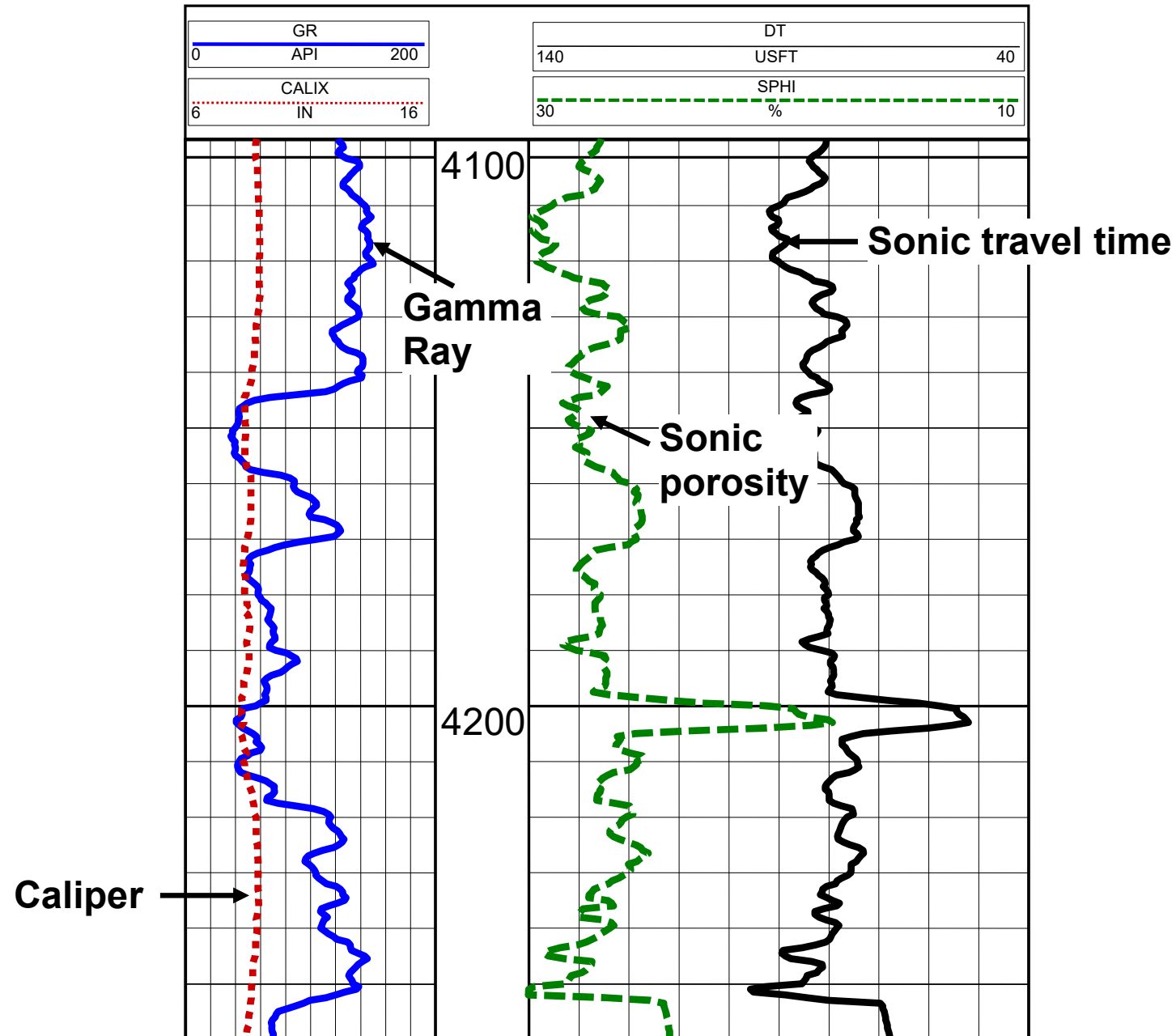
or

$$\phi_s = \phi = \frac{\Delta t_L - \Delta t_{ma}}{\Delta t_f - \Delta t_{ma}}$$

$\phi_s$  = Porosity calculated from sonic log reading, fraction  
 $\Delta t_L$  = Travel time reading from log, microseconds/ft  
 $\Delta t_{ma}$  = Travel time in matrix, microseconds/ft  
 $\Delta t_f$  = Travel time in fluid, microseconds/ ft



# ACOUSTIC (SONIC) LOG



# SONIC LOG

The response can be written as follows:

$$t_{\log} = t_{ma} (1 - \phi) + t_f \phi$$

$$\phi = \frac{t_{\log} - t_{ma}}{t_f - t_{ma}}$$

$t_{\log}$  = log reading,  $\mu\text{sec/ft}$

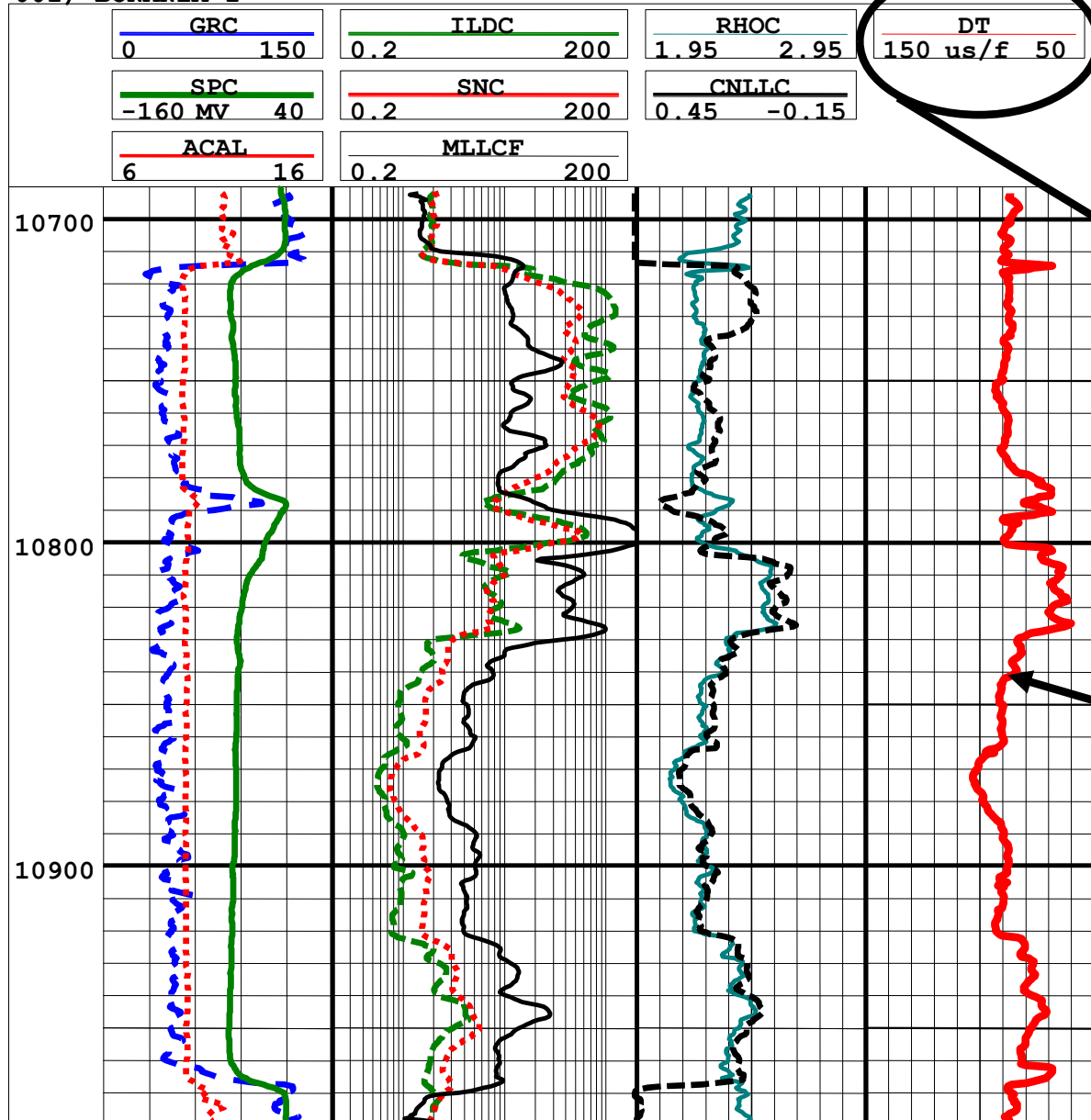
$t_{ma}$  = the matrix travel time,  $\mu\text{sec/ft}$

$t_f$  = the fluid travel time,  $\mu\text{sec/ft}$

$\phi$  = porosity

# SONIC LOG

001) BONANZA 1



Sonic  
Log

# EXAMPLE

## Calculating Rock Porosity Using an Acoustic Log

Calculate the porosity for the following intervals. The measured travel times from the log are summarized in the following table.

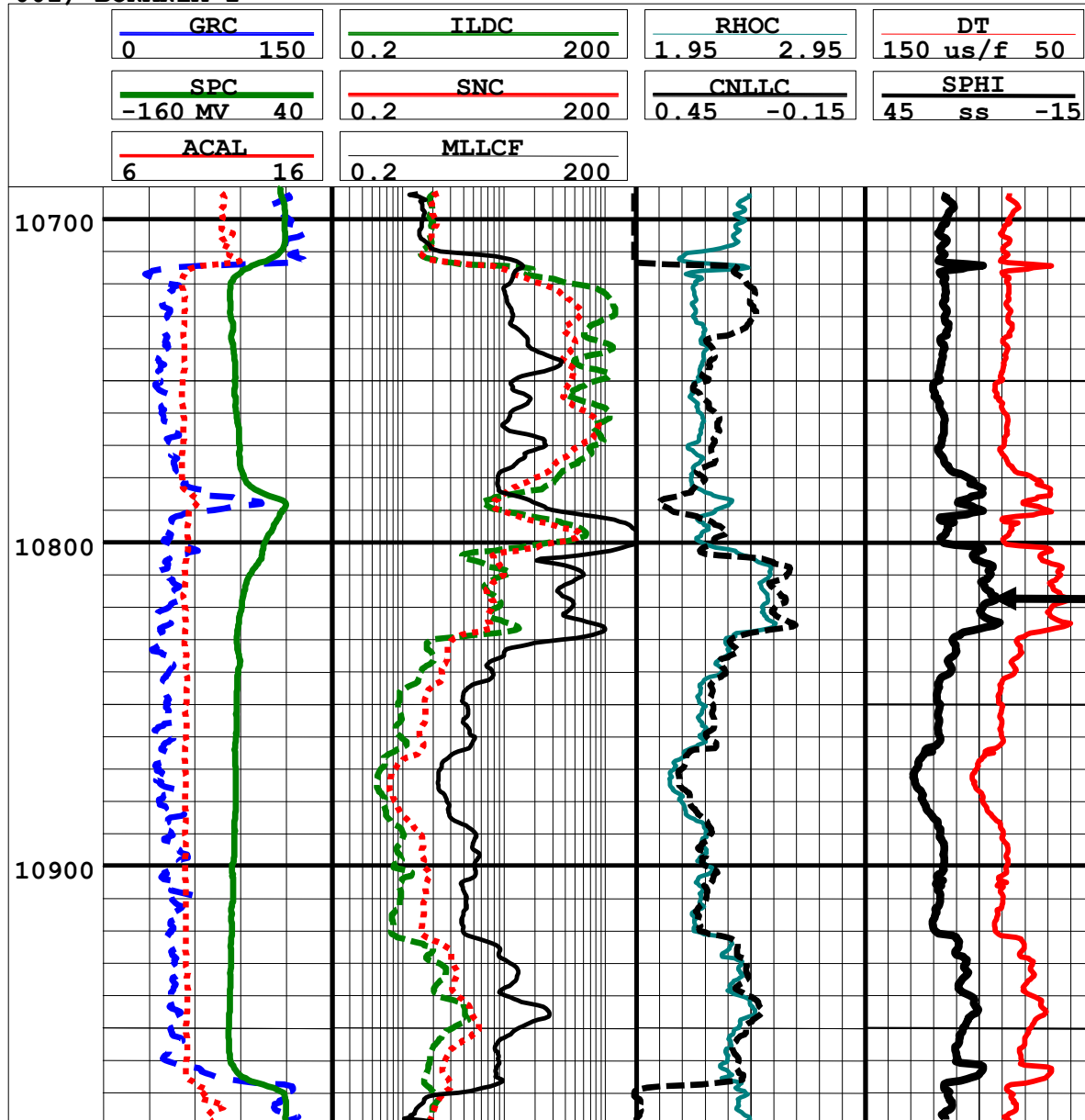
At depth of 10,820', acoustic log reads travel time of 65  $\mu\text{s}/\text{ft}$ .

Calculate porosity. Does this value agree with density and neutron logs?

Assume a matrix travel time,  $\Delta t_m = 51.6 \mu\text{sec}/\text{ft}$ . In addition, assume the formation is saturated with water having a  $\Delta t_f = 189.0 \mu\text{sec}/\text{ft}$ .

# EXAMPLE SOLUTION SONIC LOG

001) BONANZA 1



# FACTORS AFFECTING SONIC LOG RESPONSE

- Unconsolidated formations
- Naturally fractured formations
- Hydrocarbons (especially gas)
- Rugose salt sections

# RESPONSES OF POROSITY LOGS

## The three porosity logs:

- Respond differently to different matrix compositions
- Respond differently to presence of gas or light oils

## Combinations of logs can:

- Imply composition of matrix
- Indicate the type of hydrocarbon in pores

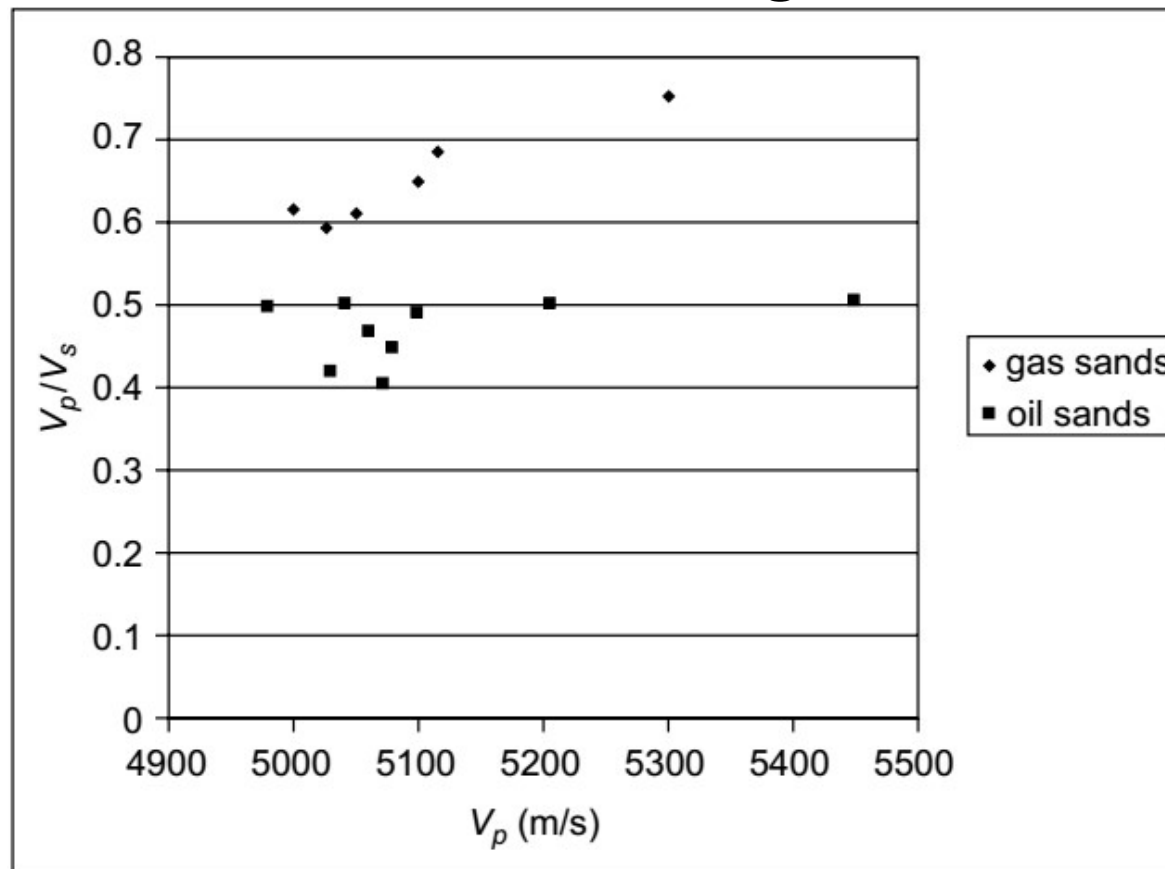


# GAS EFFECT

- Density -  $\phi$  is too high
  - Neutron -  $\phi$  is too low
  - Sonic -  $\phi$  is not significantly affected by gas
- Remember that the density log, the neutron log, and the sonic logs do not measure porosity. Rather, porosity is calculated from measurements such as **electron density**, **hydrogen index** and **sonic travel time**.
  - The calculated density porosity is too high only because in the calculation we typically don't account for the fluid density change. In other words, we assume the fluid density is 1 (or completely liquid filled) even though with gas that value is lower, which causes the calculated porosity to be too high.
  - The neutron porosity is too low because the hydrogen index or the hydrogen density of gas is lower; therefore, the liquid-filled porosity is what the neutron log sees. So when gas is present, that value is lower than the actual porosity.
  - And finally, the sonic log is not significantly affected by gas because it reads very near the wellbore and small gas saturations do not impact the overall travel time significantly.

# Identifying the Fluid Type and Contacts

- One reliable way to identify gas zones is to use the shear sonic log (if available) combined with the compressional sonic. If ratio of compressional velocity ( $V_p$ ) to sheer velocity ( $V_s$ ) is plotted against  $V_p$ , then because  $V_p$  is much more affected by gas than is  $V_s$ , a deviation will be observed in gas zones.



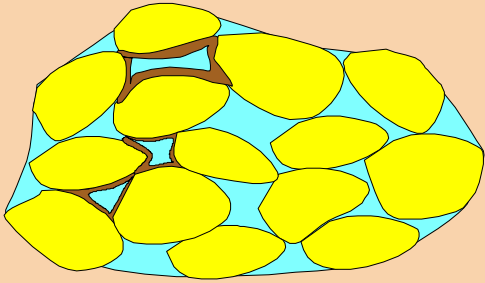
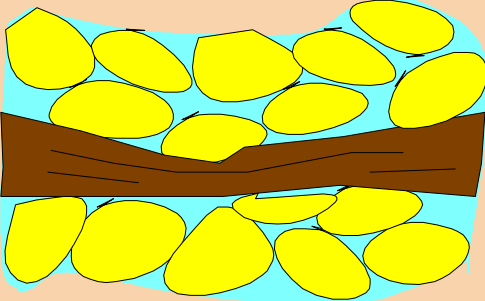
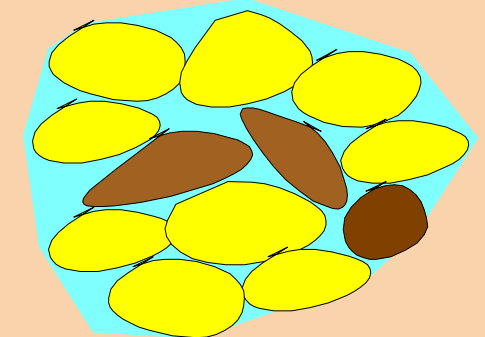
# ESTIMATING POROSITY FROM WELL LOGS

Openhole logging tools are the most common method of determining porosity:

- Less expensive than coring and may be less risk of sticking the tool in the hole
- Coring may not be practical in unconsolidated formations or in formations with high secondary porosity such as vugs or natural fractures.

If porosity measurements are very important, both coring and logging programs may be conducted so the log-based porosity calculations can be used to calibrated to the core-based porosity measurements.

# Influence Of Clay-Mineral Distribution On Effective Porosity

|                                                                                     |                                                                                                                                               |                                                                               |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|    | <p><b>Dispersed Clay</b></p> <ul style="list-style-type: none"> <li>• Pore-filling</li> <li>• Pore-lining</li> <li>• Pore-bridging</li> </ul> | <div> <div><math>\phi_e</math></div> <div>Clay Minerals</div> </div>          |
|   | <p><b>Clay Lamination</b></p>                                                                                                                 | <div> <div><math>\phi_e</math></div> <div>Detrital Quartz Grains</div> </div> |
|  | <p><b>Structural Clay</b><br/>(Rock Fragments, Rip-Up Clasts, Clay-Replaced Grains)</p>                                                       | <div> <div><math>\phi_e</math></div> <div>Clay Minerals</div> </div>          |

# Nuclear Magnetic Resonance

- Is used to derive a permeability estimate and realise fluid characterisation
- is essentially an electrical measurement, is sensitive to the quantity and distribution of free protons in the formation. Free protons occur uniquely in the fluids, so that their quantity provides another value for porosity.
- Their distribution, in small pores or large pores, leads to the determination of an average pore size and hence, through various empirical transforms, to the prediction of permeability.
- The viscosity of the fluid also affects the movement of the protons during a resonance measurement, so that the data can be interpreted to give viscosity.

# **Resistivity Logs – Lecture 6**

# Resistivity

- Resistance is the inherent property of all materials, regardless of their shape and size, to resist the flow of electric current.
- Resistivity is the measurement of resistance; the reciprocal of resistivity is conductivity  

$$\text{Resistivity} = \frac{1}{\text{Conductivity}}$$
- Units are ohm-m<sup>2</sup> / m; usually ohm-m (Ω.m)

$$R = \frac{r \times A}{L}$$

- Factors Affecting Resistivity:

- Resistivity of water
- Porosity of the formation,
- Pore geometry - tortuosity
- Lithology of the formation
- Degree of cementation, and
- Type and amount of clay in the rock.





# Resistivity Logs

- Two major applications of resistivity logs are:
  - Determination of water saturation
  - Characterization of the invasion process.
- Formation resistivity covers a broad range of magnitudes depending on porosity, water saturation, salinity and shaliness.
- The specific resistivity of the formation water is a function of salinity (concentration of dissolved salts) and temperature.
- The temperature dependence of NaCl-based water can be described by Arps' equation.

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 21.5}{T_2 + 21.5}$$

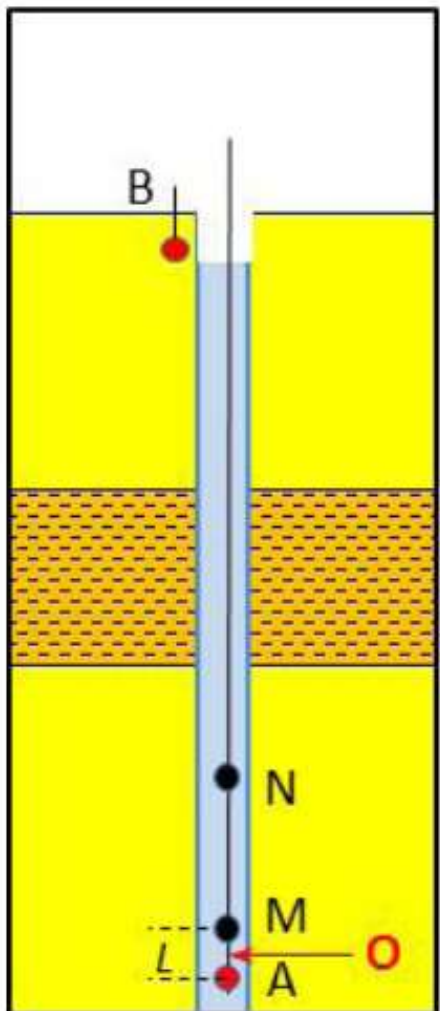
Where temperatures  $T_1$  &  $T_2$  are in °Celsius

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 6.77}{T_2 + 6.77}$$

Where temperatures  $T_1$  &  $T_2$  are in °Fahrenheit

# Early Resistivity Logging Tools

- This first resistivity tools used 4 electrodes (current electrodes A & B, voltage electrodes M & N) - The Normal tool and Gradient tool.
- Conventional tools have moderate depth of investigation (depth increases with spacing) and no sharp vertical resolution (vertical resolution decreases with spacing)

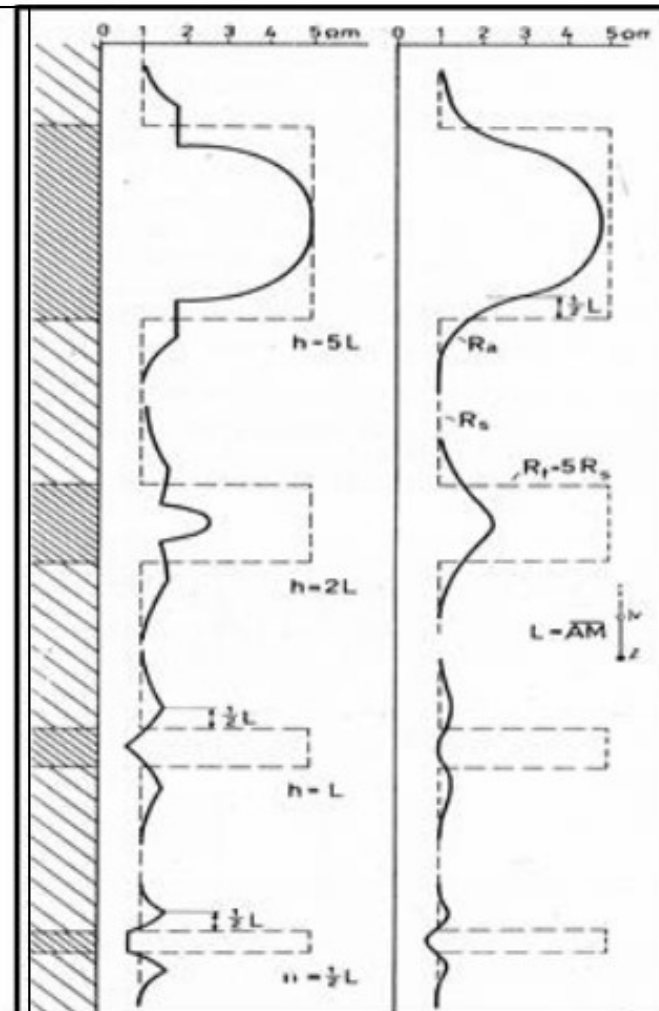


## Electrode configuration of a Normal sonde.

There are 2 standard configuration of 4-electrode normal tools

(a) Short normal with the spacing  $L=16$  inch  $\approx 0.40$  m

(a) Long normal with the spacing  $L=64$  inch  $\approx 1.60$  m



**Shape of resistivity curves (Normal) and influence of spacing  $L$  & bed thickness  $h$ .**

Left trace without mud influence, Right trace with mud influence

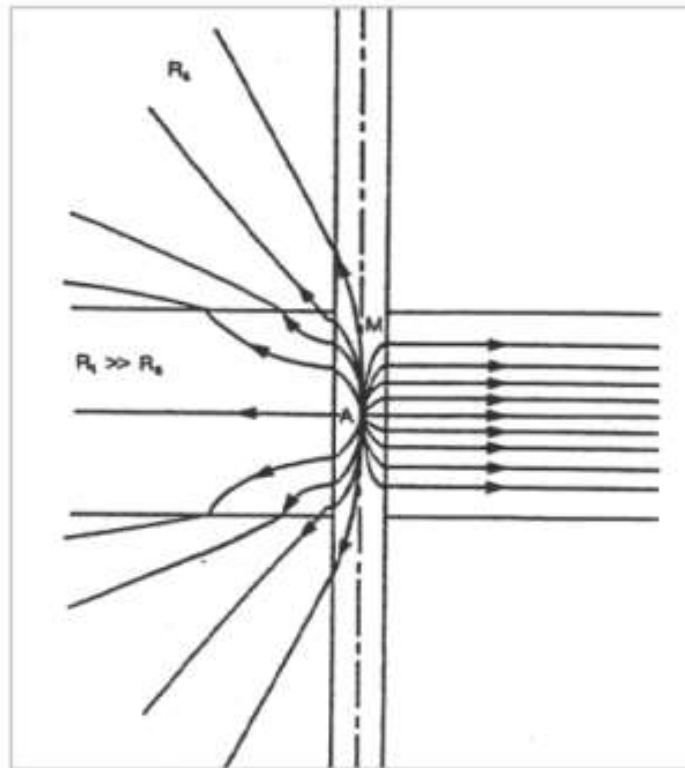
# Issues with Early Resistivity Measurements

- The ratio of bed thickness to spacing controls the logged resistivity curve. There is no sharp step at the boundary but a smoothed shape called “shoulder bed effect”.
- In highly resistive formations the current from the electrode A has the tendency to miss the formation and prefer adjacent better conductive zones and the mud column.
- Focusing electrode systems or laterologs were developed. Additional bucking or guard electrodes above and below the measuring current electrode A emit current to focus the measuring current laterally into the formation. The guard current is regulated such that there is no potential difference between additional monitoring electrodes.
- With this technique three goals are attained: (a) Focusing measuring current flowing perpendicular to the tool (b) Deep penetration of the focused current (into the invaded zone) and simultaneously (c) Sharp vertical resolution

# Issues with Early Resistivity Measurements

Non-focusing tool (normal device):

For non-focusing tools the “free current flow” avoids the highly resistive layer and takes the way of “lowest resistivity”.



Focusing tool (Laterolog):

The additional bucking current from the guard electrodes presses the measuring current into the highly resistive formation.

- Current lines in a highly resistive bed for conventional (a) non-focusing (b) focusing tools

# Resistivity logs

- Resistivity tools can be grouped into two:
  - **Electrode tools:** electrical current sent by electrodes into formation. **Requires water-base muds**
  - **Induction tools:** generate a magnetic field that induces a current in the formation (Oil-base, air, or fresh-water muds)

## Electrode Tools

- **Wireline (conveyed on line)**
  - ✓ Dual Laterolog (LLD and LLS)
  - ✓ Azimuthal tool (deep and shallow)
  - ✓ Spherically-focused
  - ✓ Micro-resistivity
- **LWD (Logging-While-Drilling)**
  - ✓ Resistivity at bit (or at least close)
  - ✓ Side-scanning electrodes

## Induction Tools

- **Wireline (conveyed on line)**
  - ✓ Dual Induction (Deep; Medium)
  - ✓ High Resolution Induction
  - ✓ Dielectric induction
  - ✓ Array Induction
  - ✓ EWR/CDR
- **LWD (Logging-While-Drilling)**
  - ✓ x

# Resistivity logs in the invaded zone

- For investigation of the invaded zone (and determination of  $R_{xo}$ ) another tool type is applied – the family of micrologs.
- The electrode system is fixed on a pad and pressed against the borehole wall. The electrodes have a very short distance (microlog 2.5cm) and are therefore characterized by:
  - Extreme vertical resolution, able to detect thin layers, but also fractures,
  - Extremely small depth of investigation ( $R_{mc}$ ,  $R_{xo}$ )
- There are unfocused systems (classic Microlog ML) as well as focused systems (e.g. Microlaterolog MLL, MicroSphericalFocusing log MSFL).

# INDUCTION LOG



# Induction Log - Introduction

- Induction log was originally developed to measure formation resistivity in boreholes containing oil-based and fresh water-based drilling muds.
- Electrode devices (conventional electric logs) do not function in non-conductive mud systems.
- Induction logging devices are focused to minimize influence of borehole and surrounding formations.
- Designed for deep investigation to determine  $R_t$ .
- New induction log devices are being developed using improved electronics, telemetry, and computer processing.

# Induction - Interpretation Goals

- **True (undisturbed) formation resistivity,  $R_t$ :**
  - True formation resistivity can be determined from the deep induction curve by applying corrections for borehole size, mud resistivity and bed thickness. The medium and shallow curves are needed for invasion correction along with the proper algorithms and/or charts. In most cases, the deep induction curve is sufficiently close to  $R_t$  thus not requiring correction.
- **Geopressure (overpressure) Detection:**
  - Deep resistivity values in the shales in a well are plotted on a reduced scale, and a trend with deep is defined. The trend is usually one of increasing resistivity with depth. An abrupt decrease in shale resistivity with depth usually indicates an increase in formation pressure.
- **Diameter of Invasion:**
  - is determined as a by-product of the correction of the Deep Induction curve using a graphical “tornado chart” (or mathematical equivalent).
- **Correlation:**
  - Curves are scanned for similarities in shape and magnitude. The logs are often displayed at scales of 1 inch per 100 feet or 2 inches per 100 feet. Correlation of resistivity is really a correlation of fluid volume and fluid salinity.

# Induction - Interpretation Goals

- **Fluid Saturation,  $S_w$ , via Archie's:**

$$S_w = \left( \frac{a \cdot R_w}{\phi^m \cdot R_t} \right)^{\frac{1}{n}}$$

- $S_w$  = formation water saturation
- $R_w$  = formation water resistivity
- $R_t$  = “true” formation resistivity (from the induction log)
- $\phi$  = porosity
- $a$  = cementation factor
- $m$  = cementation exponent
- $n$  = saturation exponent

- Archie's equation assumes that all electrical conductivity occurs in the water saturated portion of the porosity in the rock, with the rock matrix and any hydrocarbons acting as insulators. The presence of clays in the formation (“a shaly sand”) creates additional formation conductivity (a lower formation resistivity than an equivalent clean sand). In this case, Archie's equation will predict an  $S_w$  greater than is actually in the formation.
- Several “shaly sand equations have been developed to account for the effects of these clays. Commonly used are Simandoux, Dual Water, and Waxman-Smiths.

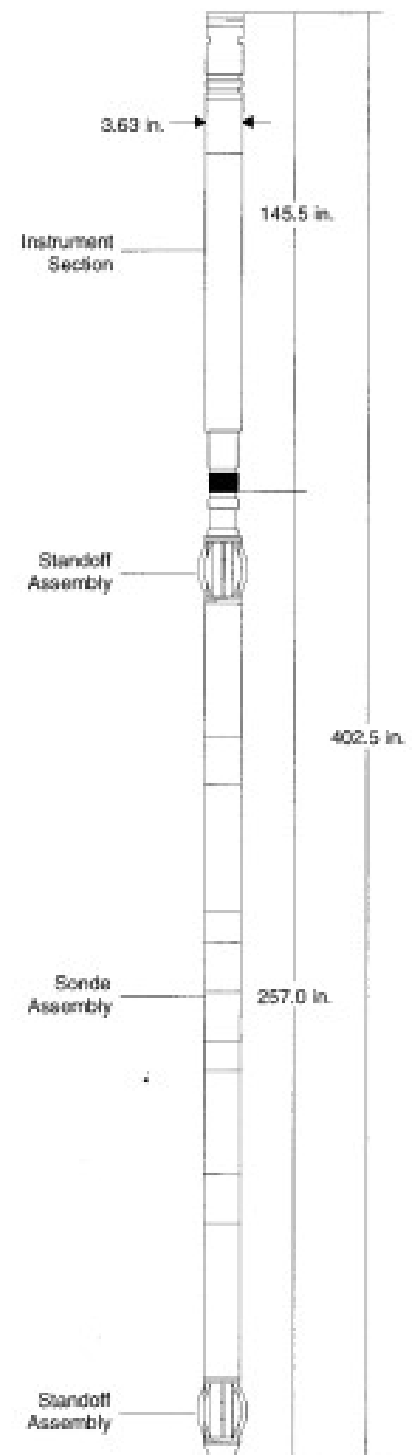
# Induction

- **Physics of Measurement**

- Transmitter coils induce an alternating current in the formation. Receiver coils sense the response of the formation, both in magnitude and phase. This response is proportional to the formation conductivity (the inverse of resistivity). Multiple transmitter and coils are used in the effort to minimize borehole and invasion effects on the tool.
- Measurements of the in-phase and out-of-phase parts of the signal are digitally recorded and the tool operates at different frequencies in order to improve its accuracy. Accuracy is further enhanced by environmental corrections done in real time.
- Array tools have many receivers, usually at small spacings, and rely on signal processing to create a common vertical resolution for all measurements.

- **Operational Constraints**

- In Open hole, the tool is run centered
- Can be run in air, water or oil-based mud
- Logging speed: 30 to 100 feet/minute depending on tool type



Array Induction  
Logging Tool

# Induction

## Fundamentals:

- **Transmitter coils:**

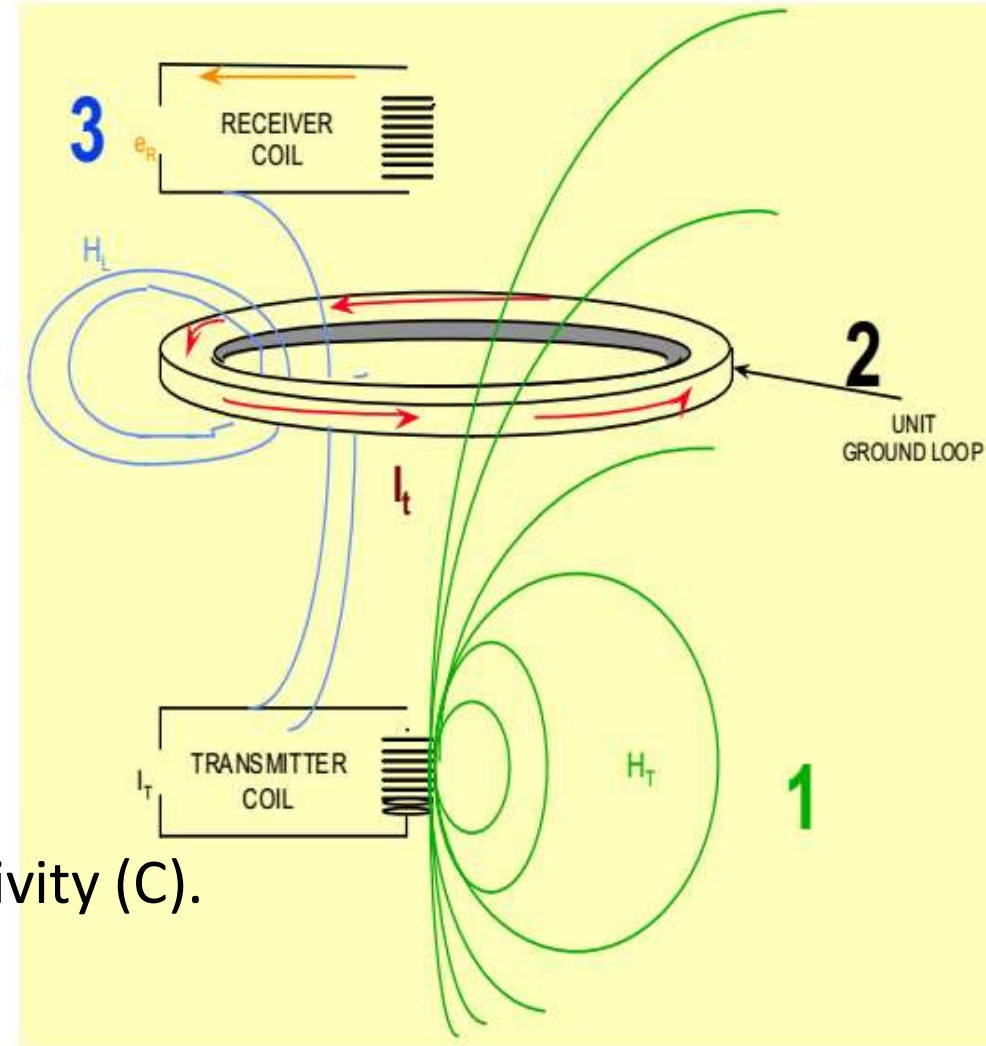
- Current ( $I_T$ ).
- Creates magnetic field ( $H_T$ ).

- **Formation currents,  $I_t$ :**

- Caused by magnetic field ( $H_T$ ).
- Create magnetic field ( $H_L$ ).

- **Receiver coils:**

- Senses magnetic field ( $H_L$ ).
- Records output voltage ( $e_R$ ).



## Estimation of Resistivity Response

- Voltage ( $e_R$ ) proportional to conductivity (C).
- Resistivity computed as:
- induction tools are focused.
- Depth of measurement depends on:
  - Frequency: Multiple frequencies
  - Number and position of coils

$$R(\text{ohm} - \text{m}) = \frac{1000}{C (\text{mmho/m})}$$

# Induction Log – Quality Control

- All three resistivity curves should stack (have the same value) in impermeable beds once they are corrected for borehole and bed thickness effects.
- Curves should stack at about 2000 ohm-m in impermeable, non-conductive beds (e.g. anhydrite) greater than 20 feet thick.
- The proper invasion profile in water-bearing permeable beds is:
  - ✓ If  $R_{mf} < R_w$  then Shallow < Medium < Deep;
  - ✓ If  $R_{mf} > R_w$  then Shallow > Medium > Deep;
- Shale values should be similar to those of nearby wells.
- Induction curves should be free of spikes; especially the high resistivity range.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Cross-check the curve character with other curves from the same logging run.

# LATEROLOG

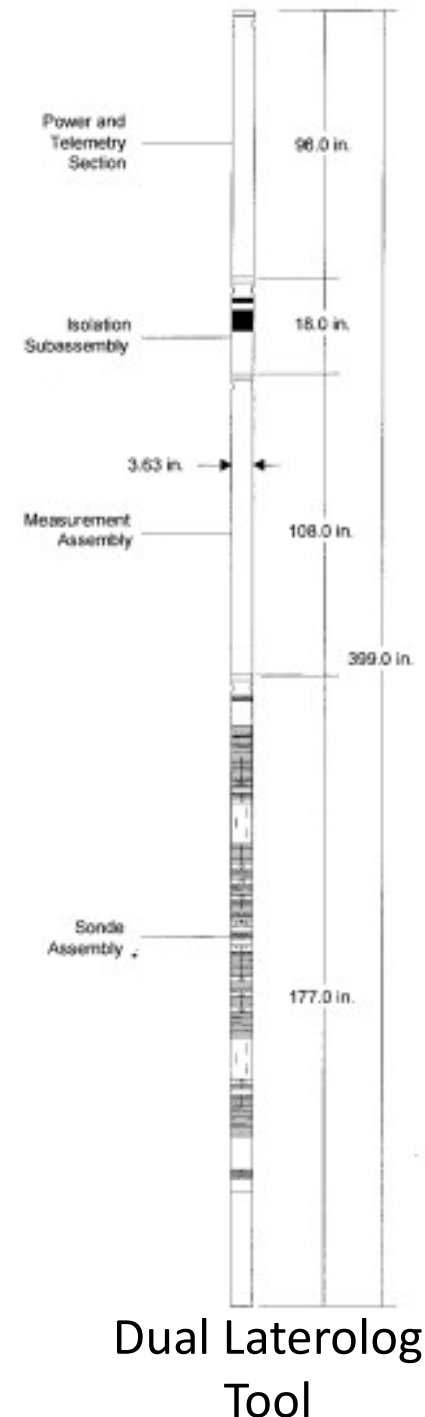


# Laterolog - Interpretation Goals

- **True (undisturbed) formation resistivity,  $R_t$ :**
  - True formation resistivity can be determined from the Deep Laterolog curve by applying corrections for borehole size and bed thickness. The Shallow Laterolog and  $R_{xo}$  curves are needed for invasion correction along with the proper algorithms and/or charts. In most cases, the Deep Laterolog curve is sufficiently close to  $R_t$  thus not requiring correction.
- **Geopressure (overpressure) Detection:** Same as for Induction Log.
- **Correlation:** Same as for Induction Log.
- **Fluid Saturation,  $S_w$ , via Archie's:** Same as for Induction Log.
- **Diameter of Invasion:**
  - is determined as a by-product of the correction of the Deep Induction curve using a graphical “tornado chart” (or mathematical equivalent). Using the chart, ratios of the Deep/Shallow and Deep/ $R_{xo}$  resistivity measurements are entered on the x and y axis, and values of invasion diameter ( $d_i$ ),  $R_t/R_{xo}$ , and  $R_t/R_{deep}$  are read from the three families of curves which comprise the chart.  $R_t$  is determined from the  $R_t/R_{deep}$  ratio and  $R_{xo}$  from  $R_t/R_{xo}$  ratio
  - Invasion can be determined qualitatively by the separation of the three resistivity curves especially if  $R_w$  and  $R_{mf}$  are significantly different.  
Deep Laterolog ~ Shallow Laterolog <  $R_{xo}$  resistivity indicates shallow invasion;

# Laterolog - Physics of Measurement

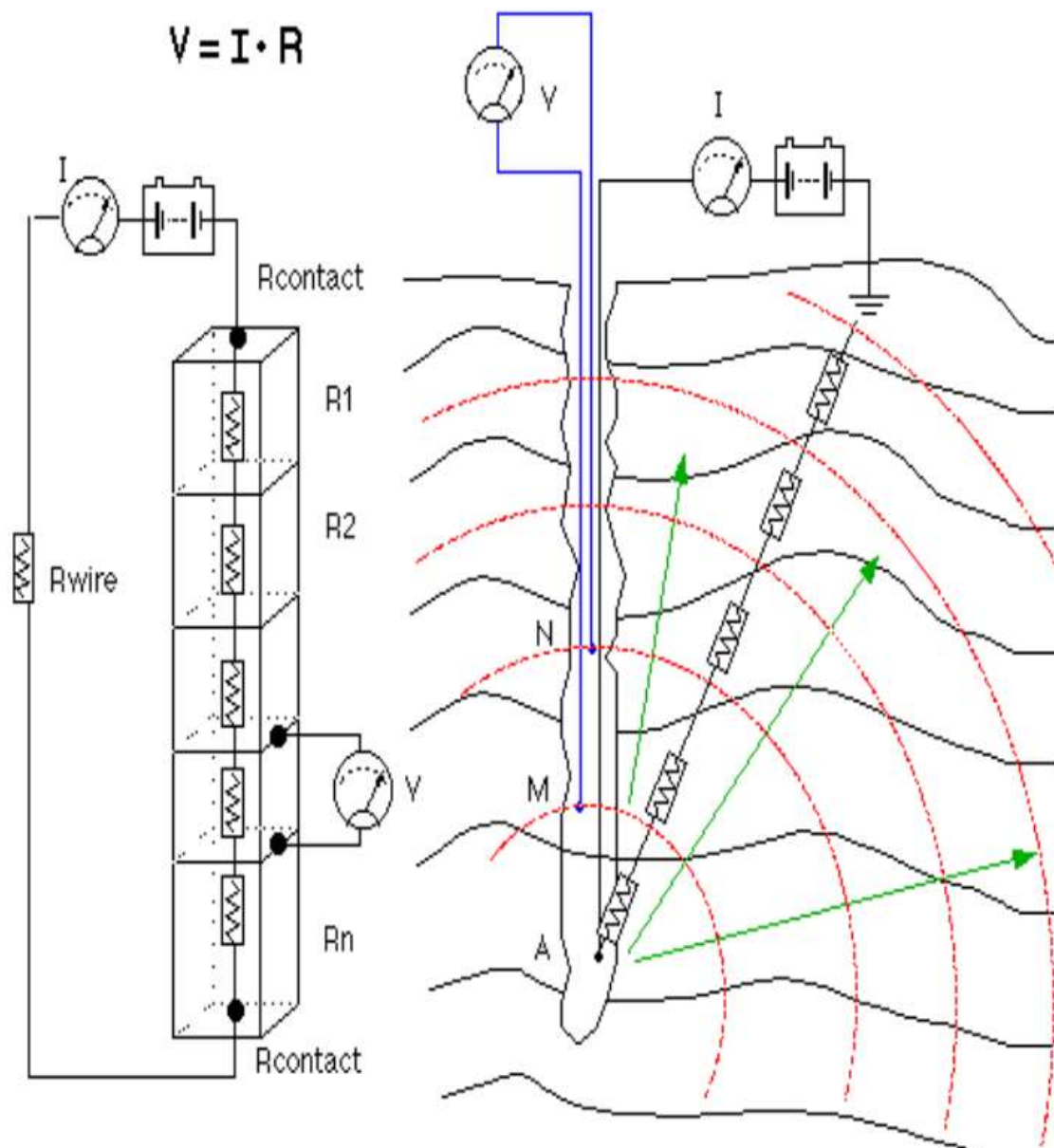
- A very low frequency current flows from the tool through the borehole, into the formation. Electrode arrays on either side of the source electrode force the measurement current into a horizontal disk-shaped pattern around the borehole. Formation resistivity is determined by monitoring the amount of current flowing from the tool.
- The tool must make contact with the formation.
- **Operational Constraints**
  - In Open hole, the tool is run centered
  - Run only in water-based mud
  - Logging speed: 100 feet/minute; slower if an Rxo tool is run in combination



# Laterolog – Principles of Electrode Tools

## Elements/Features:

- Electrode emits current,  $I$ :
  - Electrode  $A$ .
  - Green lines give current flow.
- Electrodes sense voltage,  $V$ :
  - Electrodes  $M$  and  $N$ .
  - Red lines show equipotentials.
- Formation resistivity,  $R$ :
  - $R = kV/I$ .
  - $k$  is tool constant.
- Simple electrode model ignores:
  - Current does flow up borehole.
  - Radial changes in  $R$  (invasion).



# Dual Laterolog Tool

## Deep and Shallow Current Patterns

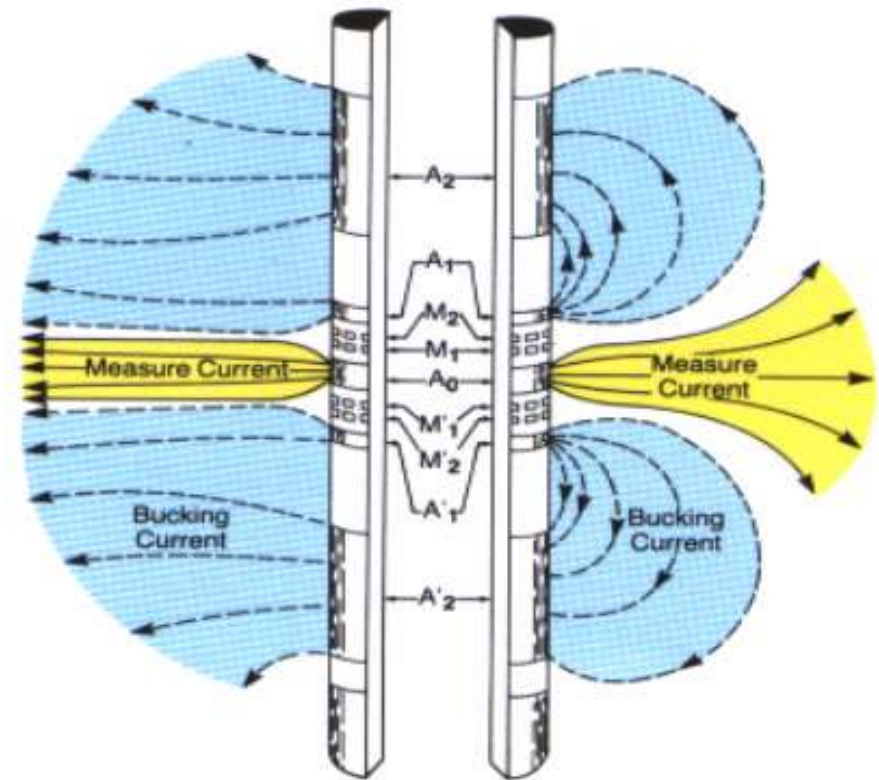
### Characteristics:

- Multiple currents:

- Measure.
- Bucking (or guard).

- Objectives:

- $R_t$ : Laterolog Deep (LLD).
- $R_{xo}$  and  $R_t$ : Laterolog Shallow (LLS).
- Eliminate borehole effect.
- Eliminate shoulder effect.



LLD

A<sub>0</sub> Measure Current Electrode  
M<sub>1</sub> } Monitoring Electrode  
M<sub>2</sub> }  
A<sub>1</sub> Bucking Current Electrode  
A<sub>2</sub> Bucking Current Electrode

LLS

A<sub>0</sub> Measure Current Electrode  
M<sub>1</sub> } Monitoring Electrode  
M<sub>2</sub> }  
A<sub>1</sub> Bucking Current Emitting Electrode  
A<sub>2</sub> Bucking Current Return Electrode

# LateroLog – Quality Control

- All three resistivity curves should stack (have the same value) in impermeable beds once they are corrected for borehole and bed thickness effects.
- The proper invasion profile in water-bearing permeable beds is:
  - ✓ If  $R_{mf} < R_w$  then Shallow < Medium < Deep;
  - ✓ If  $R_{mf} > R_w$  then Shallow > Medium > Deep;
- Shale values should be similar to those of nearby wells.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Cross-check the curve character with other curves from the same logging run.

**MICRORESISTIVITY,  $R_{x0}$**



# Microresistivity - Interpretation Goals

- **Flushed zone formation resistivity,  $R_{xo}$ :**
  - Can be determined with minimal environmental corrections, from any of the microresistivity measurements, except for the MicroLog. This value is used in Archie's equation to determine flushed zone water saturation  $S_{xo}$ , or to indicate moved fluids by comparison to the undisturbed formation resistivity,  $R_t$
- **Flushed zone water Saturation,  $S_{xo}$ , via Archie's Equation:**
  - In the flushed zone form of the Archie's equation  $R_t$  is replaced by  $R_{xo}$ , and  $R_o$  by  $R_{mf}$ , with the assumption that the original water has been replaced by drilling filtrate. Comparison of  $S_{xo}$  and  $S_w$  (using the same form of the Archie's equation) gives some qualitative indication of permeability, and the amount of hydrocarbons which will be moved during production.
- **Indication of Permeability:**
  - For MicroLogs: The micronormal resistivity is greater than the microinverse resistivity ("positive separation"). There should be mudcake, as shown in a decrease in the caliper reading.
  - For Other  $R_{xo}$  tools: Compare the reading with the resistivity from deeper reading tools. The relationship between the readings will depend on the contrast between the formation water resistivity and mud filtrate resistivity.

$$S_{xo} = \left( \frac{a \cdot R_{mf}}{\phi^m \cdot R_{xo}} \right)^{\frac{1}{n}}$$

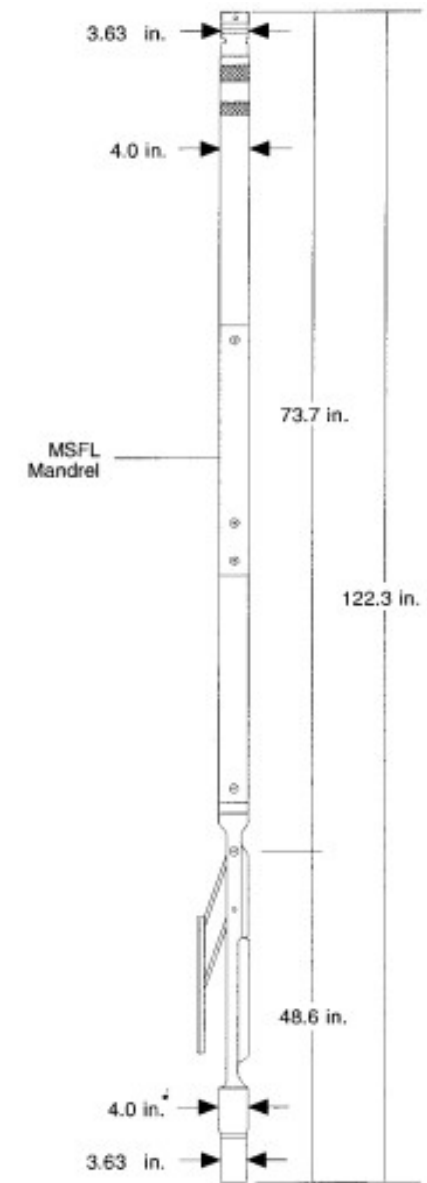


# Microresistivity - Interpretation Goals

- **Thin Bed definition:** Same as for Induction Log.
  - These tools will identify very thin beds. The bed definition can be used qualitatively to estimate the effect on deeper reading tools. Also useful for thin bed corrections to other resistivity and porosity tools
- **Fracture Identification:**
  - Rapid curve movement or “hashiness”, may be an indication of fractures as the tools see conductive mud-filled fractures alternating with less conducting beds. Rough hole may also cause the same response. Other information is required to establish presence of fractures.
- **Invasion corrections to other resistivity measurements:**
  - Rxo tools are usually run in combination with two deeper reading tools (e.g. deep and shallow laterolog, deep and medium induction log). Using the combination of three measurements, corrections can be made for invasion using “tornado chart” or equivalent algorithms.

# Microresistivity - Physics of Measurement

- Electrical current is focused into the formation by closely placed electrodes mounted on pads pressed against the borehole wall.
- Some designs like Micro Spherically Focused log, use focusing electrodes similar to the laterolog, while older designs like MicroLog, do not focus the current.
- **Operational Constraints**
  - In Open hole, the tool is run centered
  - Run only in water-based mud
  - Logging speed: 60 feet/minute;
- **Operational Constraints**
  - Mudcake corrections may need to be made if the measurements are to be used quantitatively.

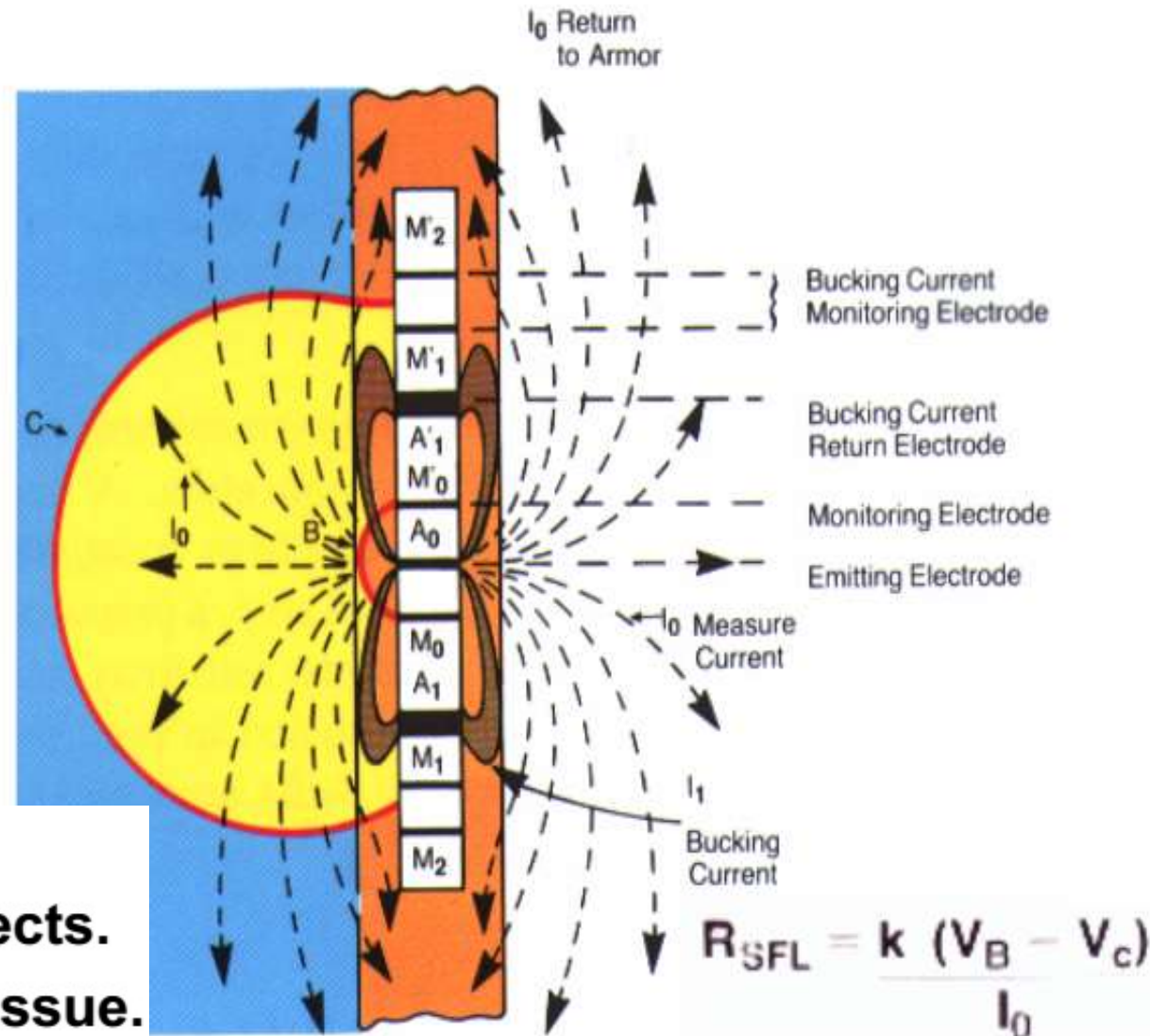


Micro Spherically  
Focused log

# Spherically Focused Resistivity Tool

## Tool Characteristics:

- SFL tool gives *shallow resistivity*.
- Usually run in combination with induction logs.
- Good tool for thin-bed detection/analysis.
- Has a different order of electrodes than the laterolog tool.
- Pad-type tools:
  - Pads reduce borehole effects.
  - Mud cake may still be an issue.
- Very shallow resistivity:
  - 2 to 5 cm profile is typical.



11.

# LateroLog – Quality Control

- All three resistivity curves should stack (have the same value) in impermeable beds once they are corrected for borehole and bed thickness effects.
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- Shale values should be similar to those of nearby wells.
- Check repeatability: curves should have the same values and character as those previous runs or repeat sections.
- Cross-check the curve character with other curves from the same logging run.

# **Principles of Log Interpretation – Lecture 4**

# Log Interpretation – Required Data

Once the section TD (Total Depth) of the hole has been reached, the petrophysicist will be expected to make an interpretation of the open hole logs that have been acquired. Prior to the log interpretation, the petrophysicist should have:

- 1) All relevant daily drilling reports, plus latest well deviation from the well, last casing depth, & mud data ( $R_m$ ,  $R_{mf}$  @ Formation Temp.)
- 2) All the latest mud-log information, including cuttings description, shows, gas reading, ROP (Rate of Penetration) and Core Data
- 3) Logs and interpretations on hand from nearby wells and regional wells penetrating the same formations, in particular where regional or field-wide values of  $m$ ,  $n$ ,  $R_w$ ,  $\rho_g$  and fluid contacts are available.
- 4) A copy of the contractor's chart book

This will allow for compilation of all information about lithology, particularly expected rock type and mineral composition, presence and type of clay/shale, pore types etc

# Log Interpretation – An Introduction

- The general goal of interpretation is the determination of:
  - Lithological profile and the detection and characterization of the potential reservoir zones with exact depth,
  - Shale content and shale type (e.g. laminated, or dispersed shale)
  - Porosity (and mineral composition),
  - Fluid saturation, water saturation, hydrocarbon saturation, moveable hydrocarbons, type of hydrocarbon (oil, gas)
  - Permeability
- Interpretation delivers a solution in two steps:
  - 1) Quick look interpretation for qualitative description of lithology and zones of interest
  - 2) Detailed quantitative analysis in terms of the goals above.



# Log Interpretation – Prior Steps

- Log interpretation requires the synthesis of logging tool response physics, geological knowledge, and auxiliary measurements or information to extract the maximum petrophysical information concerning subsurface formations.
- Knowledge of characteristics log responses, empirical and model derived equations for quantitative interpretation, integration of all available information and the skill of the interpreter is crucial for effective log evaluation.
- The first step in log evaluation is the quality control of the logs followed by correlation of these logs with the mud log and logs of surrounding wells to determine significant events, such as markers, hydrocarbon contacts etc

# Log Interpretation – Basic Quality Control

- 1) Check that the logger's TD and last casing shoe depths roughly match those from the last daily drilling report.
- 2) Check that the derrick floor elevation and ground level (or seabed) positions are correct.
- 3) Check that the log curves are on depth with each other. The tension curve can be used to identify possible zones where the tool string has become temporarily stuck, which will put the curves off depth and result in "flatlining."
- 4) Check that the caliper is reading correctly inside the casing (find out the casing ID) and that it is reading the borehole size in non-permeable zones that are not washed out.
- 5) Check the density borehole correction curve. It should not generally exceed 0.02g/cc, except in clearly washed out sections (>18in.), for which the density curve is likely to be unusable.
- 6) Check the sonic log by observing the transit time in the casing, which should read 47  $\mu\text{s}/\text{ft}$ .

# Log Interpretation – Basic Quality Control

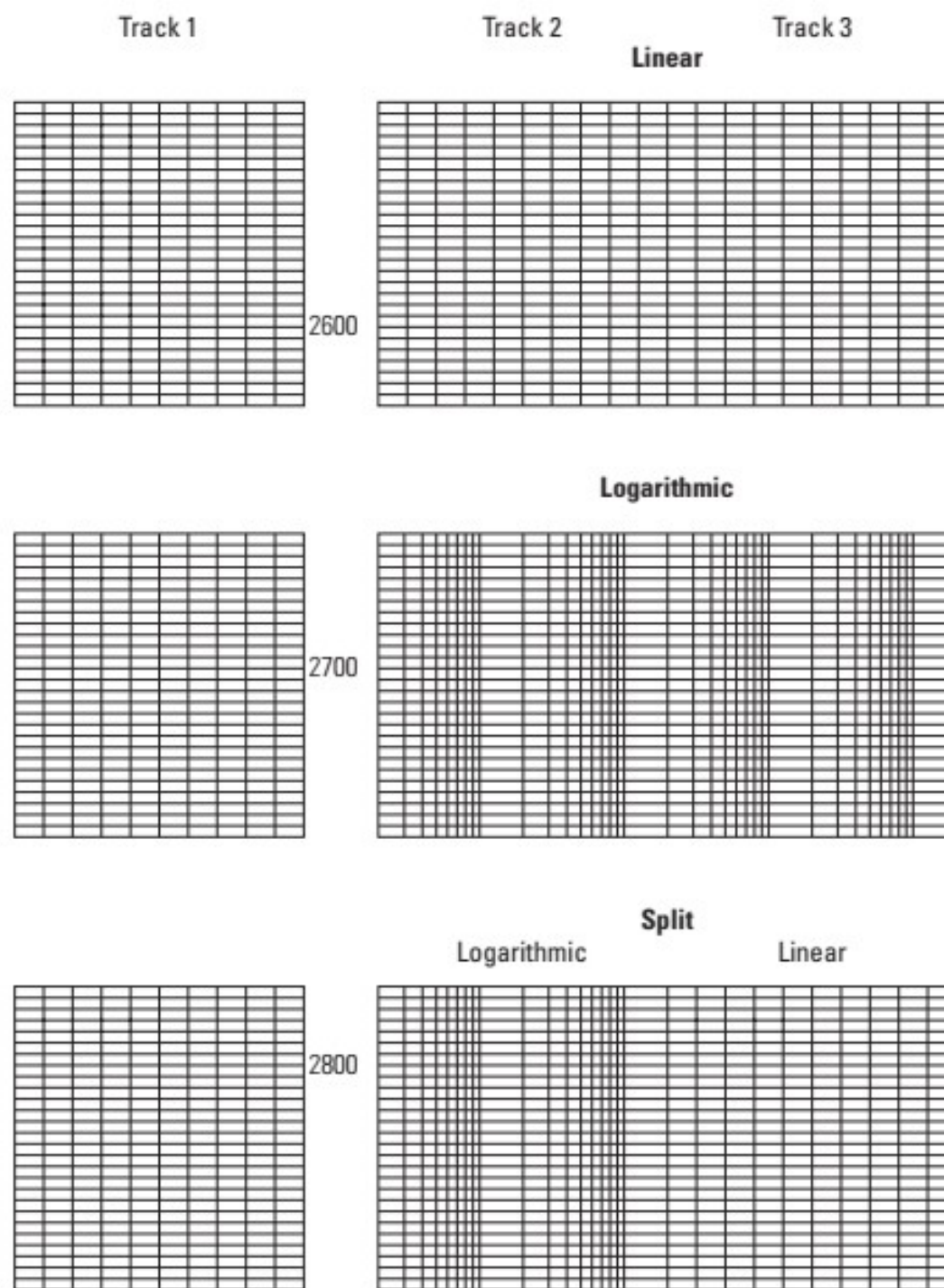
- 7) Inspect the resistivity curves. If oil-based mud (OBM) is being used, the shallow curves will usually read higher than the deep curves (except in highly gas or oil saturated zones). Likewise, with water-based mud (WBM) the shallow curves will read less than the deep curves, providing  $R_{mf} < R_w$ , or in hydrocarbon-bearing zones. In theory, the curves should overlie each other in non-permeable zones such as shales. However, in practice this is often not the case, due to either anisotropy or shoulder-bed effects.
- 8) Look out for any cycling-type behaviour on any of the curves, such as a wave pattern. This may be due to corkscrewing while drilling, causing an irregular borehole shape. However, it is necessary to eliminate any possible tool malfunction.
- 9) Check that the presentation scales on the log print are consistent with other wells or generally accepted industry norms. These are generally:
  - ☐ GR: 0–50 API
  - ☐ Caliper: 8–18"
  - ☐ Resistivity: 0.2–2000ohmm on log scale
  - ☐ Density: 1.95–2.95 g/cc (solid line)
  - ☐ Neutron:  $-0.15 \pm 0.45$  (porosity fraction) (dashed line)
  - ☐ Sonic: 140–40  $\mu\text{s}/\text{ft}$

# Information from the Log Header

- Measured log data are give mostly as a data file in a special format - Log ASCII Standard (LAS-file).
  - Hole location and name
  - Depth – Driller, Depth – Logger, Casing Depth, Bit size
  - Logged interval
  - Fluid type in the hole, Density/Viscosity, pH/Fluid Loss
  - Fluid Resistivities:  $R_m$  (mud resistivity) at measured temperature,  $R_{mf}$  (mud cake resistivity) at measured temperature,  $R_{mc}$  (mud cake resistivity) at measured temperature, Source of  $R_{mf}$ ,  $R_{mc}$
  - Temperature: Bottom hole temperature, maximum recorded temperature
  - Time since circulation

# Reading a Log

- Reading a log with ease requires familiarity with some of the standard log formats. The formats for traditional logs and most field logs are shown in Fig. LI.1 and can be seen to contain three tracks.
- A narrow column containing the depth is found between track 1 and tracks 2 and 3. The latter are contiguous.
- The top illustration shows the normal linear presentation, with the grid lines in all three tracks having linear scales each with ten divisions. The middle figure shows the logarithmic presentation for tracks 2 and 3. Four decades are drawn to accommodate the electrical measurements, which can have large dynamic ranges.
- The bottom illustration is a hybrid scale with a logarithmic grid on track 2 and a linear one in track 3. Electrical measurements that may spill over from track 2 into track 3 will still be logarithmic even though the indicated scale is linear.
- The depth is shown by the numbers in the centre track and the horizontal lines. In Fig. LI.1 the depth scale is 1/240, or 1 ft of log for 240 ft of formation. The logs have a thin horizontal line every 2 ft, a medium thick line every 10 ft, and a thick line every 50 ft.

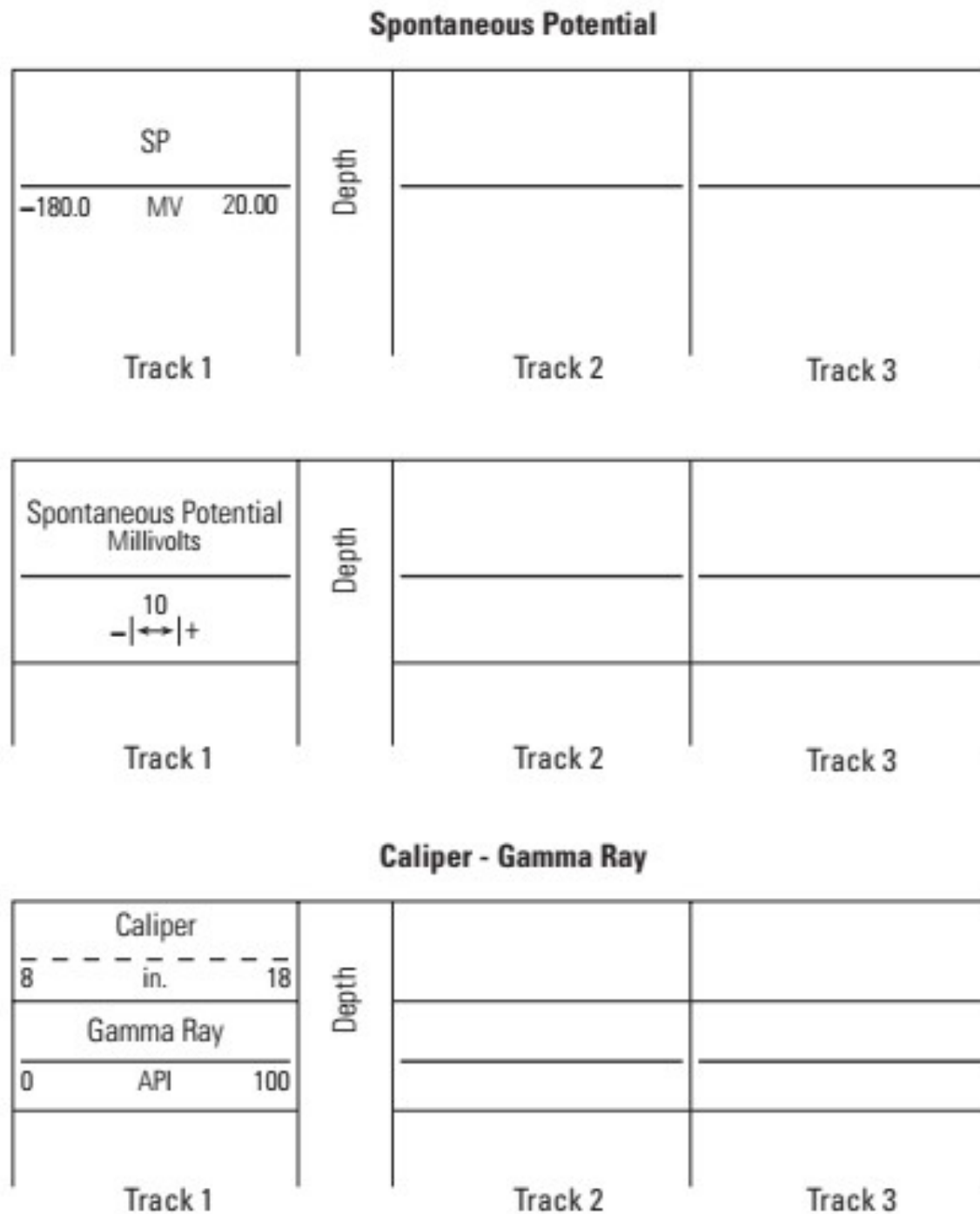


- **Fig. LI.1** Standard log presentation formats.

# Reading a Log

- Figure LI.2 shows the typical log-heading presentation for several of the basic logs that will be used. The upper two presentations show two variations for the SP, which is always presented in track 1.
- The bottom presentation shows the caliper, a one-axis measurement of the borehole diameter, and the gamma ray, which are also generally presented in track 1. Note that the SP decreases to the left. The rule given for finding clean sections was that the SP becomes less negative for increasing shale, so that deflections of the SP trace to the right will correspond to increasing shale content.
- The GR curve, as it is scaled in increasing activity (in American Petroleum Institute (API) units) to the right, will also produce curve deflections to the right for increasing shale content. Thus the two shale indicators can be expected to follow one another as the shale content varies.

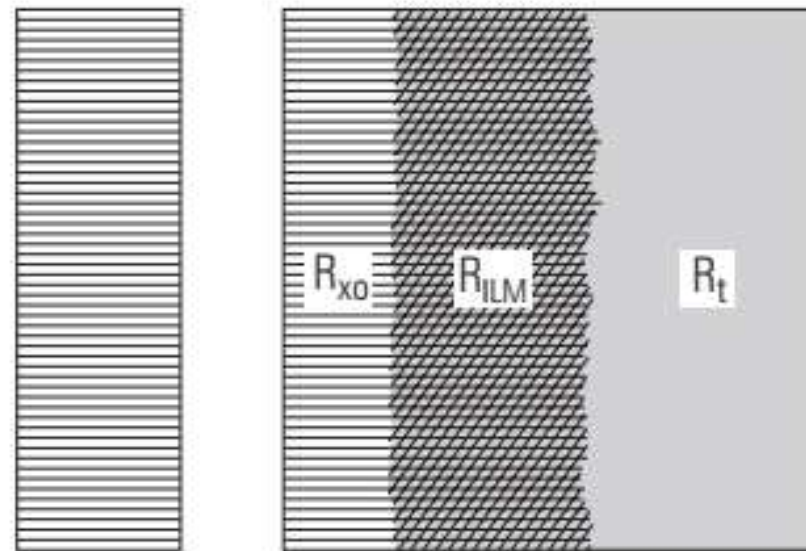
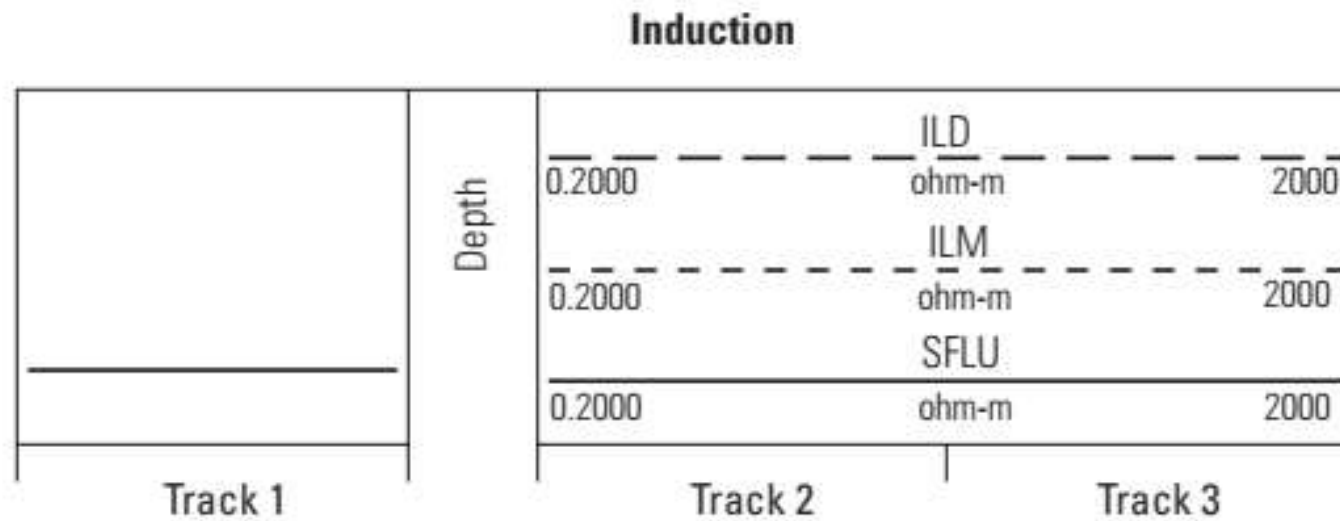




- **Fig. LI.2** Standard log presentation formats.

# Reading a Log

- Although modern resistivity tools have a larger selection of curves with different depths of investigation, the displays are similar.
- A traditional resistivity log heading is shown in Fig. LI.3, along with a schematic indication of the zones of investigation. The particular tool associated with this format is referred to as the dual induction-SFL and will normally show three resistivity traces (the units of which are ohm-m). The trace coded ILD (Induction Log Deep) corresponds to the deepest resistivity measurement and will correspond to the value of  $R_t$  when invasion is not severe. The curve marked ILM (Induction Log Medium) is an auxiliary measurement of intermediate depth of penetration and is highly influenced by the depth of invasion.
- The third curve, in this case marked SFLU (spherically focused log), is a measurement of shallow depth of investigation and reads closest to the resistivity of the invaded zone  $R_{xo}$ . By combining the three resistivity measurements, it is possible, in many cases, to compensate for the effect of invasion on the ILD reading.

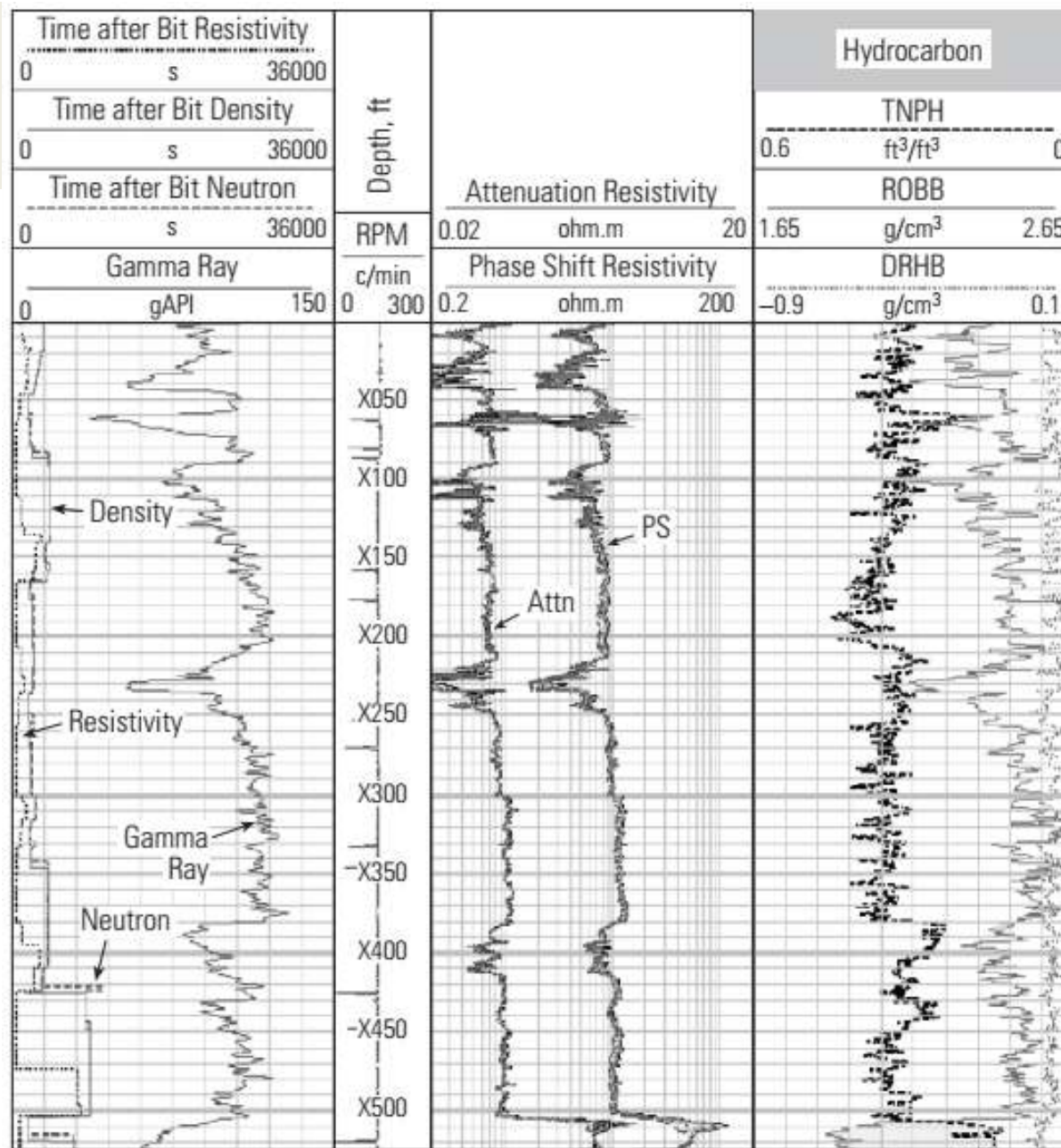


- **Fig. LI.3** The induction log heading and schematic of the formation, with three zones corresponding approximately to the simultaneous electrical measurements of different depths of investigation.

# Reading a Log

- In Fig. LI.4, three typical headings for the three types of porosity devices are indicated. The porosity is expressed as a decimal (v/v) or in porosity units (p.u.), each of which corresponds to 1% porosity. The top heading shows the format for porosities derived from neutron and density measurements simultaneously.
- In this example, although the scale may vary depending upon local usage, porosity is shown from  $-0.15$  to  $0.45$  v/v. The middle example shows an additional correction curve for the density log, which can be used to get some idea of the mudcake and rugosity of the borehole encountered during the density measurement.
- It is also common practice to present the density measurement in  $\text{g/cm}^3$  with a dynamic range of  $1 \text{ g/cm}^3$  on the density trace across the full track (or sometimes two tracks), as is done in Fig. LI.5 below. It is easy to show that a change in bulk density of  $1 \text{ g/cm}^3$  in a water-filled formation corresponds to a porosity change of about 60 p.u. Consequently the neutron trace is usually shown with a dynamic range of 60 p.u. across the density track.





• **Fig. LI.5** Example LWD log in a horizontal well. In track 1 is the familiar GR along with three curves indicating the time delay between drilling and the three types of measurements made; the tool rotation rate appears in the depth track. Track 2 - two types of resistivity measurements, each with multiple depths of investigation that overlay in this example. The third track contains the LWD versions of the neutron measurement (TNPH), the density measurement (ROBB), and the density correction (DRHB).

# Reading a Log

- The density trace is then shifted so that the zero point on the neutron trace corresponds to the density of the matrix. Two such compatible scales are in common use: one for sandstone, in which case the zero porosity point is  $2.65 \text{ g/cm}^3$  as in Fig. LI.5; the other for limestone, in which case the zero porosity point is  $2.7 \text{ g/cm}^3$ . The bottom heading of Fig. LI.4 is for the traditional sonic log with the apparent transit time  $\Delta t$  increasing to the left. In all three presentations, the format is such that increasing porosity produces curve deflections to the left.
- For the neutron and density logs, another point to be aware of is the matrix setting. This setting corresponds to a rock type assumed in a convenient pre-interpretation that establishes the porosity from the neutron and density device measurements. In both examples shown in Fig. LI.4, the matrix setting is listed as SS, which means that the rock type is taken to be sandstone. If the formations being logged are indeed sandstone, then the porosity values recorded on the logs will correspond closely to the actual porosity of the formation. However, if the actual formation matrix is different, say limestone, then the porosity values will need to be shifted or corrected in order to obtain the true porosity in this particular matrix.



# **GUIDES TO QUICK LOOK WELLSITE LOG INTERPRETATION**

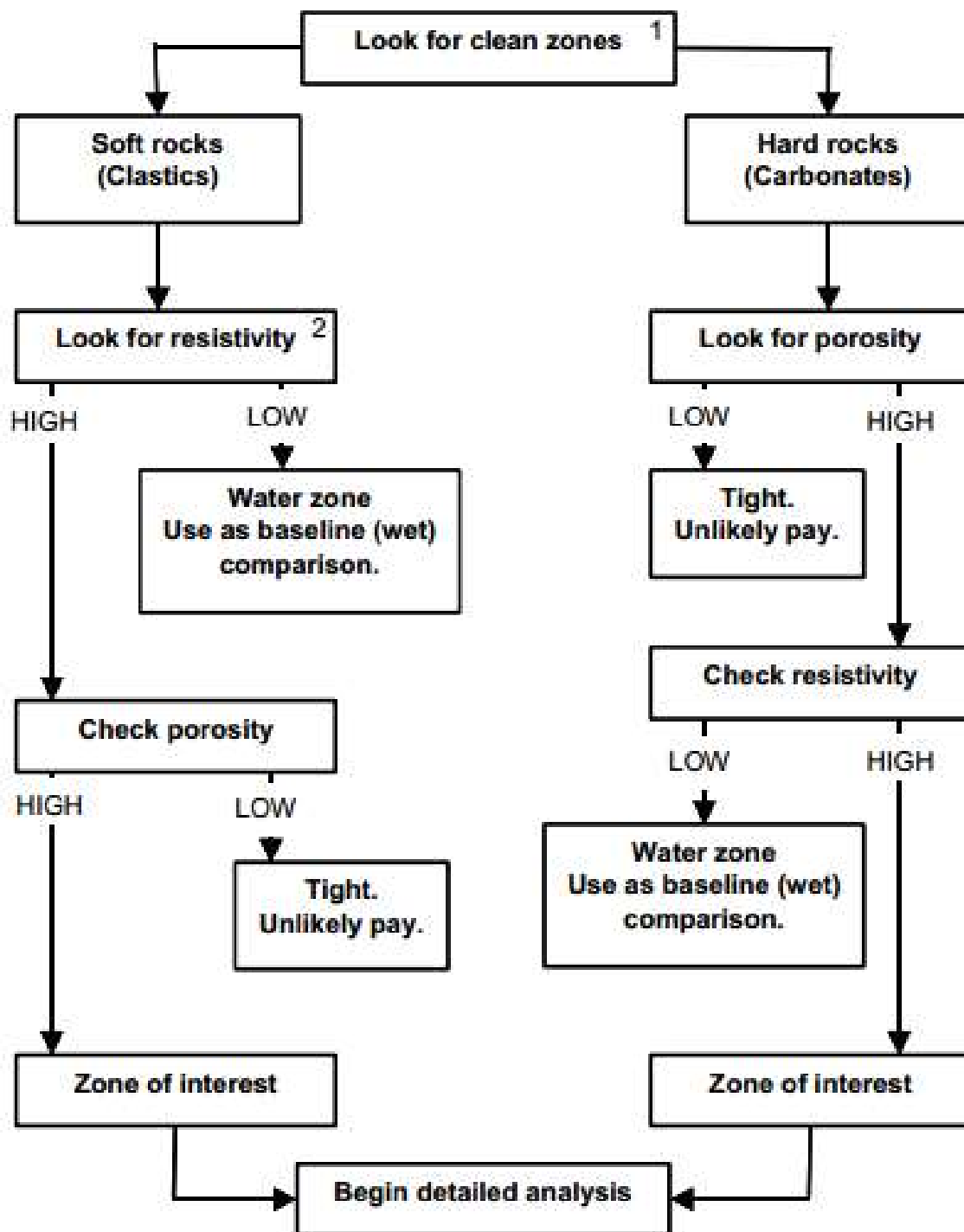
# Log Interpretation – Wellsite Evaluation

- Prior to detailed log interpretation, wellsite interpretation is undertaken. This refers to the rapid and somewhat cursory approach to scanning an available set of logging measurements, and the ability to identify and draw some conclusion about zones of possible interest.
- These zones, probably hydrocarbon-bearing, will warrant a closer and more quantitative analysis, which is possible only by the inclusion of additional knowledge and measurements.
- The three most important questions to be answered by wellsite interpretation are:
  1. Can the porous, clean formation be identified?
  2. Does the formation contain hydrocarbons?
  3. What is the formation porosity? And Saturations?
- The three questions above will enable us determine the quantity of hydrocarbons present. N.B.: Question 3 would be addressed under Quantitative Interpretation.

# Quick Look Log Interpretation

Asquith and Krygowsky (2004) recommended the following flowchart for quick evaluation to identify zones of interest:

- 1) Separate the shaly sections, select “clean sections” then
- 2) If clastics are logged, look at the (deep) resistivity first – if it is high, hydrocarbons may be present; then look for porosity (only in the case of extremely low porosity the rock with the high resistivity is tight no reservoir)
- 3) If carbonates are logged, look at the porosity first – if it is sufficient, look at the resistivity – if it is high hydrocarbons may be present.



# Interpretation

- Clastics: Because the Shales surrounding the clastic reservoirs tend to have slowly varying resistivity with depth, the interpretation process targets formation resistivity to identify zones of interest.
- Carbonates: Because the resistivity of a sequence of carbonate formations may vary over a wide range, the interpretation process targets formation porosity to identify zones of interest.
- N.B.: Zones which appear to be shales may be radioactive productive zones<sup>1</sup>. Shaly zones may produce even when having low resistivity<sup>2</sup>.

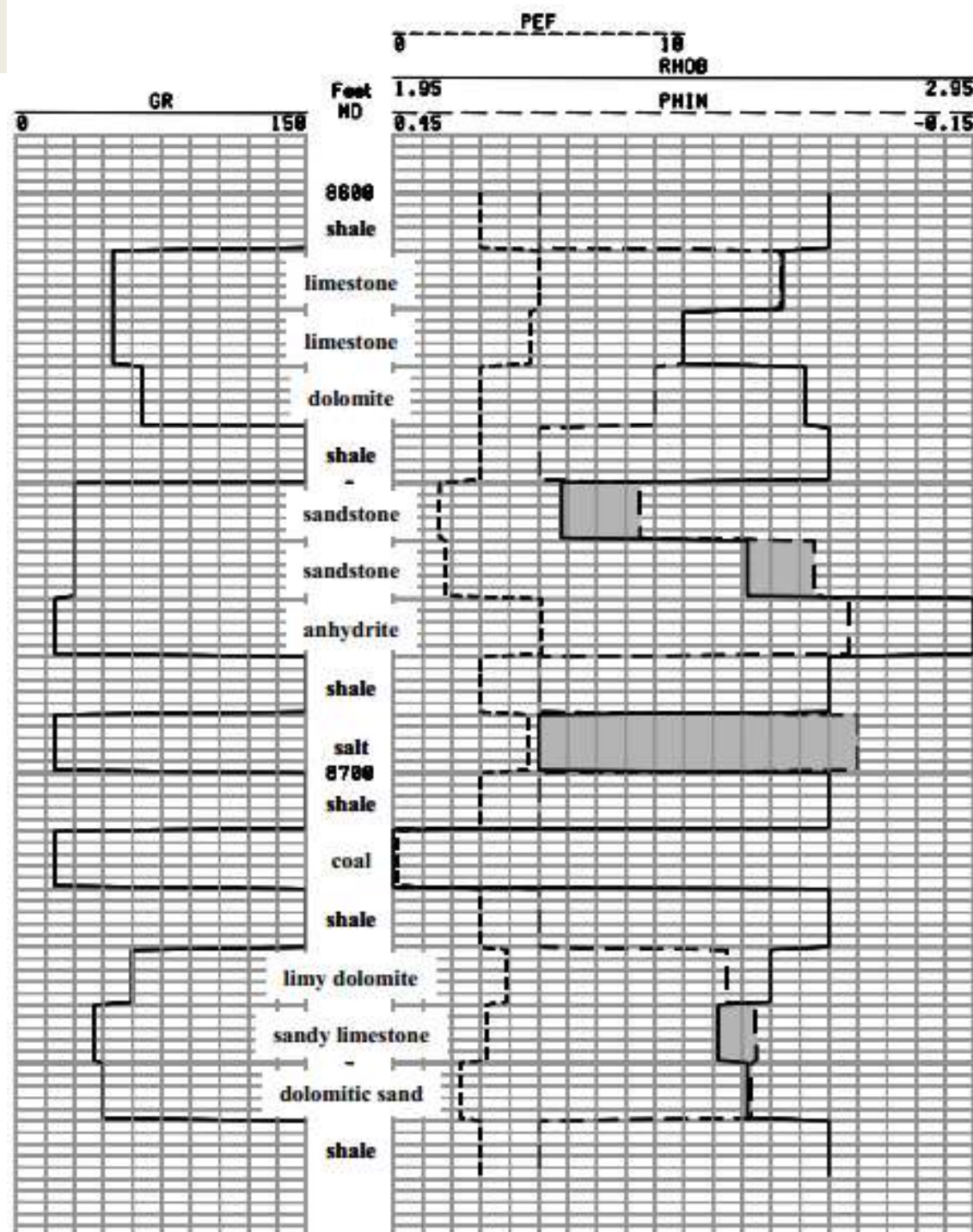
# Wellsite Evaluation – Identifying Clean Zones

- Formations with low shale content are much more likely to produce accumulated hydrocarbons. Thus the first task is to identify the zones with a low-volume fraction of shale ( $V_{shale}$ ), also known as clean zones.
- This task has traditionally been accomplished via two measurements: the gamma ray, and the spontaneous potential (SP). The qualitative behaviour of the SP (a voltage measurement reported in mV) is to become less negative with increases in formation shale content. The gamma ray signal will generally increase in magnitude according to the increase in shale content.
- In clastic reservoirs, in nearly all cases, indication of reservoir rock will correspond to a fall in the gamma ray (GR) log (except for the presence in sands of radioactive minerals).
- Other techniques have been used recently, for example the separation between the neutron and density measurements, the nuclear magnetic resonance (NMR) distribution, and elemental spectroscopy analysis.

# Wellsite Evaluation – Identifying Clean Zones

- The most reliable indicator of reservoir rock will be from the behaviour of the density/neutron logs, with the density moving to the left (lower density) and touching or crossing the neutron curve (Darling, 2003).
- Shales can be clearly identified as zones where the density lies to the right of the neutron, typically by 6 or more neutron porosity units.
- The greater the crossover between the density and neutron logs, the better the quality of the reservoir. However, gas zones will exhibit a greater crossover for a given porosity than oil or water zones.
- The following approach is recommended (Darling, 2003).
  - 1) Determine an average GR reading in clean sands ( $GR_{sa}$ ) and a value for shales ( $GR_{sh}$ ). For  $GR_{sh}$ , do not take the highest reading observed, but rather the mode of the values observed.
  - 2) Define the shale volume,  $V_{sh}$ , as  $(GR - GR_{sa}) / (GR_{sh} - GR_{sa})$ . By comparing  $V_{sh}$  with the density/neutron response, determine a value of  $V_{sh}$  to use as a cut-off. Typically 50% is used.

# Identifying Clean Zones



- While the positions on the log will vary with varying changing porosity, the relative position of the neutron and density curves will remain fairly constant with lithology.
- The Photoelectric Effect (PE) curve may be useful to resolve some ambiguities which occur with some lithologic mixtures. (N.B.: The PE curve captures low energy gamma rays returning from the formation).
- It is assumed that (a) the Neutron porosity and Density curves (or bulk density) are scaled by limestone matrix (b) formations are assumed to be clean (i.e. No shales/clays) (c) filled with liquid only (no gas present)



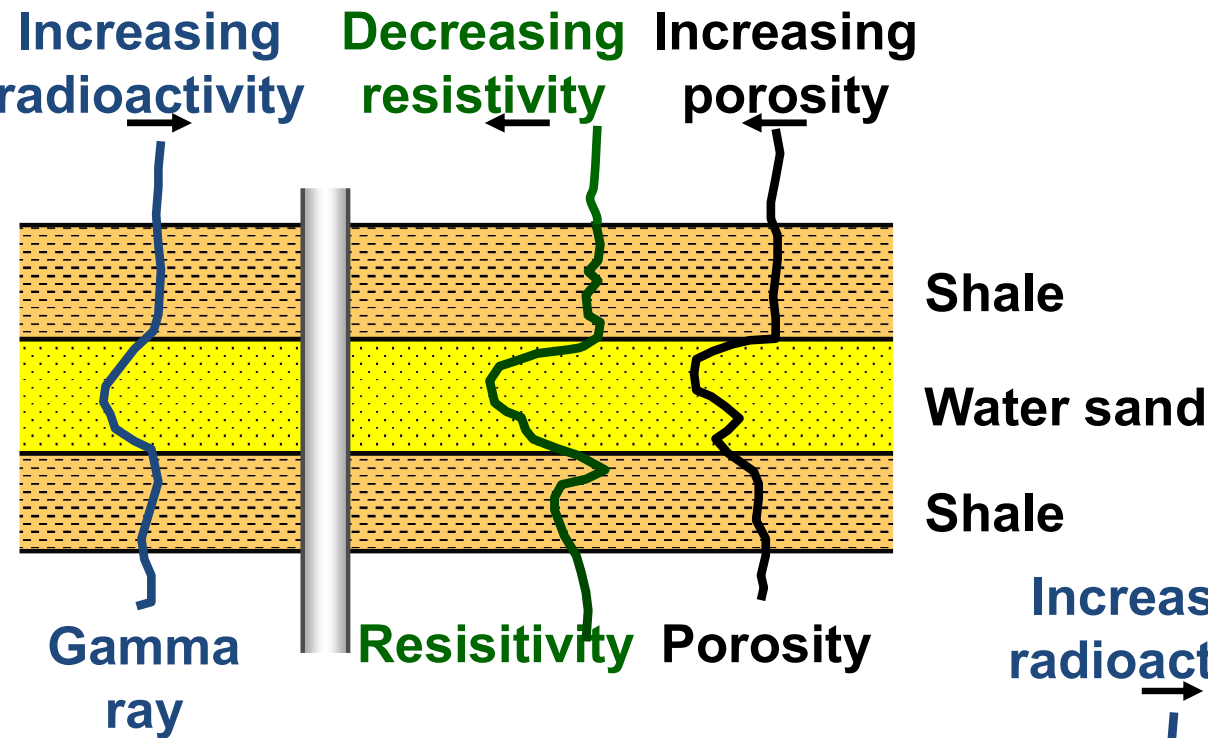
# Wellsite Evaluation – Identifying the Fluid Type

- The second step especially for clastics is to answer the question: “Does the formation contain hydrocarbons?”
- Once a porous, clean formation is identified, the analyst is faced with deciding whether it contains hydrocarbons or not. This analysis is done in quite an indirect way, using the resistivity  $R_t$  of the formation.
- Basically, if the porous formation contains conductive brine, its resistivity will be low. If, instead, it contains a sizable fraction of non-conducting hydrocarbon, then the formation resistivity will be rather large.
- The hydrocarbons may be oil or gas. The distinction is most easily made by comparing the formation density and neutron porosity measurement.

# Wellsite Evaluation – Identifying the Fluid Type

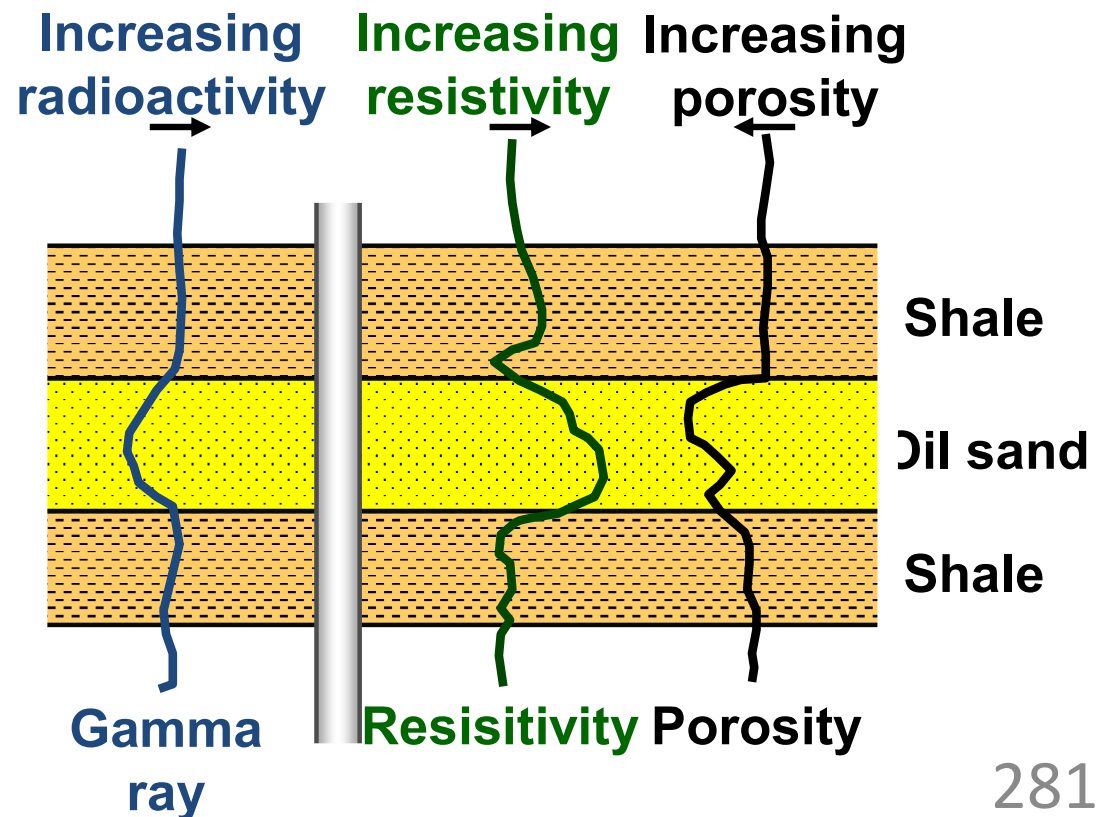
- To at least have a working assumption regarding the fluids, start by comparing the density and deepest reading resistivity log for any evidence of hydrocarbons. In the classic response, the resistivity and density (and also GR) will be seen to “tramline” (i.e., follow each other to the left or right) in water sands and to “Mae West” (i.e., be a mirror image of each other) in hydrocarbon sands (see Figure WE.2).
- However, some hydrocarbon/water zones will not exhibit such behaviour, the reasons being:
  - 1) When the formation-water salinity is very high, the resistivity may also drop in clean sands.
  - 2) In shaly sand zones having a high proportion of conductive dispersed shales, the resistivity may also fail to rise in reservoir zones.
  - 3) If the sands are thinly laminated between shales, the deep resistivity may not be able to “resolve” the sands, and the resistivity may remain low.
  - 4) If the well has been drilled with very heavy overbalance, invasion may be such as to completely mask the hydrocarbon response.
  - 5) When the formation water is very fresh (high  $R_w$ ), the resistivity may Mae West even in water-bearing zones.

# Wellsite Evaluation – Identifying the Fluid Type



**Figure WE.2**

Typical Logging Signatures in  
Water-bearing sand vs.  
Oil-bearing sand



# Wellsite Evaluation – Identifying the Fluid Type

- When either of the first two situations arises, it is very important to look at the absolute value of the deep resistivity, rather than at only the behaviour compared with the density.
- As long as a known water sand has been penetrated in the well (or a neighbouring well), a good idea of what the resistivity ought to be for a water-bearing sand should exist. If the resistivity is higher than this value, whatever the shape of the curve, then hydrocarbons should be suspected.
- There is also an effect of porosity on the resistivity. As porosity increases, the value of  $R_t$  will decrease if the water saturation remains constant.
- Gas zones will exhibit a greater density/neutron crossover than oil zones. In a very clean porous sand, any GOC can be identified on the log relatively easily.

# Wellsite Evaluation – Identifying the Fluid Type

- Another common resistivity measurement,  $R_{xo}$ , corresponds to the resistivity of the flushed zone, a region of formation close to the borehole, where drilling fluids may have invaded and displaced the original formation fluids.
- The measurement of  $R_{xo}$  is used to get some idea of the recoverability of hydrocarbons in the following way. If the value of  $R_{xo}$  is found to be the same as the value of  $R_t$ , then it is most likely that the original formation fluids are present in the flushed zone, indicating that no formation fluid displacement has taken place. However, if  $R_{xo}$  is different than  $R_t$ , then some invasion has taken place, and the fluids are movable.
- This can be taken one step further. If the ratio of  $R_{xo}$  to  $R_t$  is the same as the ratio of the water resistivities in the two zones ( $R_{mf}$  and  $R_w$ ), then the flushed and non-flushed zones have either the same quantity of hydrocarbons or none. Any hydrocarbons are unlikely to be producible in this case. If the ratio of  $R_{xo}$  to  $R_t$  is less than that of  $R_{mf}$  to  $R_w$ , then some hydrocarbons have been moved by the drilling fluid and will probably be producible.

# Significance of Saturation estimation in the flushed and un-invaded zones

- When oil is present in the flushed zone, the degree of flushing by mud filtrate can be determined from the difference between water saturations in the flushed zone ( $S_{xo}$ ) and the un-invaded zone ( $S_w$ ). The remaining oil is called residual oil ( $S_{ro} = 1.0 - S_{xo}$ ) where  $S_{ro}$  equals residual oil saturation.
- Even in hydrocarbon bearing reservoirs, there is always a layer of formation water on grain surfaces. By using water saturation data, the reservoir's hydrocarbon saturation can be determined thus: ( $S_h = 1.0 - S_w$ )
- The ratio between the un-invaded zone's water saturation ( $S_w$ ) and the flushed zone saturation ( $S_{xo}$ ) is an index of hydrocarbon moveability.

# Wellsite Evaluation – Reservoir Porosity

Recall that four logging devices yield estimates of porosity.

- In the case of the density tool, the measured parameter is the formation bulk density  $\rho_b$ . As porosity increases, the bulk density  $\rho_b$  decreases.
- The neutron tool is sensitive to the presence of hydrogen. Its reported measurement is the neutron porosity  $\varphi_n$ , which reflects the value of the formation hydrogen content.
- The acoustic tool measures the compressional wave slowness or, interval transit time  $\Delta t$  (reported in  $\mu\text{s}/\text{ft}$ ). It will increase with porosity.
- The total NMR signal depends on the amount of hydrogen and therefore increases with porosity.

**N.B.:** *In all three presentations, the format is usually such that increasing porosity produces curve deflections to the left.*



# Wellsite Evaluation – Summary Relations

| Descriptor              | Measurement                              | Functional behavior                          |               |                                               |
|-------------------------|------------------------------------------|----------------------------------------------|---------------|-----------------------------------------------|
| Clean/shaley            | SP                                       | $V_{shale} \uparrow$                         | $\rightarrow$ | SP $\uparrow$                                 |
|                         | GR                                       | $V_{shale} \uparrow$                         | $\rightarrow$ | GR $\uparrow$                                 |
| Porosity ( $\phi$ )     | Density                                  | $\phi \uparrow$                              | $\rightarrow$ | $\rho_b \downarrow$                           |
|                         | Neutron                                  | $\phi \uparrow$                              | $\rightarrow$ | $\phi_n \uparrow$                             |
|                         | Acoustic                                 | $\phi \uparrow$                              | $\rightarrow$ | $\Delta t \uparrow$                           |
| Hydrocarbon             | $R_t$                                    | $S_w \uparrow$                               | $\rightarrow$ | $R_t \downarrow$                              |
|                         |                                          | $(S_o \uparrow$                              | $\rightarrow$ | $R_t \uparrow)$                               |
|                         |                                          | $\phi \downarrow$                            | $\rightarrow$ | $R_t \uparrow$                                |
| Recoverable/<br>movable | $R_{xo}$ vs. $R_t$<br>(shallow vs. deep) | $R_{xo} = R_t$                               | $\rightarrow$ | No invasion                                   |
|                         |                                          | $R_{xo} = R_t$                               | $\rightarrow$ | If $R_{mf} = R_w$ , no<br>movable hydrocarbon |
|                         |                                          | $\frac{R_{xo}}{R_t} \neq \frac{R_{mf}}{R_w}$ | $\rightarrow$ | Moved fluid                                   |

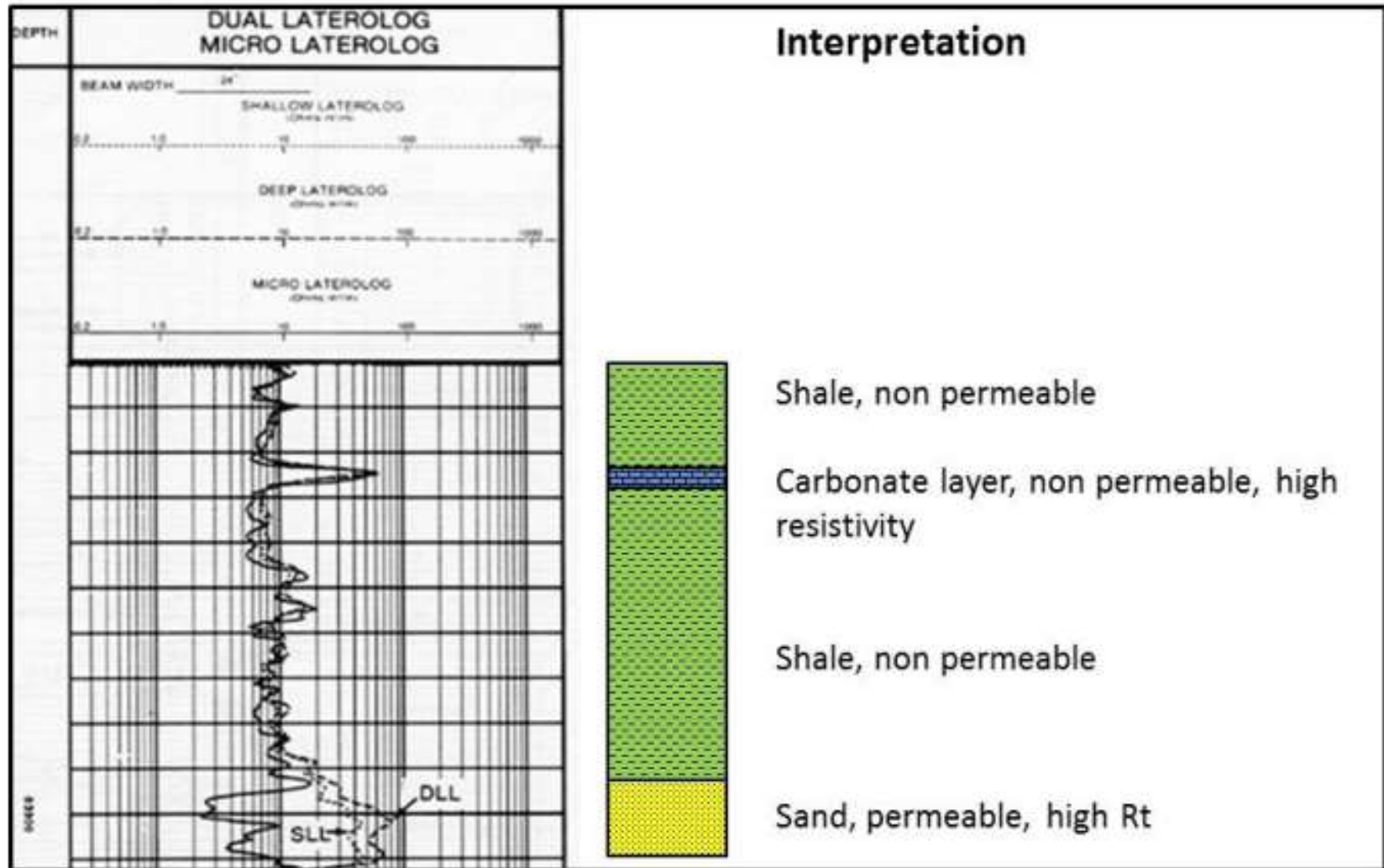
A summary of phenomenological interpretation

# Wellsite Evaluation – Permeable Bed Identification

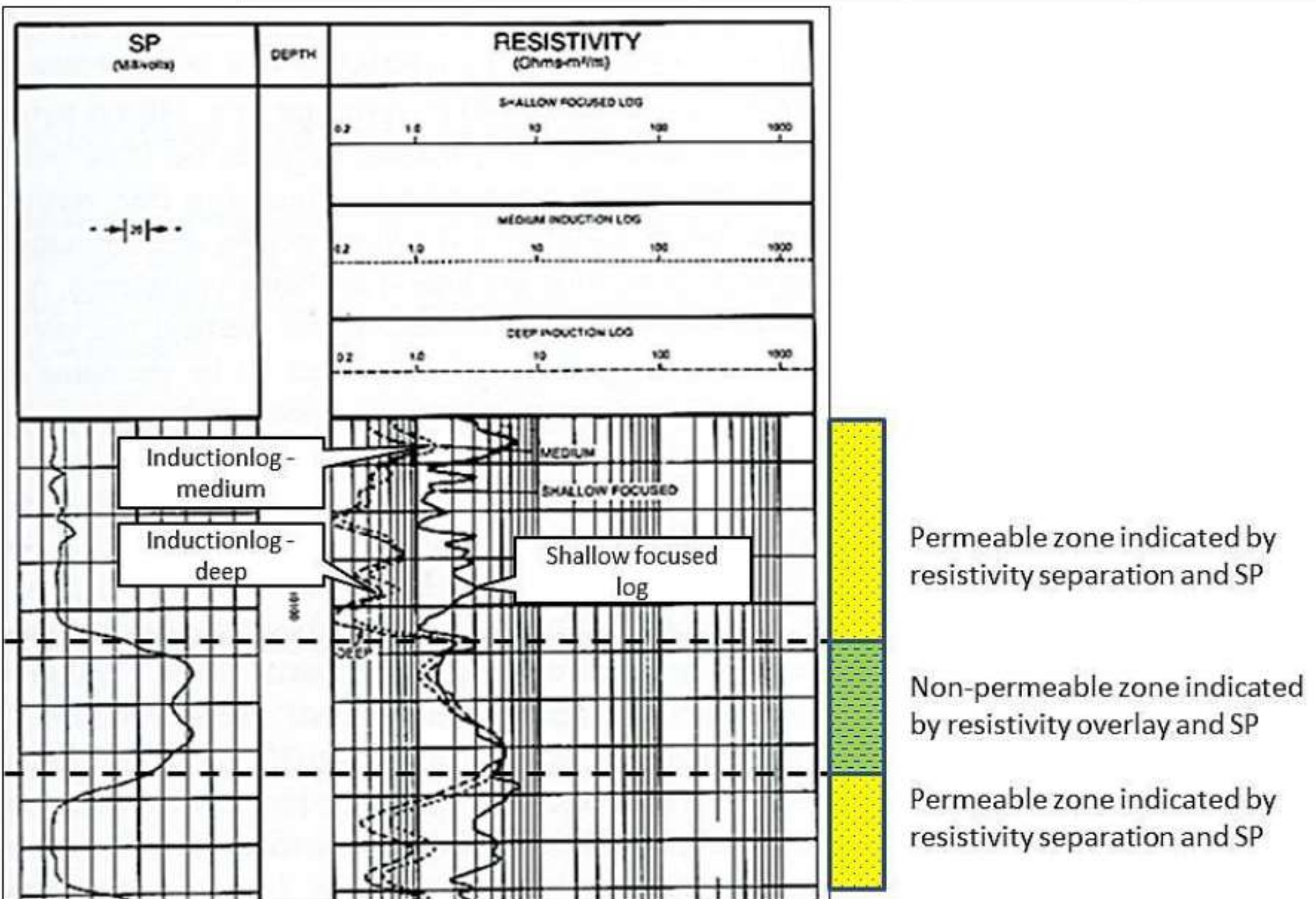
- Permeable beds can be identified quickly using the following indicators:
  - Spontaneous Potential (SP); the Diffusion Potential directly indicates a permeable formation,
  - Mud Filtrate Invasion detected by multiple resistivity measurements,
  - Presence of mudcake indicated by Caliper log,
- Note: The GR Log gives an indication related to mineral composition, but not directly to permeable and non-permeable zones
- For example Figure WE3 shows a section of sand-shale profile with  $R_{mf} > R_w$ . The three resistivity curves fit well in the non-permeable shale beds (no invasion) and separate in permeable sandstone. In the sandstone the Micro-Latero Log (MLL) – controlled by  $R_{xo}$  – shows a low resistivity compared to with the Laterolog Deep (LLD) and Laterolog Shallow (LLS) which are strongly influenced by the non-invaded zone ( $R_t$ ). Because the

# Wellsite Evaluation – Permeable Bed Identification

- Because the  $R_{MLL} < R_{DLL}$  and  $R_{SLL}$  it can be concluded that the sandstone is hydrocarbon-bearing.



- Fig. WE3** Permeable zone identification



- Separation of the resistivity logs results from an invasion effect and indicates permeable zones (sandstone), whereas a fit of the three indicates a non-permeable zone (shale)

# **Quantitative Log Evaluation Models – Lecture 5**

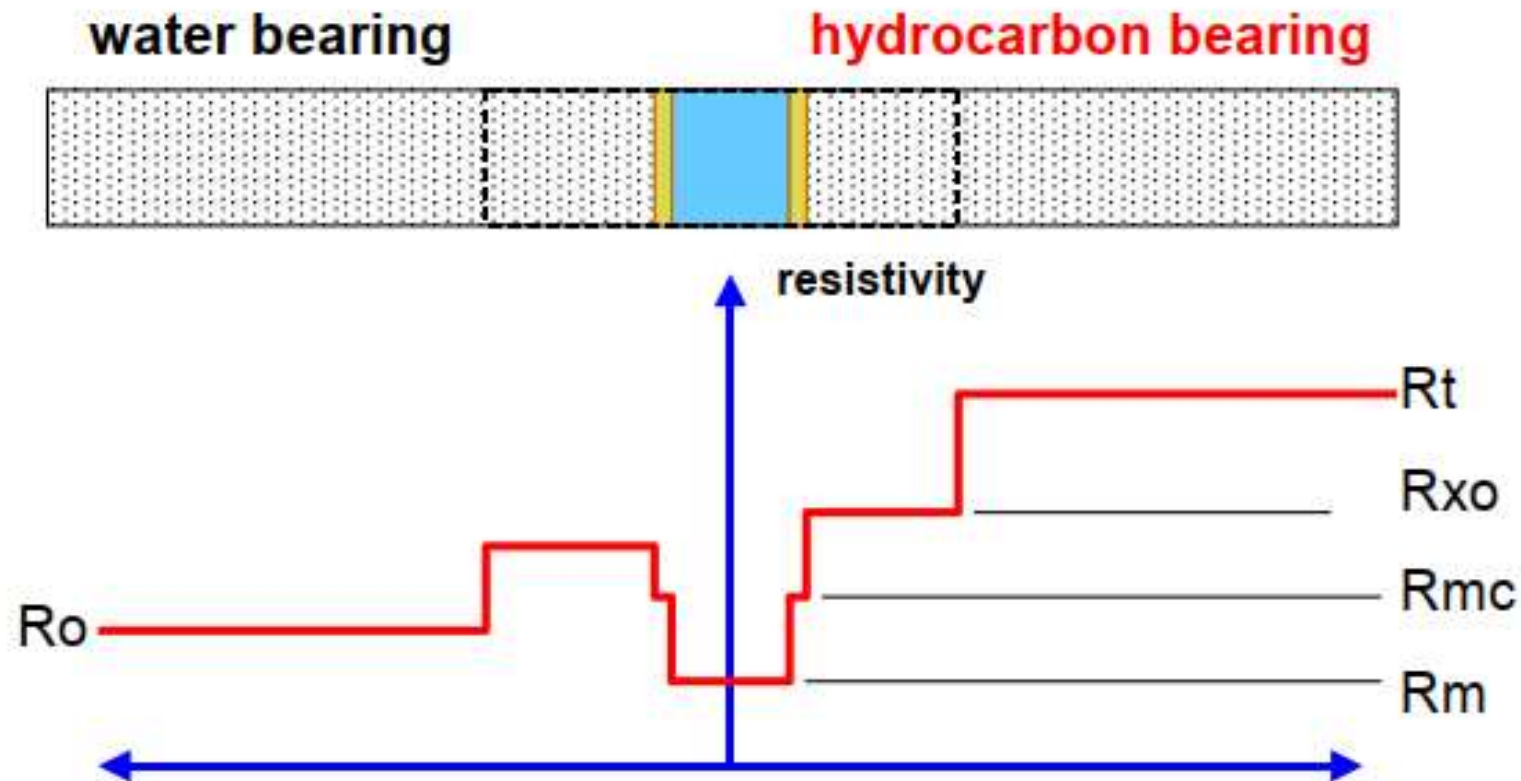
# Determination of True Resistivity

- In a water-bearing formation with low formation water resistivity  $R_w$ , the infiltration of the mud filtrate with higher resistivity  $R_{mf}$  results in an increase in rock resistivity in the infiltrated ( $R_{xo}$ ) zone compared to the non-invaded zone ( $R_o$ )
- In a hydrocarbon-bearing formation the resistivity  $R_t$  is relatively high as a result of lower water saturation  $S_w$ . Infiltration displaces a part of the original fluid (water and hydrocarbon) and results in a decrease in of rock resistivity in the infiltrated ( $R_{xo}$ ) zone compared to the non-invaded zone ( $R_o$ )
  - The  $R_{xo}$  level in the hydrocarbon-bearing formation is somewhat higher than the corresponding level in the water-saturated formation. This is the effect of a residual hydrocarbon volume in the pore space which is not totally displaced by the mud filtrate.
  - The magnitude of  $R_t$  in the hydrocarbon-bearing formation is controlled by porosity, water saturation, and formation saturation. Therefore can be low for highly saline formation water also in case of high HC saturation



# Determination of True Resistivity

radial profile of fluids  $\rightarrow$  radial profile of properties



Resistivity profile for water-bearing (left) and a HC-bearing zone



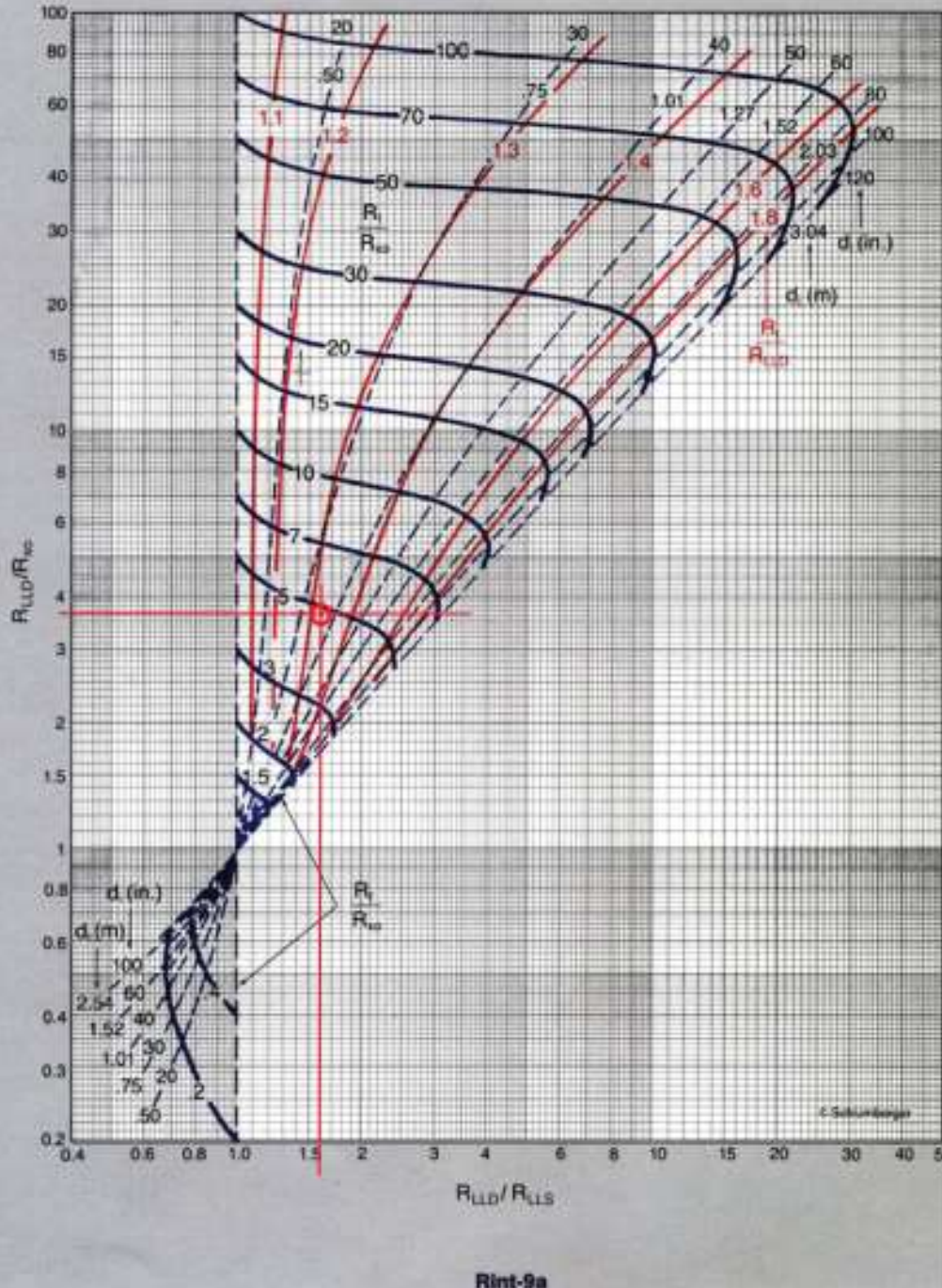
# Determination of True Resistivity

- For a “quick-look” estimate of  $R_{xo}$  a microresistivity tool is used in many cases and  $R_t$  is approximated by a “deep-reading” focused tool.
- A more exact technique is based on the graphic or numerical solution of the problem to derive “true resistivities”  $R_t$ ,  $R_{xo}$ , and  $d_i$  from a set of measured “apparent resistivities”.
- This can be demonstrated using “Tornado charts” which are the result of a forward modelling of the response answer of tools with different formation resistivity models (given in terms of  $R_t$ ,  $R_{xo}$ ,  $d_i$ ). The chart for a dual laterolog-  $R_{xo}$  device combination for thick beds and 8 inch borehole is shown on the next slide.
- The charts are plotted in logarithmic scale for resistivity ratios in general:
  - The x-axis shows the ratio of the forward calculated ratio  $R_{LLD}/R_{LLS}$  and the y-axis the forward calculated ratio  $R_{LLD}/R_{xo}$
  - The parameter of the individual curves correspond to the model input (ratio  $R_t/R_{xo}$ , ratio  $R_t/R_{LLD}$  and depth of invasion  $d_i$ ) attributed to the calculated curves.

## Dual Laterolog- $R_{xo}$ Device

DLT-B LLD—LLS— $R_{xo}$  Device

Thick Beds, 8-in. (203-mm) Hole,  
No Annulus, No Transition Zone,  
Use Data Corrected for Borehole Effect



# True Resistivity

- The chart is applied in the following steps
- 1) Take the measured resistivities  $R_{xo}$ ,  $R_{LLS}$ ,  $R_{LLD}$
- 2) Calculate the ratios  $R_{LLD}/R_{LLS}$  and  $R_{LLD}/R_{xo}$
- 3) Enter the abscissa and ordinate and plot the ratios on the chart
- 4) The intersection of the lines is the solution of the problem in terms of  $R_t/R_{xo}$ ,  $R_t/R_{LLD}$  &  $d_i$
- 5) Calculate the true resistivities  $R_t$  and  $R_{xo}$ .

See Page 59 of the main text for the example

# Det. of True Resistivity - Example

- Log Readings:

- $R_{LLD} = 16 \text{ Ohmm}; R_{LLS} = 10 \text{ Ohmm}; R_{MSFL} = 4.5 \text{ Ohmm}$

- Calculate the ratios:

- $R_{LLD} / R_{MSFL} = 3.56; R_{LLD} / R_{LLS} = 1.6;$

- At the intersection:

- $R_t / R_{LLD} = 1.33; R_t / R_{xo} = 4.7; d_i = 36 \text{ inch}$

- **Results:**

- $R_t = 21.3 \text{ Ohmm}; R_{xo} = 4.53 \text{ Ohmm}.$

- Please Note:

- The true resistivity of the non-invaded formation  $R_t$ , is distinctly higher than the measured resistivity of the deepest reading tool,  $R_{LLD}$ . This results from the influence of the lower resistivity of the invaded zone. Neglecting this effect would result in an overestimate of  $S_w$ .
  - The true resistivity of the invaded zone  $R_{xo}$  is close to the measured resistivity of the micro tool  $R_{MSFL}$ .

# Porosity from a single log

- Porosity as a fundamental reservoir property can be determined indirectly from Gamma-Gamma-Density log, Neutron log, and Acoustic -/Sonic log.
- Porosity calculation requires the matrix and fluid properties as expressed in the equations as (left side with Greek symbols, right side with Greek symbols) additional input.

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_{fl}}$$

$$PHI = \frac{RHO_{ma} - RHOB}{RHO_{ma} - RHO_{fl}}$$

$$\phi = \frac{\phi_N - \phi_{N,ma}}{\phi_{N,fl} - \phi_{N,ma}}$$

$$PHI = \frac{NPFI - NPFI_{ma}}{NPFI_{fl} - NPFI_{ma}}$$

$$\phi = \frac{\Delta t_{ma} - \Delta t}{\Delta t_{ma} - \Delta t_{fl}}$$

$$PHI = \frac{DT - DT_{ma}}{DT_{fl} - DT_{ma}}$$

$\rho_b(RHOB)$ ,  
 $\phi_N(NPFI)$ ,  
 $\Delta t(DT)$ ,  
 are the measured  
 (logged) data

# Porosity from a single log

- Matrix properties can be determined from:
  - Known lithology/mineral composition; in this case representative values are recommended. The most critical is matrix slowness.
  - Core laboratory measurements (for example grain density)
  - Cross plot techniques

| Component | $\rho_{ma}, \rho_{fl}$ in g/cm <sup>3</sup> | $\phi_{N,ma}, \phi_{N,fl}$ | $\Delta t_{ma}, \Delta t_{fl}$ in $\mu$ s/m | $\Delta t_{ma}, \Delta t_{fl}$ in $\mu$ s/ft |
|-----------|---------------------------------------------|----------------------------|---------------------------------------------|----------------------------------------------|
| Quartz    | 2.65                                        | - 0.02 ... - 0.04          | 180                                         | 55                                           |
| Calcite   | 2.71                                        | 0                          | 156                                         | 47.4                                         |
| Dolomite  | 2.87                                        | 0.02.....0.04              | 143                                         | 43.5                                         |
| Water     | 1.00                                        | 1.00                       | 620                                         | 189                                          |

Recommended properties for matrix and fluid

# Porosity from a single log – invasion rules

- Invasion tends to force gas out of the formation and replace it with borehole fluid. Porosity calculation from NEU & DEN measurement requires fluid properties. The influence of the two porosity logs depends on their radial sensitivities and their depths of investigation (DOI)
- If the invasion fluid front gets deep into the reservoir, the NEU & DEN Porosity measurements approach the true porosity for the assumption of mud filtrate density and the Neutron response as fluid properties.
- For shallow invasion (shallow with respect to the depth of investigation) the tools' responses are spatially weighted averages of the invaded and non-invaded regions of the formation. The problem in deriving the porosity in the presence of shallow invasion comes from the fact that the neutron and density logging devices have different radial response:
  - The 50% DOI of the thermal neutron porosity tool is 6 to 12 inches (15 to 30 cm).
  - The 50% DOI of the density tool is about 2 to 3 inches (5 to 8 cm).



# Porosity from a single log – invasion rules

- In the presence of borehole invasion into a gas-bearing reservoir, some specific conditions must be considered.
  - When the invasion front is greater than 12 inches (30 cm), both tools see only water-filled (mud filtrate) formations and the two porosity estimates agree and read true porosity.
  - When the invasion front is less than 12 inches (30 cm) but greater than 6 inches (15 cm), the density tool on sees the invaded formation, while the neutron tool is sensitive to both the invaded and the non-invaded region. Under these conditions, the density porosity estimate is the true value while the Neutron porosity estimate is still low.
  - Below 6 inches (15 cm) of invasion, both tools are sensitive to both the invaded and non-invaded regions. Thus, for a certain range of invasion, accurate determination of formation porosity becomes very difficult. A Root Mean Square (RMS) is recommended for gas reservoirs:

$$\phi_{ND,gas} = \sqrt{\frac{\phi_N^2 + \phi_D^2}{2}} \approx \frac{\phi_N}{3} + 2 \cdot \frac{\phi_D}{3}$$



# Porosity and Mineral Composition

## – Multiple Porosity Methods

- Two or more independent methods for porosity are applied to determine an exact porosity value. Combined interpretation makes derivation of porosity and rock composition possible. The combination of DEN and NEU Log is applied most frequently.
- If the lithology (mineralogy) and also the water saturation can be assumed for a formation, then porosity can be calculated with both methods. If both independent calculations deliver the same porosity it may be concluded that:
  - 1) The used interpretation model (limestone/calcite & water) is correct.
  - 2) The calculated porosity is correct.
- If the calculated porosities are different, the conclusion is that the used interpretation model (matrix mineralogy, fluid) is not correct. The conflict of the two answers can be resolved by derivation of representative mineral composition and fluid type using any of these practical techniques.
  - 1) Overlay Technique
  - 2) Cross plot technique
  - 3) Numerical solution of a system of response equations

# Porosity and Mineral Composition

## – Overlay technique

- Two porosity curves are plotted in the same log trace and referenced to an identical porosity scale and a defined rock model (matrix and fluid). If for example, Neutron porosity (limestone-calibrated tool) and Density-derived porosity (calculated under assumption of calcite as matrix with density  $2.71 \text{ g/cm}^3$ ) are plotted then in a:
  - Water-saturated limestone both tools read same porosity,
  - Sandstone or dolomite both tools read a somewhat different porosity,
  - Shaly section neutron reads higher porosity than density resulting from clay-bound water,
  - Gas zone neutron reads lower porosity than density resulting from the lower hydrogen concentration measured by the neutron log.
- Important for overlay analysis is to understand the scales for both methods (whether identical scales are used or increasing Neutron porosity and decreasing density). If the formation is composed of different lithologies and this overlay technique is applied, it is more comfortable to convert or transform both porosity logs to a sandstone matrix and expect a fit for both logs in such cases.

# Porosity and Mineral Composition

## – Crossplot technique (Clean Matrix)

- While overlays present the logs of a section, cross plots technique combines individual data sets of defined depth steps. For the combination of Neutron and Density log, the Neutron readings from the LAS file are plotted on the x-axis and the measured Gamma-Gamma-Density in reverse on the y-axis for each depth step.
- Plots start in the lower left corner with the “matrix-points” and go up to the upper right corner to the “water-point” with linear scaled porosity. For practical use only the part for porosity  $< 0.45$  (45%) is plotted:
  - The coordinates  $x = \text{NPHI} = 0.0$  and  $y = \text{RHOB} = 2.71 \text{ g cm}^{-3}$  describe a dense (no porosity) calcite and therefore is called “matrix-point limestone”.
  - Increasing porosity (pore fluid water) moves the position linearly upward from this matrix point in the direction of a “fluid-point water” at  $x = \text{NPHI} = 1.0$  and  $y = \text{RHOB}_{\text{fl}} = \text{RHOB} = 1.0 \text{ g cm}^{-3}$ . The connecting straight line represents a pure limestone with linear scaled porosity.
  - Different rocks (sandstone/quartz, dolomite) have a matrix point defined by matrix density and matrix neutron response.
- The result are three lines for limestone, dolomite and sandstone. The deviation from a straight line of the dolomite and sandstone curve is contributed for better approximation and depends on tool characteristics.

# Porosity and Mineral Composition

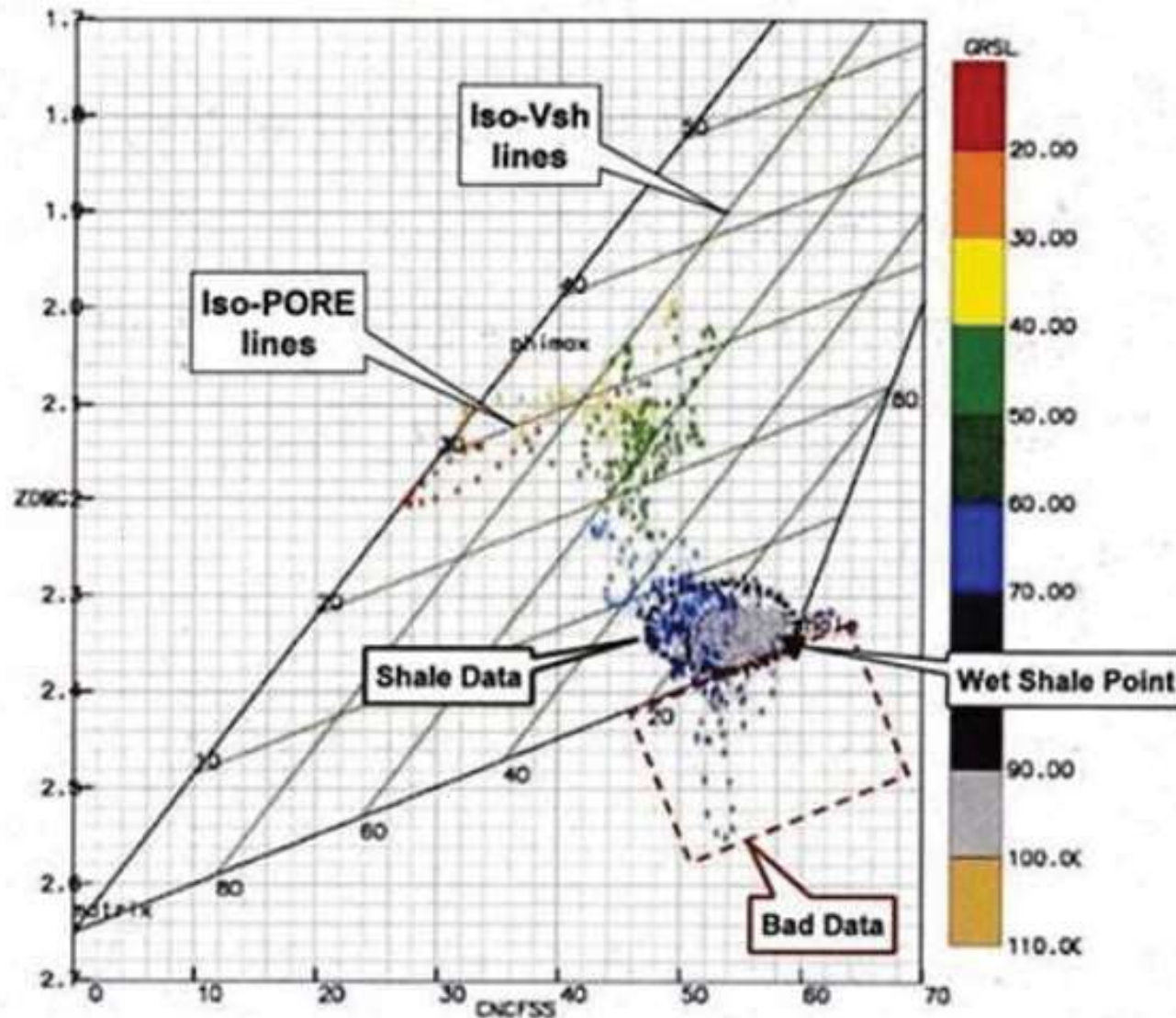
## – Crossplot technique (Clean Matrix)

- Cross plots allow an estimate of mineralogy (lithology) and the determination of a consistent porosity. Measured data are plotted into the designated curve set. The position of the data point delivers the correct porosity and the mineral type or composition.
- Deviations and problems occur for the Neutron-Density-Crossplot if shale or gas is present:
  - Gas in the formation shifts the points towards low neutron porosity, because gas has a lower hydrogen concentration compared with water and in most cases the Neutron log “reads” the non-invaded zone .
  - Shale content shifts the points from the sand line towards dolomite, because apparent Neutron porosity increases (clay-bound water effect). For shaly sand a specific cross plot can be designed with a sand point, a wet-shale point and a water point. This is a component of shaly sand analysis.
- A helpful modification of the NEU-DEN Cross plot is the implementation of a third dimension: individual points are colour coded by their gamma activity. If the colour indicates low shale content (low gamma), then the point represents a clean rock (in sandstone, limestone or dolomite region). If the colour indicates high shale content and the point plots, for example, in the porous dolomite region then this is not a highly porous dolomite but probably a shaly sand.

# Porosity and Mineral Composition

## – Crossplot technique (Shaly Sand)

- A specific application is the interpretation of shaly sands. In a NEU-DEN-Crossplot data points of the clean sandstone are arranged along the sandstone line.



With increasing shale content the data points of the clean sandstone are shifted to higher NEU porosity (clay-bound water effect). If data points are colour coded with Gamma intensity (GR) this effect is visible as demonstrated below.



# Porosity and Mineral Composition

## – Crossplot technique (Shaly Sand)

- The distribution of all data points allows picking the “wet-shale” point at the highest GR point (pure shale). It characterises the shale including the clay bound water. The wet shale density could be extracted on the vertical axis and the wet shale Neutron porosity (neutron response wet shale) on the horizontal axis.
- The volumetric rock model (below) describes the porosity and density in terms of effective and shale porosity.



Shaly sand model

# Porosity and Mineral Composition

## – Crossplot technique (Shaly Sand)

- The equations for (measured) Neutron porosity  $\phi_N$  and bulk density  $\rho_b$  can be solved for shale content.

$$V_{sh} = \frac{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_b) - (\phi_N - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}$$

- Effective porosity:

$$\phi_{eff} = \frac{(\phi_N - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_b)}{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}$$

Where

- $\phi_{N,fl}$  is the neutron response (hydrogen index) of the fluid
- $\phi_{N,ma}$  is the neutron response (hydrogen index) of the matrix
- $\phi_{N,sh}$  is the neutron response (hydrogen index) of the wet shale
- $\rho_{fl}$  is the density of the fluid
- $\rho_{ma}$  is the density of the matrix
- $\rho_{sh}$  is the density of the wet shale



# Porosity and Mineral Composition

## – Generalized solution for a composite matrix

- Porosity and volume fractions of rock components (minerals) can be determined very elegantly mathematically by formulation of a system of linear equations. The response equations for a combination of Neutron porosity and Gamma-gamma-density in case of a (clean) carbonate section consisting of limestone (calcite) and dolomite are:

$$- \rho_b = RHOB = 2.71 * V_{\text{calcite}} + 2.87 * V_{\text{dolomite}} + 1.00 * \phi$$

$$- \phi_N = NPHI = 0 * V_{\text{calcite}} + 0.02 * V_{\text{dolomite}} + 1.00 * \phi$$

- The volume balance is:  $1 = V_{\text{calcite}} + V_{\text{dolomite}} + \phi$
- This represents a system of 3 linear equations which can be solved for the 3 unknowns  $V_{\text{calcite}}$ ,  $V_{\text{dolomite}}$ ,  $\phi$ . The mathematical solution can be calculated comfortably by a “matrix (math) inversion”. The response equations in the matrix notation are:

$$\begin{bmatrix} \rho \\ \phi_N \\ 1 \end{bmatrix} = \begin{bmatrix} 2.71 & 2.87 & 1.00 \\ 0 & 0.02 & 1.00 \\ 1.00 & 1.00 & 1.00 \end{bmatrix} \cdot \begin{bmatrix} V_{\text{calcite}} \\ V_{\text{dolomite}} \\ \phi \end{bmatrix} \quad \text{or } \mathbf{M} = \mathbf{R}\mathbf{V}$$

**M** matrix - measured properties, = **R** matrix – response,

**V** matrix – volume fractions

# Porosity and Mineral Composition

## – Generalized solution for a composite matrix

- The solution is  $\mathbf{V} = \mathbf{R}^{-1}\mathbf{M}$  where  $\mathbf{R}^{-1}$  is the inverse matrix:

$$\begin{bmatrix} V_{calcite} \\ V_{dolomite} \\ \phi \end{bmatrix} = \begin{bmatrix} -5.099 & -9.209 & 14.818 \\ 5.203 & 8.377 & -14.100 \\ -0.104 & 0.833 & 0.282 \end{bmatrix} \cdot \begin{bmatrix} \rho \\ \phi_N \\ 1 \end{bmatrix}$$

- The set of linear equations can be generalized for a rock with  $n$  different materials components (solids, fluids). For any property  $g$  with a linear response equation, the measured magnitude is:

$$g_m = \sum_i^n V_i \cdot g_{m,i}$$

Where:

- $g_m$  is the property, measured with a method  $m$ ,
- $g_{m,i}$  is the property (related to method  $m$ ), of the component  $i$ ,
- $V_i$  is the volume fraction of component  $i$ ,

Additionally, there is the volume balance equation  $\sum_i V_i = 1$

# Porosity and Mineral Composition

## – Generalized solution for a composite matrix

- This is a system of  $m + 1$  independent linear equations; it can be solved for  $n = m + 1$  components (deterministic solution). The example has demonstrated this techniques for two methods (Neutron and Gamma-Gamma). The system can be expanded by implemented of the Photo-electric (PE) log, Acoustic/Sonic log (if the linear principle of the time-average equation can be applied). In case of 4 independent measurements the system can be solved deterministically for  $4 + 1 = 5$  volume fractions (porosity and 4 minerals).
- The calculation of rock composition and porosity based on this set of linear equations:
  - The measured magnitude of the property
  - The properties of the expected components (rock model)
  - The volume fraction of the components
- The definition of the expected rock composition (rock model) is a very important step. Valuable input for this step comes from geological knowledge of the investigated formation, core data with respect to the minerals, and cuttings data with respect to the minerals. Particularly in carbonates with mixture mineralogy some minerals with very strong effect on measured properties (e.g. high density of anhydrite) can be present.

# Early Quantitative Models

- Archie's Equation (1942): describes the correlation between specific electrical resistivities (measured properties), porosity, water saturation (derived properties) and empirical parameters.
- Wyllie's Equation (1956)
- Coates equation (1974, 1999)

# Clean Rocks – Archie's (1942) Equations

- In a clean porous rock (for example sandstone without clay and/or other conductive solid components) the formation water is the only conductive component because matrix (quartz) and HC or air in the pore spaces are insulators. In this case the specific resistivity of the rock is proportional to the specific resistivity of the pore water.
- The specific resistivity of the formation water is a function of salinity (concentration of the salts) and temperature.
- The proportionality between rock resistivity and formation water resistivity is the fundamental idea of the Archie's (1942) equations which describe the relationship of rock resistivity to porosity and water saturation.
- For a water saturated porous rock ( $S_w = 1$ ) the specific rock resistivity  $R_o$  is controlled by the resistivity of the pore water thus  $R_w$  thus:  
$$R_o = F * R_w$$

# Clean Rocks – Archie's (1942) Equations

- For a water saturated porous rock ( $S_w = 1$ ) the specific rock resistivity  $R_o$  is controlled by the resistivity of the pore water thus  $R_w$  thus:  
$$R_o = F * R_w$$

- The introduced formation resistivity factor  $F$  depends on porosity:  
$$R_o = (1/\phi^m) * R_w$$

- Where the exponent  $m$  is the cementation exponent whose value varies with grain size, grain size distribution and the complexity of the paths between the pores i.e. Tortuosity ( $m$  is in the order of 2 for sandstones).

- For partial water saturation  $S_w$ , the specific rock resistivity  $R_t$  increases with decreasing water saturation. Archie (1942) formulated this by introducing a “Resistivity Index  $I$ ” and referenced the rock resistivity  $R_t$  at any water saturation to the rock resistivity at total water resistivity  $R_o$  thus:

$$R_t = I * R_o = R_o * (S_w)^{-n}$$

Where  $n$  is the empirical saturation exponent

# Clean Rocks – Archie's (1942) Equations

- In the plot  $\log I$  versus water saturation  $S_w$  with logarithmically scaled axis, this is also represented by a straight line with the slope  $n$ .
- Thus, the formation resistivity factor  $F$  expresses the magnification of the rock resistivity (compared with water resistivity) as a result of non-conductive solid minerals.
- The resistivity index  $I$  expresses the magnification of the rock resistivity (compared with water resistivity) as a result of non-conductive fluid components (oil, gas).
- The relationships deliver Archie's fundamental equation for water saturation calculation

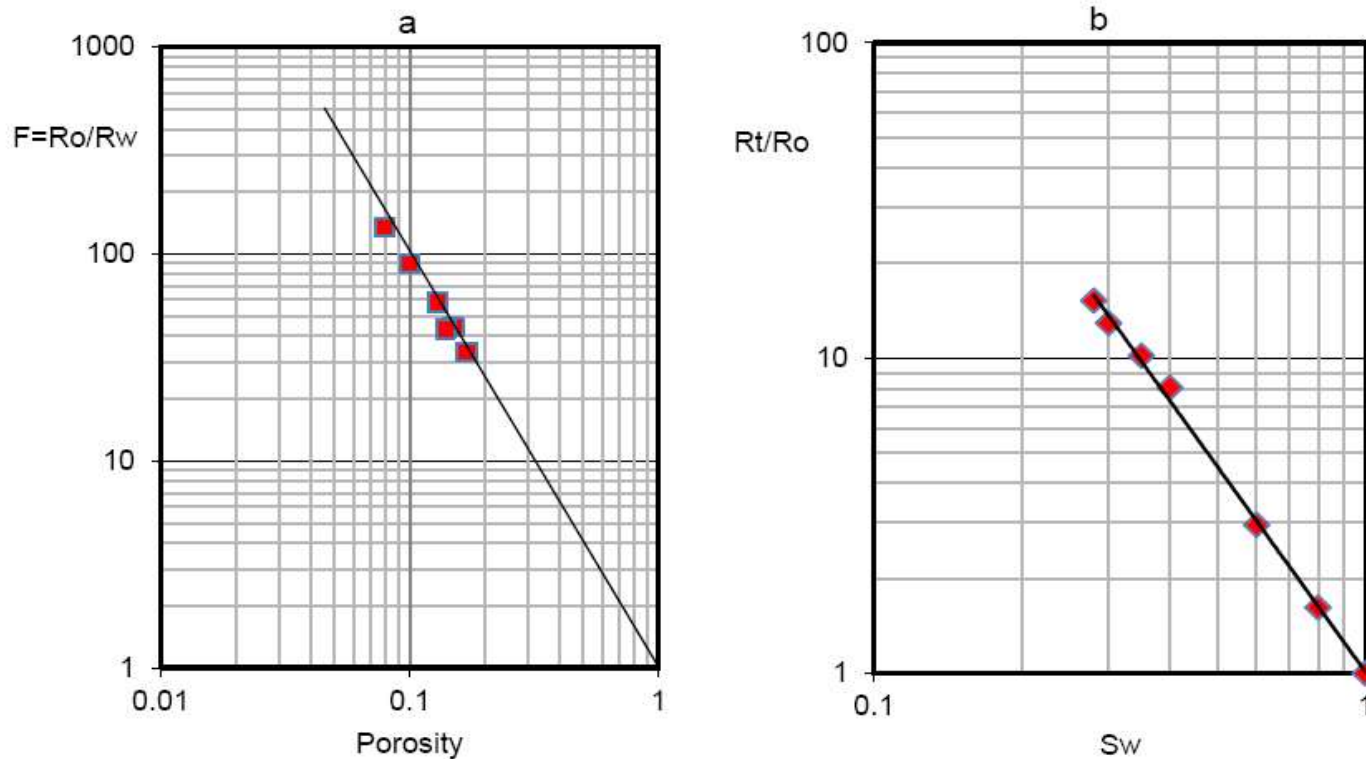
$$S_w = \left( \frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left( \frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

- The two Archie exponents  $m$  and  $n$  are empirical rock parameters controlled by the geometry of the pore network and the subject of core analysis.



# Clean Rocks – Archie's (1942) Equations

- In a plot of  $F$  versus porosity  $\phi$  with the logarithmically scaled axis this is represented by a straight line with the slope  $m$ .



- In general, it was recognized that  $m$  increased with the tortuosity of the electric path through the pore space.

*Archie's relationships, experimental data from Darling (2005)*

# Clean Rocks – Archie's (1942) Equations

- In many cases (and if no other information is available)  $m \approx n \approx 2$  is recommended for a first orientation. In this case the equation results in:

$$S_w = \sqrt{\frac{R_o}{R_t}} = \frac{1}{\phi} \cdot \sqrt{\frac{R_w}{R_t}}$$

- The equation shows that for saturation calculation, the following input is necessary.
  - The measured formation resistivity  $R_t$
  - The formation resistivity at total water saturation  $R_o$ , or
  - The porosity  $\phi$  and water resistivity  $R_w$
- The formation resistivity  $R_o$  at  $S_w = 1$  can be:
  - Estimated for a quick look interpretation from a water zone in the same reservoir
  - Calculated from Archie's equation with water resistivity  $R_w$ , an independent porosity determination, and the exponent  $m$

# Clean Rocks – Archie's (1942) Equations

- In a permeable porous formation Archie's equation can be applied
  - On the non-invaded zone in order to determine the water saturation  $S_w$  from formation resistivity  $R_t$ .  $(1 - S_w) = S_{hy}$  gives hydrocarbon saturation.

$$S_w = \left( \frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left( \frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}} \qquad S_{xo} = \left( \frac{R_{mf}}{R_{xo}} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

- On the invaded zone in order to determine the mud filtrate saturation  $S_{xo}$  from the resistivity of the invaded zone  $R_{xo}$  using mud filtrate resistivity  $R_{xo}$  as resistivity of the conductive fluid with the saturation equation.
- The difference  $(1 - S_{xo}) = S_{hy, non-mov}$  gives the non-movable (residual) hydrocarbon saturation (part of the original fluid which is not displaced by infiltration). The fluid displacement is an indication of fluid mobility.

# Shaly Rocks

- The presence of shale results in a decrease of reservoir quality (porosity, permeability) and “disturbs” traditional interpretation algorithms and techniques – particularly determination of water saturation using Archie’s equation.
- Shale in the formation is a second conductivity component in addition to the formation water. “Shale conductivity” results in interface effects of clay minerals and adjacent electrolyte (formation water).
- Frequently, this effect is described by the Cation Exchange Capacity (CEC) of clay. Neglecting this conductivity contribution of shale to the measured conductivity ( $=\text{resistivity}^{-1}$ ) results in an overestimate of the calculated water saturation.
- For a correct  $S_w$ -calculation from resistivity logs this second conductivity effect therefore must be eliminated.

# Shaly Rocks

- The conductivity contribution of shale/clay depends on:
  - Shale Content (Volume fraction) , derived from e.g. Gamma Log
  - Shale Type (Clay Mineralogy) controlling the “shale conductivity”
  - Distribution of shale in the formation (laminar shale or dispersed shale)
- Standard methods for a shale content estimate are: SP-log, Gamma-log, Neutron-Density-Combination.
- Calculation of the shale volume is the **first** step of quantitative interpretation. The calculated shale content is used for:
  - General Reservoir Description
  - Volumetric Rock Model
  - Various “shale corrections”
- Various “shaly sand equations” have been derived in order to model the different types of shale in a rock.

# Laminated Shaly Sand – The Poupon Equation

- For the simple case of laminated shaly sand Poupon et al. (1954) derived an equation. The laminated shaly sand represents an alternating sand-shale layering with the thickness of the individual layers in the millimetre range. Resistivity tools cannot resolve individual layers but measure the effect of a parallel conductor assuming current lines are parallel to the layering.



- The conductivity (the inverse of resistivity) of this parallel conductor model is:

Where:

- $V_{sh-lam}$  is the (laminar) shale content,
- $R_{sh}$  is the shale resistivity
- $R_{sd}$  is the sand resistivity

$$\frac{1}{R_t} = \frac{1 - V_{sh-lam}}{R_{sd}} + \frac{V_{sh-lam}}{R_{sh}}$$

# Laminated Shaly Sand – The Poupon Equation

- Application of this equation has two steps:

- 1) Calculation of the resistivity of the sand fraction:

$$R_{sd} = (1 - V_{sh-lam}) \cdot \left( \frac{1}{R_t} - \frac{V_{sh-lam}}{R_{sh}} \right)^{-1}$$

- 2) Calculation of the water saturation for the sand fraction using Archie's equation

$$S_w = \left[ \frac{R_w}{R_{sd}} \cdot \frac{1}{\phi^m} \right]^{\frac{1}{n}}$$

- Thus, as input the following parameters are necessary:

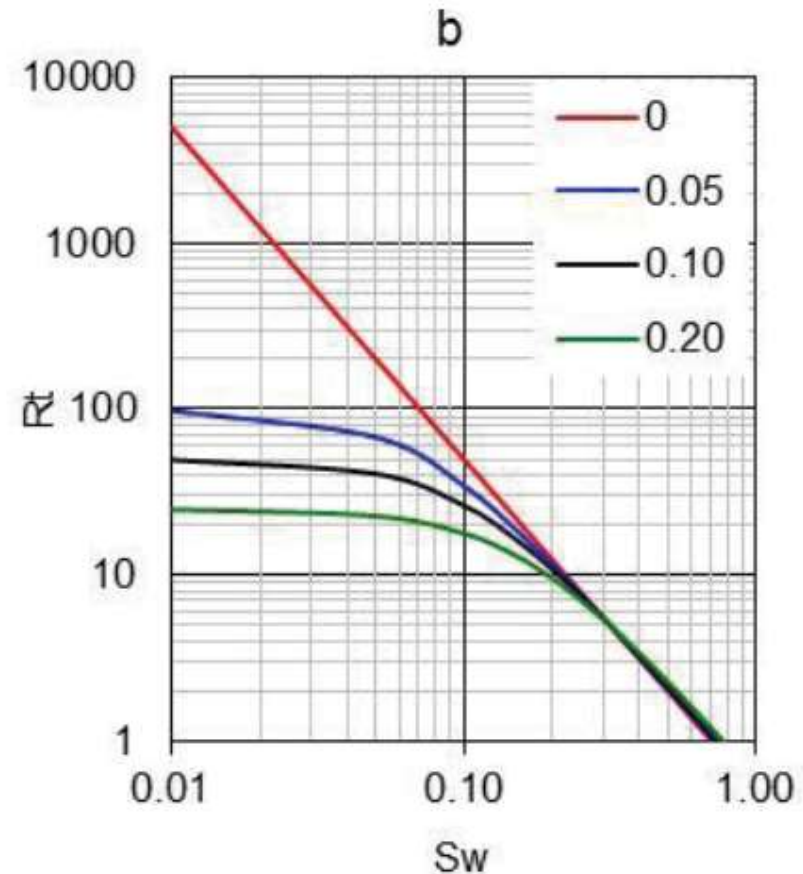
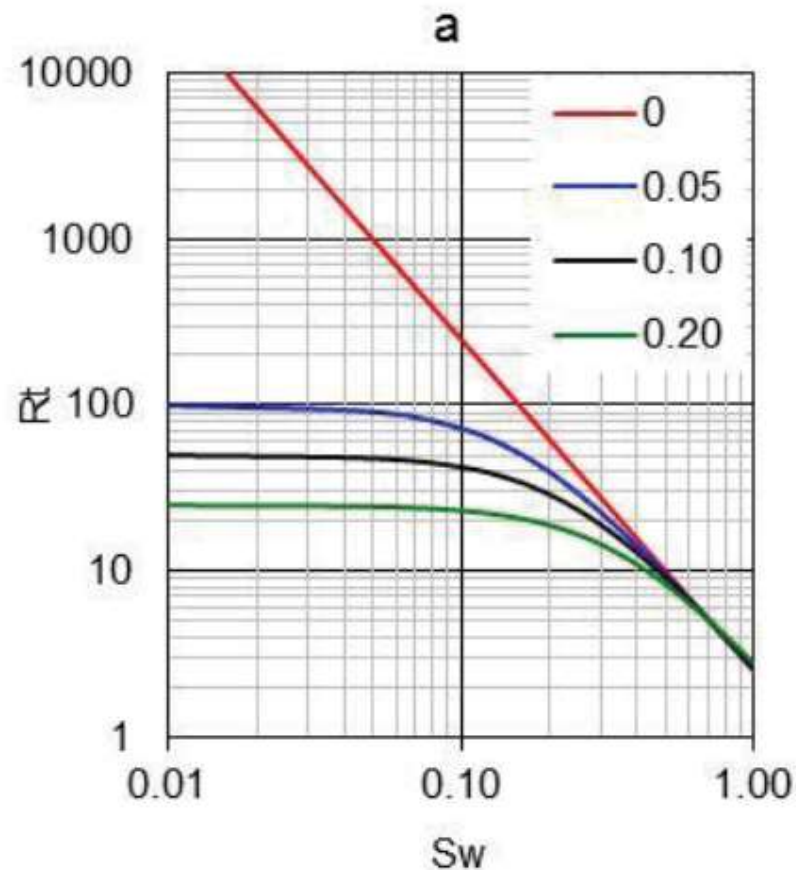
|                                       |          |                                       |
|---------------------------------------|----------|---------------------------------------|
| $R_t$ from the log measurement,       | $R_{sh}$ | resistivity of an adjacent shale bed, |
| $R_w$ formation water resistivity,    | $V_{sh}$ | shale content e.g. from Gamma log,    |
| $\phi$ porosity (from porosity logs), | $m, n$   | Archie exponents                      |

- We must ascertain that the clay is laminated (possibly using the Thomas-Stieber plot to discriminate between dispersed & laminated clay)



# Effect of Shale for Laminated Shaly Sand

- The effect of shale is very strong at low water saturation and increases with increasing water saturation.



- Formation Resistivity  $R_t$  as a function of water saturation  $S_w$  for four different laminar shale contents  $V_{sh}$ . Porosity = 0.2,  $m=n=2$ ,  $R_{sh}=5$  Ohm m, (a)  $R_w = 0.10$  Ohm m, (b)  $R_w = 0.02$  Ohm m,

# Laminated Shaly Sand – Example Calculation

- Input data:  $R_t = 4 \text{ Ohm m}$ ,  $R_{sh} = 1.5 \text{ Ohm m}$ ,  $R_w = 0.02 \text{ Ohm m}$ ,  $V_{sh} = 0.20$ ,  $\phi = 0.18$ ,  $m = n = 2$ ,  $F = 31$ .
- Wrong Calculation: Neglect  $V_{sh}$  & apply Archie's equation directly

$$S_w = \left( \frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}} = \sqrt{\frac{0.02}{4} \cdot 31} = 0.39$$

- Correct Calculation: Consider  $V_{sh}$  & apply Poupon's equation

$$S_w = \left[ R_w \cdot \left( \frac{1}{R_t} - \frac{V_{sh}}{R_{sh}} \right) \cdot \frac{1}{\phi^m} \cdot \frac{1}{1 - V_{sh}} \right]^{\frac{1}{n}} = \left[ 0.02 \cdot \left( \frac{1}{4} - \frac{0.2}{1.5} \right) \cdot 31 \cdot \frac{1}{1 - 0.2} \right]^{\frac{1}{2}} = 0.30$$

- Conclusion: The shale gives a contribution to the rock conductivity & decreases the resistivity. Shale therefore have an effect which acts in the same direction as water (in our example as 9% water). Neglecting the shale results in a non-realistically high water saturation (or low hydrocarbon saturation).

# Dispersed Shaly Sand – The Models

- If shale is distribution dispersed in the pore spaces then the “shale conductivity” must be placed in the pores at the same location as the pore water.
- There are several models:

| SN | Model                                                | Application                                         |
|----|------------------------------------------------------|-----------------------------------------------------|
| 1  | Waxman and Smits (1967)                              | Water-bearing rock;<br>Hydrocarbon-bearing rock     |
| 2  | The Dual-Water Model (Clavier et. al, 1977, 1984)    |                                                     |
| 3  | (Simandoux, 1965) modified by Bardon and Pied (1969) | Structural and dispersed type of shale distribution |
| 4  | The Indonesian Equation (Poupon and Lavaux, 1971)    | Shaly formations with fairly fresh water            |
| 5  |                                                      |                                                     |
| 6  |                                                      |                                                     |

# Dispersed Shaly Sand – Waxman & Smits

- Waxman and Smits (1967) developed a model with the “architecture” of Archie’s equation and replaced water conductivity ( $C_w = R_w^{-1}$ ) by the sum ( $C_w + B \cdot Q_v$ ), where the second term expresses the shale conductivity component.
- For water saturated ( $S_w = 1$ ) porous rocks results:

With the terms

$$\frac{1}{R_o} = C_o = \frac{1}{F^*} \cdot (C_w + B \cdot Q_v)$$

$\phi$  porosity,

$S_w$  water saturation,  $F^*$  formation factor of the shaly sand,

$C_t$  conductivity of the shaly sand,

$C_w$  conductivity of the formation water,

$B$  equivalent conductivity of sodium clay-exchange cations,

$Q_v$  shalyness factor (cation exchange capacity per unit volume, meq cm<sup>-3</sup>).

# Dispersed Shaly Sand – Waxman & Smits

- For an oil- or gas- bearing rock the Waxman-Smit eq. is:

$$\frac{1}{R_t} = C_t = \frac{S_w^n}{F^*} \cdot \left( C_w + \frac{B \cdot Q_v}{S_w} \right)$$

- And water saturation  $S_w$  results (with  $n=2$ ) as

$$S_w = \left[ \frac{F^* \cdot R_w}{R_t} + \left( \frac{B \cdot Q_v \cdot R_w}{2} \right)^2 \right]^{-\frac{1}{2}} - \left( \frac{B \cdot Q_v \cdot R_w}{2} \right)$$

- $B \cdot Q_v$  must be determined at the laboratory and depends on temperature.

# Dispersed Shaly Sand – Dual-Water Model

- The Dual-Water Model is based on the concept of two types of water:
  - Free water in the pore space  $wf$ ,
  - Water near the clay surface (bound water)  $bw$ .
- Both conductivities again form a parallel circuit:

With the terms

$$C_t = \frac{1}{R_t} = \frac{\phi^m \cdot S_{wf}^n}{a} \cdot \left[ C_w + \frac{S_{wb}}{S_{wf}} \cdot (C_{wb} - C_{wf}) \right]$$

|          |                          |          |                           |
|----------|--------------------------|----------|---------------------------|
| $S_{wf}$ | free water saturation,   | $S_{wb}$ | bound water saturation,   |
| $C_{wf}$ | free water conductivity, | $C_{wb}$ | bound water conductivity, |
| $C_t$    | rock conductivity,       |          |                           |

# Dispersed Shaly Sand – Simandoux Equation

- The modified Simandoux equation is recommended for structural and dispersed type of shale distribution.

$$\frac{1}{R_t} = C_t = \frac{\phi^m}{R_w} \cdot S_w^n + V_{sh} \cdot C_{sh} \cdot S_w$$

- For  $n = 2$  results:

$$S_w = \frac{1}{2} \cdot \frac{R_w}{\phi^m} \cdot \left[ \sqrt{4 \cdot \frac{\phi^m}{R_w \cdot R_t} + \left( \frac{V_{sh}}{R_{sh}} \right)^2} - \frac{V_{sh}}{R_{sh}} \right]$$



# Dispersed Shaly Sand – Indonesia Equation

- The Indonesia equation (Poupon and Levaux, 1971) is recommended for shaly formations with fairly fresh water.

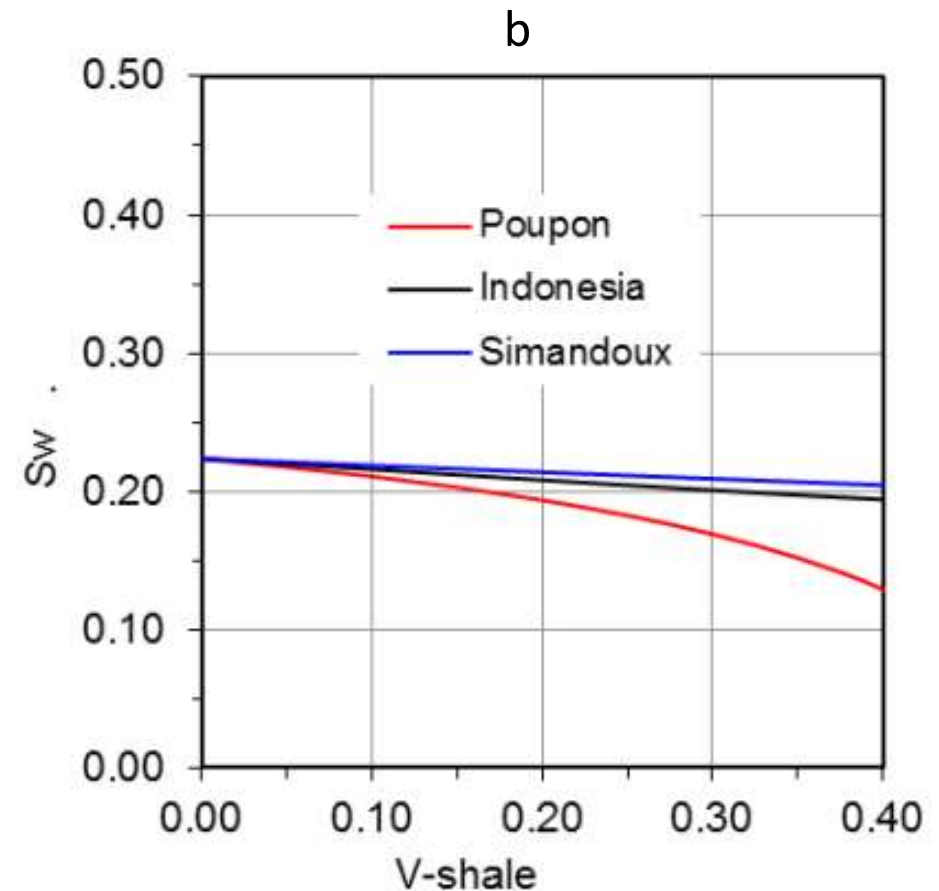
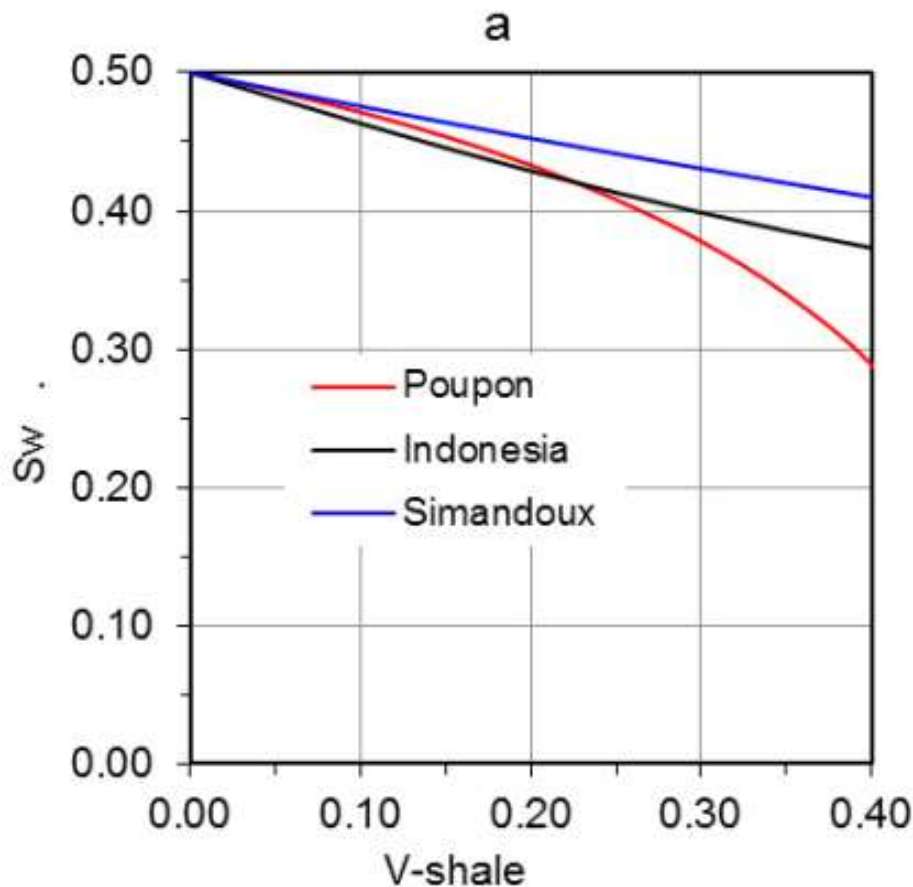
$$C_t = \frac{C_W}{F} \cdot S_W^2 + 2 \cdot \sqrt{\frac{C_W \cdot C_{sh}}{F} \cdot V_{sh}^{2-V_{sh}}} \cdot S_W^2 + V_{sh}^{2-V_{sh}} \cdot C_{sh} \cdot S_W^2$$

- Or in a simplified form for  $V_{sh} \leq 0.5$ :

$$C_t = \frac{C_W}{F} \cdot S_W^2 + 2 \cdot \sqrt{\frac{C_W \cdot C_{sh}}{F} \cdot V_{sh}} \cdot S_W^2 + V_{sh} \cdot C_{sh} \cdot S_W^2$$

# Effect of Shale on Sw-calculation for three shaly sand equations

- The effect of shale is very strong at low water saturation and increases with increasing water saturation.

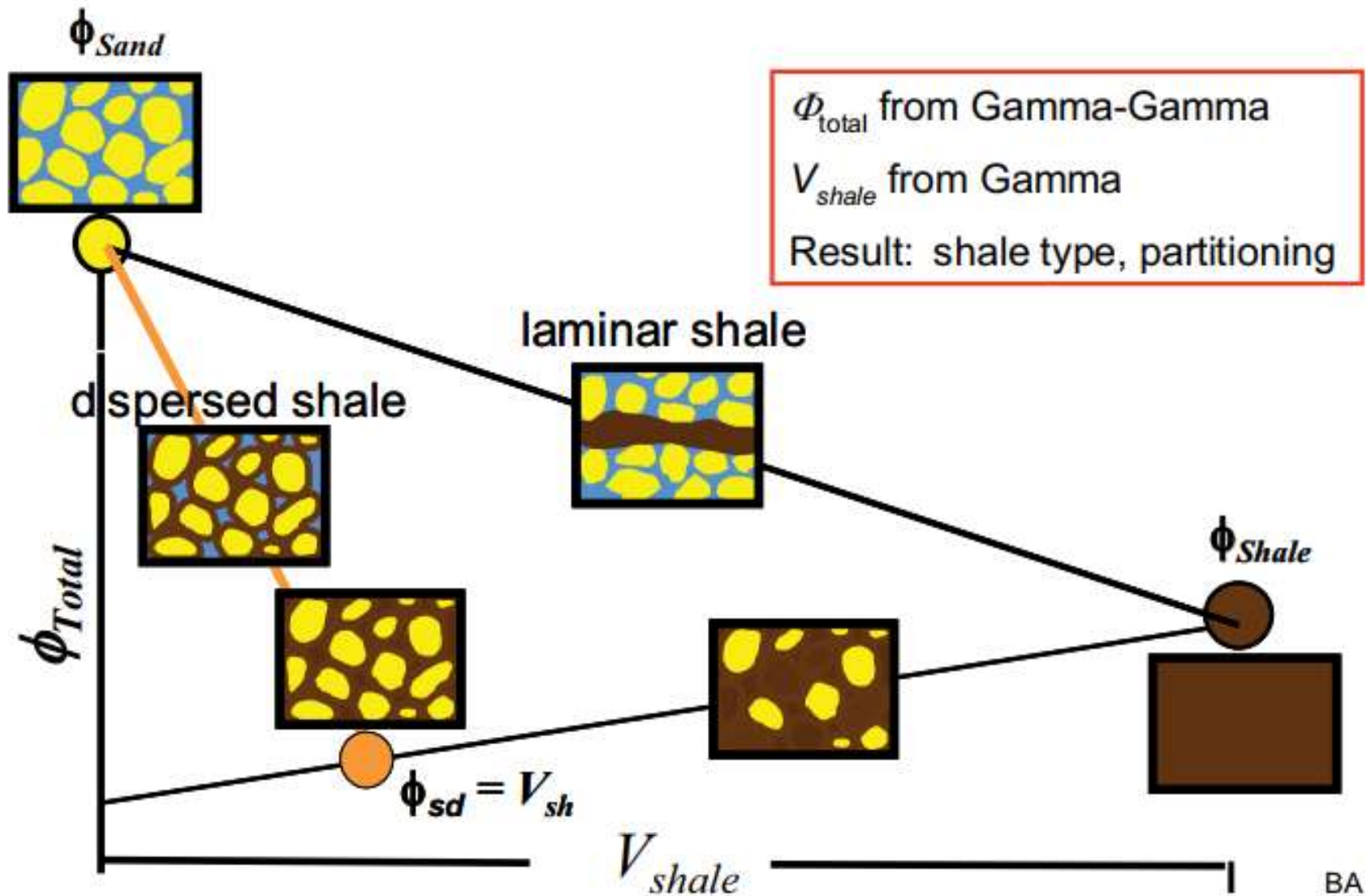


- Water saturation versus shale content  $R_t = 10$  Ohm m Porosity = 0.2,  $m=n=2$ ,  $R_{sh}=5$  Ohm m, (a)  $R_w = 0.10$  Ohm m, (b)  $R_w = 0.02$  Ohm m,

# Thomas-Stieber Crossplot for Shale Type Investigation

- Shale type investigation depends on:
  - Shale content ( $V_{sh}$ )
  - Shale and distribution type ( $V_{sh-lam}$ ,  $V_{sh-disp}$ ),
  - Shale properties ( $R$ -shale,  $B Q_v$ )
- Thomas and Stieber (1975) propose a cross plot technique to determine the type of shale distribution using only log data. They plot a “porosity tool” (Density log) and a “shale volume tool” (Gamma log). The plot demonstrates that:
  - Laminated shaly sand plots along the connection of the two “corner values” for clean sand and pure shale
  - Dispersed shaly sand plots starting at the clean sand point and by filling the pores with clay it reaches the end if  $V_{shale} = \phi$  and all pores are filled.

# Thomas-Stieber Crossplot for Shale Type Investigation



# Early Quantitative Models

- Coates equation (1974, 1999): Permeability in millidarcy (md) is derived as follows:

$$k_{Coates} = \left( \frac{\phi}{C} \right)^m \cdot \left( \frac{BVM}{BVI} \right)^n$$

where

$\phi$  is porosity in percent,

$BVM$  is bulk volume fluid movable,

$BVI$  is bulk volume fluid non-movable (irreducible),

$m, n$  are empirical parameters and approximately equal to 4 and 2,

$C$  is also an empirical parameter generally between 6 and 15.

# Early Quantitative Models

- Coates equation reflects the philosophy of Archie's equation with a simple, physically understandable structure (based on a capillary model) and empirical parameters covering the complicated geometry and structure of the pore space in a fascinating way.
- It implies that laboratory core data is needed to derive empirical parameters for a specific formation i.e. tools need to be calibrated with laboratory core data

# **PGG 413 EXAMPLE A – An oil bearing (clean) sandstone**

Dr. O. Olamigoke

*Source: Basic Well Logging & Formation Evaluation 1<sup>st</sup> Edition  
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# Goals of this exercise

- Lithological profile with description of reservoirs
- Compilation of input parameters for quantitative interpretation
- Determination of porosity
- Determination of water saturation; derivation of hydrocarbon saturation
- Determination of moveable and non-moveable oil saturation

# Basic Information needed in Quantitative Log Interpretation

- **Lithology:** In quantitative analysis, there are several reasons why it is important to know the lithology of the zone. (i) Porosity logs require a lithology or matrix constant before a zone's porosity can be calculated (ii) The formation factor, used in Archie's water saturation equation vary with lithology
- **Temperature of the formation:** Formation Temperature ( $T_f$ ) is also important in log analysis because the resistivities of the drilling mud ( $R_m$ ) and mud filtrate ( $R_f$ ), and the formation water ( $R_w$ ) vary with temperature.

# Data Provided

**Table 1:** Input data provided

| Parameter                                    | Magnitude              |
|----------------------------------------------|------------------------|
| $R_{mf}$ – measured at 90 °F (32 °C)         | 0.336 Ohm m            |
| Formation temperature                        | 178 °F (81 °C)         |
| Matrix density                               | 2.65 g/cm <sup>3</sup> |
| Fluid density                                | 1 g/cm <sup>3</sup>    |
| Transit time (matrix)                        | 52 $\mu$ s/ft          |
| Transit time (fluid)                         | 190 $\mu$ s/ft         |
| Archie's parameters (from core measurements) | $m = n = 2$            |

# Definition of Terms

- The index PE (Photoelectric Effect) strongly depends on the elemental composition and is therefore a typical characteristic of various minerals.
- The maximum difference between the most negative and the most positive level is called Static Spontaneous Potential (SSP)
- Dol – Depth of Interest

# Equations Needed

- Arps Equation: The temperature dependence of NaCl-based water can be described by Arps' equation (Arps, 1953)

- $F = (1/\phi^m)$

$$R_w(T_2) = R_w(T_1) \cdot \frac{T_1 + 21.5}{T_2 + 21.5}$$

- $R_o = (1/\phi^m) * R_w$

- $R_t = R_o * (S_w)^{-n}$

$$S_w = \left( \frac{R_o}{R_t} \right)^{\frac{1}{n}} = \left( \frac{R_w}{R_t} \cdot \frac{1}{\phi^m} \right)^{\frac{1}{n}}$$

$$SSP = k_{SP} \cdot \log \left( \frac{R_{mf}}{R_w} \right)$$

# Log Plot

- Fig. 1 shows the plotted logs. The original limestone-calibrated Neutron porosity was transformed into sandstone-referenced Neutron porosity (NPHISS) because the reservoir is sandstone.

# Log Tracks

| Track        |             | Log description                          | unit              |
|--------------|-------------|------------------------------------------|-------------------|
| First track  | CAL (IN)    | Caliper                                  | Inch              |
|              | DRHO (PU)   | Correction for Density                   | g/cm <sup>3</sup> |
| Second track | PEF (BARN)  | Photoelectric cross section              | b/e               |
|              | GR (GAPI)   | Gamma log                                | API               |
|              | SP (MV)     | Spontaneous Potential                    | mV                |
| Third track  | RT (OHMM)   | Formation Resistivity                    | Ohm-m             |
|              | RT (OHMM)   | Resistivity of invaded zone              | Ohm-m             |
|              | RT (OHMM)   | Resistivity Micro Spherical Focusing Log | Ohm-m             |
| Fourth track | RHOB (GC)   | Bulk Density                             | g/cm <sup>3</sup> |
|              | NPHISS (DE) | Neutron Density (Sandstone converted)    | fraction          |
| Fifth track  | DT (μs/FT)  | Slowness                                 | μs/ft             |

The original limestone-calibrated Neutron porosity was transformed into sandstone-referenced Neutron porosity (NPHSS) because the reservoir is sandstone.



# Quick look Interpretation – QA/QC

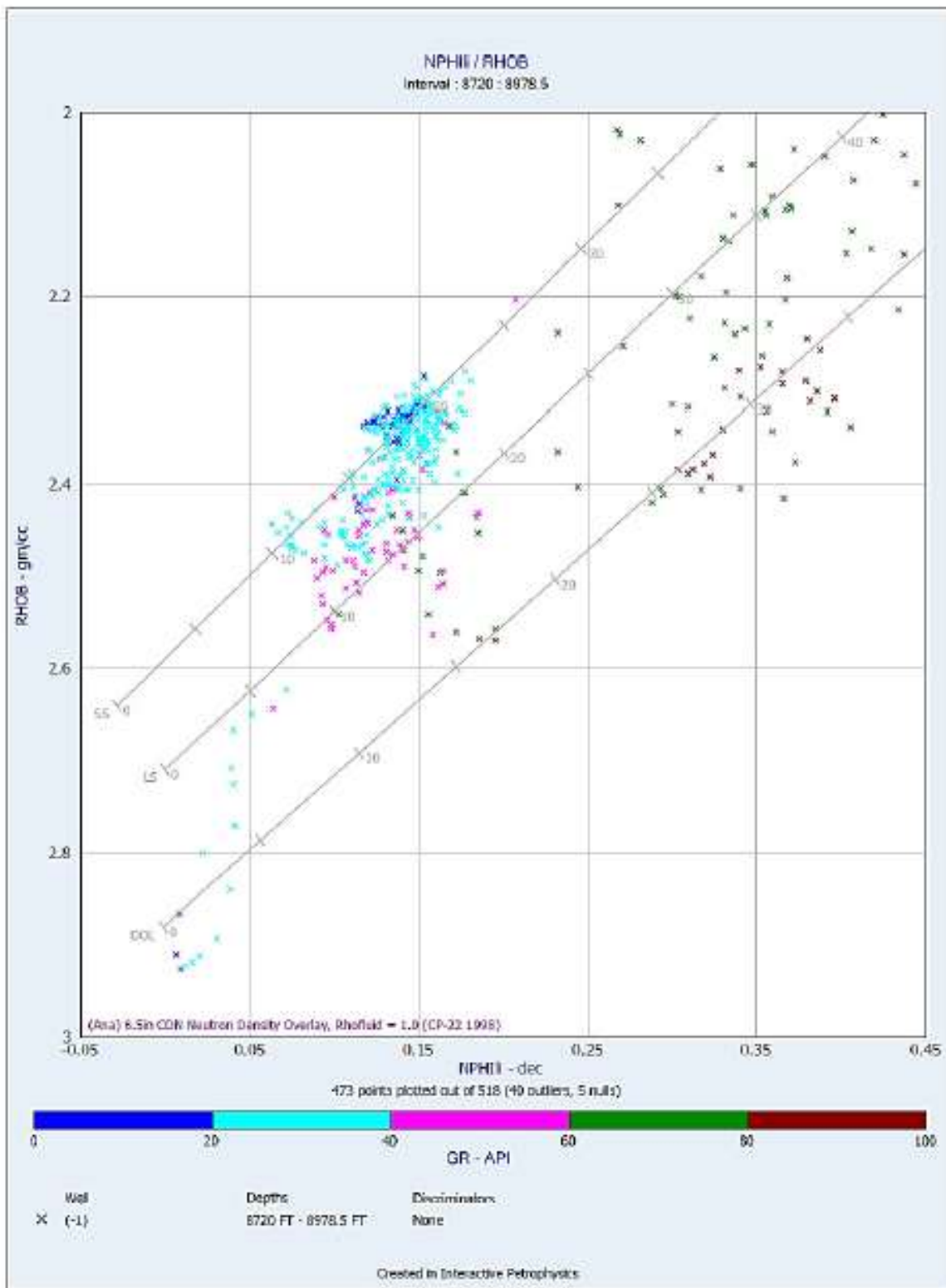
- The Caliper log (first track) shows in the uppermost part – particularly from 8745 to 8785 ft – a caliper enlargement (“bad hole”)
- A look at the Gamma log and SP log indicates this section as shale with some stability problems. The correction term for density-derived porosity DRHO is very high (up to -0.23) and does not allow any further quantitative density interpretation.

# Quick look Interpretation - Lithology

The “lithology logs” GR and SP clearly separate sand and shaly zones of the section. The following profile can be derived:

- Shale zone from top to 8784 ft; bad hole.
- The zone from 8784 to 8795 ft is characterized by correct Caliper, low GR, PEF of about 5 b/e, extremely high resistivity, very low NPHI and extremely high RHOB (data in LAS-file show  $> 2.9 \text{ g/cm}^3$ ), and low DT. All these indications – particularly the density – are typical of anhydrite. It forms a cap rock covering the reservoir.
- Sand zone (reservoir) from 8795 to 8910 ft with shaly/silty interlayers at 8822 to 8826 ft, 8876 to 8879 ft – PEF confirms a (quartz) and with 1.8 p/e. The log plot shows Neutron porosity in the porosity column. NPHISS is plotted from 0 to 0.5 and density RHOB from  $2.65 \text{ g/cm}^3$  to  $1.83 \text{ g/cm}^3$ . Thus, the fit of Neutron and Density Porosity is indicative of sandstone.
- The following zone is a shaly sand (increase of GR)

# pretation



- The Neutron-Density cross-plot (Fig. 2) confirms lithological units.
- Most data points are close to the sandstone line (porosity 15 .... 20%); data points in the upper right part are influenced by the shale section and caliper.
- Note that there are also some data points in the lower left corner indicating anhydrite.

# Quick look Interpretation - Resistivity

- For the following quick-look interpretation of resistivity, it is important to note that the reservoir is characterized by a nearly constant porosity of about 20% - it is a homogeneous reservoir.
- The resistivity  $R_t$  below 8855 ft is very low (0.02 – 0.04 Ohm-m); this indicates a very high salinity and a strong contrast to the mud filtrate resistivity (therefore SSP is very high – about 200 mV)
- From a depth of 8800 ft upwards, the resistivity  $R_t$  increases by a factor of about 10.  $R_t$  is still below  $R_{xo}$  ( $R_{xo}$  is controlled by low saline mud filtrate). This increase of  $R_t$  must be originated by a decrease of water saturation (because porosity is constant). This part is probably the transition zone. From 8834 ft to 8795 ft, the reservoir is above transition zone.
- The nearly constant level of  $R_{xo}$  (and MSFL) confirms reservoir homogeneity and clearly indicates invasion as a result of permeability.

# Compilation of input parameters for quantitative interpretation

## Shale content from GR log:

- Minimum and maximum magnitude (average for the reservoir): GRmin = 16 API; GRmax = 110 API. Note: Particular GRmax cannot be taken from the shale zone of this plot because of bad hole conditions!

## Fluid parameter for resistivity analysis:

- Calculation of Rmf at formation temperature 178 °F:
  - Rmf at 90 °F is 0.366 Ohm-m, conversion with Arps equation results in
  - $$R_{mf,178} = 0.336 \cdot \frac{90 + 6.77}{178 + 6.77} = 0.176 \text{ Ohmm.}$$
- Determination of Rw at formation temperature 178 °F:
  - This fundamental property for calculation of water saturation will be derived from resistivity in the water zone and from SP log analysis

# Compilation of input parameters for quantitative interpretation

## Calculation of $R_w$ from water zone:

- Within the water zone, the resistivity equals  $R_o$ . For a selected part (8890 ... 8900 ft) it is on the average 0.3 Ohm-m, for this part porosity is 0.2.
- Thus, since  $R_o = (1/\phi^m) * R_w$ ,  $R_w = R_o / (1/\phi^m)$
- $R_w = (0.3) / (1/(0.2)^2) = 0.012$  Ohm-m

## Calculation of $R_w$ from SP:

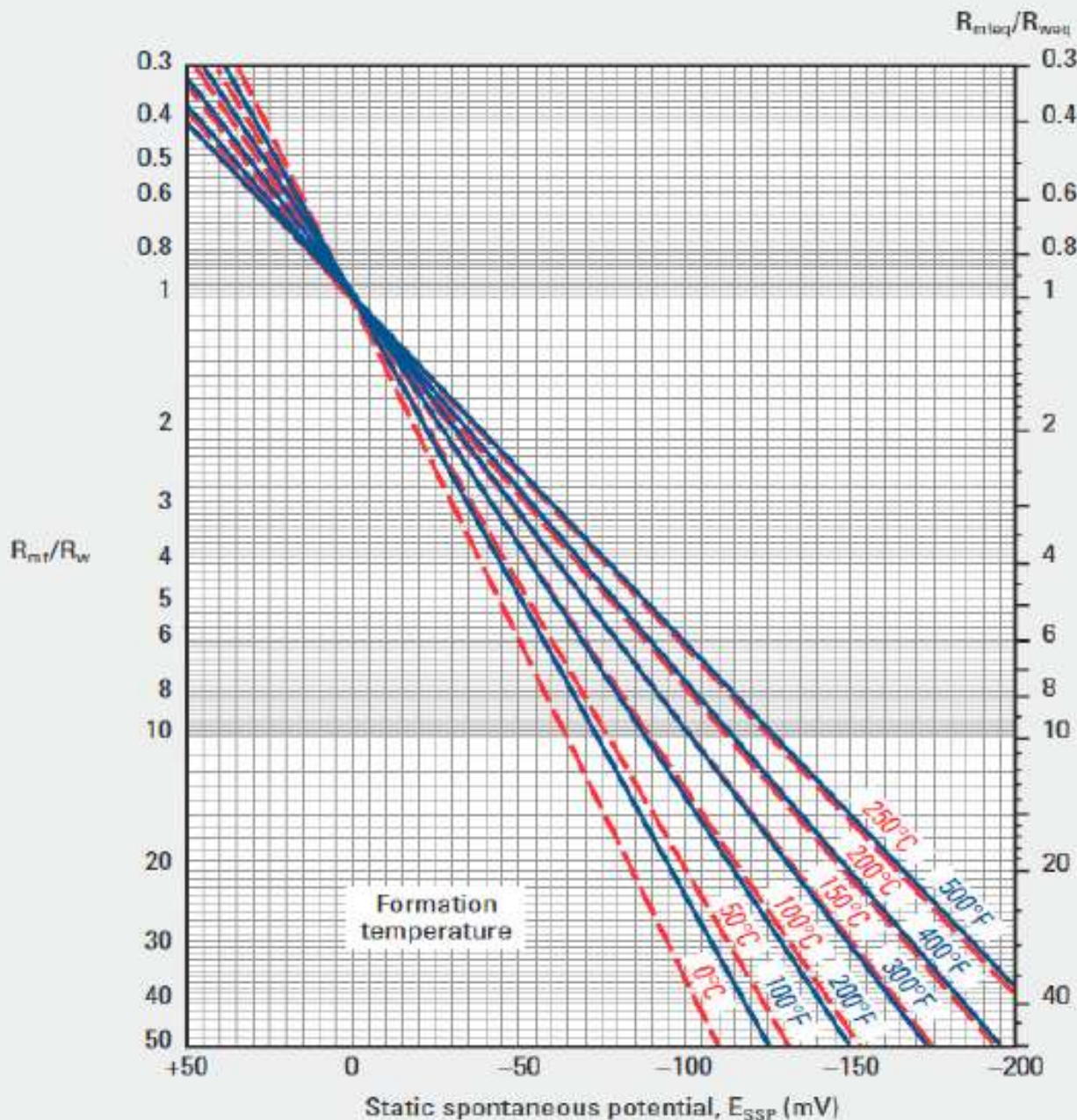
- The Static Spontaneous Potential (SSP) is given by:

$$SSP = k_{SP} \cdot \log\left(\frac{R_{mf}}{R_w}\right)$$

- Where the factor  $k_{SP}$  is mainly controlled by temperature, this dependence is presented in the monogram in Figure 3.



# Compilation of input parameters for quantitative interpretation



- Figure 3: Determination of  $R_w$  from SSP, Schlumberger Chart Book, 2000.



# Compilation of input parameters for quantitative interpretation

## Calculation of $R_w$ from SP:

- The log shows a maximum SSP difference is -100 mV. At the formation temperature of 178 °F, the circle in Figure 3 results in a ratio:

$$\frac{R_{mf}}{R_w} = 15 \quad R_w = \frac{0.176}{15} = 0.012 \text{ Ohmm}$$

- This is in confidence with the value from the water zone and confirms the very high salinity of the formation water.

# Quantitative interpretation for 4 reservoir zones

- Table 3 shows the measured data (from the LAS-file) for 4 depths. The calculation is demonstrate step by step for the depth of 8804 ft. For the other 3 depths results are given in Table 4.

| Depth | GR   | DT   | RHOB  | NPHI_SS | RT    | RX0   |
|-------|------|------|-------|---------|-------|-------|
| 8804  | 18.5 | 79.6 | 2.321 | 0.188   | 3.818 | 6.837 |
| 8820  | 26.7 | 84.9 | 2.291 | 0.226   | 1.242 | 6.025 |
| 8839  | 20.7 | 74.7 | 2.328 | 0.163   | 0.635 | 6.735 |
| 8859  | 16.7 | 76.9 | 2.325 | 0.189   | 0.272 | 5.327 |
|       |      |      |       |         |       |       |

Table 3: Log data from selected depths (from LAS file). The Resistivity values in the LAS file and on the log are inconsistent.

# Quantitative interpretation for depth 8804ft: Shale Content

Property: Shale Content

$$V_{sh} = \frac{GR_{DOI} - GR_{MIN}}{GR_{MAX} - GR_{MIN}} = \frac{18.5 - 16}{92 - 16} = 0.041 \text{ (Shale Content is 4.1\%)}$$

- The low shale content (note that the linear equation delivers the maximum shale content) is not implemented in the following calculations. Particularly for the resistivity interpretation the very low shale content and high water conductivity allow application of Archie's equation.

# Quantitative interpretation for depth 8804ft: Porosity

## Property: Porosity

- Porosity from Density log. (Shale Content is 4.1%)

$$DPHI = \frac{\rho_{ma} - \rho_{DOI}}{\rho_{ma} - \rho_{fl}} = \frac{2.65 - 2.31}{2.65 - 1.00} = 0.206$$

- Porosity from Neutron log: NPHI = 0.2
- Average Porosity: (DPHI + NPHI)/2 = 0.203
- For comparison, Acoustic log porosity confirms the value

$$DTPHI = \frac{\Delta t_{log} - \Delta t_{ma}}{\Delta t_{fl} - \Delta t_{ma}} = \frac{79.6 - 52}{190 - 52} = 0.2$$

# Quantitative interpretation for depth

## 8804ft: Saturation

Property: Saturation,  $S_w$

- Formation Factor:

$$F = \frac{1}{\phi^m} = \frac{1}{(0.2)^2} = 25$$

- Formation resistivity if completely filled with water,  $R_o$ :

$$R_o = F * R_w = 25 \times 0.012 = 0.3 \text{ Ohmm}$$

- The measured resistivity is distinctly higher because non-conducting oil is present.

- $S_w = \sqrt{\frac{R_o}{R_t}} = \sqrt{\frac{0.3}{3.818}} = 0.28$  water saturation = 28%
- oil saturation = 72%

# Quantitative interpretation for depth

## 8804ft: Non-moveable oil in the invaded zone

Property: Non-moveable oil in the invaded zone

- Saturation with respect to mud filtrate is:

$$S_{xo} = \sqrt{\frac{F * R_{mf}}{R_{xo}}} = \sqrt{\frac{25 * 0.176}{6.837}} = 0.802$$

- The invaded zone is 80% saturated with mud filtrate (it has displaced the original fluid). But there is still 20% non-displaced (non-movable) oil in the pore space
- The reservoir at 8804ft has 20% porosity and water saturation of 28%.
- The oil saturation in the non-invaded zone is 72%,
- Non-moveable oil =  $1 - S_{xo} = 20\%$  of the oil are non-moveable,
- $S_{xo} - S_w = 80\% - 28\% = 52\%$  are moveable

# **PGG 413 EXAMPLE B – Shaly Sand Profile**

Dr. O. Olamigoke

*Source: Basic Well Logging & Formation Evaluation 1<sup>st</sup> Edition  
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# Goals of this exercise

- Evaluation of the shale type and determination of the “wet shale parameters”
- Application of a shaly sand equation and comparison with formal application of Archie’s equation (neglecting shaliness).
- Demonstration of the effect of shaliness on a precise determination of water saturation

# Data Provided & Log Databases

| Parameter                                    | Magnitude              |
|----------------------------------------------|------------------------|
| $R_{mf}$ – at Formation Temperature          | 0.110 Ohm m            |
| $R_w$ – at Formation Temperature             | 0.065 Ohm m            |
| Matrix density                               | 2.65 g/cm <sup>3</sup> |
| Fluid density                                | 1 g/cm <sup>3</sup>    |
| Archie's parameters (from core measurements) | m = 1.95<br>n = 2      |

| Track                 | Log               | Log description                                   | unit              |
|-----------------------|-------------------|---------------------------------------------------|-------------------|
| 2 <sup>nd</sup> track | CAL (IN)          | Caliper                                           | Inch              |
| 3 <sup>rd</sup> track | GR (GAPI)         | Natural Gamma log                                 | API               |
| 4 <sup>th</sup> track | Rxo, Rt<br>(Ohmm) | Processed Resistivity from LLD, LLS, ILD,<br>MSFL | Ohm-m             |
| 5 <sup>th</sup> track | RHOB (GC)         | Bulk Density                                      | g/cm <sup>3</sup> |
|                       | NPHISS (DE)       | Neutron Density (Sandstone converted)             | fraction          |

# Log Plot

- Fig. 2 shows the log plots for Example B. The right scale for Neutron porosity (sandstone-referenced) NPHISS) corresponds to RHOB= 2.65 g/cm<sup>3</sup> which indicates the same porosity scale assuming a sandstone.

Data set from LAS-file

| Depth m | GR API | Rt Ohm m | RHOB g/cm <sup>3</sup> | NPHI-SS |
|---------|--------|----------|------------------------|---------|
| 1914.8  | 35.6   | 7.68     | 2.123                  | 0.382   |

# Step 1 – Selection of zones of interest (Quick Look Interpretation)

A visual interpretation of the log shows:

- 1860 – 1907 m: Shale with  $GR \approx 60$  API, caliper enlarged. The fit of the two resistivities (no invasion) and the separation/crossover of NEU & DEN logs confirm shale.
- 1907 – 1966 m: More sandy and shaly layers alternate. Shaly layers are indicated by high GR and separation/crossover of NEU & DEN logs. The sandy layers with low GR, fit of NEU & DEN logs and high  $R_t$  are probably oil-bearing layers. In the figure only four layers are selected. N.B.: High  $R_t$  indicates HC (or less water) and the fit of NEU & DEN logs indicates oil (gas would show a crossover  $NEU < DEN$ ).
- Below 1966 m: Sand (low shale content), water-bearing. The low Resistivity ILD indicates water saturation. Note the fit of NEU (sandstone referenced) & DEN log.

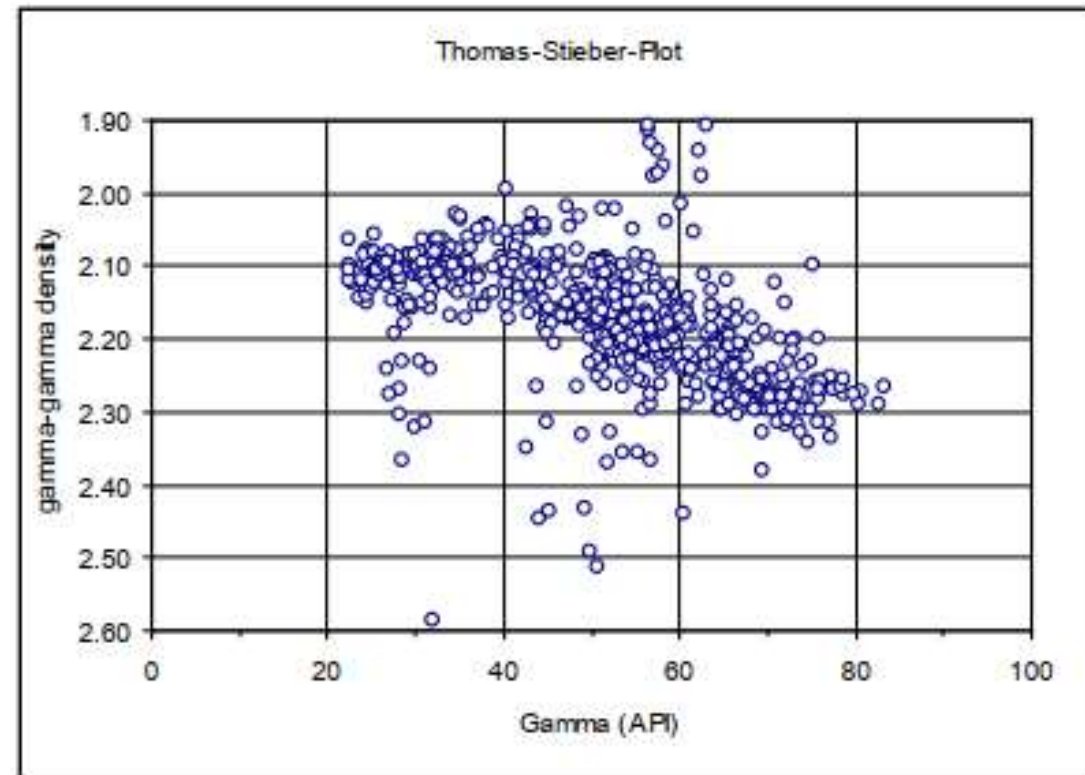
# Step 2 – Determination of Shale Properties

Because reservoir zones are shaly, a determination of shale properties as input for shaly sand interpretation is necessary. Relevant shale parameters are:

- a) Shale content,
- b) Type of Shale (laminar or dispersed)
- c) Shale properties: Shale resistivity, wet shale neutron porosity, wet shale density

Shale content is calculated from GR in two steps:

- 1) Calculation of “Gamma-Ray Index” with  $GR$ ,  $min = 20$  API and  $max = 83$  API,
- 2) Application of the Larionov equation for older rocks

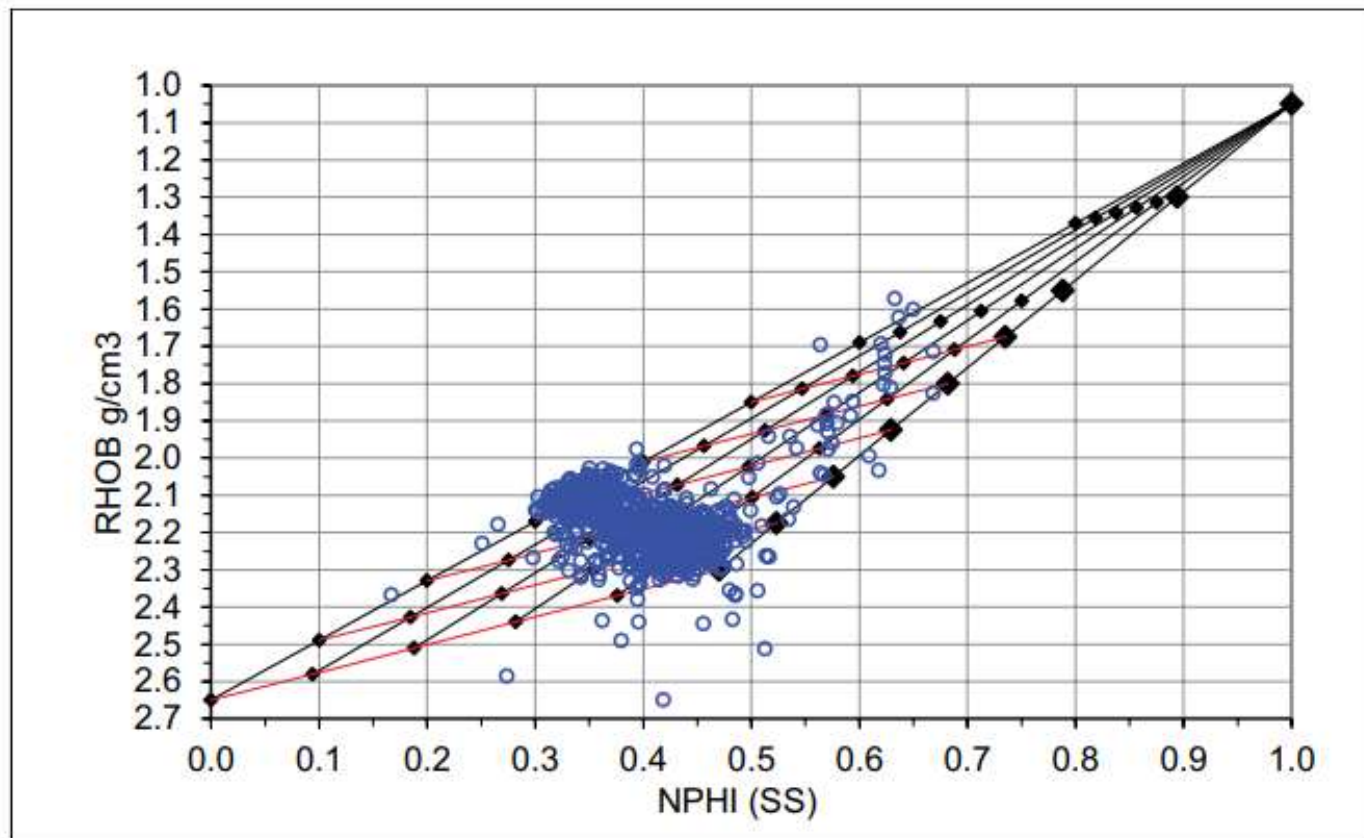


A Thomas-Stieber plot indicates laminated shaly sand.

$$I_{GR} = \frac{36.5 - 20}{83 - 20} = 0.247$$

$$V_{sh} = 0.33 \left[ \left( 2^{2 \cdot I_{GR}} \right) - 1.0 \right] = 0.135 \quad 360$$

# Step 2 – Determination of Shale Properties



Properties of the wet shale are taken from a NEU-DEN crossplot. Data points at the lower right corner of the cloud represent pure shale with the properties of the wet shale point :  $NPHI - \text{Shale} = 0.47$  and  $RHO - \text{Shale} = 2.30 \text{ g/cm}^3$ .

$R_{sh} = 1.8 \text{ Ohm m}$  is used for shale resistivity (taken from a thick shale bed outside this section because the measured shale resistivity in the upper part is reduced by the caliper effect).

# Step 3 – Effective Porosity

The Neutron-Density combination solved for effective porosity gives:

$$\phi_{eff} = \frac{(\phi_N - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_b)}{(\phi_{N,fl} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{sh}) - (\phi_{N,sh} - \phi_{N,ma}) \cdot (\rho_{ma} - \rho_{fl})}$$

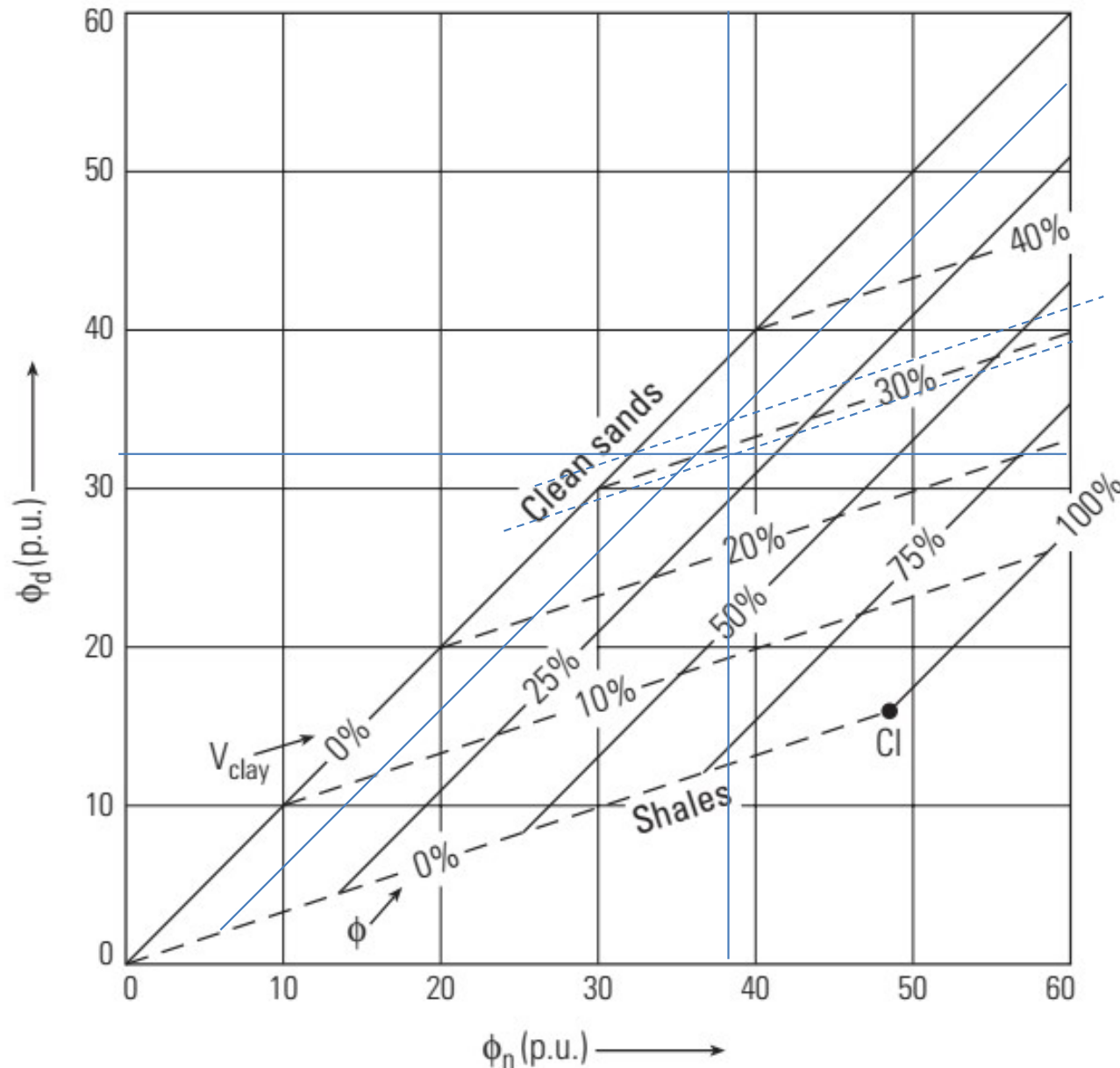
$$\phi_{eff} = \frac{(0.382 - 0) \cdot (2.65 - 2.3) - (0.47 - 0) \cdot (2.65 - 2.123)}{(1 - 0) \cdot (2.65 - 2.3) - (0.47 - 0) \cdot (2.65 - 1)} = 0.267897$$

N.B.

- Alternatively, use the equation for Porosity corrected for shaliness,  $\phi_{shcor} = \phi_{log} - V_{sh} \cdot \phi_{sh} = 0.382 - 0.135 \cdot 0.47 = 0.31855$ , which is equivalent to the porosity obtainable from the density log  $DPHI = (2.65 - 2.123) / (2.65 - 1.00) = 0.31939$ .
- The value obtained in Schon(2015) for effective porosity (0.287) is obtainable from the neutron-density cross plot in the presence of shale. (See next slide)



# Step 3 – Effective Porosity



- The value obtained in Schon(2015) for effective porosity (0.287) is obtainable from the neutron-density cross plot in the presence of shale

# Compilation of input parameters for quantitative interpretation

## Shale content from GR log:

- Minimum and maximum magnitude (average for the reservoir): GRmin = 16 API; GRmax = 110 API. Note: Particular GRmax cannot be taken from the shale zone of this plot because of bad hole conditions!

## Fluid parameter for resistivity analysis:

- Calculation of Rmf at formation temperature 178 °F:
  - Rmf at 90 °F is 0.366 Ohm-m, conversion with Arps equation results in
  - $$R_{mf,178} = 0.336 \cdot \frac{90 + 6.77}{178 + 6.77} = 0.176 \text{ Ohmm.}$$
- Determination of Rw at formation temperature 178 °F:
  - This fundamental property for calculation of water saturation will be derived from resistivity in the water zone and from SP log analysis

# Compilation of input parameters for quantitative interpretation

## Calculation of $R_w$ from water zone:

- Within the water zone, the resistivity equals  $R_o$ . For a selected part (8890 ... 8900 ft) it is on the average 0.3 Ohm-m, for this part porosity is 0.2.
- Thus, since  $R_o = (1/\phi^m) * R_w$ ,  $R_w = R_o / (1/\phi^m)$
- $R_w = (0.3) / (1/(0.2)^2) = 0.012$  Ohm-m

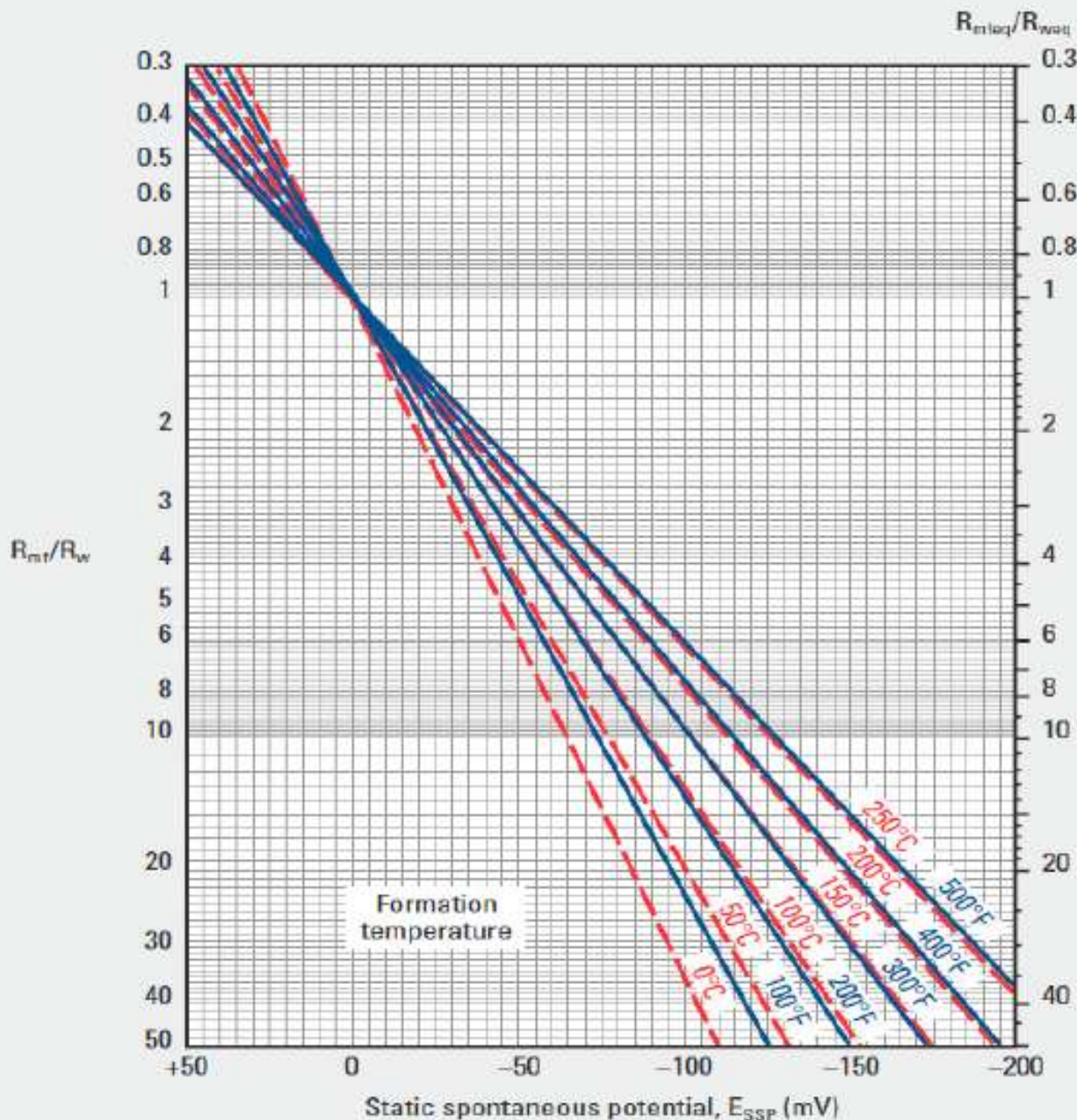
## Calculation of $R_w$ from SP:

- The Static Spontaneous Potential (SSP) is given by:

$$SSP = k_{SP} \cdot \log\left(\frac{R_{mf}}{R_w}\right)$$

- Where the factor  $k_{SP}$  is mainly controlled by temperature, this dependence is presented in the monogram in Figure 3.

# Compilation of input parameters for quantitative interpretation



- Figure 3: Determination of  $R_w$  from SSP, Schlumberger Chart Book, 2000.

# Compilation of input parameters for quantitative interpretation

## Calculation of $R_w$ from SP:

- The log shows a maximum SSP difference is -100 mV. At the formation temperature of 178 °F, the circle in Figure 3 results in a ratio:

$$\frac{R_{mf}}{R_w} = 15 \quad R_w = \frac{0.176}{15} = 0.012 \text{ Ohmm}$$

- This is in confidence with the value from the water zone and confirms the very high salinity of the formation water.

# Quantitative interpretation for 4 reservoir zones

- Table 3 shows the measured data (from the LAS-file) for 4 depths. The calculation is demonstrate step by step for the depth of 8804 ft. For the other 3 depths results are given in Table 4.

| Depth | GR   | DT   | RHOB  | NPHI_SS | RT    | RX0   |
|-------|------|------|-------|---------|-------|-------|
| 8804  | 18.5 | 79.6 | 2.321 | 0.188   | 3.818 | 6.837 |
| 8820  | 26.7 | 84.9 | 2.291 | 0.226   | 1.242 | 6.025 |
| 8839  | 20.7 | 74.7 | 2.328 | 0.163   | 0.635 | 6.735 |
| 8859  | 16.7 | 76.9 | 2.325 | 0.189   | 0.272 | 5.327 |
|       |      |      |       |         |       |       |

Table 3: Log data from selected depths (from LAS file). The Resistivity values in the LAS file and on the log are inconsistent.

# Quantitative interpretation for depth 8804ft: Shale Content

Property: Shale Content

$$V_{sh} = \frac{GR_{DOI} - GR_{MIN}}{GR_{MAX} - GR_{MIN}} = \frac{18.5 - 16}{92 - 16} = 0.041 \text{ (Shale Content is 4.1\%)}$$

- The low shale content (note that the linear equation delivers the maximum shale content) is not implemented in the following calculations. Particularly for the resistivity interpretation the very low shale content and high water conductivity allow application of Archie's equation.



# Quantitative interpretation for depth 8804ft: Porosity

## Property: Porosity

- Porosity from Density log. (Shale Content is 4.1%)

$$DPHI = \frac{\rho_{ma} - \rho_{DOI}}{\rho_{ma} - \rho_{fl}} = \frac{2.65 - 2.31}{2.65 - 1.00} = 0.206$$

- Porosity from Neutron log: NPHI = 0.2
- Average Porosity:  $(DPHI + NPHI)/2 = 0.203$
- For comparison, Acoustic log porosity confirms the value

$$DTPHI = \frac{\Delta t_{log} - \Delta t_{ma}}{\Delta t_{fl} - \Delta t_{ma}} = \frac{79.6 - 52}{190 - 52} = 0.2$$

# Quantitative interpretation for depth

## 8804ft: Saturation

Property: Saturation,  $S_w$

- Formation Factor:

$$F = \frac{1}{\phi^m} = \frac{1}{(0.2)^2} = 25$$

- Formation resistivity if completely filled with water,  $R_o$ :

$$R_o = F * R_w = 25 \times 0.012 = 0.3 \text{ Ohmm}$$

- The measured resistivity is distinctly higher because non-conducting oil is present.

- $S_w = \sqrt{\frac{R_o}{R_t}} = \sqrt{\frac{0.3}{3.818}} = 0.28$  water saturation = 28%
- oil saturation = 72%

# Quantitative interpretation for depth

## 8804ft: Non-moveable oil in the invaded zone

Property: Non-moveable oil in the invaded zone

- Saturation with respect to mud filtrate is:

$$S_{xo} = \sqrt{\frac{F * R_{mf}}{R_{xo}}} = \sqrt{\frac{25 * 0.176}{6.837}} = 0.802$$

- The invaded zone is 80% saturated with mud filtrate (it has displaced the original fluid). But there is still 20% non-displaced (non-movable) oil in the pore space
- The reservoir at 8804ft has 20% porosity and water saturation of 28%.
- The oil saturation in the non-invaded zone is 72%,
- Non-moveable oil =  $1 - S_{xo} = 20\%$  of the oil are non-moveable,
- $S_{xo} - S_w = 80\% - 28\% = 52\%$  are moveable